

CHEMICAL ENGINEERING DESIGN

PRINCIPLES, PRACTICE AND ECONOMICS
OF PLANT AND PROCESS DESIGN

THIRD EDITION

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RAY SINNOTT



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Preface to the Third Edition

This book was originally written by Ray Sinnott as Volume 6 of the “Chemical Engineering” series edited by Coulson and Richardson. It was intended to be a standalone design textbook for undergraduate design projects that would supplement the other volumes in the Coulson and Richardson series. In 2008 we published the first edition of *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design* as an adaptation of Coulson and Richardson Volume 6 for the North American market. Some older sections of the book were updated, and references to laws, codes, and standards were changed to an American rather than British basis; however, the general layout and philosophy of the book remained unaltered.

The first edition of this book was widely adopted, and I received a great deal of valuable feedback from colleagues on both the strengths and weaknesses of the text in the context of a typical North American undergraduate curriculum. The experiences and frustrations of my students at Northwestern University and comments from coworkers at Honeywell UOP also helped suggest areas where the book could be improved. The second edition rearranged the material substantially to make the book more valuable to students and industrial practitioners by incorporating new material to address obvious gaps, while eliminating some material that was dated or repetitive of foundation classes. The book was divided into two parts. Part I: Process Design covers the topics that are typically taught in a lecture class on process design and economics. The broad themes of Part I are flowsheet development, economic analysis, safety and environmental impact, and optimization. Part II: Plant Design contains chapters on equipment design and selection that can be used as supplements to a lecture course. These chapters contain step-by-step methods for designing most unit operations, together with many worked examples, and should become essential references for students when they begin working through their design projects or face design problems early in their industrial career. The third edition retains the same format and has been updated throughout to reflect legislative changes, the most recent codes and standards, and updated information on process and equipment costs.

Chemical engineers work in a very diverse set of industries, and many of these industries have their own design conventions and specialized equipment. I have attempted to include examples and problems from a broad range of process industries, but where space or my lack of expertise in the subject has limited coverage of a particular topic, references to specialized texts are provided.

This book draws on Ray Sinnott’s and my experience of the industrial practice of process design, as well as our experience teaching design at the University of Wales Swansea, University of Manchester, Northwestern University, and National University of Singapore. Since the book is intended to be used in practice and not just as a textbook, our aim has been to describe the tools and methods that are most widely used in industrial process design. We have deliberately avoided describing idealized conceptual methods that have not yet gained wide currency in industry. The reader can find good descriptions of these methods in the research literature and in more academic textbooks.

Standards and codes of practice are an essential part of engineering, and the relevant North American standards are cited. The codes and practices covered by these standards will be applicable to other countries. They will be covered by equivalent national standards in most developed countries, and in some cases the relevant British, European, or international standards have also been cited. Brief summaries of important U.S. and Canadian safety and environmental legislation have been given in the relevant chapters. The design engineer should always refer to the original source references of laws, standards, and codes of practice, as they are updated frequently.

Most industrial process design is carried out using commercial design software. Extensive reference has been made to commercial process and equipment design software throughout the book. Many of the commercial software vendors provide licenses of their software for educational purposes at nominal fees. I strongly believe that students should be introduced to commercial software at as early a stage in their education as possible. The use of academic design and costing software should be discouraged. Academic programs usually lack the quality control and support required by industry, and the student is unlikely to use such software after graduation. All computer-aided design

tools must be used with some discretion and engineering judgment on the part of the designer. This judgment mainly comes with experience, but I have tried to provide helpful tips on how to best use computer tools.

Ray wrote in the preface to the first edition of his book: *"The art and practice of design cannot be learned from books. The intuition and judgment necessary to apply theory to practice will come only from practical experience."* As with all our previous editions, I hope that this latest edition will make it easier for readers to begin acquiring that experience.

Gavin Towler

How To Use This Book

This book has been written primarily for students on undergraduate courses in chemical engineering and has particular relevance to their senior design projects. It should also be of interest to new graduates working in industry who find they need to broaden their knowledge of unit operations and design. Some of the earlier chapters of the book can also be used in introductory chemical engineering classes and by other disciplines in the chemical and process industries.

PART I: PROCESS DESIGN

Part I has been conceived as an introductory course in process design. The material can be covered in 20 to 30 lecture hours, and presentation slides are available to qualified instructors in the supplementary material available at www.elsevier.com/books-and-journals/book-companion/9780128211793. Chapter 1 is a general overview of process design and contains an introductory section on product design. Chapter 2, Chapter 3, Chapter 4, Chapter 5, and Chapter 6 address the development of a process flowsheet from initial concept to the point where the designer is ready to begin estimating capital costs. Chapter 2 covers the selection of major unit operations and also addresses design for revamps and modification of conventional flowsheets. Chapter 3 introduces utility systems and discusses process energy recovery and heat integration. Chapter 4 provides an introduction to process simulation and shows the reader how to complete process material and energy balances. Chapter 5 covers those elements of process control that must be understood to complete a process flow diagram and identify where pumps and compressors are needed in the flowsheet. The selection of materials of construction can have a significant effect on plant costs, and this topic is addressed in Chapter 6. The elements of process economic analysis are introduced in Chapter 7, Chapter 8, and Chapter 9. Capital cost estimation is covered in Chapter 7. Operating costs, revenues, and price forecasting are treated in Chapter 8. Chapter 9 concludes the economics section of the book with a brief introduction to corporate finance, a description of economic analysis methods, and a discussion on project selection criteria used in industry. Chapter 10 examines the role of safety considerations in design and introduces the methods used for process hazard analysis. Chapter 11 addresses site design and environmental impact. Part I concludes with a discussion of optimization methods in Chapter 12.

PART II: PLANT DESIGN

Part II contains a more detailed treatment of design methods for common unit operations. Chapter 13 provides an overview of equipment design and is also a guide to the following chapters. Chapter 14 discusses the design of pressure vessels and provides the necessary background for the reader to be able to design reactors, separators, distillation columns, and other operations that must be designed under pressure vessel codes. Chapter 15 covers the design of mixers and reactors, with an emphasis on the practical mechanical layout of reactors. Chapter 16 and Chapter 17 address fluid phase separations. Multistage column separations (distillation, absorption, stripping, and extraction) are described in Chapter 17, while other separation processes, such as adsorption, membrane separation, decanting, crystallization, precipitation, ion exchange, and chromatography, are covered in Chapter 16. Chapter 18 examines the properties of granular materials and introduces the processes used for storing, conveying, mixing, separating, heating, drying, and altering the particle size distribution of solids. Chapter 19 covers all aspects of the design of heat-transfer equipment, including plate exchangers, air coolers, fired heaters, and direct heat transfer to vessels, as well as design of shell and tube heat exchangers, boilers, and condensers. Chapter 20 addresses the design of plant hydraulics and covers design and selection of pumps, compressors, piping systems, and control valves. The material in Part II can be used to provide supplementary lectures in a design class or as a supplement to foundation

courses in chemical engineering. The chapters have also been written to serve as a guide to selection and design, with extensive worked examples, so that students can dip into individual chapters as they face specific design problems when working on a senior year design project.

SUPPLEMENTARY MATERIAL

Many of the calculations described in the book can be performed using spreadsheets. Templates of spreadsheet calculations and equipment specification sheets are available in Microsoft Excel format online and can be downloaded from www.elsevier.com/books-and-journals/book-companion/9780128211793. An extensive set of design problems are included in the Appendices, which are also available at the same website.

Additional supplementary material, including Microsoft PowerPoint presentations to support most of the chapters and a full solutions manual, are available only to instructors, by registering at the Instructor section on <https://educate.elsevier.com/9780128211793>.

Acknowledgments

As stated in the preface, after launching the first edition of this book I received a great deal of very valuable feedback from students and colleagues. I have tried to make good use of this feedback in the second and third editions. Particular thanks are due to John Baldwin, Michael Caracotsios, Elizabeth Carter, Dan Crawl, Mario Eden, Mahmoud El-Halwagi, Igor Kourkine, Harold Kung, Justin Notestein, A.V. Purushothaman, Matthew Realff, Tony Rogers, Warren Seider, Farzin Shemirani, Bill Wilcox, and Dave Wegerer, all of whose suggestions I have gratefully incorporated. Many further improvements were suggested during the review phase, and I would like to thank Mark James, Barry Johnston, Ken Joung, Yoshiaki Kawajiri, Peg Stine, Ross Taylor, and Andy Zarchy for their thoughtful reviews and input. Rajeev Gautam and Ben Christolini allowed me to pursue this project and make use of Honeywell UOPs extensive technical resources. As always, many colleagues at Honeywell UOP, AIChE, and CACHE and students and colleagues at Northwestern and National University of Singapore have shared their experience and given me new insights into chemical engineering design and education.

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I am grateful to Aspen Technology Inc., Cost Engineering Consultancy and Honeywell Inc. for permission to include the screenshots that were generated using their software to illustrate the process simulation and costing examples. The material safety data sheet in Appendix I is reproduced with permission of Fischer Scientific Inc. Aspen Plus®, Aspen Process Economic Analyzer, Aspen Capital Cost Estimator, Aspen Kbase, Aspen ICARUS, and all other AspenTech product names or logos are trademarks or registered trademarks of Aspen Technology Inc. or its subsidiaries in the United States and/or in other countries. All rights reserved.

The supplementary material contains images of processes and equipment from many sources. I would like to thank the following companies for permission to use these images: Alfa-Laval, ANSYS, Aspen Technology, Bete Nozzle, Bos-Hatten Inc., Chemineer, Dresser, Dresser-Rand, Enardo Inc., Honeywell, Komax Inc., Riggins Company, Tyco Flow Control Inc., United Valve Inc., Honeywell UOP LLC, and The Valve Manufacturers Association.

Stephen Merken led the Elsevier team in developing this book and provided much useful editorial guidance. I would also like to thank Beth LoGiudice and Kamatchi Madhavan for their excellent work in managing all the stages of production and printing.

The biggest debt that I must acknowledge is to my coauthor, Ray Sinnott. Although Ray was not involved in writing this edition, it is built on the foundation of his earlier work, and his words can be found in every chapter. I hope I have remained true to Rays philosophy of design and have preserved the strengths of his book. It was necessary for me to remove some older material to make space for new sections in the book, and I hope that Ray will forgive these changes. Needless to say, I am entirely responsible for any deficiencies or errors that have been introduced.

My regular job at Honeywell UOP keeps me very busy, and I worked on this book in the evenings and on the weekends, so it would not have been possible without the love and support of my wife, Xiaona, and our children Miranda, Jimmy, Andrew and Michael.

*Gavin P. Towler
Inverness, Illinois*

1

Introduction to design

KEY LEARNING OBJECTIVES

- How design projects are carried out and documented in industry, including the formats used for design reports
- Why engineers in industry use codes and standards in design
- Why it is necessary to build margins into a design
- Methods used by product design engineers to translate customer needs into product specifications

1.1 Introduction

This chapter is an introduction to the nature and methodology of the design process and its application to the design of chemical products and manufacturing processes.

1.2 Nature of design

This section is a general discussion of the design process. The subject of this book is chemical engineering design, but the methodology described in this section applies equally to other branches of engineering.

Chemical engineering has consistently been one of the highest-paid engineering professions. There is a demand for chemical engineers in many sectors of industry, including the traditional process industries: chemicals, polymers, fuels, foods, pharmaceuticals, and paper, as well as other sectors such as electronic materials and devices, consumer products, mining and metals extraction, biomedical implants, and power generation.

The reason that companies in such a diverse range of industries value chemical engineers so highly is the following:

Starting from a vaguely defined problem statement such as a customer need or a set of experimental results, chemical engineers can develop an understanding of the important underlying physical science relevant to the problem and use this understanding to create a plan of action and set of detailed specifications, which if implemented, will lead to a predicted financial outcome.

The creation of plans and specifications and the prediction of the financial outcome if the plans are implemented is the activity of chemical engineering design.

Design is a creative activity, and as such can be one of the most rewarding and satisfying activities undertaken by an engineer. The design does not exist at the start of the project. The designer begins with a specific objective or

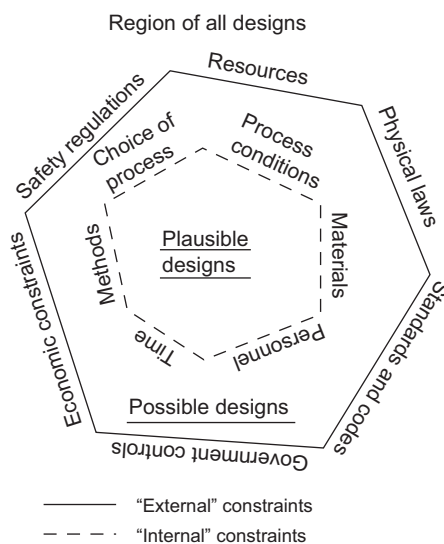


FIG. 1.1 Design constraints.

customer need in mind, and by developing and evaluating possible designs, arrives at the best way of achieving that objective, be it a better chair, a new bridge, or for the chemical engineer, a new chemical product or production process.

When considering possible ways of achieving the objective the designer will be constrained by many factors, which will narrow down the number of possible designs. There will rarely be just one possible solution to the problem, just one design. Several alternative ways of meeting the objective will normally be possible, even several best designs, depending on the nature of the constraints.

These constraints on the possible solutions to a problem in design arise in many ways. Some constraints will be fixed and invariable, such as those that arise from physical laws, government regulations, and engineering standards. Others will be less rigid and can be relaxed by the designer as part of the general strategy for seeking the best design. The constraints that are outside the designer's influence can be termed *the external constraints*. These set the outer boundary of possible designs, as shown in Fig. 1.1. Within this boundary there will be a number of plausible designs bounded by the other constraints, the internal constraints, over which the designer has some control; this can include choice of process, choice of process conditions, materials, and equipment.

Economic considerations are obviously a major constraint on any engineering design: plants must make a profit. Process costing and economics are discussed in Chapters 7, 8, and 9.

Time will also be a constraint. The time available for completion of a design will usually limit the number of alternative designs that can be considered.

The stages in the development of a design, from the initial identification of the objective to the final design, are shown diagrammatically in Fig. 1.2. Each stage is discussed in the following sections.

Fig. 1.2 shows design as an iterative procedure. As the design develops, the designer will become aware of more possibilities and more constraints and will be constantly seeking new data and evaluating possible design solutions.

1.2.1 The design objective (the need)

All design starts with a perceived need. In the design of a chemical product or process, the need is the public need for the product, creating a commercial opportunity, as foreseen by the sales and marketing organization. Within this overall objective the designer will recognize subobjectives: the requirements of the various units that make up the overall process.

Before starting work, the designer should obtain as complete, and as unambiguous, a statement of the requirements as possible. If the requirement (need) arises from outside the design group, from a customer or from another department, then the designer will have to elucidate the real requirements through discussion. It is important to distinguish between the needs that are "must-haves" and those that are "should-haves." The "should-haves" are

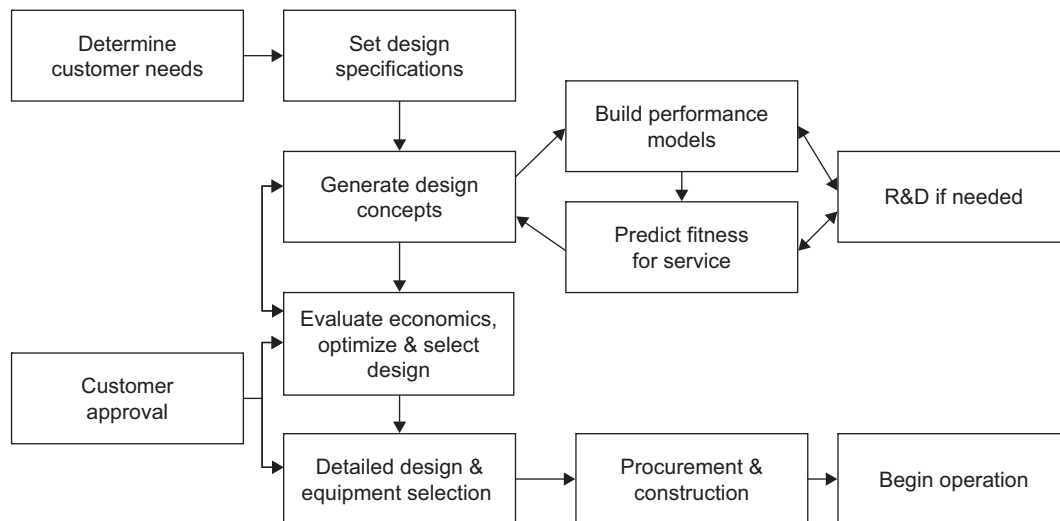


FIG. 1.2 The design process.

those parts of the initial specification that may be thought desirable but that can be relaxed if necessary as the design develops. For example, a particular product specification may be considered desirable by the sales department but may be difficult and costly to obtain, and some relaxation of the specification may be possible, producing a saleable but cheaper product. Whenever possible, the designer should always question the design requirements (the project and equipment specifications) and keep them under review as the design progresses. It is important for the design engineer to work closely with the sales or marketing department or with the customer directly to have as clear as possible an understanding of the customer's needs.

When writing specifications for others, such as for the mechanical design or purchase of a piece of equipment, the design engineer should be aware of the restrictions (constraints) that are being placed on other designers. A well-thought-out, comprehensive specification of the requirements for a piece of equipment defines the external constraints within which the other designers must work.

1.2.2 Setting the design basis

The most important step in starting a process design is translating the customer need into a design basis. The design basis is a more precise statement of the problem that is to be solved. It will normally include the production rate and purity specifications of the main product, together with information on constraints that will influence the design, such as:

1. The system of units to be used.
2. The national, local, or company design codes that must be followed.
3. Details of raw materials that are available.
4. Information on potential sites where the plant might be located, including climate data, seismic conditions and infrastructure availability. Site design is discussed in detail in [Chapter 11](#).
5. Information on the conditions, availability, and price of utility services such as fuel gas, steam, cooling water, process air, process water, and electricity that will be needed to run the process.

The design basis must be clearly defined before design can begin. If the design is carried out for a client, then the design basis should be reviewed with the client at the start of the project. Most companies use standard forms or questionnaires to capture design basis information. An example template is given in Appendix G and can be downloaded in Microsoft Excel format from the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

1.2.3 Generation of possible design concepts

The creative part of the design process is the generation of possible solutions to the problem for analysis, evaluation, and selection. In this activity most designers largely rely on previous experience, their own and that of others. It is doubtful if any design is entirely novel. The antecedence of most designs can usually be easily traced. The first motor cars were clearly horse-drawn carriages without the horse, and the development of the design of the modern car can be traced step by step from these early prototypes. In the chemical industry, modern distillation processes have developed from the ancient stills used for rectification of spirits, and the packed columns used for gas absorption have developed from primitive, brushwood-packed towers. So, it is not often that a process designer is faced with the task of producing a design for a completely novel process or piece of equipment.

Experienced engineers usually prefer the tried and tested methods rather than possibly more exciting but untried novel designs. The work that is required to develop new processes, and the cost, are usually underestimated. Commercialization of new technology is difficult and expensive, and few companies are willing to make multimillion-dollar investments in technology that is not well proven (a phenomenon known in industry as “*me third*” syndrome). Progress is made more surely in small steps; however, when innovation is wanted, previous experience, through prejudice, can inhibit the generation and acceptance of new ideas (known as “*not invented here*” syndrome).

The amount of work, and the way it is tackled, will depend on the degree of novelty in a design project. Development of new processes inevitably requires much more interaction with researchers and collection of data from laboratories and pilot plants.

Chemical engineering projects can be divided into three types, depending on the novelty involved:

1. Modifications and additions to an existing plant, usually carried out by the plant design group. Projects of this type represent about half of all the design activity in industry.
2. New production capacity to meet growing sales demand and the sale of established processes by contractors. Repetition of existing designs, with only minor design changes, including designs of vendor's or competitor's processes carried out to understand whether they have a compellingly better cost of production. Projects of this type account for about 45% of industrial design activity.
3. New processes developed from laboratory research, through pilot plant to a commercial process. Even here, most of the unit operations and process equipment will use established designs. This type of project accounts for less than 5% of design activity in industry.

The majority of process designs are based on designs that previously existed. The design engineer very rarely sits down with a blank sheet of paper to create a new design from scratch, an activity sometimes referred to as “*process synthesis*.” Even in industries such as pharmaceuticals, where research and new product development are critically important, the types of processes used are often based on previous designs for similar products, so as to make use of well-understood equipment and smooth the process of obtaining regulatory approval for the new plant.

The first step in devising a new process design will be to sketch out a rough block diagram showing the main stages in the process and to list the primary function (objective) and the major constraints for each stage. Experience should then indicate what types of unit operations and equipment should be considered. The steps involved in determining the sequence of unit operations that constitutes a process flowsheet are described in [Chapter 2](#).

The generation of ideas for possible solutions to a design problem cannot be separated from the selection stage of the design process; some ideas will be rejected as impractical as soon as they are conceived.

1.2.4 Fitness testing

When design alternatives are suggested, they must be tested for fitness for purpose. In other words, the design engineer must determine how well each design concept meets the identified need. In the design of chemical plants, it is usually prohibitively expensive to build several designs to find out which one works best. Instead, the design engineer builds a mathematical model of the process, usually in the form of computer simulations of the process, reactors, and other key equipment. In some cases, the performance model may include a pilot plant or other facility for predicting plant performance and collecting the necessary design data. In other cases, the design data can be collected from an existing full-scale facility or can be found in the chemical engineering literature.

The design engineer must assemble all of the information needed to model the process so as to predict its performance against the identified objectives. For process design, this will include information on possible processes, equipment performance, and physical property data. Sources of process information are reviewed in [Chapter 2](#).

Many design organizations will prepare a basic data manual containing all the process “know-how” on which the design is to be based. Most organizations will have design manuals covering preferred methods and data for the more frequently used design procedures. The national standards are also sources of design methods and data. They are also design constraints, as new plants must be designed in accordance with national standards and regulations. If the necessary design data or models do not exist, then research and development work is needed to collect the data and build new models.

Once the data have been collected and a working model of the process has been established, the design engineer can begin to determine equipment sizes and costs. At this stage it will become obvious that some designs are uneconomical and they can be rejected without further analysis. It is important to make sure that all of the designs that are considered are fit for the service (i.e., meet the customer’s “must-have” requirements). In most chemical engineering design problems this comes down to producing products that meet the required specifications. A design that does not meet the customer’s objective can usually be modified until it does so, but this always adds extra costs.

1.2.5 Economic evaluation, optimization, and selection

Once the designer has identified a few candidate designs that meet the customer objective, then the process of design selection can begin. The primary criterion for design selection is usually economic performance, although factors such as safety and environmental impact may also play a strong role. The economic evaluation usually entails analyzing the capital and operating costs of the process to determine the return on investment, as described in [Chapters 7, 8, and 9](#).

The economic analysis of the product or process can also be used to optimize the design. Every design will have several possible variants that make economic sense under certain conditions. For example, the extent of process heat recovery is a trade-off between the cost of energy and the cost of heat exchangers (usually expressed as a cost of heat exchange area). In regions where energy costs are high, designs that use a lot of heat exchange surface to maximize recovery of waste heat for reuse in the process will be attractive. In regions where energy costs are low, it may be more economical to burn more fuel and reduce the capital cost of the plant. Techniques for energy recovery are described in [Chapter 3](#). The mathematical techniques that have been developed to assist in the optimization of plant design and operation are discussed briefly in [Chapter 12](#).

When all of the candidate designs have been optimized, the best design can be selected. Very often, the design engineer will find that several designs have very close economic performance, in which case the safest design or that which has the best commercial track record will be chosen. At the selection stage an experienced engineer will also look carefully at the candidate designs to make sure that they are safe, operable, and reliable and to ensure that no significant costs have been overlooked.

1.2.6 Detailed design and equipment selection

After the process or product concept has been selected, the project moves on to detailed design. Here the detailed specifications of equipment such as vessels, exchangers, pumps, and instruments are determined. The design engineer may work with other engineering disciplines, such as civil engineers for site preparation, mechanical engineers for design of vessels and structures, and electrical engineers for instrumentation and control.

Many companies engage specialist engineering, procurement, and construction (EPC) companies, commonly known as *contractors*, at the detailed design stage. The EPC companies maintain large design staffs who can quickly and competently execute projects at relatively low cost.

During the detailed design stage there may still be some changes to the design, and there will certainly be ongoing optimization as a better idea of the project cost structure is developed. The detailed design decisions tend to focus mainly on equipment selection, though, rather than on changes to the flowsheet. For example, the design engineer may need to decide whether to use a U-tube or a floating-head exchanger, as discussed in [Chapter 19](#), or whether to use trays or packing for a distillation column, as described in [Chapter 17](#).

1.2.7 Procurement, construction, and operation

When the details of the design have been finalized, the equipment can be purchased and the plant can be built. Procurement and construction are usually carried out by an EPC firm unless the project is very small. Because they work on many different projects each year, the EPC firms are able to place bulk orders for items such as piping, wire, valves, etc., and can use their purchasing power to get discounts on most equipment. The EPC companies also have a great deal of experience in field construction, inspection, testing, and equipment installation. They can therefore normally contract to build a plant for a client cheaper (and usually also quicker) than the client could build it on their own.

Finally, once the plant is built and readied for start-up, it can begin operation. The design engineer will often then be called upon to help resolve any start-up issues and teething problems with the new plant.

1.3 The organization of a chemical engineering project

The design work required in the engineering of a chemical manufacturing process can be divided into two broad phases.

Phase 1: Process design, which covers the steps from the initial selection of the process to be used through to the issuing of the process flowsheets and includes the selection, specification, and chemical engineering design of equipment. In a typical organization, this phase is the responsibility of the process design group, and the work is mainly done by chemical engineers. The process design group may also be responsible for the preparation of the piping and instrumentation diagrams.

Phase 2: Plant design, including the detailed mechanical design of equipment; the structural, civil, and electrical design; and the specification and design of the ancillary services. These activities will be the responsibility of specialist design groups, having expertise in the whole range of engineering disciplines.

Other specialist groups will be responsible for cost estimation and the purchase and procurement of equipment and materials.

The sequence of steps in the design, construction, and start-up of a typical chemical process plant is shown diagrammatically in Fig. 1.3, and the organization of a typical project group is shown in Fig. 1.4. Each step in the design process will not be as neatly separated from the others as is indicated in Fig. 1.3, nor will the sequence of events be as clearly defined. There will be a constant interchange of information between the various design sections as the design develops, but it is clear that some steps in a design must be largely completed before others can be started.

A project manager, often a chemical engineer by training, is usually responsible for the coordination of the project, as shown in Fig. 1.4.

As was stated in Section 1.2.1, the project design should start with a clear specification defining the product, capacity, raw materials, process, and site location. If the project is based on an established process and product, a full specification can be drawn up at the start of the project. For a new product, the specification will be developed from an economic evaluation of possible processes, based on laboratory research, pilot plant tests, and product market research. Techniques for new product design are discussed in Section 1.8.

Some of the larger chemical manufacturing companies have their own project design organizations and carry out the whole project design and engineering, and possibly construction, within their own organization. More usually, the design and construction, and possibly assistance with start-up, are subcontracted to one of the international EPC firms.

The technical “know-how” for the process could come from the operating company or could be licensed from the contractor or a technology vendor. The operating company, technology provider, and contractor will work closely together throughout all stages of the project.

On many modern projects, the operating company may well be a joint venture between several companies. The project may be carried out between companies based in different parts of the world. Good teamwork, communications, and project management are therefore critically important in ensuring that the project is executed successfully.

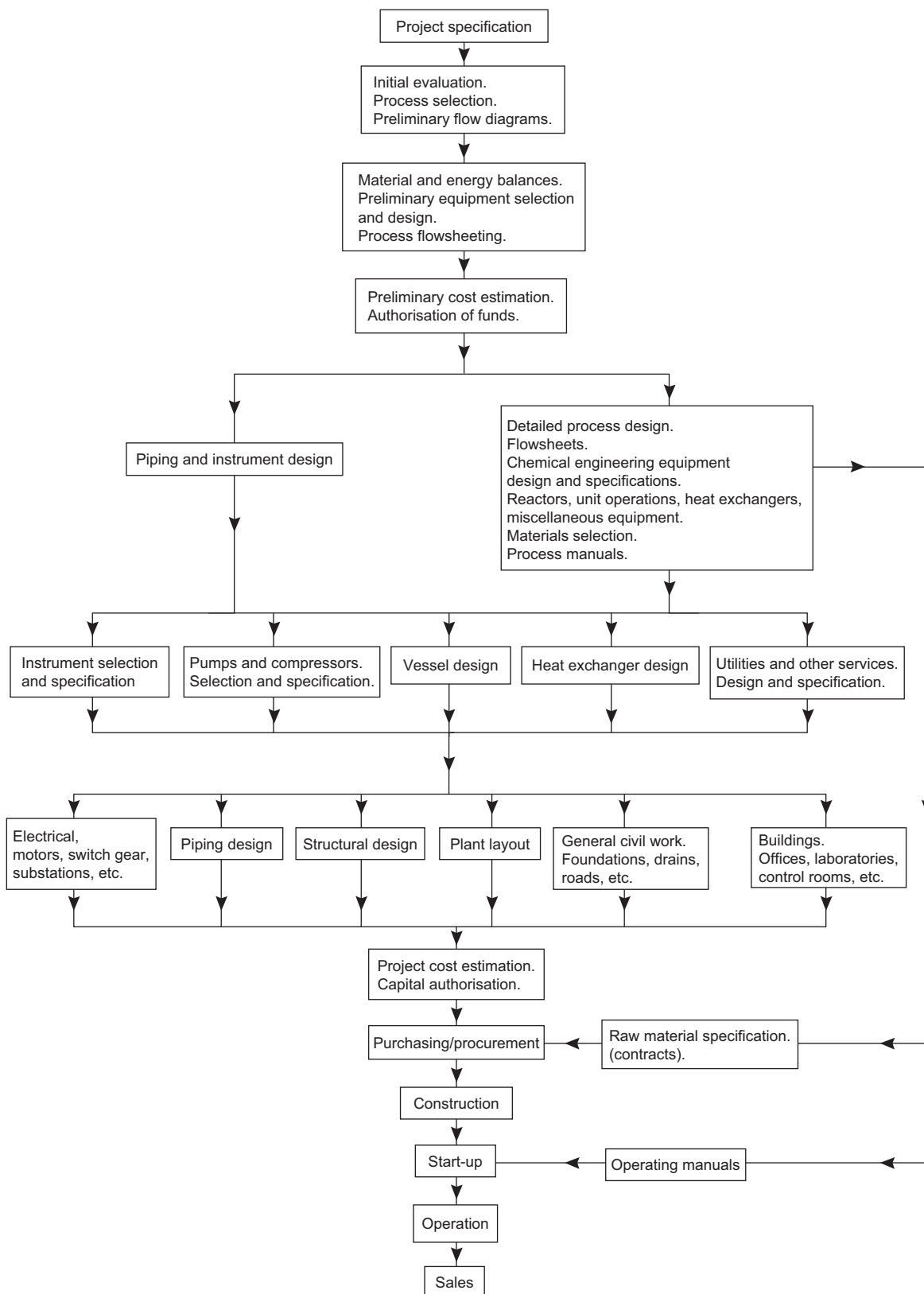


FIG. 1.3 The structure of a chemical engineering project.

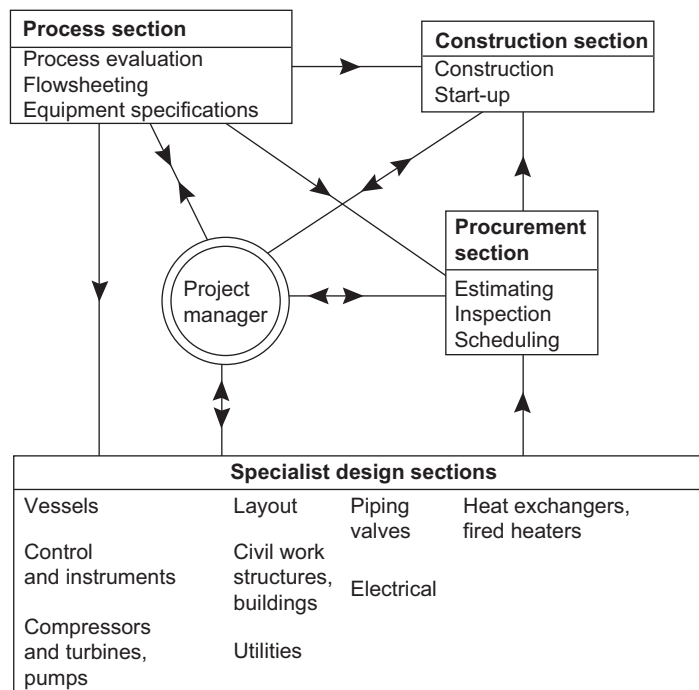


FIG. 1.4 Project organization.

1.4 Project documentation

As shown in Fig. 1.4 and described in Section 1.3, the design and engineering of a chemical process requires the cooperation of many specialist groups. Effective cooperation depends on effective communications, and all design organizations have formal procedures for handling project information and documentation. The project documentation will include:

1. General correspondence within the design group and with:
 - government departments
 - equipment vendors
 - site personnel
 - the client
2. Calculation sheets:
 - design calculations
 - cost estimates
 - material and energy balances
3. Drawings:
 - flowsheets
 - piping and instrumentation diagrams
 - layout diagrams
 - plot/site plans
 - equipment details
 - piping diagrams (isometrics)
 - architectural drawings
 - design sketches
4. Specification sheets:
 - the design basis
 - feed and product specifications
 - an equipment list
 - sheets for equipment, such as heat exchangers, pumps, heaters, etc.

5. Health, safety, and environmental information:
 - materials safety data sheets (MSDS forms)
 - HAZOP or HAZAN documentation (see [Chapter 10](#))
 - emissions assessments and permits
6. Purchase orders:
 - quotations
 - invoices

All documents are assigned a code number for easy cross-referencing, filing, and retrieval.

1.4.1 Design documents

Calculation sheets

The design engineer should develop the habit of setting out calculations so that they can be easily understood and checked by others. It is good practice to include on calculation sheets the basis of the calculations and any assumptions and approximations made in sufficient detail for the methods, as well as the arithmetic, to be checked. Design calculations are normally set out on standard sheets. The heading at the top of each sheet should include the project title and identification number, the revision number and date, and most importantly, the signature (or initials) of the person who checked the calculation. A template calculation sheet is given in Appendix G and can be downloaded in Microsoft Excel format from the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Drawings

All project drawings are normally drawn on specially printed sheets with the company name, project title and number, drawing title and identification number, drafter's name, and person checking the drawing clearly set out in a box in the bottom-right corner. Provision should also be made for noting on the drawing all modifications to the initial issue.

Drawings should conform to accepted drawing conventions, preferably those laid down by the national standards. The symbols used for flowsheets and piping and instrument diagrams are discussed in [Chapters 2 and 5](#). Computer-aided design (CAD) methods are used to produce the drawings required for all aspects of a project: flowsheets, piping and instrumentation, and mechanical and civil work. Although the released versions of drawings are usually drafted by a professional, the design engineer will often need to mark up changes to drawings or make minor modifications to flowsheets, so it is useful to have some proficiency with the drafting software.

Specification sheets

Standard specification sheets are normally used to transmit the information required for the detailed design, or purchase, of equipment items, such as heat exchangers, pumps, columns, pressure vessels, etc.

As well as ensuring that the information is clearly and unambiguously presented, standard specification sheets serve as checklists to ensure that all the information required is included.

Examples of equipment specification sheets are given in Microsoft Excel format in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. These specification sheets are referenced and used in examples throughout the book. Standard worksheets are also often used for calculations that are commonly repeated in design.

Process manuals

Process manuals are usually prepared by the process design group to describe the process and the basis of the design. Together with the flowsheets, they provide a complete technical description of the process.

Operating manuals

Operating manuals give the detailed, step-by-step instructions for operation of the process and equipment. They would normally be prepared by the operating company personnel, but may also be issued by a contractor or technology licensor as part of the technology transfer package for a less experienced client. The operating manuals are used for operator instruction and training and for the preparation of the formal plant operating instructions.

1.4.2 Design reports

Design reports are used as a means of organizing, recording, and communicating the information developed during a design project. The format of the report depends on the function of the design project. A technoeconomic analysis of a new product or process might require a strong focus on marketing and commercial aspects of the project and less technical detail, whereas a basic engineering design package that is to be used to generate a $\pm 10\%$ cost estimate will require substantial information on equipment designs but needs no financial analysis whatsoever.

When writing a design report, the design engineer should begin by thinking about the needs of the audience that will be using the report. Information is usually conveyed in the form of tables and charts as much as possible, with brief descriptions in the text when necessary. Most design reports are compiled from flow diagrams, specification sheets, and standard templates for economic analysis so that the technical information that users require is easily accessible. The written portion of the report is usually brief and is limited to an explanation of the key design features, assumptions, decisions, and recommendations. The following examples illustrate some of the different report formats that are commonly used in industry, and the final example discusses a suitable format for university design projects.

Example 1.1 Technoeconomic analysis

This type of report is used to summarize a preliminary technical and economic analysis of a proposed new product or process technology. Such a report might be written by an engineer working in product or process development or by a consulting company that has been asked to assess a new product or manufacturing route. This type of report is also often written as an assessment of a competitor's technology or in an effort to understand a supplier's cost structure. The purpose of the report is to provide sufficient technical and economic analysis of the process to determine whether it is economically attractive and to understand the costs of production, often in comparison with a conventional alternative. In addition to describing the technology and determining the cost of production, the report should review the attractiveness of the market and assess the risks inherent in practicing the technology. A sample contents list with guidance on each section is given in [Table 1.1](#).

TABLE 1.1 Technoeconomic analysis

1. Executive summary (1- to 2-page summary of overall findings and recommendations, including highlights of financial analysis)
2. Technology description
2.1. Process chemistry (describe the feeds, reaction mechanism, catalyst, reaction conditions, how important by-products are formed)
2.2. Process specification (brief description of the process, including block flow diagram)
3. Commercial analysis
3.1. Product applications (major end-use markets, competing products, legislative issues)
3.2. Competitor assessment (market shares, competitor strengths, weaknesses, regional/geographic factors)
3.3. Existing and planned capacity (how much and where, include plants that make feed or consume product if these have an impact on project viability—usually presented as a table)
3.4. Market forecast (estimate growth rate, future price trends, regional variations in market)
3.5. Project location criteria (discuss the criteria for locating a new plant, market issues, legislative factors, etc. [see Chapter 11])
4. Economic analysis
4.1. Pricing basis (forecasting method, price, and/or margin assumptions)
4.2. Investment analysis (explain the basis for the capital cost estimate, e.g., factorial estimate based on equipment design, curve cost estimate, etc. [see Chapter 7])
4.3. Cost of production analysis (breakdown of the cost of production of product, usually presented as a table showing variable and fixed cost components [see Chapter 8])
4.4. Financial analysis (evaluation of project profitability, usually presented as standard tables [see Chapter 9])
4.5. Sensitivity analysis (discuss the financial impact of varying key assumptions, such as prices, plant capacity, investment cost, construction schedule [see Chapter 9])
5. Risk analysis
5.1. Process hazard analysis summary (summary of critical safety issues in the design, issues raised during process hazard analysis)
5.2. Environmental impact assessment summary (summary of critical environmental issues)
5.3. Commercial risk assessment (discuss business risks inherent in the investment)
6. Appendices
6.1. Process flow diagram
6.2. Equipment list and capital cost summary

Example 1.2 Technical proposal

A technical proposal document is intended to convey the information needed to make a technology selection. When a company has decided to build a new plant, they will often invite several engineering or licensing firms to submit proposals for the plant design. Although the proposal does not contain a complete design, there must be sufficient technical information for the customer to be able to select between the proposed design and the competitor's proposals. Often, the customer will specify the contents and section headings of the proposal to ensure that all proposals follow the same format. Because the customer has already completed their own market analysis, this information is not required. Similarly, the plant capacity and location have usually already been specified. Instead, the focus of the report is on conveying the unique features of the design, the basis for selecting these features, and the proof that these features have worked in actual practice. A sample contents list is given in [Table 1.2](#).

TABLE 1.2 Technical proposal

1. Executive summary
 - 1.1. Proposed technology (brief description of the process, including block flow diagram)
 - 1.2. Benefits and advantages (summarize key advantages relative to competing technologies)
2. Proposal basis
 - 2.1. Processing objectives (restate the design problem)
 - 2.2. Feedstocks (describe available feedstocks, grades, quality issues)
 - 2.3. Product grades (give product specifications, usually as tables or reference to ASTM specifications)
 - 2.4. Processing options (describe technical alternatives evaluated)
3. Proposed technology
 - 3.1. Process description (more detailed process description)
 - 3.2. Reactor selection (what reactor type is recommended, why it was selected, and how it was designed)
 - 3.3. Catalyst selection recommendations (what catalysts are recommended and why)
 - 3.4. Key equipment recommendations (describe any critical unit operations and explain what was selected and how it was designed, key specifications, etc.)
 - 3.5. Pilot plant and commercial experience (describe any work that proves that the proposed design will operate as described)
4. Technical and economic assessment
 - 4.1. Estimated raw materials consumption (usually a table)
 - 4.2. Estimated utility consumption (usually a table giving breakdowns for each utility [see [Chapter 3](#)])
 - 4.3. Estimated manpower requirements (how many operators are needed per shift)
 - 4.4. Estimated cost of production (breakdown of the cost of production of product, usually presented as a table showing variable and fixed cost components [see [Chapter 8](#)])
 - 4.5. Estimated installed capital cost (breakdown by plant section of the plant capital cost estimate)
5. Process flow diagrams
6. Preliminary equipment specification sheets
7. Typical plot plan

Example 1.3 Basic engineering design

A basic engineering design report (BEDR) is often used at the end of the process design phase to collect and review information before beginning the plant design phase and detailed design of equipment, piping, plot layout, etc. The purpose of the BEDR is to ensure that all the information necessary for detailed design has been assembled, reviewed, and approved so as to minimize errors and rework during detailed design. The BEDR also serves as a reference document for the detailed design groups and provides them with stream flows, temperatures, pressures, and physical property information. One of the most important functions of a basic engineering design report is to document the decisions and assumptions made during the design and the comments and suggestions made during design review meetings. These are often documented as separate sections of the report so that other engineers who later join the project can understand the reasons why the design evolved to its current form. A sample contents list for a basic engineering design report is given in [Table 1.3](#).

TABLE 1.3 Basic engineering design

1. Process description and basis
1.1. Project definition (customer, location, key feeds, and products)
1.2. Process description (brief description of process flowsheet and chemistry, including block flow diagrams)
1.3. Basis and scope of design (plant capacity, project scope, design basis table)
2. Process flow diagrams
3. Mass and energy balances
3.1. Base case stream data (stream temperature and pressure, mass flow and molar flow of each component in all streams, stream mass and molar composition, and total stream mass and molar flow, usually given as tables)
3.2. Modified cases stream data (same data for each variant design case, for example, winter/summer cases, start of run/end of run, different product grades, etc.)
3.3. Base case physical property data (physical properties required by detailed design groups, such as stream density, viscosity, thermal conductivity, etc.)
4. Process simulation (description of how the process was simulated and any differences between the simulation model and process flow diagram that detailed design groups need to understand)
5. Equipment list
6. Equipment specifications
6.1. Pressure vessels
6.2. Heaters
6.3. Heat exchangers
6.3.1. Tubular
6.3.2. Air cooled
6.4. Fluid handling equipment
6.4.1. Pumps
6.4.2. Compressors
6.5. Solid handling equipment
6.6. Drivers
6.6.1. Motors
6.6.2. Turbines
6.7. Unconventional or proprietary equipment
6.8. Instrumentation
6.9. Electrical specifications
6.10. Piping
6.11. Miscellaneous
7. Materials of construction (what materials are to be used in each section of the plant and why they were selected, often presented as a table or as a marked-up version of the process flow diagram)
8. Preliminary hydraulics (pump-and-line calculations of pressure drop used as a basis for sizing pumps and compressors [see Chapter 20])
9. Preliminary operating procedures (describe the procedures for plant start-up, shut-down, and emergency shut-down)
10. Preliminary hazard analysis (description of major materials and process hazards of the design [see Chapter 10])
11. Capital cost estimate (breakdown of capital cost, usually for each piece of equipment plus bulks and installation, usually given as a table or list)
12. Heat integration and utilities estimate (overview of any pinch analysis or other energy optimization analysis, composite curves, table giving breakdown of utility consumption and costs [see Chapter 3])
13. Design decisions and assumptions (description of the most significant assumptions and selection decisions made by the designers, including references to calculation sheets for alternatives that were evaluated and rejected)
14. Design review documentation
14.1. Meeting notes (notes taken during the design review meeting)
14.2. Actions taken to resolve design review issues (description of what was done to follow up on issues raised during the design review)
15. Appendices
15.1. Calculation sheets (calculations to support equipment selection and sizing, numbered and referenced elsewhere in the report)
15.2. Project correspondence (communications between the design team, marketing, vendors, external customers, regulatory agencies, and any other parties whose input influenced the design)

Example 1.4 Undergraduate design project

Senior-year design projects have a range of objectives, but these always include demonstrating proficiency in engineering design and economic evaluation. More technical information is needed than in Example 1.1, whereas more commercial and marketing analysis is needed than in Examples 1.2 and 1.3, so none of the report formats used in industry is ideal. A reasonable approach is to use the format of Example 1.1 and include the material listed in Example 1.3 as appendices. For shorter classes, or when there is insufficient time to develop all the information listed in Example 1.3, some of the sections of [Table 1.3](#) can be omitted.

1.5 Codes and standards

The need for standardization arose early in the evolution of the modern engineering industry; Whitworth introduced the first standard screw thread to give a measure of interchangeability between different manufacturers in 1841. Modern engineering standards cover a much wider function than the interchange of parts. In engineering practice, they cover:

1. Materials, properties, and compositions
2. Testing procedures for performance, compositions, and quality
3. Preferred sizes; for example, tubes, plates, sections, etc.
4. Methods for design, inspection, and fabrication
5. Codes of practice for plant operation and safety

The terms *standard* and *code* are used interchangeably, though *code* should really be reserved for a code of practice covering, say, a recommended design or operating procedure, and *standard* for preferred sizes, compositions, etc.

All of the developed countries, and many of the developing countries, have national standards organizations responsible for the issue and maintenance of standards for the manufacturing industries and for the protection of consumers. In the United States, the government organization responsible for coordinating information on standards is the National Institute of Standards and Technology (NIST); standards are issued by federal, state, and various commercial organizations. The principal ones of interest to chemical engineers are those issued by the American National Standards Institute (ANSI), the American Petroleum Institute (API), the American Society for Testing Materials (ASTM), the American Society of Mechanical Engineers (ASME) (pressure vessels and pipes), the National Fire Protection Association (NFPA) (safety), the Tubular Exchanger Manufacturers Association (TEMA) (heat exchangers), and the Instrumentation, Systems and Automation Society (ISA) (process control). Most Canadian provinces apply the same standards used in the United States. The preparation of the standards is largely the responsibility of committees of persons from the appropriate industry, the professional engineering institutions, and other interested organizations.

The International Organization for Standardization (ISO) coordinates the publication of international standards. The European countries used to each maintain their own national standards, but these are now being superseded by common European standards.

Lists of codes and standards and copies of the most current versions can be obtained from the national standards agencies or by subscription from commercial websites such as IHS (www.ihs.com).

As well as the various national standards and codes, the larger design organizations will have their own (in-house) standards. Much of the detail in engineering design work is routine and repetitious, and it saves time and money, and ensures conformity between projects, if standard designs are used whenever practicable.

Equipment manufacturers also work to standards to produce standardized designs and size ranges for commonly used items, such as electric motors, pumps, heat exchangers, pipes, and pipe fittings. They will conform to national standards, where they exist, or to those issued by trade associations. It is clearly more economic to produce a limited range of standard sizes than to have to treat each order as a special job.

For the designer, the use of a standardized component size allows for the easy integration of a piece of equipment into the rest of the plant. For example, if a standard range of centrifugal pumps is specified, the pump dimensions will be known, and this facilitates the design of the foundation plates, pipe connections, and selection of the drive motors: standard electric motors would be used.

For an operating company, the standardization of equipment designs and sizes increases interchangeability and reduces the stock of spares that must be held in maintenance stores.

Though there are clearly considerable advantages to be gained from the use of standards in design, there are also some disadvantages. Standards impose constraints on the designer. The nearest standard size will normally be selected on completing a design calculation (rounding up), but this will not necessarily be the optimum size, though; as the standard size will be cheaper than a special size, it will usually be the best choice from the point of view of initial capital cost. The design methods given in the codes and standards are, by their nature, historical and do not necessarily incorporate the latest techniques.

The use of standards in design is illustrated in the discussion of pressure vessel design in [Chapter 14](#) and the description of heat exchanger design in [Chapter 19](#). Relevant design codes and standards are cited throughout the book.

1.6 Design factors (design margins)

Design is an inexact art; errors and uncertainties arise from uncertainties in the design data available and in the approximations necessary in design calculations. Experienced designers include a degree of overdesign known as a *design factor*, *design margin*, or *safety factor* to ensure that the design that is built meets product specifications and operates safely.

In mechanical and structural design, the design factors that are used to allow for uncertainties in material properties, design methods, fabrication, and operating loads are well established. For example, a factor of around 4 on the tensile strength, or about 2.5 on the 0.1% proof stress, is normally used in general structural design. The recommended design factors are set out in the codes and standards. The selection of design factors in mechanical engineering design is illustrated in the discussion of pressure vessel design in [Chapter 14](#).

Design factors are also applied in process design to give some tolerance in the design. For example, the process stream average flows calculated from material balances are usually increased by a factor, typically 10%, to give some flexibility in process operation. This factor will set the maximum flows for equipment, instrumentation, and piping design. Where design factors are introduced to give some contingency in a process design, they should be agreed within the project organization and clearly stated in the project documents (drawings, calculation sheets, and manuals). If this is not done, there is a danger that each of the specialist design groups will add its own “factor of safety,” resulting in gross and unnecessary overdesign. Companies often specify design factors in their design manuals.

When selecting the design factor, a balance has to be made between the desire to make sure the design is adequate and the need to design to tight margins to remain competitive. Greater uncertainty in the design methods and data requires the use of bigger design factors.

1.7 Systems of units

Most of the examples and equations in this book use SI units; however, in practice the design methods, data, and standards that the designer will use are often only available in the traditional scientific and engineering units. Chemical engineering has always used a diversity of units, embracing the scientific CGS and MKS systems and both the American and British engineering systems. Those engineers in the older industries will also have had to deal with some bizarre traditional units, such as degrees Twaddle or degrees API for density and barrels for quantity. Although almost all of the engineering societies have stated support for the adoption of SI units, this is unlikely to happen worldwide for many years. Furthermore, much useful historic data will always be in the traditional units, and the design engineer must know how to understand and convert this information. In a globalized economy, engineers are expected to use different systems of units even within the same company, particularly in the contracting sector, where the choice of units is at the client's discretion. Design engineers must therefore have a familiarity with SI, metric, and customary units, and a few of the examples and many of the exercises are presented in customary units.

It is usually best practice to work through design calculations in the units in which the result is to be presented, but if working in SI units is preferred, data can be converted to SI units, the calculation made, and the result converted to whatever units are required. Conversion factors to the SI system from most of the scientific and engineering units used in chemical engineering design are given in Appendix D, which is at the end of this book as well as in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Some license has been taken in the use of the SI system in this volume. Temperatures are given in degrees Celsius ($^{\circ}\text{C}$); degrees Kelvin are only used when absolute temperature is required in the calculation. Pressures are often given in bar (or atmospheres) rather than in Pascals (N/m^2), as this gives a better feel for the magnitude of the pressures. In design calculations the bar can usually be taken as equivalent to an atmosphere, whatever definition is used for atmosphere. The abbreviations bara and barg are often used to denote bar absolute and bar gauge, analogous to psia and psig when the pressure is expressed in pound force per square inch. When bar is used on its own, without qualification, it is normally taken as absolute.

For stress, N/mm^2 have been used, as these units are now generally accepted by engineers, and the use of a small unit of area helps indicate that stress is the intensity of force at a point (as is pressure). The corresponding traditional unit for stress is the ksi, or thousand pounds force per square inch. For quantity, kmol are generally used in preference to mol, and for flow, kmol/h instead of mol/s, as this gives more sensibly sized figures, which are also closer to the more familiar lb/h.

For volume and volumetric flow, m^3 and m^3/h are used in preference to m^3/s , which gives ridiculously small values in engineering calculations. Liters per second are used for small flow rates, as this is the preferred unit for pump specifications.

Plant capacities are usually stated on an annual mass flow basis in metric tons per year. Unfortunately, the literature contains a variety of abbreviations for metric tons per year, including tonnes/y, metric tons/y, MT/y (also kMTA = thousand metric tons per year), mtpy, and the correct term, t/y. The nonstandard abbreviations have occasionally been used, as it is important for design engineers to be familiar with all of these terms. The unit t denotes a metric ton of 1000 kg. In this book the unit ton is generally used to describe a short ton or U.S. ton of 2000 lb (907 kg) rather than a long ton or UK ton of 2240 lb (1016 kg), although some examples use long tons. The long ton is closer to the metric ton. A thousand metric tons is usually denoted as a kiloton (kt); the correct SI unit gigagram (Gg) is rarely used.

In the United States, the prefixes M and MM are often used to denote thousand and million, which can be confusing to anyone familiar with the SI use of M as an abbreviation for *mega* ($\times 10^6$). This practice has generally been avoided, except in the widely used units MMBtu (million British thermal units) and the common way of abbreviating \$1 million as \$1 MM.

Most prices have been given in U.S. dollars, denoted US\$ or \$, reflecting the fact that the data originated in the United States.

Where, for convenience, other than SI units have been used on figures or diagrams, the scales are also given in SI units, or the appropriate conversion factors are given in the text. Where equations are presented in customary units, a metric equivalent is generally given.

Some approximate conversion factors to SI units are given in Table 1.4. These are worth committing to memory to give some feel for the units for those more familiar with the traditional engineering units. The exact conversion factors are also shown in the table. A more comprehensive table of conversion factors is given in Appendix D.

TABLE 1.4 Approximate conversions between customary units and SI units

Quantity	Customary Unit	SI Unit	
		Approx.	Exact
Energy	1 Btu	1 kJ	1.05506
Specific enthalpy	1 Btu/lb	2 kJ/kg	2.326
Specific heat capacity	1 Btu/lb $^{\circ}$ F	4 kJ/kg $^{\circ}$ C	4.1868
Heat transfer coeff.	1 Btu/ft 2 h $^{\circ}$ F	6 W/m 2 $^{\circ}$ C	5.678
Viscosity	1 centipoise	1 mNs/m 2	1.000
	1 lb $_f$ /ft h	0.4 mNs/m 2	0.4134
Surface tension	1 dyne/cm	1 mN/m	1.000
Pressure	1 lbf/in 2 (psi)	7 kN/m 2	6.894
	1 atm	1 bar	1.01325
		10 5 N/m 2	
Density	1 lb/ft 3	16 kg/m 3	16.0185
	1 g/cm 3	1 kg/m 3	
Volume	1 US gal	3.8 $\times 10^{-3}$ m 3	3.7854 $\times 10^{-3}$
Flow-rate	1 US gal/min	0.23 m 3 /h	0.227

Note:

1 US gallon = 0.84 imperial gallons (UK)

1 barrel (oil) = 42 US gallons $\approx$ 0.16 m 3 (exact 0.1590)

1 kWh = 3.6 MJ

1.8 Product design

The design of new chemical products goes through the same stages described in Section 1.2 and illustrated in Fig. 1.2. The successful introduction of a new product usually requires not only the design of the product itself but also the design of the plant that will make the product. In the process industries the conception and development of new chemical products are often led by chemists, biologists, pharmacists, food scientists, or electrical or biomedical engineers; however, chemical engineers can be involved from the earliest stages and will certainly be engaged in designing the manufacturing process and developing the first estimates of the cost of production and capital investment required.

The launch of a new product always has high commercial risk. The new product must meet a customer need and outperform the existing alternatives. Customers may have multiple requirements of the product, and these requirements may not be stated in a way that is easy to relate to technical specifications. The company that introduces the product needs to build market share and command a high enough price to ensure that the investment in research, development, and a new plant can be justified.

Most of the engineering work that is done in launching a new product goes into the design of the manufacturing process, but considerable care must be taken to ensure that the commercial risks have also been properly addressed. Consequently, in new product design much more attention is paid to the steps of understanding customer preferences, translating these needs into product specifications, and market testing to ensure fitness for service.

This section introduces some of the methods that are used for product development in the process industries and that may be useful to chemical engineers engaged in new product design. Vast quantities of books on innovation and new product design have been published in the general engineering and business literature. Among the best are those by Cooper (2017), Ulrich et al. (2020), and Cooper and Edgett (2009). Product design books aimed specifically at chemical engineers have been written by Cussler and Moggridge (2011) and Seider et al. (2016).

1.8.1 New Chemical Products

Chemical engineers work in many industries and may be engaged in designing all kinds of products, but for the purposes of this chapter the discussion will be limited to new products that are based on the application of novel chemistry, biology, or materials science. These can be broadly categorized as new molecules, new formulations, new materials, and new equipment and devices.

New molecules

The process industries produce and consume a surprisingly large number of distinct chemical species. Under the Toxic Substances Control Act of 1976 (TSCA) (15 U.S.C. 2601 *et seq.*), the U.S. Environmental Protection Agency (EPA) regulates the manufacture, import, and export of 83,000 chemicals. The European Chemicals Agency (ECHA) was established in 2006 under the European Regulation, Evaluation, Authorization and Restriction of Chemicals (REACH) regulation, with the goal of registering all chemicals in use in Europe. At the time of writing, 143,000 chemicals have been submitted to ECHA for pre-registration. The infinite possibilities of organic chemistry ensure that we will never run out of new molecular species to test for any given application.

New molecules are often commercialized in high-value applications such as specialty chemicals, additives, and active pharmaceutical ingredients (APIs). New molecules may also be needed when use of an existing chemical is restricted for safety or environmental reasons. For example, chlorinated hydrocarbons were phased out as refrigerants and propellants under the Montreal Protocol after concern that they caused ozone depletion. The fluorocarbon compounds that replaced them are in turn likely to be replaced due to concerns about their high global-warming potential as greenhouse gases.

Various methods are used to identify new molecules for an application. Optimization of computer models based on molecular simulation or group contribution methods may provide insights into molecular structures that give desired properties. More often, chemists will look at variants on known molecules, for example, by addition, removal, or substitution of methyl-, ethyl-, phenyl- or other substituent groups. The chemists will also use their knowledge of synthesis routes to propose compounds that are easier to prepare in high yield using known chemical pathways and starting from available feeds. The same is true for biologically derived compounds, where the biochemist or genetic engineer will attempt to isolate enzymes or strains that maximize the yield of the target molecule.

New formulations

Almost all process industry products sold to the general public are formulations made from multiple chemicals. Examples range from pharmaceuticals, cosmetics, healthcare products, fragrances, foods, and beverages to paints, adhesives, fuels, and cleaning products. Every household contains a multitude of mixtures of products.

The prevalence of formulated products arises directly from the need to meet multiple customer requirements. You can wash your hands quite effectively using linear alkylbenzene sulfonate (a surfactant), but you probably prefer it to be blended into a gel that smells nice, has an attractive color, and provides some antibacterial action. The same surfactant would also be quite suitable for washing your car, clothes, dishes, carpets, hair, and toilet, but in each case specific user requirements lead to a different formulated product.

Formulated products are usually produced in blending plants. In some simple cases the feed compounds are just mixed together and sent to a packaging line. More commonly, the mixing and blending operations must be carefully designed to ensure (or prevent) emulsification and guarantee uniform product properties. Formulation plants are also often designed to produce a range of different products tailored to different market segments, in which case the plant must be designed to switch between products with minimal downtime and product wastage.

The blend composition of a formulated product is designed to meet the customer needs in a cost-effective manner that provides an adequate profit margin for the manufacturer. Where possible, manufacturers seek to substitute expensive components with cheaper materials that have the same effect; however, marketing and brand management can sometimes be used to justify using more expensive materials. For example, “natural” compounds derived from agricultural products can often be effectively marketed to replace cheaper synthetic alternatives.

Consumer products are highly regulated and carry high potential liability risks because of the large number of end users. These factors place additional constraints on the product designers. Extensive product safety testing must be carried out when new chemicals are introduced into consumer product formulations.

New materials

Chemical engineers play a leading role in the manufacture of polymers, synthetic fibers, composite materials, papers, films, electronic materials, catalysts, and ceramics. The properties of these materials are often determined as much by the manufacturing process as by the chemical composition. For example, multiple grades of polyethylene can be produced, with very different properties, depending on the production route and distribution of molecular weight in the polymer.

New product development in the manufacturing industries is often based on materials substitution. Injection-molded or film-blown polymers are usually a cheaper substitute for metal, wood, or glass components that require more labor-intensive casting or machining. Many chemical engineers work on tailoring the properties of engineering materials such as polymers, resins, and composites to optimize the material to particular end applications.

The development of new materials applications requires close collaboration with the end user of the material. Most of the product specifications will be based on physical properties such as strength, elasticity, hardness, etc., and flow properties that affect ease of manufacture, but resistance to chemicals, solvents, oxidation, and corrosion can also be important factors.

New equipment and devices

Many sensors, medical devices, and power systems are based on chemical or biological processes. If a device requires sound understanding of kinetics and transport processes, chemical engineers will probably be involved in its design. Chemical engineers also play an important role in the design of new proprietary equipment for the processing industries and are frequently involved in the design and customization of equipment such as dryers, crystallizers, membrane units, and other proprietary separation devices.

Device manufacture usually involves the assembly of multiple subcomponents, and the production line methods that are used are very different from the methods used in the process industries. Evaluating the production costs of manufactured devices requires familiarity with industrial engineering methods and is beyond the scope of this book.

1.8.2 Understanding customer needs

The first step in new product development is to find out what customers want and are prepared to pay for. If the new product is not better than existing alternatives in some way, then it will be difficult to build market share and generate a return on the investment. If new features are added, they must be of value to the customer; otherwise, the new product will not be differentiated from the existing alternatives. One of the roles of the marketing group in a company is to develop an understanding of customer requirements and willingness to spend and use this understanding to guide new product development teams.

The level of market research that is needed depends on the nature of the product and the homogeneity of the customer base. In some cases, the customers may all have very similar needs. For example, when UOP developed a renewable jet fuel based on hydrotreated vegetable oils, it was clear that the product must meet all the standard ASTM specifications for jet fuels. More often, however, the customers fall into different groups, known as *market segments*, each with different requirements. The product development team must consider the needs of each segment and determine whether a product can be designed to meet the needs of several segments or whether it is necessary to develop customized products for each segment.

It is important to distinguish between proximate and ultimate customers when carrying out market research. Many chemical products are sold to other manufacturers (proximate customers) who then incorporate the chemical product into their own products to sell to end users (ultimate customers). Some product features may be very valuable to the proximate customer while having little value to the ultimate customer. Improving the processability, handling, storage, or safety properties of a product will make it easier and potentially cheaper to use, but may have little effect on its end-use application. For example, a paint composition with a faster drying time may be attractive to an automobile manufacturer, but will not be noticed by the customer who buys the car.

Many methods have been developed for market research. Interviews and customer conferences can be used when the number of customers is small or when a representative sample group can be assembled. When the customer base is large and diverse, manufacturers use surveys and focus groups. The questions that are posed in market research studies must be carefully formulated so as to not only discover customer preferences but also identify latent needs that are not met by the existing products. Ulrich et al. (2020) suggest the following generic questions that can be used in interviews or focus groups:

- When and why do you use this product?
- What do you like about the existing products?
- What do you dislike about the existing products?
- What issues do you consider when purchasing the product?
- What improvements would you make to the product?

In addition to finding customer needs, good market research studies determine the relative importance of different needs and the willingness of the customer to pay for certain features. As the new product undergoes development, it may be necessary to repeat the market research to validate the product concept and test how well it meets customer expectations.

1.8.3 Developing product specifications

The needs stated by customers in the marketing study are usually not expressed in terms of technical product specifications. The design team must translate these needs into measurable properties of the product and then set a target value or range for each property. Product specifications must reflect all of the following factors:

- Product safety and regulatory requirements
- Potential liability concerns
- Fitness for purpose
- Customer needs and preferences
- Marketing advantages
- Maximization of profit margin

When setting specifications, it is important to remember that a specification should tell you *what* the product does, but not *how* it does it. For example, a customer need for a beverage such as a milk shake is to have the right “mouth feel.” One way to accomplish this might be by setting a specification on viscosity. The design team could then modify the recipe to meet the viscosity specification in many different ways. It would not be as effective to set a specification on xanthan gum concentration, as this presupposes the use of a particular thickener and overconstrains the design of the product.

Regulations and standards can be important sources of specifications. If a product is subject to regulation, then all the regulated specifications must be met, and new features can only be introduced if they do not require regulation or have obtained the necessary approval. Product safety, disposal, and environmental impact considerations can also lead to specifications that may not have been articulated by the customers. It is also important for the design team to consider potential product liability. The fact that a product is not currently regulated does not mean that it is safe,

and if there are concerns about public health or safety, then these should be raised and properly evaluated so that the company can assess the potential for future litigation.

Quality function deployment

A method that is widely used in translating customer needs into specifications is *Quality Function Deployment* (QFD) (Hauser and Clausing, 1988). Several variations of the QFD method have been developed, but all are based on the concept of relating customer needs to product specifications and comparing the proposed product to the existing competitors.

A QFD analysis is set out as a table or matrix and is usually carried out using a spreadsheet. Examples of simple QFD tables are given in Figs. 1.5 and 1.6. The first column lists the customer needs identified by the market research study. Each customer need is assigned a priority or importance, P , which is usually an integer on a 1 to 10 scale, based on the customer feedback. In some versions of the method a measure or metric is assigned to each customer need; however, this is not always necessary. The design team then lists all the product specifications that they envision and enters each specification as a column in the table. The team assigns a score, s , to how strongly each specification affects each customer need. A typical scoring scale might be 3 = critical, 2 = strong, 1 = weak, and 0 = no impact. The scores are multiplied by the corresponding customer priority and summed to give an overall relative importance of each specification, which is entered at the bottom of each column:

$$\text{Relative importance of specification } i = \sum_j P_j s_{ij} \quad (1.1)$$

Where: P_j = customer priority assigned to need j

s_{ij} = score for how well specification i meets need j

In some cases, additional columns are added to the right of the table for the existing competing products, as shown in Fig. 1.5. Each existing product can be assigned a score, c , for how well it meets each customer need, using the same scoring scale used for the specifications. These scores can also be multiplied by the corresponding customer priority and summed to give an indication of the relative strength of the existing products.

The QFD exercise has several uses. It helps the design team identify which specifications correlate most strongly with each customer need, and hence focuses effort on the aspects of the product that customers value most. If none of the specifications has a high score against a particular need, then it can highlight the need for new features or specifications. It can help identify strengths and weaknesses in competitors' products and identify which specifications must be adjusted to give superior performance to the competition. Lastly, it can help identify specifications that have an impact on multiple customer needs and potentially lead to trade-offs between different customer desires.

Customer needs	Priority	Specification 1	Specification 2	Specification 3	Specification 4	Specification 5	Specification 6	Competing product 1	Competing product 2	Competing product 3	Competing product 4	Competing product 5
Need 1	P_1	s_{11}	s_{21}	s_{31}	s_{41}	s_{51}	s_{61}	c_{11}	c_{21}	etc.		
Need 2	P_2	s_{12}	s_{22}	s_{32}	s_{42}	s_{52}	s_{62}	c_{12}	c_{22}			
Need 3	P_3	s_{13}	s_{23}	s_{33}	s_{43}	s_{53}	s_{63}	c_{13}	c_{23}			
etc.												
Relative importance		$\sum P_j s_{ij}$	etc.									

P_j = priority assigned to need j by customer

s_{ij} = score for how well specification i meets need j

c_{ij} = score for how well competing product i meets need j

FIG. 1.5 QFD table.

	Priority	Abrasive content	Fluoride content	Non-sugar sweetener	Flavor content	Viscosity modifier	Solid thickener	Antiseptic content	Bleach content
Customer needs									
Cleans teeth	8	3	0	0	0	0	0	1	0
Removes plaque	9	3	0	0	0	0	0	0	0
Whitens teeth	5	0	0	0	0	0	0	0	3
Tastes fresh	6	0	0	3	3	1	1	2	1
Freshens breath	7	0	0	0	3	0	0	2	1
Squeezes out right	5	2	0	0	0	3	3	0	0
Not gritty	6	3	0	0	0	0	2	0	0
Strengthens teeth	8	0	3	0	0	0	0	0	0
Prevents gingivitis	9	0	0	0	0	0	0	3	0
Relative Importance		79	24	18	39	21	33	61	28

FIG. 1.6 Completed QFD for toothpaste.

A simplified example of a QFD analysis is given in Example 1.5. More information on details of the method is given in the book by [Ulrich et al. \(2020\)](#) and the article by [Hauser and Clausing \(1988\)](#). The QFD method has become very widely used as part of the Six Sigma methodology; see [Pyzdek and Keller \(2018\)](#) for more on Six Sigma.

Example 1.5. QFD analysis

Complete a QFD analysis to determine the important specifications for a toothpaste product.

Solution

One possible solution is shown in [Fig. 1.6](#). A market survey (with a very limited set of customers) identified the following customer needs for toothpaste: cleans teeth, removes plaque, whitens teeth, tastes fresh, freshens breath, squeezes out right, not gritty, strengthens teeth, and prevents gingivitis. These are entered in the first column, with the relative priorities listed in the second column.

Some possible product specifications are then listed as additional columns. These include abrasive content, fluoride content, non-sugar sweetener, flavor content, viscosity modifier, solid thickener, antiseptic content, and bleach content. Note that these specifications do not specify the use of a particular bleach, sweetener, flavor, etc., so the designers might be able to meet several specifications using the same compound.

The scores are then entered for each specification. For example, the abrasive content is critical for “cleans teeth” and “removes plaque” (score 3 in both cases) but has no effect on “whitens teeth,” “tastes fresh,” or “freshens breath” (score 0). The abrasive content can have a strong effect on how the paste squeezes (score 2) and can have a critical impact on “not gritty” (score 3). Note that in this last case, the impact is negative and the customer desire for a particular mouth feel in the product is somewhat at odds with improving product performance.

The relative importance of the specification is then calculated as the priority-weighted sum of the scores, using Equation 1.1.

$$\begin{aligned}
 \text{Relative importance for abrasive content} &= 8(3) + 9(3) + 5(2) + 6(3) \\
 &= 24 + 27 + 10 + 18 = 79
 \end{aligned}$$

Scores are then assigned to how well every other specification meets each need until the table is completed.

Reviewing the completed table, we can see that all of the specifications have a critical impact on at least one of the customer needs and some have an impact on several needs. The abrasive content clearly has a strong impact on product performance and also on “not gritty,” so one conclusion of the QFD study might be to focus on examining different abrasive materials or different particle size distributions of abrasive so as to attempt to strike a better balance between these conflicting needs.

1.8.4 Fitness testing

As the design team develops potential product concepts, they will need to test each concept to determine how well it meets the desired specifications. In the cases of new molecules and new materials, testing will usually consist of synthesizing the material and carrying out experiments to determine its properties. For new equipment and formulations, more extensive prototyping and customer validation of the benefits of the design may be needed.

Prototype testing

Engineers build prototypes to address several different aspects of new product development:

- If new features are introduced in the design, it may be necessary to build a prototype to test these features and make sure that they work properly and safely.
- When a product is assembled from many components, it may be necessary to build a prototype to ensure that all the components work together properly when integrated as a system.
- The assembly of a prototype helps the designers understand the manufacturing process for the final product and can highlight features of the design that will make manufacturing easy or difficult. Prototyping is thus an important step in *design for manufacture*.
- In the design of formulated products, the manufacturer will often want to evaluate whether a component can be substituted with a cheaper material that has similar properties. It may be necessary to prepare alternative versions of the formulation with each component so that they can be tested side by side for properties and customer acceptance.
- A prototype can be used as a communication device to demonstrate features of a design. It can therefore be used to validate design features with potential customers or with management and hence confirm the marketing advantages of the new design.

Prototypes can take many forms, depending on the product type and stage of development. In the early stages of product development, conceptual or computer models are widely used. Working models of subcomponents are usually easier to test than full products; however, a full physical working model or exact recipe must usually be created for final product testing. Note that the activity of prototyping is not restricted to equipment and devices: testing different formulations of shampoo or cookie dough accomplishes the same goals.

Before a prototype is built, the design team should have a clear idea of the purpose of the prototype and the testing or experiments for which it will be used. Engineers from the manufacturing plant should be engaged as part of the development team to ensure that manufacturability concerns are flushed out and addressed. Several iterations of prototyping may need to be planned before a final product design can be selected.

Safety and efficacy testing

One of the most rigorous new product testing processes is the procedure used for obtaining approval from the U.S. Food and Drug Administration (FDA) for new medicines. The evaluation process is designed to ensure both the safety and efficacy of new drugs. If a company believes it has developed a new molecule with a therapeutic application, then it must go through the following steps:

- *Preclinical trials*: Initial testing on enzymes or cells in a laboratory, followed by animal tests usually on at least two species.
- *Phase I studies*: Testing on a small number of healthy volunteers (often medical students!) to ensure the treatment is safe.
- *Phase II studies*: Testing on patients who have the same disease or condition that is to be treated to evaluate efficacy at different dose levels.
- *Phase III studies*: Testing on a large number (hundreds or thousands) of patients who have randomly been assigned either the drug or a placebo to validate the efficacy of the treatment and assess possible side effects and interactions with other medications.

The results of the clinical trials are reviewed by an independent FDA panel to determine if the benefits of treatment outweigh the risks posed by any observed side effects. The entire process typically takes over 8 years and can cost more than \$800 million (DiMasi et al., 2003; FDA, 2006). Even when the product is approved, the manufacturer must still submit to FDA inspections to ensure that quality control procedures are adequate and the production facility complies with FDA current good manufacturing practices (cGMP). Additional information on GMP requirements is given in the discussion of bioreactor quality control in Section 15.9.8.

1.9 References

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1.10 Nomenclature

		Dimensions in \$MLT\theta\$
c_{ij}	Score for how well competing product i meets customer need j	—
P_j	Priority assigned by customer to need j	—
s_{ij}	Score for how well specification i meets customer need j	—

1.11 Problems

- Develop project plans for the design and construction of the following processes. Use Fig. 1.2 as a guide to the activities that must occur. Estimate the overall time required from launching the project to the start of operation.
 - A petrochemical process using established technology to be built on an existing site.
 - A process for full-scale manufacture of a new drug, based on a process currently undergoing pilot plant trials.
 - A novel process for converting cellulose waste to fuel products.
 - A spent nuclear fuel reprocessing facility.
 - A solvent recovery system for an electronics production facility.
- You are the project manager of a team that has been asked to complete the design of a chemical plant up to the stage of design selection. You have three engineers available (plus yourself), and the work must be completed in 10 weeks. Develop a project plan and schedule of tasks for each engineer. Be sure to allow sufficient time for equipment sizing, costing, and optimization. What intermediate deliverables would you specify to ensure that the project stays on track?
- You are part of a product design team that has been asked to develop a low-calorie chocolate chip cookie dough.
 - Poll your classmates to determine customer requirements.
 - Carry out a QFD analysis to map the customer requirements into product specifications

Process flowsheet development

KEY LEARNING OBJECTIVES

- How to read and draw a process flow diagram (PFD)
- When to design a batch process or a continuous process
- Factors to consider when adopting or improving commercially proven technology
- How to develop a flowsheet for a revamp design
- How to synthesize a flowsheet for an entirely new process
- How to review a flowsheet and check for completeness and errors

2.1 Introduction

This chapter covers the preparation and presentation of the process flowsheet, also known as the *process flow diagram* (PFD). The flowsheet is the key document in process design. It shows the arrangement of the equipment selected to carry out the process, the stream connections, the stream flow rates and compositions, and the operating conditions. It is a diagrammatic model of the process. Chemical engineers in industry are usually very proficient at reading PFDs and use it as the primary means of transmitting and recording process information.

The flowsheet is used by specialist design groups as the basis for their designs. These include piping, instrumentation, and equipment design and plant layout. It is also used by operating personnel for the preparation of operating manuals and operator training. During plant start-up and subsequent operation, the flowsheet forms a basis for the comparison of operating performance with design. If the plant is later revamped to new specifications, the PFD of the original plant is the starting point for the revamp design.

Several types of PFDs are used by chemical engineers, depending on the level of detail required. A simple block flow diagram can be used to give a rough idea of the overall process flow structure and may be useful when giving a presentation. A full PFD should include all of the process vessels and equipment and show all the process and utility flow lines. A full heat and material balance of the process showing the composition, flow rate, and temperature of every stream is usually included in or with a PFD. The PFD also indicates the location of every control valve, as control valves play an important role in determining the pressure balance of the process and hence in the sizing of pumps and compressors. A piping and instrumentation diagram (P&ID) is a more detailed version of the PFD that also includes information on ancillary instruments and valves, sampling and drain lines, start-up and shut-down systems, and pipe sizes and metallurgy. The P&ID is used in detailed design and safety analysis.

This chapter presents an overview of how to read and draw flowsheets and discusses how the unit operations of a process are selected to form the basic process flow structure. [Chapter 3](#) discusses energy flows within a process and describes heat and power recovery methods that are used to make processes more energy-efficient. [Chapter 4](#) covers the use of commercial process simulation tools to generate the heat and material balances for the flowsheet, and [Chapter 5](#) introduces the elements of process control that must be understood to fill in the control systems on the PFD.

2.2 Flowsheet presentation

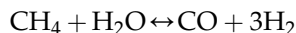
As the process flowsheet is the definitive document on the process, the presentation must be clear, comprehensive, accurate, and complete. This section describes how to read and draw PFDs.

2.2.1 Block diagrams

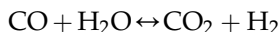
A block diagram is the simplest form of flow diagram. Each block can represent a single piece of equipment or a complete stage in the process. Block diagrams are useful for showing simple processes. For complex processes, their use is limited to showing the overall process, broken down into its principal stages.

Block diagrams are useful for representing a process in a simplified form in reports, textbooks, and presentations but have limited use as engineering documents. The stream flow rates and compositions can be shown on the diagram adjacent to the stream lines when only a small amount of information is to be shown or can be tabulated separately.

[Fig. 2.1](#) shows a block flow diagram of a steam reforming process for making hydrogen from methane. The methane feed enters on the left and is mixed with steam and preheated in the convective section of a fired heater. The steam–methane mixture then passes through the reactor tubes in the radiant section of the heater, where the steam reforming reaction takes place:



The products from the steam reformer are sent to a shift reactor. The shift reactor increases the amount of hydrogen in the product by allowing the water–gas shift reaction to re-equilibrate at a lower temperature:



The shift reactor products are then further cooled and scrubbed in an absorber to remove carbon dioxide before being sent to a pressure-swing adsorption process that separates hydrogen from carbon dioxide, unconverted methane, and water vapor.

Block diagrams are often drawn using simple graphics programs such as Microsoft Visio or PowerPoint. Block flow diagrams are not restricted to using boxes to represent process operations and often use the same flowsheet symbols used in PFDs. Generally, any flowsheet sketch that does not show all the required process equipment should be classified as a block flow diagram.

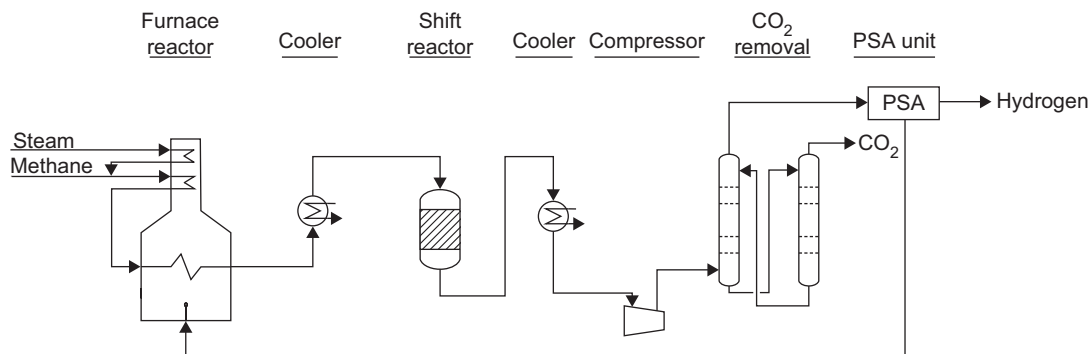


FIG. 2.1 Block flow diagram of steam reforming process for hydrogen.

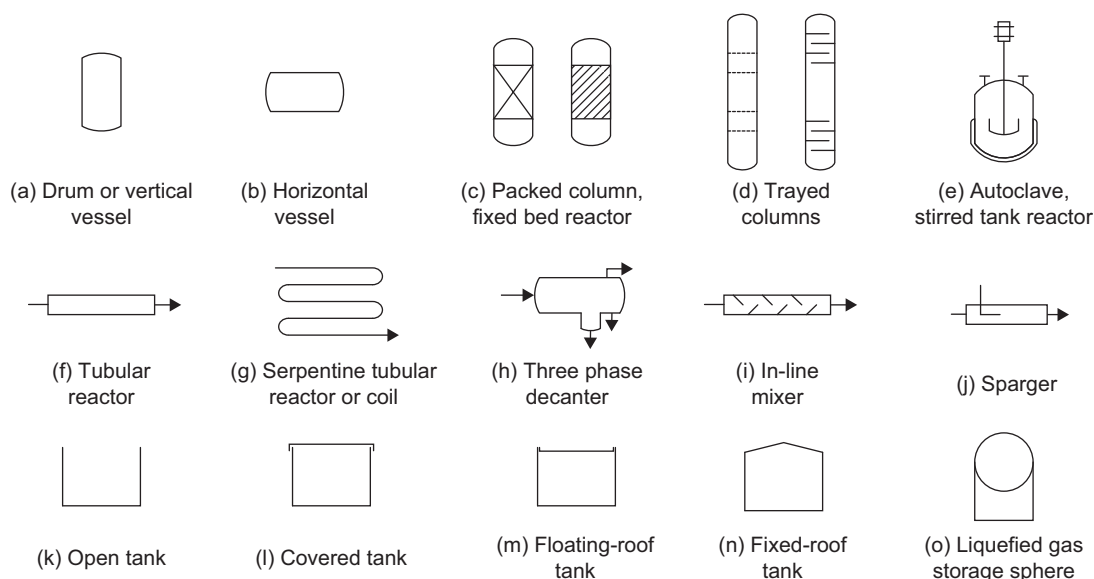


FIG. 2.2 PFD symbols for reactors, vessels, mixers, and tanks

2.2.2 PFD symbols

On the detailed flowsheets used for design and operation, the equipment is normally drawn in a stylized pictorial form. For tender documents or company brochures, actual scale drawings of the equipment are sometimes used, but it is more usual to use a simplified representation. There are several international standards for PFD symbols, but most companies use their own standard symbols, as the cost of converting all of their existing drawings would be excessive. ISO 10628 is the international standard for PFD drawing symbols. Most European countries have adopted ISO 10628 as their standard, but very few North American companies apply this standard, and there is currently no U.S. standard for PFD symbols. The symbols given in British Standard, BS 1553 (1977) "Graphical Symbols for General Engineering" Part 1, "Piping Systems and Plant" are more typical of those in common use. The professional edition of Microsoft Visio contains a library of PFD icons that includes the ISO 10628 symbols, as well as symbols commonly used in the United States and Canada. Examples of standard symbols are given in Appendix A, which is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

Fig. 2.2 shows symbols that are used for reactors, mixers, vessels, and tanks. Fig. 2.3 shows symbols used for heat transfer equipment. Fig. 2.4 provides symbols for fluid-handling equipment, and Fig. 2.5 gives symbols for solids-handling operations. Some general symbols that are used in combination with other symbols are shown in Fig. 2.6.

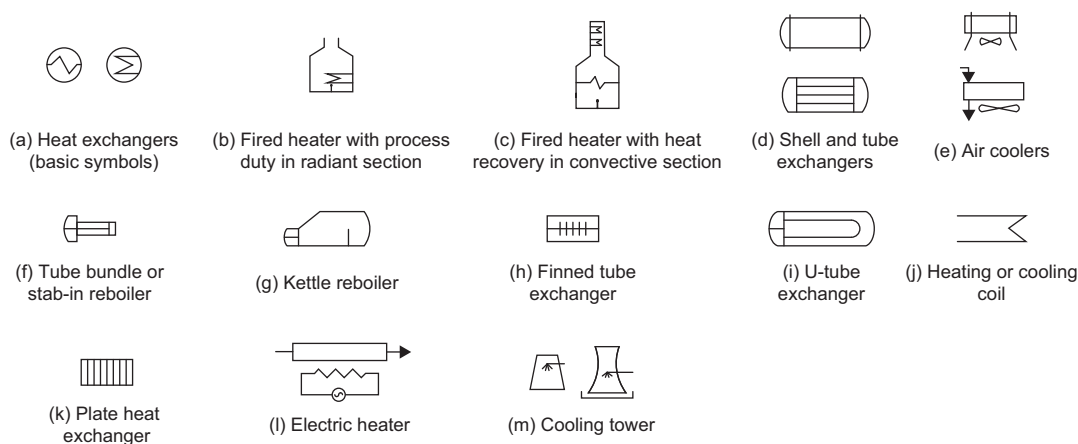


FIG. 2.3 PFD symbols for heat transfer equipment

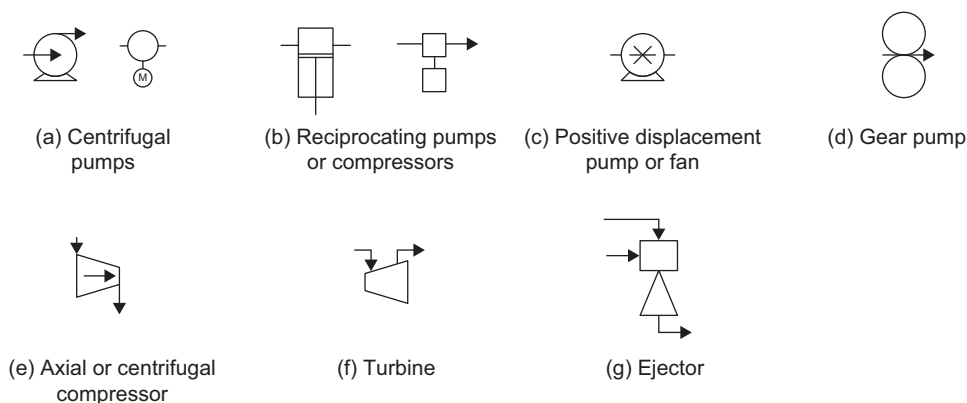


FIG. 2.4 PFD symbols for fluid-handling equipment

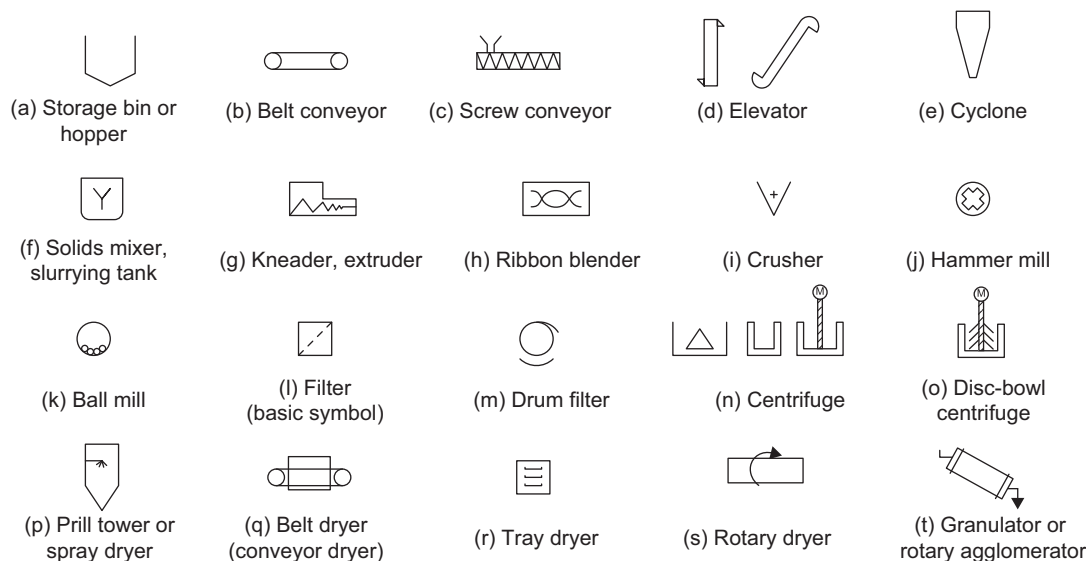


FIG. 2.5 PFD symbols for solids-handling equipment

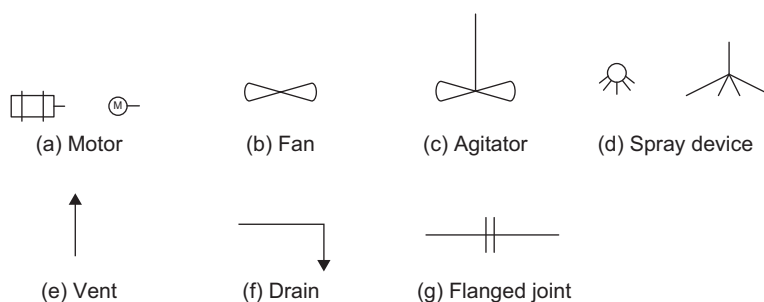


FIG. 2.6 General PFD symbols used with other symbols

The symbols that are used for process instruments, valves, and controllers are given in the section on P&ID diagram symbols in [Chapter 5](#). The operation and design of the different types of equipment illustrated in these figures are described in Part II of this book.

Note that some types of equipment have generic symbols as well as symbols that describe a particular equipment type. If the wrong symbol is selected, this can cause confusion for other engineers who read the flowsheet. For example, [Fig. 2.2\(i\)](#) shows an in-line mixer, which would be used downstream of a T-junction to ensure rapid mixing

of two liquid streams. Fig. 2.5(f) shows a solids mixer or blender that would be used to mix solids into a liquid. Fig. 2.6(c) is the symbol for a propeller agitator that might be used in a mixing tank. All of these symbols could be referred to as a *mixer*, but the designer's intention is obviously different in each case. Some abbreviations that are commonly used in PFDs are listed in the nomenclature at the end of this chapter.

2.2.3 Presentation of stream flow rates

The data on the flow rate of each individual component, on the total stream flow rate, and the percentage composition can be shown on the flowsheet in various ways. The simplest method, suitable for simple processes with few pieces of equipment, is to tabulate the data in blocks alongside the process stream lines, as shown in Fig. 2.7. Only a limited amount of information can be shown this way, and it is difficult to make neat alterations or to add more data.

A better method for the presentation of data on flowsheets is shown in Figs. 2.8 and 2.9. In this method each stream line is numbered and the data are tabulated at the bottom of the sheet. Alterations and additions can be easily made. This is the method generally used by professional design offices. A typical commercial flowsheet is shown in Fig. 2.10. Guide rules for the layout of this type of flowsheet presentation are given in Section 2.2.5.

2.2.4 Information to be included

The amount of information shown on a flowsheet will depend on the custom and practice of the particular design office. The list given here has therefore been divided into essential items and optional items. The essential items must always be shown; the optional items add to the usefulness of the flowsheet but are not always included.

Essential information

1. Always show *all* process equipment, including feed and product storage and equipment used for transporting fluids and solids.
2. Always indicate the location of process control valves.
3. For stream composition, either:
 - i. Tabulate the flow rate of each individual component, kg/h, which is preferred,
 - ii. Or give the stream composition as a weight fraction.
4. Total stream flow rate, kg/h.
5. Stream temperature, degrees Celsius preferred.
6. Nominal operating pressure (the required operating pressure).

Optional information

1. Molar percentage composition and/or molar flow rates.
2. Stream enthalpy, kJ/h.
3. Physical property data, with mean values for the stream, such as:
 - i. Density, kg/m³
 - ii. Viscosity, mN s/m²
4. Stream name: a brief one- or two-word description of the nature of the stream, for example, "ACETONE COLUMN BOTTOMS"

The convention of showing control valves on a PFD is not followed by all companies, but is recommended as a best practice. The presence of a control valve always causes pressure drop and may create a requirement for additional pumps or compressors. Showing the control valves can help identify locations where the designer may have overlooked the need for a pump.

Effective cooperation depends on effective communications, and all design organizations have formal procedures for handling project information and documentation. The project documentation will include:

1. General correspondence within the design group and with:
 - i. government departments
 - ii. equipment vendors
 - iii. site personnel
 - iv. the client

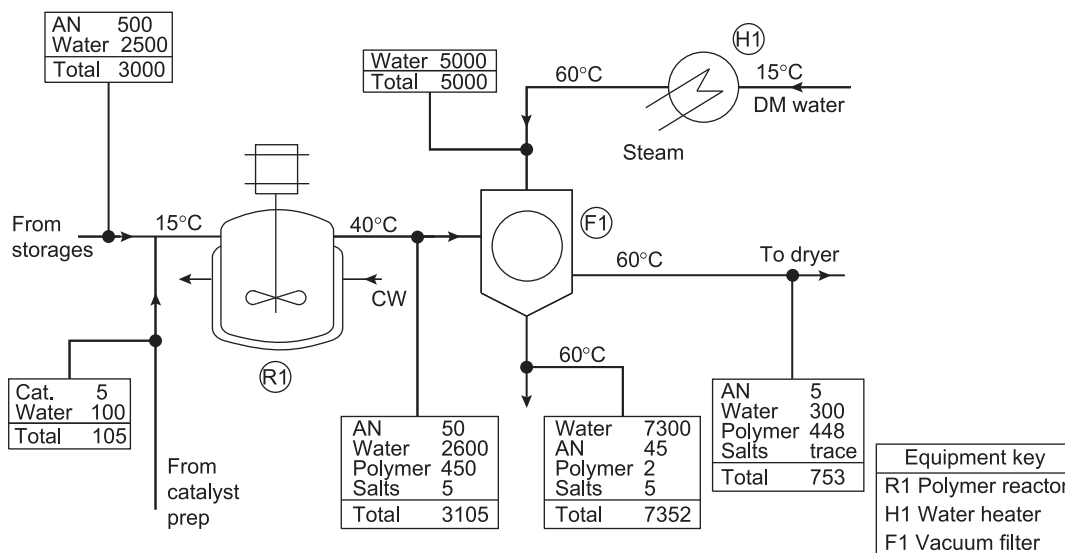


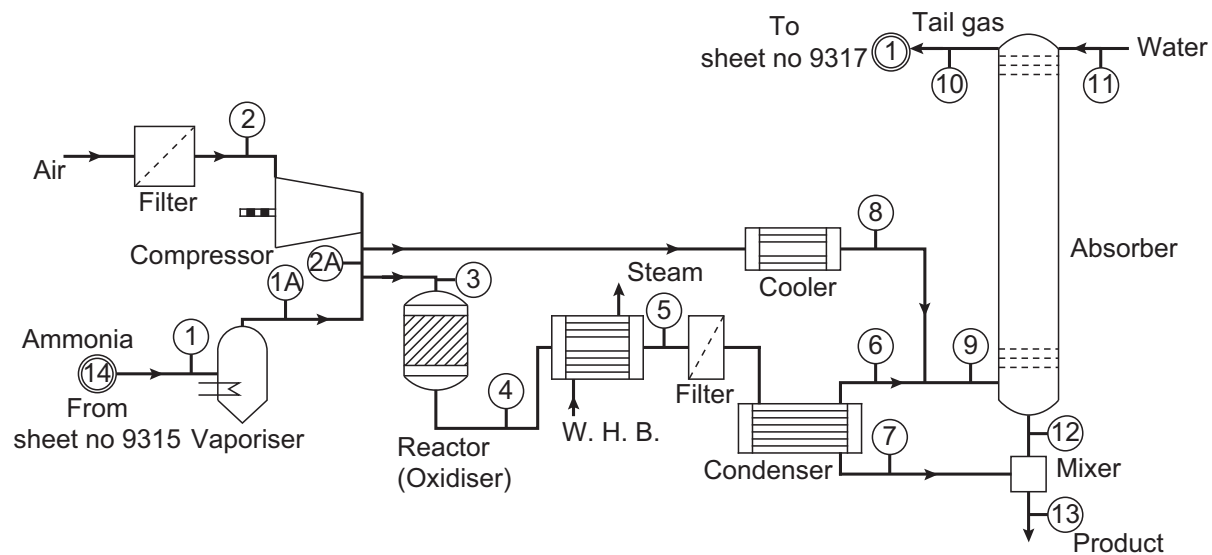
FIG. 2.7 Flowsheet: Polymer production.

2. Calculation sheets:
 - i. design calculations
 - ii. cost estimates
 - iii. material and energy balances
3. Drawings:
 - i. flow sheets
 - ii. piping and instrumentation diagrams
 - iii. layout diagrams
 - iv. plot/site plans
 - v. equipment details
 - vi. piping diagrams (isometrics)
 - vii. architectural drawings
 - viii. design sketches
4. Specification sheets:
 - i. the design basis
 - ii. feed and product specifications
 - iii. an equipment list
 - iv. sheets for equipment, such as heat exchangers, pumps, heaters, etc.
5. Health, safety, and environmental information:
 - i. materials safety data sheets (MSDS forms)
 - ii. HAZOP or HAZAN documentation (see [Chapter 10](#))
 - iii. emissions assessments and permits
5. Purchase orders:
 - i. quotations
 - ii. invoices

All documents are assigned a code number for easy cross-referencing, filing, and retrieval.

2.2.5 Layout

The sequence of the main equipment items shown symbolically on the flowsheet follows that of the proposed plant layout. Some license must be exercised in the placing of ancillary items, such as heat exchangers and pumps, or the layout will be too congested. The aim should be to show the flow of material from stage to stage as it will occur and to give a general impression of the layout of the actual process plant.



Flows kg/h pressures nominal

Line no. Stream component	1 Ammonia feed	1A Ammonia vapor	2 Filtered air	2A Oxidiser air	3 Oxidiser feed	4 Oxidiser outlet	5 W.H.B. outlet	6 Condenser gas	7 Condenser acid	8 Secondary air	9 Absorber feed	10 Tail(2) gas	11 Water feed	12 Absorber acid	13 Product acid	C & R Construction Inc
NH ₃	731.0	731.0	—	—	731.0	Nil	—	—	—	—	—	—	—	—	—	Nitric acid 60 percent
O ₂	—	—	3036.9	2628.2	2628.2	935.7	(935.7) ⁽¹⁾	275.2	Trace	408.7	683.9	371.5	—	Trace	Trace	100,000 t/y
N ₂	—	—	9990.8	8644.7	8644.7	8668.8	8668.8	8668.8	Trace	1346.1	10,014.7	10,014.7	—	Trace	Trace	Client BOP chemicals
NO	—	—	—	—	—	1238.4	(1238.4) ⁽¹⁾	202.5	—	—	202.5	21.9	—	Trace	Trace	SLIGO
NO ₂	—	—	—	—	—	Trace	(?) ⁽¹⁾	967.2	—	—	967.2	(Trace) ⁽¹⁾	—	Trace	Trace	Sheet no. 9316
HNO ₃	—	—	—	—	—	Nil	Nil	—	850.6	—	—	—	—	1704.0	2554.6	
H ₂ O	—	—	Trace	—	—	1161.0	1161.0	29.4	1010.1	—	29.4	26.3	1376.9	1136.0	2146.0	
Total	731.0	731.0	13,027.7	11,272.9	12,003.9	12,003.9	12,003.9	10,143.1	1860.7	1754.8	11,897.7	10,434.4	1376.9	2840.0	4700.6	
Press bar	8	8	1	8	8	8	8	8	1	8	8	1	8	1	1	Dwg by Date
Temp. °C	15	20	15	230	204	907	234	40	40	40	40	25	25	40	43	Checked 25/7/1980

FIG. 2.8 Flowsheet: Simplified nitric acid process.

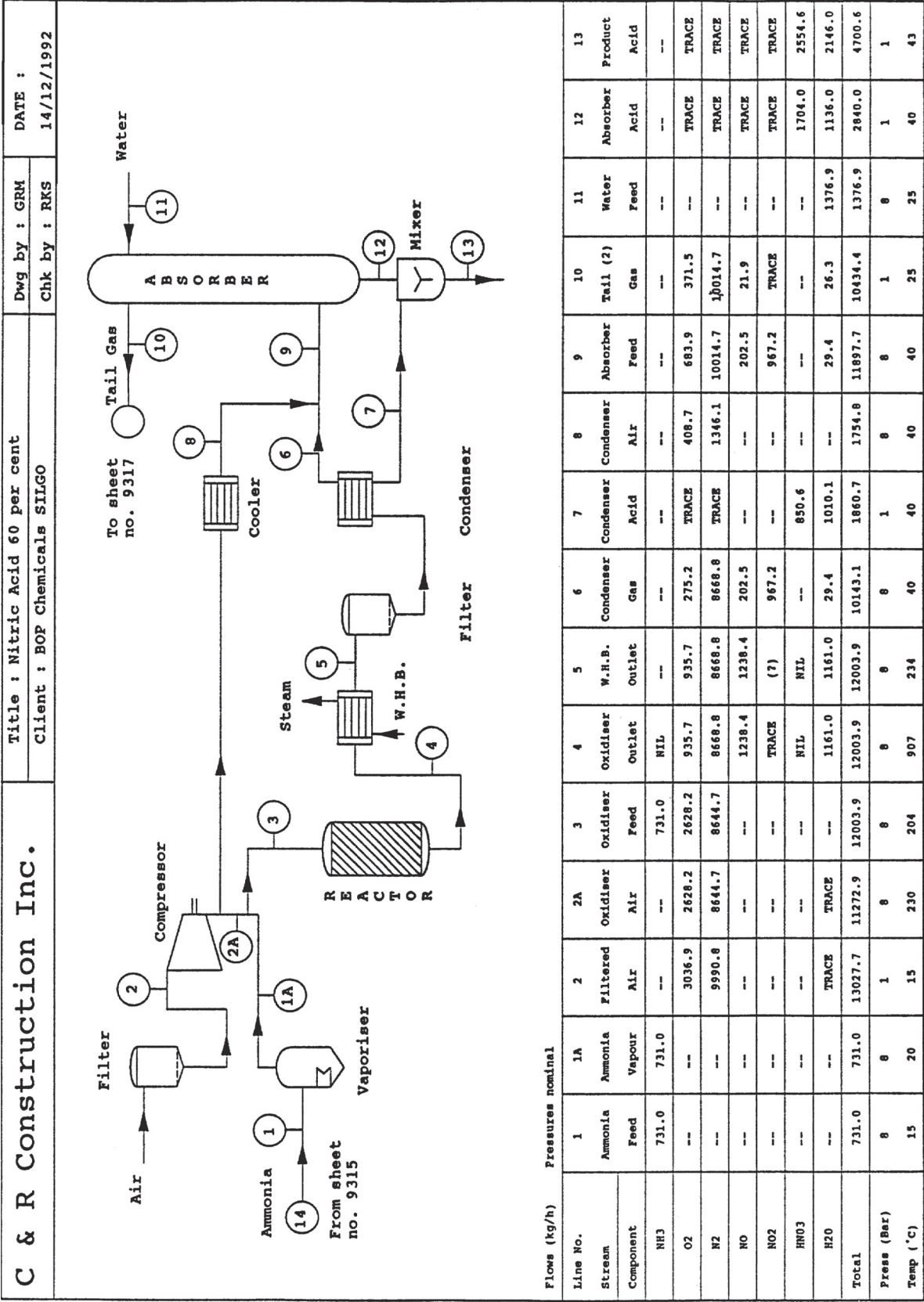


FIG. 2.9 Alternative presentation.

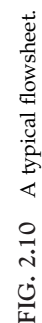


FIG. 2.10 A typical flowsheet.

The equipment should be drawn approximately to scale. Again, some license is allowed for the sake of clarity, but the principal equipment items, such as reactors, vessels, and columns, should be drawn roughly in the correct proportion. Ancillary items can be drawn out of proportion. For a complex process with many process units, several sheets may be needed, and the continuation of the process streams from one sheet to another must be clearly shown. One method of indicating a line continuation is shown in Fig. 2.8; those lines that are continued over to another drawing are indicated by a double concentric circle around the line number, and the continuation sheet number is written below. An alternative method is to extend lines to the side of the page and then indicate the drawing sheet on which the line is continued.

The overall process flow will usually be from left to right across the diagram, with recycle streams flowing right to left. Streams should be shown entering and leaving vessels in a manner that is consistent with the stream phase, density, and mode of operation. When a vessel has multiple feeds or products, their arrangement in the diagram should correspond to the physical arrangement intended by the designer. For example, the gas feed to an absorption column should enter below the bottom tray, the liquid feed should enter on the top tray, gas product should be shown leaving the top of the vessel, and liquid product should be shown leaving the bottom of the vessel.

The equipment should be well spaced out so that streams can be labeled without the drawing becoming cluttered. It is better to use several continuation sheets than to try to fit everything on one page.

The table of stream flows and other data can be placed above or below the equipment layout. Normal practice is to place it below. The components should be listed down the left side of the table, as in Figs. 2.8 and 2.9. For a long table, it is good practice to repeat the list at the right side so the components can be traced across from either side.

The stream line numbers should follow consecutively from left to right of the layout as far as is practicable, so that when reading the flowsheet it is easy to locate a particular line and the associated column containing the data.

All the process stream lines shown on the flowsheet should be numbered and the data for the stream given. On a large flowsheet, the designers sometimes use different series of numbers for different plant sections; for example, beginning the stream numbering at 100 for feed preparation, 200 for reaction, 300 for separation, and 400 for purification. This can be helpful in quickly tracing a stream to a section of the plant. There is always a temptation to leave out the data on a process stream if it is clearly just formed by the addition of two other streams, as at a junction, or if the composition is unchanged when flowing through a process unit, such as a heat exchanger; this temptation should be avoided. What may be clear to the process designer is not necessarily clear to the others who will use the flowsheet. Complete, unambiguous information on all streams should be given, even if this involves some repetition. The purpose of the flowsheet is to show the function of each process unit, even when the function has no discernible impact on the mass and energy balance.

2.2.6 Precision of data

The total stream and individual component flows do not normally need to be shown to a high precision on the process flowsheet; three or four significant figures are all that is usually justified by the accuracy of the flowsheet calculations and will typically be sufficient. The flows should, however, balance to within the precision shown. If a stream or component flow is so small that it is less than the precision used for the larger flows, it can be shown to a greater number of places if its accuracy justifies this and the information is required. If the composition of a component is very low but is specified as a process constraint, as, say, for an effluent stream or product quality specification, it can be shown in parts per million (ppm). Imprecise small flows are best shown as "TRACE."

A trace quantity should not be shown as zero, nor should the space in the tabulation be left blank, unless the process designer *is sure* that it has no significance. The process designer should be aware that if the space in the data table is left blank opposite a particular component, the quantity may be assumed to be zero by the specialist design groups, who take their information from the flowsheet. Trace quantities can be important. Only a trace of an impurity is needed to poison a catalyst, and trace quantities can determine the selection of the materials of construction; see Chapter 6.

2.2.7 Basis of the calculation

It is good practice to show on the flowsheet the basis used for the flowsheet calculations. This includes the operating hours per year, the reaction and physical yields, and the datum temperature used for energy balances. It is also helpful to include a list of the principal assumptions used in the calculations. This alerts the user to any limitations that may have to be placed on the flowsheet information.

If the amount of information that needs to be presented is excessive, it can be summarized in a separate document that is referenced on the flowsheet.

In some cases, mass and energy balances are prepared for multiple scenarios. These might include winter and summer operating conditions, start of catalyst life and end of catalyst life, manufacture of different products or product grades, etc. Usually these different scenarios are shown as several tables on the same flowsheet, but occasionally different flowsheets are drawn for each case.

2.2.8 Batch processes

Flowsheets drawn up for batch processes normally show the quantities required to produce one batch. If a batch process forms part of an otherwise continuous process, it can be shown on the same flowsheet, providing a clear break is made when tabulating the data between the continuous and batch sections (i.e., the change from kg/h to kg/batch).

A continuous process may include batch make-up of minor reagents, such as the catalyst for a polymerization process. Batch flows into a continuous process are usually labeled “Normally no flow” and show the flow rates that will be obtained when the stream is flowing. It is these instantaneous flow rates that govern the equipment design rather than the much lower time-averaged flow rates.

2.2.9 Utilities

To avoid cluttering up the flowsheet, it is not normal practice to show the utility (service) headers and lines on the process flowsheet. The utility connections required on each piece of equipment should be shown and labeled, for example, “CTW” for cooling tower water. The utility requirements for each piece of equipment should be tabulated on the flowsheet. A glossary of common abbreviations used for utility streams in PFDs is given in the nomenclature at the end of this chapter. Utility systems are described in more detail in [Chapter 3](#).

2.2.10 Equipment identification

Each piece of equipment shown on the flowsheet must be identified with a code number and name. The identification number (usually a letter and some digits) is normally that assigned to a particular piece of equipment as part of the general project control procedures and is used to identify it in all the project documents.

If the flowsheet is not part of the documentation for a project, then a simple, but consistent, identification code should be devised. The easiest code is to use an initial letter to identify the type of equipment, followed by digits to identify the particular piece. For example, H — heat exchangers, C — columns, R — reactors. Most companies have a standard convention that should be followed, but if there is no agreed standard, then the key to the code should be shown on the flowsheet.

2.2.11 Flowsheet drafting programs

Most design offices use drafting software for the preparation of flowsheets and other process drawings. With drafting software, standard symbols representing the process equipment, instruments, and control systems are held in files, and these symbols are called up as required when drawing flowsheets and piping and instrumentation diagrams. Final flowsheet drawings are usually produced by professional drafters, who are experienced with the drafting software and conventions, rather than by the design engineer. The design engineer has to provide the required numbers, sketch the flowsheet, and review the final result.

Although most process simulation programs feature a graphical user interface (GUI) that creates a drawing that resembles a PFD, printouts of these drawings should not be used as actual PFDs. The unit operations shown in the process simulation usually do not exactly match the unit operations of the process. The simulation may include dummy items that do not physically exist and may omit some equipment that is needed in the plant but is not part of the simulation.

2.3 The anatomy of a chemical manufacturing process

This section describes the basic components of chemical processes and discusses how designers select between batch and continuous processes. The effects of reactor yield and selectivity on flowsheet structure are examined and used to illustrate how flowsheets become complex when there are multiple feeds and products.

2.3.1 Components of a chemical process

The basic components of a typical chemical process are shown in Fig. 2.11, in which each block represents a stage in the overall process for producing a product from the raw materials. Fig. 2.11 represents a generalized process; not all the stages will be needed for any particular process, and the complexity of each stage will depend on the nature of the process. Chemical engineering design is concerned with the selection and arrangement of the stages and the selection, specification, and design of the equipment required to perform the function of each stage.

Stage 1. Raw material storage

Unless the raw materials (also called *feed stocks* or *feeds*) are supplied as intermediate products (intermediates) from a neighboring plant, some provision will have to be made to hold several days', or weeks', worth of storage to smooth out fluctuations and interruptions in supply. Even when the materials come from an adjacent plant, some provision is usually made to hold inventory for a few hours, or even days, to decouple the processes. The storage required depends on the nature of the raw materials, the method of delivery, and what assurance can be placed on the continuity of supply. If materials are delivered by ship (tanker or bulk carrier) several weeks' worth of stocks may be necessary, whereas if they are received by road or rail in smaller lots, less storage will be needed.

Stage 2. Feed preparation

Some purification and preparation of the raw materials will usually be necessary before they are sufficiently pure, or in the right form, to be fed to the reaction stage. For example, acetylene generated by the carbide process contains arsenic and sulfur compounds and other impurities, which must be removed by scrubbing with concentrated sulfuric acid (or other processes) before it is sufficiently pure for reaction with hydrochloric acid to produce dichloroethane. Feed contaminants that can poison process catalysts, enzymes, or microorganisms must be removed. Liquid feeds need to be vaporized before being fed to gas-phase reactors, and solids may need crushing, grinding, and screening. Solid feeds may also need to be weighed and mixed into slurries or solutions so that they can be brought to process pressure and easily mixed with other components.

The feed preparation stage always includes means for getting the feeds out of storage and into the process. Liquids are usually pumped out of storage through control valves that regulate the feed flow rate. Gases and vapors may need compression if the storage is not pressurized. Solids are conveyed from storage using a variety of equipment, described in Chapter 18.

Biological processes require careful feed preparation. The growth media and any other fluids that are fed to the cell culture must be sterile to prevent unwanted organisms from entering the process. Sterilization is usually accomplished by heating the feed to a high temperature and keeping it hot for long enough to kill unwanted organisms, then cooling the feed to the desired temperature for the reactor. The preparation of biological reactor feeds is discussed in more detail in Section 15.9.

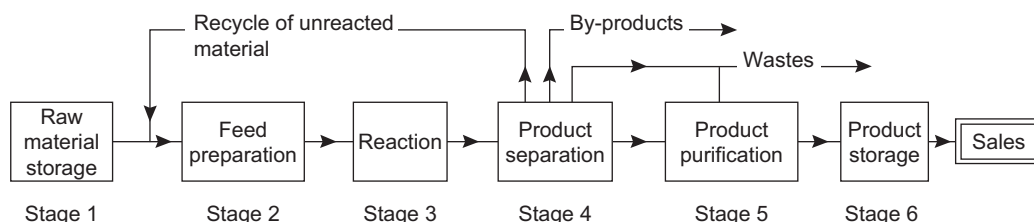


FIG. 2.11 Anatomy of a chemical process.

Stage 3. Reaction

The reaction stage is the heart of a chemical manufacturing process. In the reactor the raw materials are brought together under conditions that promote the production of the desired product; almost invariably, some by-products will also be formed, either through the reaction stoichiometry, by side reactions, or from reactions of impurities present in the feed. Reactor design is discussed in [Chapter 15](#).

Stage 4. Product separation

After the reactor(s), the products and by-products are separated from any unreacted material. If in sufficient quantity, the unreacted material will be recycled to the reaction stage or to the feed purification and preparation stage. The by-products may also be separated from the products at this stage and may undergo further processing for recovery or sale. In most chemical processes there are multiple reaction steps, each followed by one or more separation steps.

Stage 5. Purification

Before sale, the main product will often need purification to meet the product specifications. If produced in economic quantities, the by-products may also be purified for sale. For by-products, there will always be an economic trade-off between purifying the by-product for sale or disposing of it as recycle or waste.

Stage 6. Product storage

Some inventory of finished product must be held to match production with sales. Provision for product packaging and transport is also needed, depending on the nature of the product. Liquids are normally dispatched in drums and in bulk tankers (road, rail, and sea); solids in sacks, cartons, or bales.

The amount of stock that is held will depend on the nature of the product and the market.

Ancillary processes

In addition to the main process stages shown in [Fig. 2.11](#), provision must be made for the supply of the utilities needed, such as process water, cooling water, compressed air, and steam. The design of utility systems is discussed in [Chapter 3](#).

2.3.2 Continuous and batch processes

Continuous processes are designed to operate 24 hours a day, 7 days a week, throughout the year. Some downtime will be allowed for maintenance and, in some processes, for catalyst regeneration. The plant attainment or operating rate is the percentage of the available hours in a year that the plant operates and is usually between 90% and 95%.

$$\text{Attainment \%} = \frac{\text{hours operated}}{8760} \times 100 \quad (2.1)$$

A typical design basis would assume 8000 operating hours per year.

Batch processes are designed to operate intermittently, with some, or all, of the process units being frequently shut down and started up. It is quite common for batch plants to use a combination of batch and continuous operations. For example, a batch reactor may be used to feed a continuous distillation column.

Continuous processes will usually be more economical for large-scale production. Batch processes are used when some flexibility is wanted in production rate or product specifications.

The advantages of batch processing are:

- Batch processing allows production of multiple different products or different product grades in the same equipment.
- In a batch plant, the integrity of a batch can be preserved as it moves from operation to operation. This can be useful for quality control purposes.
- The production rate of batch plants is flexible, as there are no turn-down issues when operating at low output.
- Batch plants are easier to clean and maintain sterile operation.
- Batch processes are easier to scale up from chemist's recipes.
- Batch plants have low capital for small production volumes. The same piece of equipment can often be used for several unit operations.

The drawbacks of batch processing are:

- The scale of production is limited.
- It is difficult to achieve economies of scale by going to high production rates.
- Batch-to-batch quality can vary, leading to high production of waste products or off-spec product.
- Recycle and heat recovery are harder, making batch plants less energy efficient and more likely to produce waste by-products.
- Asset utilization is lower for batch plants, as the plant almost inevitably is idle part of the time.
- Batch plants are more labor-intensive, and so the fixed costs of production are much higher for batch plants on a \$/unit mass of product basis.

Choice of continuous versus batch production

Given the higher fixed costs and lower plant utilization of batch processes, batch processing usually only makes sense for products that have high value and are produced in small quantities. Batch plants are commonly used for:

- Food products
- Pharmaceutical products such as drugs, vaccines, and hormones
- Personal care products
- Blended products with multiple grades, such as paints, detergents, etc.
- Specialty chemicals

Even in these sectors, continuous production is favored if the process is well understood, the production volume is large, and the market is competitive.

2.3.3 Effect of reactor conversion and yield on flowsheet structure

It is important to distinguish between conversion and yield. Conversion has to do with reactants; yield with products.

Conversion

Conversion is a measure of the fraction of the reagent that reacts. To optimize reactor design and minimize by-product formation, the conversion of a particular reagent is often less than 100%. If more than one reactant is used, the reagent on which the conversion is based must be specified.

Conversion is defined by the following expression:

$$\begin{aligned}\text{Conversion} &= \frac{\text{amount of reagent consumed}}{\text{amount supplied}} \\ &= \frac{(\text{amount in feed stream}) - (\text{amount in product stream})}{(\text{amount in feed stream})}\end{aligned}\quad (2.2)$$

This definition gives the total conversion of the particular reagent to all products.

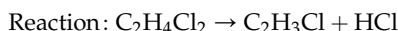
Example 2.1

In the manufacture of vinyl chloride (VC) by the pyrolysis of dichloroethane (DCE), the reactor conversion is limited to 55% to reduce carbon formation, which fouls the reactor tubes.

Calculate the quantity of DCE fed to the reactor to produce 5000 kg/h VC.

Solution

Basis: 5000 kg/h VC (the required quantity).



Molar weights: DCE 99, VC 62.5

$$\text{kmol/h VC produced} = \frac{5000}{62.5} = \underline{\underline{80}}$$

From the stoichiometric equation, 1 kmol DCE produces 1 kmol VC. Let X be DCE feed in kmol/h:

$$\text{Percent conversion} = 55 = \frac{80}{X} \times 100$$

$$X = \frac{80}{0.55} = \underline{\underline{145.5 \text{ kmol/h}}}$$

In this example, the small loss of DCE to carbon and other products has been neglected. All the DCE reacted has been assumed to be converted to VC.

Selectivity

Selectivity is a measure of the efficiency of the reactor in converting reagent to the desired product. It is the fraction of the reacted material that was converted into the desired product. If no by-products are formed, then the selectivity is 100%. If side reactions occur and by-products are formed, then the selectivity decreases. Selectivity is always expressed as the selectivity of feed A for product B and is defined by the equation:

$$\begin{aligned} \text{Selectivity} &= \frac{\text{moles of B formed}}{\text{moles of B that could have been formed if all A reacted to give B}} \\ &= \frac{\text{moles of B formed}}{\text{moles of A consumed} \times \text{stoichiometric factor}} \end{aligned} \quad (2.3)$$

Stoichiometric factor = moles of B produced per mole of A reacted in the reaction stoichiometric equation

Selectivity is usually improved by operating the reactor at low conversion. At high conversion, the reactor has low concentrations of at least one reagent and high concentrations of products, so reactions that form by-products are more likely to occur.

Reagents that are not converted in the reactor can be recovered and recycled. Reagents that become converted to by-products usually cannot be recovered, and the by-products must be purified for sale or else disposed as waste (see Section 8.2.3). The optimum reactor conditions thus usually favor low reactor conversion to give high selectivity for the desired products when all of these costs are taken into account.

Yield

Yield is a measure of the performance of a reactor or plant. Several different definitions of yield are used, and it is important to clearly state the basis of any yield numbers. This is often not done when yields are quoted in the literature, and judgment must be used to decide what was intended.

The yield of product B from feed A is defined by:

$$\text{Yield} = \frac{\text{moles of B formed}}{\text{moles of A supplied} \times \text{stoichiometric factor}} \quad (2.4)$$

For a reactor, the yield is the product of conversion and selectivity:

$$\begin{aligned} \text{Reaction yield} &= \text{Conversion} \times \text{Selectivity} \\ &= \frac{\text{moles A consumed}}{\text{moles A supplied}} \times \frac{\text{moles B formed}}{\text{moles A consumed} \times \text{stoichiometric factor}} \end{aligned} \quad (2.5)$$

With industrial reactors, it is necessary to distinguish between “reaction yield” (chemical yield), which includes only chemical losses to side products, and the overall “reactor yield,” which also includes physical losses, such as losses by evaporation into vent gas.

If the conversion is near 100% it may not be worth separating and recycling the unreacted material; the overall process yield would then include the loss of unreacted material. If the unreacted material is separated and recycled, the overall process yield *taken over the reactor and separation step* would include any physical losses from the separation step.

Plant yield is a measure of the overall performance of the plant and includes all chemical and physical losses.

Plant yield (applied to the complete plant or any stage):

$$= \frac{\text{moles of product produced}}{\text{moles of reagent supplied to the process} \times \text{stoichiometric factor}} \quad (2.6)$$

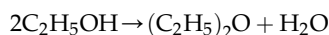
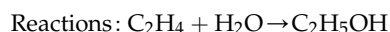
Where more than one reagent is used or product produced, it is essential that the product and reagent to which the yield refers are clearly stated.

The plant yield of B from A is the product of the reactor selectivity of feed A for product B and the separation efficiency (recovery) of each separation step that handles product B or reagent A. As a useful check, the plant yield should be greater than the reactor yield if a separation and feed recycle scheme has been implemented. If the feed recovery and recycle system were 100% efficient, the plant yield would approach the reactor selectivity.

Example 2.2

In the production of ethanol by the hydrolysis of ethylene, diethyl ether is produced as a by-product. A typical feed stream composition is 55% ethylene, 5% inerts, 40% water, and product stream: 52.26% ethylene, 5.49% ethanol, 0.16% ether, 36.81% water, and 5.28% inerts. Calculate the selectivity of ethylene for ethanol and for ether.

Solution



Basis: 100 moles feed (easier calculation than using the product stream)

Note: The flow of inerts will be constant, as they do not react, and it can thus be used to calculate the other flows from the compositions.

Feed stream

ethylene	55 mol
inerts	5 mol
water	40 mol

Product stream

$$\text{ethylene} = \frac{52.26}{5.28} \times 5 = 49.49 \text{ mol}$$

$$\text{ethanol} = \frac{5.49}{5.28} \times 5 = 5.20 \text{ mol}$$

$$\text{ether} = \frac{0.16}{5.28} \times 5 = 0.15 \text{ mol}$$

$$\text{Amount of ethylene reacted} : = 55.0 - 49.49 = 5.51 \text{ mol}$$

$$\text{Selectivity of ethylene for ethanol} : = \frac{5.20}{5.51 \times 1.0} \times 100 = \underline{\underline{94.4\%}}$$

As 1 mol of ethanol is produced per mol of ethylene, the stoichiometric factor is 1.

$$\text{Selectivity of ethylene for ether} : = \frac{0.15}{5.51 \times 0.5} \times 100 = \underline{\underline{5.44\%}}$$

The stoichiometric factor is 0.5, as 2 mol of ethylene produce 1 mol of ether.

Note that the conversion of ethylene to all products is given by:

$$\begin{aligned} \text{Conversion} &= \frac{\text{mols fed} - \text{mols out}}{\text{mols fed}} = \frac{55 - 49.49}{55} \times 100 \\ &= \underline{\underline{10 \text{ percent}}} \end{aligned}$$

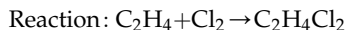
The selectivity based on water could also be calculated but is of no real interest, as water is relatively inexpensive compared with ethylene. Water is clearly fed to the reactor in considerable excess.

The yield of ethanol based on ethylene is:

$$\text{Reaction yield} = \frac{5.20}{55 \times 1.0} \times 100 = \underline{\underline{9.45\%}}$$

Example 2.3

In the chlorination of ethylene to produce DCE, the conversion of ethylene is reported as 99.0%. If 94 mol of DCE are produced per 100 mol of ethylene reacted, calculate the selectivity and the overall yield based on ethylene. The unreacted ethylene is not recovered.

Solution

The stoichiometric factor is 1.

$$\begin{aligned}\text{Selectivity} &= \frac{\text{moles DCE produced}}{\text{moles ethylene reacted} \times 1} \times 100 \\ &= \frac{94}{100} \times 100 = \underline{94\%}\end{aligned}$$

$$\text{Overall yield (including physical losses)} = \frac{\text{moles DCE produced}}{\text{moles ethylene fed} \times 1} \times 100$$

Ninety-nine moles of ethylene are reacted for 100 moles fed, so

$$\text{Overall yield} = \frac{94}{100} \times \frac{99}{100} = \underline{93.1\%}$$

Note that we get the same answer by multiplying the selectivity (0.94) and conversion (0.99).

The principal by-product of this process is trichloroethane.

Effect of conversion, selectivity, and yield on flowsheet structure

Very few processes produce the desired product in stoichiometric yield with no by-products and no equilibrium limitations.

If the desired reaction is limited by equilibrium between the feeds and products, the reaction will not proceed to 100% conversion in the reactor, and it will be necessary to separate the product from unreacted feed components. It will usually be economically attractive to recover the unconverted feeds either to the reactor section or to the feed preparation section of the plant.

In most chemical processes the designer must also address the formation of by-products through unwanted reactions, in which case the selectivity of the feed for the desired product is less than 100%. The presence of nonselective reactions can have a number of undesirable effects on process economics. The most important impact of by-product formation is that by-products represent a loss of potential product. Because feedstock costs are usually the main component of the overall cost of production, low selectivity can have a strong negative impact on process economics. The by-products must be separated from the desired product, causing additional complexity and cost in the separation section. If the by-products have value, they can be purified and sold, but this adds more equipment to the process. If the by-products are not worth recovering, then in some cases they can be recycled within the process and converted back to feed or products. These recycles also add cost and complexity to the process. If the by-products cannot be sold or recycled, they must be disposed of as waste streams. Additional processing steps may be needed to bring the waste stream to a safe condition for discharge or disposal.

Because of the high costs of dealing with by-products, most processes are operated under conditions that maximize reactor selectivity. This often means operating at low conversion and accepting large recycles of feeds. Alternatively, a cheaper feed may be used in excess, so that a high conversion of the more expensive feed can be achieved at high selectivity, as discussed in the following section.

Use of excess reagent

In industrial reactions the components are seldom fed to the reactor in exact stoichiometric proportions. A reagent may be supplied in excess to promote the desired reaction; to maximize the use of an expensive reagent; or to ensure complete reaction of a reagent, as in combustion.

The percentage excess reagent is defined by the following equation:

$$\text{Percent excess} = \frac{\text{quantity supplied} - \text{stoichiometric}}{\text{stoichiometric quantity}} \times 100 \quad (2.7)$$

It is necessary to state clearly to which reagent the excess refers.

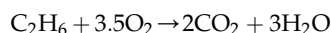
Example 2.4

To ensure complete combustion, 20% excess air is supplied to a furnace burning natural gas. The gas composition (by volume) is methane 95% and ethane 5%.

Calculate the moles of air required per mole of fuel.

Solution

Basis: 100 mol gas, as the analysis is volume percentage.



$$\text{Stoichiometric moles of O}_2 \text{ required} = 95 \times 2 + 5 \times 3.5 = \underline{207.5}$$

$$\text{With 20\% excess, moles of O}_2 \text{ required} = 207.5 \times 1.2 = 249$$

$$\text{Moles of air (21\% O}_2) = 249 \times 100/21 = 1185.7$$

$$\text{Air per mole of fuel} = 1185.7 / 100 = \underline{11.86 \text{ mol}}$$

Sources of conversion, selectivity, and yield data

If there is minimal by-product formation, then the reactor costs (volume, catalyst, heating, etc.) can be traded off against the costs of separating and recycling unconverted reagents to determine the optimal reactor conversion. More frequently, the selectivity of the most expensive feeds for the desired product is less than 100%, and by-product costs must also be taken into account. The reactor optimization then requires a relationship between reactor conversion and selectivity, not just for the main product but for all the by-products that are formed in sufficient quantity to have an impact on process costs.

In simple cases, when the number of by-products is small, it may be possible to develop a mechanistic model of the reaction kinetics that predicts the rate of formation of the main product and by-products. If such a model is fitted to experimental data over a suitably wide range of process conditions, then it can be used for process optimization. The development of reaction kinetics models is discussed in Section 15.3 and is described in most reaction engineering textbooks. See, for example, [Levenspiel \(1998\)](#), [Froment et al. \(2010\)](#), [Fogler \(2016\)](#), and [Rawlings and Eckerdt \(2013\)](#).

In cases where the reaction quickly proceeds to equilibrium, the yields are easily estimated as the equilibrium yields. Under these circumstances, the only possibilities for process optimization are to change the temperature, pressure, or feed composition so as to obtain a different equilibrium mixture. The calculation of reaction equilibrium is easily carried out using commercial process simulation programs, as described in Section 4.5.1.

When the number of components or reactions is too large, or the mechanism is too complex to deduce with statistical certainty, then response surface models can be used instead. Methods for the statistical design of experiments can be applied, reducing the amount of experimental data that must be collected to form a statistically meaningful correlation of selectivity and yield to the main process parameters. See [Montgomery \(2019\)](#) for a good introduction to the statistical design of experiments.

In the early stages of design, the design engineer will often have neither a response surface nor a detailed mechanistic model of the reaction kinetics. Few companies are prepared to dedicate a laboratory or pilot plant and the necessary staff to collecting reaction kinetics data until management has been satisfied that the process under investigation is economically attractive. A design is thus needed before the necessary selectivity and yield data set has been collected. Under such circumstances, the design engineer must select the optimal reactor conditions from whatever data are available. This initial estimate of reactor yield may come from a few data points collected by a chemist or taken from a patent or research paper. The use of data from patents is discussed in Section 2.4.1.

For the purposes of completing a design, only a single estimate of reactor yield is needed. Additional yield data taken over a broader range of process conditions give the designer greater ability to properly optimize the design. In process synthesis projects, one purpose of the design may be to set yield targets for a research team, as described in Section 2.6.1.

2.3.4 Recycles and purges

Processes in which a flow stream is returned (recycled) to an earlier stage in the processing sequence are common. If the conversion of a valuable reagent in a reaction process is appreciably less than 100%, the unreacted material is usually separated and recycled.

Separation processes can also be a source of recycles. The return of reflux to the top of a distillation column is an example of a recycle process in which there is no reaction.

The presence of recycle streams makes the calculation of process material and energy balances more difficult. Without recycle, the material balances on a series of processing steps that can be carried out sequentially, taking each unit in turn, with the calculated flows out of one unit becoming the feeds to the next. If a recycle stream is present, then at the point where the recycle is returned, the flow will not be known, as it will depend on downstream flows not yet calculated. Without knowing the recycle flow, the sequence of calculations cannot be continued to the point where the recycle flow can be determined.

Two approaches to the solution of recycle problems are possible:

1. **The cut and try ("tear") method.** The recycle stream flows can be estimated and the calculations continued to the point where the recycle is calculated. The estimated flows are then compared with those calculated, and a better estimate is made. The procedure is continued until the difference between the estimated and the calculated flows is within an acceptable tolerance.
2. **The formal algebraic method.** The presence of recycle implies that some of the mass balance equations must be solved simultaneously. The equations are set up with the recycle flows as unknowns and solved using standard methods for the solution of simultaneous equations.

With simple problems that have only one or two recycle loops, the calculation can often be simplified by the careful selection of the basis of calculation and the system boundaries. This is illustrated in Example 2.5.

The solution of more complex material balance problems involving several recycle loops is discussed in Chapter 4.

Example 2.5

The block diagram in Fig. 2.12 shows the main steps in the balanced process for the production of vinyl chloride from ethylene. Each block represents a reactor and several other processing units. The main reactions are:

Block A, chlorination

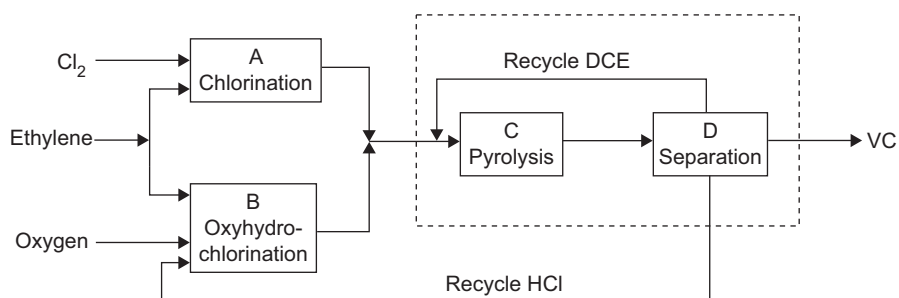
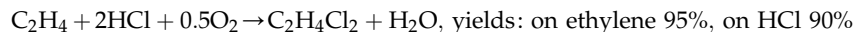


FIG. 2.12 Block flow diagram of balanced process for vinyl chloride (VC).

Block B, oxyhydrochlorination**Block C, pyrolysis**

The HCl from the pyrolysis step is recycled to the oxyhydrochlorination step. The flow of ethylene to the chlorination and oxyhydrochlorination reactors is adjusted so that the production of HCl is in balance with the requirement. The conversion in the pyrolysis reactor is limited to 55%, and the unreacted DCE is separated and recycled.

Using the yields given, and neglecting any other losses, calculate the flow of ethylene to each reactor and the flow of DCE to the pyrolysis reactor, for a production rate of 12,500 kg/h VC.

Solution

Molecular weights: vinyl chloride 62.5, DCE 99.0, HCl 36.5.

$$\text{VC per hour} = \frac{12,500}{62.5} = 200 \text{ kmol/h}$$

Draw a system boundary around each block, enclosing the separation section (block D) and the DCE recycle within the boundary of step C, as shown in Fig. 2.12.

Let the flow of ethylene to block A be X and to block B be Y and the HCl recycle be Z .

Then the total moles of DCE produced = $0.98X + 0.95Y$, allowing for the yields, and the moles of HCl produced in block C

$$= (0.98X + 0.95Y)0.995 = Z \quad (a)$$

Consider the flows to and from block B. The yield of DCE based on HCl is 90%, so the moles of DCE produced:

$$= 0.5 \times 0.90Z$$

Note: The stoichiometric factor is 0.5 (2 mol HCl per mol DCE).

The yield of DCE based on ethylene is 95%, so

$$0.5 \times 0.9Z = 0.95Y$$

$$Z = 0.95 \times 2Y/0.9$$

Substituting for Z into equation (a) gives

$$Y = (0.98X + 0.95Y)0.995 \times \frac{0.9}{2 \times 0.95} \quad (b)$$

$$Y = 0.837X$$

Turning to block C, total VC produced = $0.99 \times$ total DCE, so

$$0.99(0.98X + 0.95Y) = 200 \text{ kmol/h}$$

Substituting for Y from equation (b) gives $X = \underline{113.8 \text{ kmol/h}}$

$$\text{and } Y = 0.837 \times 113.8 = \underline{95.3 \text{ kmol/h}}$$

HCl recycle from equation (a)

$$Z = (0.98 \times 113.8 + 0.95 \times 95.3)0.995 = \underline{201.1 \text{ kmol/h}}$$

$$\text{Note: Overall yield on ethylene} = \frac{200}{(113.8 + 95.3)} \times 100 = \underline{96 \text{ percent}}$$

The total flow of DCE from blocks A and B is $\frac{200}{0.99} = 202 \text{ kmol/h}$, but this does not include the recycle. Because the conversion is 55%, if the recycle flow is R , then $202 / (202 + R) = 0.55$; hence the total flow of DCE to the pyrolysis reactor is $202 + R = 202 / 0.55 = 367.3 \text{ kmol/h}$

Purge

It is usually necessary to bleed off a portion of a recycle stream to prevent the buildup of unwanted material. For example, if a reactor feed contains inert components or by-products that are not separated from the recycle stream in the separation units, these inerts would accumulate in the recycle stream until the stream eventually consisted entirely of inerts. Some portion of the stream must be purged to keep the inert level within acceptable limits. A continuous purge would normally be used. Under steady-state conditions:

$$\text{Loss of inert in the purge} = \text{Rate of feed of inerts into the system}$$

The concentration of any component in the purge stream is the same as that in the recycle stream at the point where the purge is taken off. So the required purge rate can be determined from the following relationship:

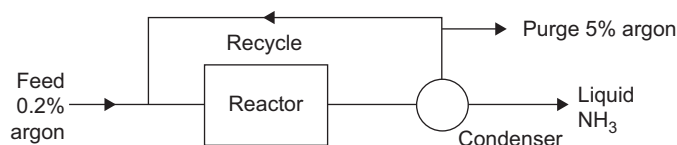
$$\begin{aligned} & [\text{Feed stream flow} - \text{rate}] \times [\text{Feed stream inert concentration}] \\ &= [\text{Purge stream flow} - \text{rate}] \times [\text{Specified(desired) recycle inert concentration}] \end{aligned}$$

Example 2.6

In the production of ammonia from hydrogen and nitrogen, the conversion, based on either raw material, is limited to 15%. The ammonia produced is condensed from the reactor (converter) product stream, and the unreacted material is recycled. If the feed contains 0.2% argon (from the nitrogen separation process), calculate the purge rate required to hold the argon in the recycle stream below 5.0%. Percentages are by volume.

Solution

Basis: 100 moles feed (purge rate will be expressed as moles per 100 mol feed, as the production rate is not given).
Process diagram:



Volume percentages are taken as equivalent to mol%.

Argon entering system with feed = $100 \times 0.2/100 = 0.2$ mol.

Let purge rate per 100 mol feed be F .

Argon leaving system in purge = $F \times 5/100 = 0.05 F$.

At the steady state, argon leaving = argon entering

$$0.05F = 0.2$$

$$F = \frac{0.2}{0.05} = \underline{4}$$

Purge required: 4 mol per 100 mol feed.

Bypass

A flow stream may be divided and some part diverted (bypassed) around some units. This procedure is often used to control stream composition or temperature.

Material balance calculations on processes with bypass streams are similar to those involving recycle, except that the stream is fed forward instead of backward. This usually makes the calculations easier than with recycle.

2.4 Selection, modification, and improvement of commercially proven processes

Engineers in industry do not usually design a new process from scratch if a commercially proven alternative is available. Companies usually seek to avoid the extra costs and risks inherent in technology commercialization. New molecules are usually made using adaptations of processes that have been shown to work for similar compounds. Even when a brand-new process is contemplated, the design team will usually also prepare a conventional design for comparison.

The use of a proven basic flow scheme does not eliminate innovation from the design. Several alternative designs may already be in commercial practice, each optimized around different feeds, catalysts, or reactor concepts. The design team must evaluate the different designs and optimize each to the local design basis to select the best. The commercial processes may need modification to make the desired product or by-products or to process an unusual feed material. It may be possible to improve the existing technology by substituting one or more unit operations, by using better catalysts or enzymes, by deploying improved separation or reactor technology, or by using different solvents to reduce environmental impact. The scale of production may also cause changes to the flowsheet; for example, if a large new plant requires reactors or separation columns to be constructed in parallel trains.

This section discusses factors a design team should consider when developing a flowsheet based on a commercially proven technology. The special case of developing a flowsheet for the revamp of an existing plant is treated in Section 2.5.

2.4.1 Sources of information on manufacturing processes

This section gives a brief overview of sources of information on commercial processes that can be found in the open literature.

The chemical process industries are competitive, and the information that is published on commercial processes is restricted. The articles on particular processes published in the technical literature and in textbooks invariably give only a superficial account of the chemistry and unit operations used. They lack the detailed information on reaction kinetics, process conditions, equipment parameters, and physical properties that is needed for process design. The information that can be found in the general literature is, however, useful in the early stages of a project, when searching for possible process routes. It is often sufficient for a flowsheet of the process to be drawn up and a rough estimate of the capital and production costs made.

The most comprehensive collection of information on manufacturing processes is probably the *Encyclopedia of Chemical Technology* edited by Kirk and Othmer (2001), which covers the whole range of chemical and associated products. An abridged version of the Kirk–Othmer encyclopedia is also available in paperback (Kirk, 2007). The latest version of the Kirk–Othmer encyclopedia is available through the Wiley online library at <http://onlinelibrary.wiley.com>. Another encyclopedia covering manufacturing processes is that edited by McKetta (2001). Several books have also been published that give brief summaries of the production processes used for the commercial chemicals and chemical products. The best known of these is probably Shreve's book on the chemical process industries, now updated by Austin and Basta (1998). Comyns (2019) lists named chemical manufacturing processes, with references.

The extensive German reference work on industrial processes, *Ullman's Encyclopedia of Industrial Technology*, is now available in an English translation, Ley, C., (Ed.) (2002).

Specialized texts have been published on some of the more important bulk industrial chemicals, such as that by Miller (1969) on ethylene and its derivatives; these are too numerous to list but should be available in the larger reference libraries and can be found by reference to the library catalogue. Meyers (2016) gives a good introduction to the processes used in oil refining. Kohl and Nielsen (1997) provide an excellent overview of the processes used for gas treating and sulfur recovery.

Many of the references cited here are available in electronic format from Knovel. Most companies and universities have Knovel subscriptions. Access to Knovel is also available to members of professional societies such as the American Institute of Chemical Engineers (AIChE).

Books quickly become outdated, and many of the processes described are obsolete, or at best obsolescent. More up-to-date descriptions of the processes in current use can be found in the technical journals. The journal *Hydrocarbon Processing* publishes an annual review of petrochemical processes, which was entitled *Petrochemical Developments* and is now called *Petrochemicals Notebook*; this gives flow diagrams and brief process descriptions of new process developments.

Patents

Patents can be a useful source of information, but some care is needed in extracting information from them. To obtain a patent, an inventor is legally obliged to disclose the best mode of practice of the invention; failure to do so could render the patent invalid if it were contested. Most patents therefore include one or more examples illustrating how the invention is practiced and differentiating it from the prior art. The examples given in a patent often give an indication of the process conditions used, though they are frequently examples of laboratory preparations rather than of the full-scale manufacturing process. Many process patents also include examples based on computer simulations, in which case the data should be viewed with suspicion. When using data from patents, it is important to carefully read the section that describes the experimental procedure to be sure that the experiments were run under appropriate conditions.

A patent gives its owner the right to sue anyone who practices the technology described in the patent claims without a license from the patent owner. Patent attorneys generally try to write patents to claim broad ranges of process conditions so as to maximize the range of validity and make it hard for competitors to avoid the patent by making a slight change in temperature, pressure, or other process parameters. Very often, a patent will say something along the lines of “the reaction is carried out at a temperature in the range 50° to 500 °C, more preferably in the range 100° to 300 °C and most preferably in the range 200° to 250 °C.” It is usually possible to use engineering judgment to determine the optimal conditions from such ranges. The best conditions will usually be at or near the upper or lower end of the narrowest defined range. The examples in the patent will often indicate the best operating point.

Patents can be downloaded for free from the website of the U.S. Patent Office, www.uspto.gov. The U.S. PTO website also has limited search capability. The entire U.S. PTO collection is also available at www.google.com/patents. Most large companies subscribe to more sophisticated patent search services such as Derwent Innovation (www.derwentinnovation.com), PatBase (www.patbase.com), or GetthePatent (www.getthepatent.com).

Several guides have been written to help engineers understand the use of patents for the protection of inventions and as sources of information; such as those by Auger (1992) and Gordon et al. (2017).

Consultants

Engineers in industry often hire specialist consulting firms to prepare analyses of commercial technology. Consultants can be used to provide an impartial assessment of a competitor's or vendor's process. Some consulting firms such as SRI and Nexant regularly publish assessments of the technology available for making different chemicals. These assessments are based on flowsheets and design models that the consultants have developed from information that they gathered from the literature and from direct contact with the technology suppliers.

Some caution is needed when working with consultants. The client must carry out due diligence to ensure that the consultant is truly impartial and does not bias their analysis. The client should also cross-check the information provided by the consultant against recent patents and publications to ensure that the consultant is working from the latest information.

Vendors

Technology suppliers and contractors will sometimes make design information available to a client in the hope of securing a sale. If a project team requires information to make a technology selection, the technology vendors may be willing to supply edited PFDs (for example, with stream flows blanked out or with some information missing), reactor yields, or even designs from a similar plant at a smaller production scale. More detailed information is usually provided by vendors when bidding on a contract for a project that has a high likelihood of going forward.

2.4.2 Factors considered in process selection

Once the design team has assembled information on the alternative commercial processes, they will usually need to carry out substantial customization and optimization of the designs before a selection can be made.

The information given in the open literature is usually restricted to block flow diagrams and (occasionally) reactor yields. The first step is usually to complete a full PFD and mass and energy balance of the process. These can be used for preliminary sizing and costing of the main process equipment to obtain an estimate of the required capital investment, as described in Chapter 7. The feed and product flow rates and energy consumption can be used to estimate the costs of production, as described in Chapter 8. The economic analysis methods introduced in Chapter 9

can then be applied to determine the overall project economics and choose which design gives the best overall economic performance according to the criteria established by the company.

If one process flowsheet has a particular cost advantage, this will usually become clear in the economic analysis. Factors such as feedstock or fixed-cost advantages that can be very important in selecting *between projects* are usually less important when selecting *between flowsheets* within a given project. The selection between flowsheets is usually influenced more by process yields, energy consumption, and capital requirements, and hence is sensitive to catalyst, organism or enzyme performance, and process design and optimization.

In an industrial context, technology vendors or engineering, procurement, and construction (EPC) contractors will often supply detailed PFDs and material and energy balances to a client when invited to bid on a project. Some diligence is always needed in checking the information in proposals and verifying performance claims against the actual performance of recent plants built by the vendor.

Although an economic analysis is always carried out, it is usually not the sole criterion for technology selection. Some other important factors are described next.

Freedom to practice

Freedom to practice is a legal concept that arises from patent law. If a process, catalyst, enzyme, genetically modified organism, or chemical route is patented, it can only legally be used under license from the patent holder. If another company were to use the technology without a license, they would be infringing on the patent, and the patent holder could sue to stop the use and demand damages.

Determination of freedom to practice usually requires the expert advice of patent attorneys. In rapidly evolving new fields it can be difficult to assess, as patent applications typically are not published until 1 to 2 years after they are filed, so a decision may be made to proceed with building a plant before the intellectual property space can be completely mapped. Another complication is that competing technology vendors may have overlapping patents or patents that appear to block features of each other's designs.

All patents are only valid for a fixed term; in the United States at the time of writing this is 20 years from the date the patent was filed. When a patent has expired, anyone is free to practice the technology. Care must still be taken to check that the original technology developer has not made more recent improvements that are still under patent protection. Many conventional processes are no longer protected by patents and can be bought from EPC companies without paying a royalty or license fee.

When a customer licenses technology from a vendor, the vendor will usually indemnify the customer against patent infringement. This means the technology vendor asserts that they have ownership of the technology and freedom to practice and that they will help the customer fight any patent infringement suits brought by their competitors. Technology vendors sometimes minimize the potential for such lawsuits by forming cross-licensing agreements.

Safety and environmental performance

All commercially practiced technologies should meet or exceed the minimum legally acceptable safety standards, but some older processes may no longer have acceptable environmental performance.

An economic analysis will usually not distinguish whether one process is safer or more environmentally acceptable than another. The methods described in [Chapters 10 and 11](#) can be used to make an assessment of process safety and environmental impact.

When assessing commercial technology, visits to existing sites and reviews of their safety and operational performance can also be helpful.

Government and international restrictions

Governments sometimes place restrictions on companies that can influence technology selection. It is fairly common for nationally owned companies in developing countries to be required to maximize use of indigenous technology, equipment, and parts so as to stimulate the development of local engineering industries and reduce hard currency outflows. This may lead a company to develop its own version of an older technology rather than working with a technology vendor or major international company that can supply the latest technology.

International sanctions can also play an important role in process flowsheet selection. Sanctions can disqualify some companies from offering to supply technology and reduce the set of options available. Sanctions can also restrict the availability of feedstocks. During the 1970s and 1980s, South African companies developed many processes for making chemicals from coal in response to the international sanctions aimed at ending apartheid that restricted their ability to purchase crude oil.

Experience and reliability

One of the critical factors in selecting a commercially proven technology is the extent and diversity of operating experience that has been established. If a process has been widely adopted and proven in many locations by different operating companies, then it is likely to be easy to apply in a new plant. A technology that has only been built once or twice may still experience “teething troubles” and be more difficult to implement.

As more operating experience is gained, the company also gains a better understanding of the reliability of the process. If a particular section of the flowsheet or piece of equipment is known to cause reliability problems, this may create a need to modify the equipment design or even make changes to the flowsheet.

2.4.3 Modification and improvement of established processes

All designs evolve over time. Engineers make modifications to improve process economics, safety, reliability, and environmental impact. Most changes will be minor, such as addition of instrumentation or substitution of equipment; however, significant changes in the flowsheet are sometimes needed.

Modifications that are made to an existing commercial plant are known as *revamp designs* and are addressed in Section 2.5. This section describes techniques for modifying an established process for use in a new plant.

Modifications to improve process economics

Improvements in process economics usually come from reduced capital investment or improved cost of production. Designers seeking improvements in process economics usually start by completing a PFD of the existing design and determining the current estimated capital investment and cost of production (see [Chapters 7, 8, and 9](#)). The following tactics can then be applied:

- **Improve reactor selectivity and process yield.** Feedstock costs are usually more than 80% of the cost of production, so improving yields gives the biggest impact on process economics. Improved yields usually require the development of more selective catalysts, enzymes, or organisms, or a more effective reactor design, but sometimes a more efficient separation scheme or better purification of a feed or recycle will also improve yield.
- **Improve process energy efficiency.** Energy costs are usually next largest after feedstock costs for chemicals produced on a large scale. Energy costs can be reduced by improving process energy efficiency. Several different approaches to improving process energy use are described in [Chapter 3](#).
- **Improve process fixed costs.** Fixed costs are usually second to feedstock costs in small-scale processes used for fine chemicals and pharmaceuticals manufacture. Fixed costs are described in Section 8.5. Fixed costs can be reduced by making the process more continuous and less labor-intensive and by increasing the plant attainment of batch processes.
- **Reduce capital investment.** Design engineers look for pieces of equipment that can be combined or eliminated to reduce capital cost. In batch plants, this is often done by carrying out several steps in the same piece of equipment. For example, the feed can be charged to a reactor, heated in the reactor, reacted, cooled down, and the product crystallized before pumping out the product as a slurry and repeating the process.
- **Reduce working capital.** Working capital is described in Section 9.2.3. Working capital can be reduced by decreasing inventories of raw materials, work in progress, and consumables. Making a process more continuous and using fewer different solvents in a process both lead to a reduction in working capital.

It can be seen that some of these suggestions contradict each other; for example, “make batch plants more continuous” but “carry out more operations in the same piece of equipment.” Lists of design guidelines (known as *heuristic rules*) often contain apparent contradictions. The designer must either choose which rule is most appropriate to the case under consideration using experience and judgment or else carry out a full design and costing of both alternatives. Heuristic rules are discussed further in the context of process synthesis in Section 2.6.4.

Modifications to improve plant safety

Plants can be made more inherently safe by reducing inventories of hazardous materials by making vessels and other plant equipment smaller; substituting less hazardous materials for feeds, solvents, and intermediates; eliminating explosive mixtures and exothermic reactions; eliminating use of operations that are open to the atmosphere; minimizing worker exposure to chemicals; and other methods discussed in [Chapter 10](#).

Improvements in the safety of a design can be quantified using the methods for risk assessment described in Section 10.8.

Modifications to improve plant reliability

When a plant has been in operation for a few years the operators will have a good idea of which plant sections or pieces of equipment lead to the most operational problems, require the most maintenance, and cause the most unplanned shutdowns.

Reliability problems are usually caused by equipment failures. The most common problems are usually experienced with solids-handling equipment, rotating equipment such as pumps and compressors, heat exchangers that are prone to fouling, and instruments and valves. Sometimes, specification of a more reliable piece of equipment can solve a process reliability problem. More often, a flowsheet change is needed, such as designing with two or more pieces of equipment in parallel so that the plant can continue to operate while one is taken offline for repair or cleaning. This approach is commonly applied for pumps, which are relatively cheap and prone to stalling in operation.

Corrosion, erosion, and plugging caused by corrosion products can be major contributors to poor reliability. Methods to address corrosion in design are described in [Chapter 6](#).

Modifications to improve environmental impact

Many conventional processes were first designed over 40 years ago, when different environmental laws and standards applied. Existing plants may have been modified by addition of end-of-pipe systems for reducing environmental impact; however, changes in the process flowsheet can sometimes achieve the same or better environmental performance at lower cost.

Modifications that are typically used to improve environmental impact include:

- Use of new catalysts, enzymes, or organisms that have better selectivity for the desired product and consequently lead to less waste formation.
- Optimization of reactor design to give better mixing or heat transfer and hence improve reactor selectivity and reduce by-product formation.
- Elimination of solvents or other consumables that become degraded to waste products by the process.
- Elimination of materials that have high environmental impact, such as halogenated solvents, mercury, endocrine disruptors, and compounds that persist in the environment.
- Adoption of closed-loop recirculating gas systems instead of once-through gas flow, thus reducing volatile organic compound (VOC) emissions. For example, [Fig. 2.13\(a\)](#) shows a once-through dryer in which the drying gas is vented or sent to a flare, potentially leading to VOC emissions. [Fig. 2.13\(b\)](#) shows an alternative design in which a blower is used to circulate the gas. The hot gas leaving the dryer is cooled to allow solvent to be condensed and recovered. There is much less potential for VOC emissions in the closed-loop design, and the consumption of solvent is also reduced.
- Substitution of chemicals with materials that have reduced environmental impact. For example, the cheapest way to neutralize waste sulfuric acid is to react it with lime (CaO) to form gypsum (CaSO_4), which is inert and can be sent to a landfill. Instead, if ammonia is used to neutralize the acid, the product will be ammonium sulfate, which can be used as a fertilizer.

Methods for analyzing and reducing the environmental impact of a process are described in [Chapter 11](#).

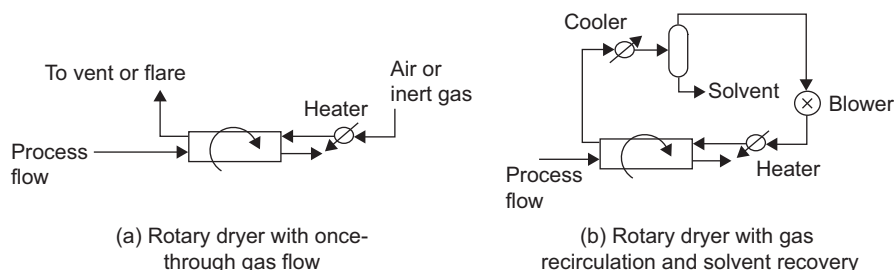


FIG. 2.13 Dryer gas circulation designs.

2.5 Revamps of existing plants

Flowsheet development for plant revamps is a specialized subject in its own right. Revamp design is rarely taught in universities, as revamp studies require access to an operating plant and the data it produces.

Revamps generally fall into two categories. *Debottlenecking* projects are carried out to increase the production rate of a plant while making the same product. *Retrofit* projects are carried out to change the design of a plant to handle different feeds; make different products; exploit better reactor, catalyst, or separation technology; or improve plant safety or environmental impact in response to new regulatory requirements.

2.5.1 Flowsheet development in revamp projects

Fig. 2.14 gives an overall work process for developing a revamp design flowsheet. One of the critical requirements of a revamp project is always to minimize project cost by maximizing reuse of existing equipment. The revamped flowsheet therefore always requires compromises between desired objectives and what can be obtained with the equipment available.

Many features of a revamped flowsheet will be different from the flowsheet of a corresponding new plant. For example, in a revamp it may make sense to add a second distillation column in parallel to an existing column rather than tearing down the existing column and building a new larger one. The use of two small columns in parallel would not be contemplated in a new design. The development of a revamped flowsheet thus requires a lot of information on the performance of existing equipment so that the equipment can be rerated or modified for a role in the new flowsheet. When the existing equipment cannot be upgraded, the designer must find the cheapest method to add new capacity or augment the existing capacity.

Once the revamped flowsheet has been completed, the designers can assess the costs of the new components that must be added. The cost of revamping a plant should always be compared with the cost of building a new plant from scratch. The revamp will usually be a cheaper method for adding small increments of capacity, but for larger capacity increases, a new unit will become more attractive.

The steps in developing a revamped process flowsheet are described in the following sections. Although there is a great deal of retrofit and revamp activity in the chemical industry, particularly in regions such as the United States

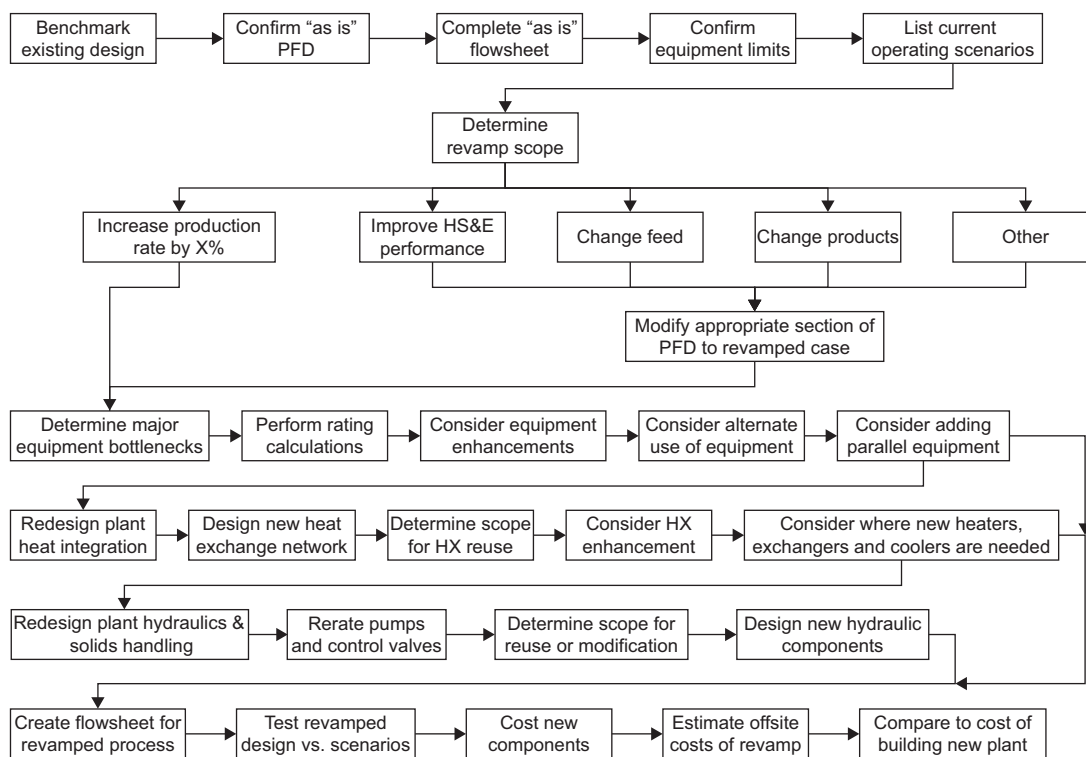


FIG. 2.14 Steps in revamp design.

and European Union, where the industry has been long-established, the authors are not aware of any comprehensive reference works on the subject. The books by Briggs et al. (1997) and Douglas (1988) contain short sections on revamp design. Process revamps for energy savings are discussed in detail by Zhu (2014).

2.5.2 Major equipment debottlenecking

In a revamp design, the capacity of existing equipment determines whether additional equipment must be added in series or parallel, and hence plays a major role in determining the revamped design flowsheet.

Most major equipment is initially specified with a design factor or margin of 10% to 20%; see Section 1.6. This overdesign allows for errors in the design data and methods, but also creates some room for potential expansion of capacity. When a plant is considered for revamp, some of the equipment may still be operating below its full capacity.

The general procedure for equipment debottlenecking follows the steps shown in Fig. 2.14. Once a mass and energy balance has been established for the existing plant, a simulation model of the equipment can be built. The model can then be tested under the proposed new process conditions to determine if the equipment is fit for the new service. For equipment that is difficult to model (for example, centrifuges, fired heaters, and dryers) a specialist or the original equipment vendor may need to be consulted. After establishing the maximum capacity that the equipment can attain while maintaining specifications, modifications to improve capacity can be considered. If it is not possible to satisfy the desired process duty with modifications to the existing equipment, then the cheapest means of adding capacity must be established. This may include complete replacement of the original equipment, with reuse of the original equipment elsewhere in the process.

Some specific examples of techniques for equipment debottlenecking are given next. Revamp of heat transfer equipment is discussed in Section 2.5.3, and revamp of hydraulic and solids handling equipment is described in Section 2.5.4.

Reactor debottlenecking

Reactors are designed with a specified residence time that has been determined to give a desired conversion. For reactors that use a fixed bed of catalyst, this is usually expressed instead as a space velocity:

$$\tau = \frac{V}{v} \quad (2.8)$$

$$SV = \frac{v}{V_{cat}} \quad (2.9)$$

where: τ = residence time
 SV = space velocity
 V = reactor volume
 V_{cat} = fixed bed catalyst volume
 v = volumetric flow rate

Space velocity is usually given on an hourly basis and defined on the basis of gas phase flow (GHSV = gas hourly space velocity), liquid phase flow (LHSV = liquid hourly space velocity), or total mass flow (WHSV = weight hourly space velocity = kg/h feed per kg of catalyst inventory). Any consistent set of units can be used for residence time and space velocity.

Equations 2.8 and 2.9 clearly show that an increase in flow rate must lead to a proportional change in either volume, residence time, or space velocity. Adding volume usually requires building additional reactors, so unless the reactors are very inexpensive, the revamp design will focus on ways to reduce residence time or increase space velocity while trying to maintain the same conversion, if possible, so as to minimize the impact on the separation and recycle sections of the plant.

An increase in space velocity or decrease in residence time can be obtained by increasing temperature; backing off on conversion; reducing the concentration of diluents; or using a more active catalyst, enzyme, or organism. Increasing temperature and reducing diluents or solvents (if any are present) will generally lead to worse selectivity and increased cost elsewhere in the process. Many fixed-bed catalytic processes are operated on a temperature cycle, where the reactor temperature is slowly increased over a 1- to 10-year period to compensate for catalyst deactivation and the catalyst is then replaced at the end of the cycle. In such cases, raising the temperature shortens the catalyst run length and requires more frequent plant shutdown. Reducing the reactor conversion also creates additional cost

elsewhere in the process as the amount of unreacted feed recycle is increased. Improving the catalyst performance is often the least expensive way to boost capacity, and the availability of new catalysts often sets the scope for revamp projects.

An additional problem with revamping fixed-bed catalytic reactors is the effect of reactor pressure drop. The pressure drop across a packed bed is proportional to the flow rate squared, so pressure drop increases rapidly as flow rate is increased. Approaches that can be taken to reduce reactor pressure drop include rearranging series reactors into parallel flow (Fig. 2.15) and converting down-flow reactors to radial flow (Fig. 2.16). (A more detailed drawing of a radial flow reactor is given in Chapter 15; see Fig. 15.29). With packed beds that are in up-flow, care must be taken to avoid fluidizing the catalyst in the revamped design. If the up-flow velocity is close to the minimum fluidization velocity, then the reactor should be converted to down-flow or replaced with a larger reactor. Sizing of packed bed reactors is discussed in more detail in Section 15.7.3.

When it is necessary to add reactor capacity, a technique that is widely used is to add a pre-reactor to the existing reactor sequence, as in Fig. 2.17. Because a pre-reactor typically runs at low conversion, it can be operated under conditions that would not normally be good for selectivity; for example, at higher temperature or with less solvent or diluent. This makes the pre-reactor more volume-efficient than the existing reactor sequence, without compromising on overall selectivity.

When a detailed model of the reaction kinetics, including side reactions, is available, more complex reactor networks can be designed that give improved selectivity and yield of desired products. A revamp project can then add reactor capacity to bring the reactor section performance closer to that of the ideal reactor network. The design of reactors and reactor networks is discussed in more detail in Chapter 15.

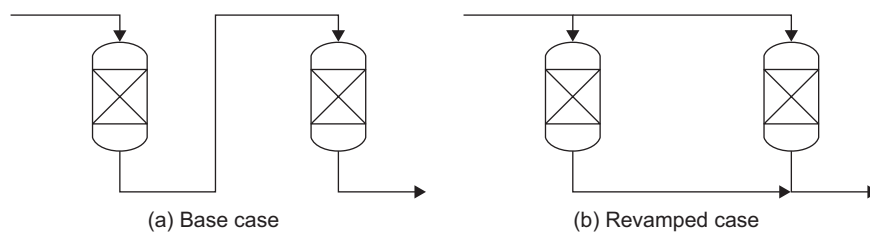


FIG. 2.15 Series to parallel reactor revamp.

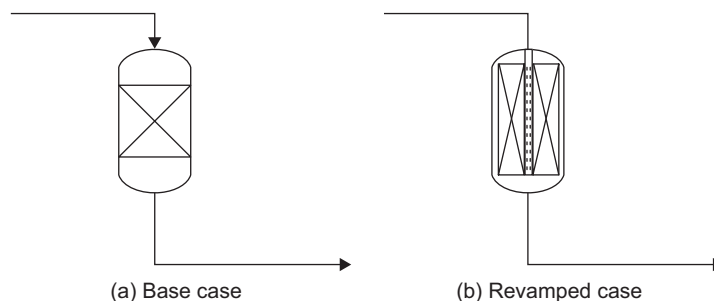


FIG. 2.16 Down-flow to radial-flow reactor revamp.

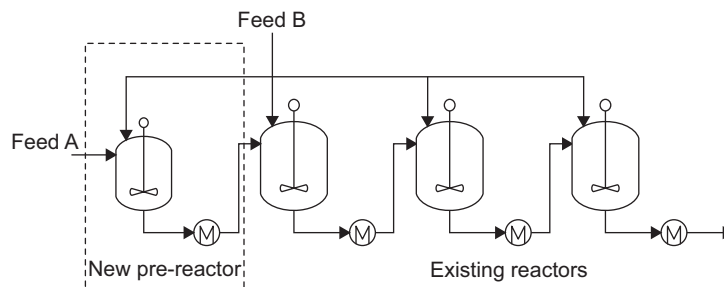


FIG. 2.17 Reactor revamp using a pre-reactor.

Separation column debottlenecking

The capacity of separation columns is usually limited by column hydraulics; see Section 17.13. If the feed flow rate increases, the vapor rate in the column increases proportionately, and at some point the column will flood and become inoperable. Two approaches can be taken to obtain more capacity:

1. Increase the open area for vapor flow to delay the onset of flooding.
2. Increase the number of stages in the column by using high-efficiency trays or trays that allow a closer tray spacing, so that the reflux ratio, and hence vapor rate, can be reduced.

Both of these methods are used by separation tray and packing vendors, and there are many proprietary designs of high-efficiency, high-capacity trays and packing on the market. When revamping a column, the common practice is to contact the tray and packing vendors, who will then supply an estimate of how many trays must be replaced to achieve a desired capacity. It is often not necessary to re-tray the entire column. The detailed design of distillation columns is discussed in [Chapter 17](#).

When it is necessary to add capacity to a distillation column and the use of high-capacity trays is not sufficient, a prefractionator scheme is sometimes used, as illustrated in [Fig. 2.18](#). The prefractionator makes a preliminary separation of the feed that reduces the reflux requirements of the main column. A revamped prefractionator is usually provided with its own reboiler and condenser to avoid increasing the load on the main column reboiler and condenser.

Another common tactic in debottlenecking separation sections is to reuse the existing distillation columns in a different application. If a plant has three or more columns, then a revamp can be carried out by building one new column to replace the largest of the existing columns, revamping the largest old column to replace the second largest, etc. In this context, largest refers to the largest diameter, which governs vapor rate and capacity. If the columns do not have sufficient height for the new application, they can be re-trayed with high-efficiency internals or combined in series. In some cases, particularly for relatively short low-pressure columns, it may even make sense to add height to a column by welding on a new top section.

2.5.3 Revamp of heat exchange networks

The heat exchangers, heaters, and coolers of a plant will cause many of the bottlenecks to plant expansion. When a plant is revamped to a new purpose, such as changed feed or products, the existing heat recovery system will no longer be optimal and may no longer make good sense. The common practice in major revamps is to complete the revamp design of the other major equipment first, then address the heaters, coolers, and exchangers subsequently, as shown in [Fig. 2.14](#).

The design of heat recovery systems and heat-exchanger networks is covered in [Chapter 3](#). [Chapter 19](#) addresses the detailed design of heat exchangers, heaters, and coolers. Revamp of a complex heat-exchange system should

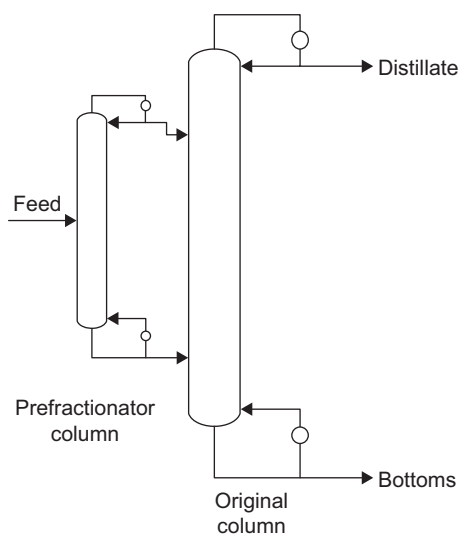


FIG. 2.18 Distillation column revamp using a prefractionator.

always be treated as a network problem rather than by revamping each item individually, as the lowest-cost solution will always be that obtained by optimizing the system as a whole.

The revamp of heat-exchanger networks has been the subject of much research, and very effective techniques and software for heat-exchange network revamp have been developed. The *network pinch* method developed by Asante and Zhu (1997) is now the most widely used method in the industry. This method has been automated (Zhu and Asante, 1999) and is used by most of the companies that offer heat integration consulting services. Smith (2016) gives a concise overview of the network pinch approach; see also Zhu (2014).

For simple processes with only a few heaters and coolers, the tactics described in the following sections can be used.

Heat exchangers

For a heat exchanger:

$$Q = UA\Delta T_m \quad (2.10)$$

and

$$Q = m_i C_{p,i} \Delta T_i \quad (2.11)$$

where : Q = heat transferred per unit time, W

U = the overall heat-transfer coefficient, W/m^2K

A = heat-transfer area, m^2

ΔT_m = the mean temperature difference, the temperature driving force, $^{\circ}C$

m_i = the mass flow rate of stream i , kg/s

$C_{p,i}$ = the specific heat capacity of stream i , J/kgK

ΔT_i = the change in temperature of stream i for a stream that undergoes only sensible heat changes, $^{\circ}C$

Increasing the flow rate increases the required duty and therefore requires an increase in heat transfer coefficient, area, or effective temperature difference.

The correlations that are used to predict heat transfer coefficients for sensible heat transfer are usually proportional to $Re^{0.8}$, where Re is the Reynolds number, which is proportional to flow rate. Hence, the process-side transfer coefficient increases almost in ratio to the increase in flow unless the exchanger is boiling or condensing the process stream. For heaters and coolers, it may therefore be possible to reuse the exchanger in the same service if an increase in utility-side heat transfer coefficient or a change in utility temperature can make up the rest of the required duty. This is topic explored in Example 2.7.

One of the first and most important steps in revamping a heat exchange system is to benchmark the current system and estimate the heat transfer coefficients that are currently being obtained. If these are substantially lower than expected from the original design of the process, this may indicate fouling, plugging, or other problems that should be addressed during the revamp.

There are several proprietary methods for enhancing the performance of tubular exchangers. Tube inserts such as hiTRAN, TURBOTAL, and Spirelf can be used to increase turbulence and tube-side heat transfer coefficient. Low-fin tubes can be used to increase shell-side effective area (Wolverine, 1984; see also Section 19.14). Reboiling and condensing coefficients can be increased by use of UOP High Flux or High Cond tubing.

Plate exchangers usually do not require enhancement methods. Gasketed plate exchangers are very easy to revamp, as more plates can simply be added to the exchanger; see Section 19.12. Welded plate exchangers are not amenable to the use of inserts or to expansion by adding plates.

Heaters and coolers

The techniques described for heat exchangers apply equally well to steam or oil heaters and water coolers. Decreasing cooling water return temperature (by increasing cooling water flow rate) and raising hot oil temperature are widely used methods in plant revamps.

Fired heaters are usually difficult to revamp and require the involvement of heater and burner specialists. If there is space in the heater, additional tubes can be added. Similarly, if there is space in the convective section, it can be used for pre-heat to off-load some of the furnace duty. In some cases, the addition of improved burners will allow more uniform heating and higher average tube-wall heat flux. Fired heater design is discussed in more detail in Section 19.17.

Air coolers (Section 19.16) are also difficult to revamp. The designer does not have the ability to specify a lower ambient temperature! Common air cooler revamps include:

- Adding more banks of tubes and installing more powerful fans.
- Adding water-spray systems to increase cooling capacity on the hottest days.
- Water-spray systems are effective, but can increase air-side fouling over time.

Example 2.7

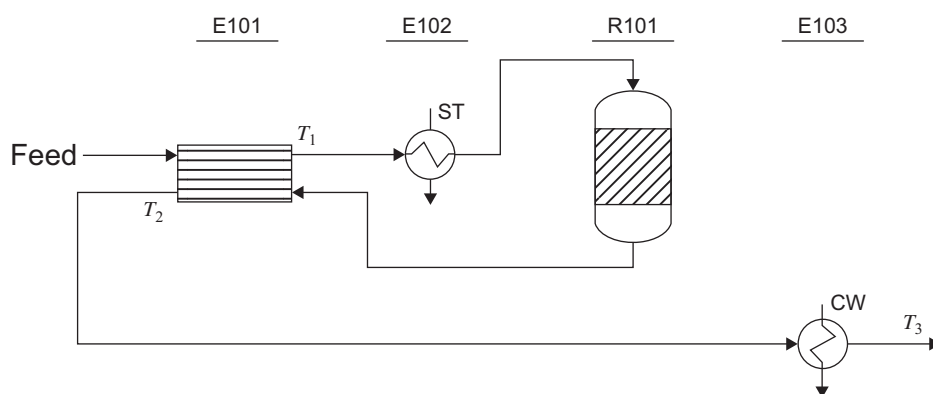
Fig. 2.19 shows a simple heat-exchange system. A feed stream is heated by heat exchange in a plate exchanger and then further heated in a steam heater before entering a fixed-bed reactor. The product from the reactor is cooled in the plate exchanger and then further cooled using cooling water. Exchanger specifications and current performance are given in the figure. Propose modifications to the system to allow a revamp to 50% higher capacity.

Solution

Start with the plate heat exchanger, E101. A 50% increase in flow through the exchanger would give a $(1.5)^2 = 2.25$ factor increase in exchanger pressure drop. If this was acceptable, then assuming that the heat transfer coefficient is proportional to $Re^{0.8}$:

$$\text{New heat transfer coefficient} = 350 \times (1.5)^{0.8} = 484 \text{ W/m}^2\text{K}$$

Assuming the reactor operates at the same outlet temperature, we can make heat balances for the base case and revamp case.



Exchanger	E101	E102	E103
Type	Plate	S&T	S&T
Duty (kW)	800	400	200
Hot side T in ($^{\circ}\text{C}$)	140	180	60
Hot side T out ($^{\circ}\text{C}$)	60	180	40
Cold side T in ($^{\circ}\text{C}$)	40	120	25
Cold side T out ($^{\circ}\text{C}$)	120	160	35
F_i factor	1	1	0.92
Hot side heat transfer coefficient ($\text{W/m}^2\text{K}$)	700	2000	700
Cold side heat transfer coefficient ($\text{W/m}^2\text{K}$)	700	500	700
Overall heat transfer coefficient ($\text{W/m}^2\text{K}$)	350	400	350
Area (m^2)	114	27.5	31.7
ΔT_{lm}	20	36.4	19.6

FIG. 2.19 Heat-exchange system for Example 2.7.

Base case:

$$Q_{101} = 800 \times 10^3 = m_f C_{p,f} (120 - 40) = m_p C_{p,p} (140 - 60)$$

$$m_f C_{p,f} = m_p C_{p,p} = 10^4$$

Where subscript f denotes feed and p denotes product.

Revamp case:

$$Q_{101} = 1.5 m_f C_{p,f} (T_1 - 40) = 1.5 m_p C_{p,p} (140 - T_2)$$

$$T_1 = 180 - T_2$$

and

$$Q_{101} = UA \Delta T_m = 484 \times 114 \times (140 - T_1)$$

So

$$1.5 \times 10^4 (T_1 - 40) = 484 \times 114 \times (140 - T_1)$$

hence

$$T_1 = 118.6^\circ \text{C}$$

$$T_2 = 61.4^\circ \text{C}$$

So we only lose 1.4 degrees of heat exchange as long as the increase in pressure drop that results from the higher flow rate is acceptable.

Now look at the steam heater, E102. In the revamp case:

$$Q_{102} = 1.5 \times 10^4 (160 - 118.6) = 621 \text{ kW}$$

The heat transfer coefficient is only increased on the process (cold) side:

$$\text{New cold - side coefficient} = 500 \times (1.5)^{0.8} = 691.6 \text{ W/m}^2\text{K}$$

$$\text{New overall heat transfer coefficient} = \left((2000)^{-1} + (691.6)^{-1} \right)^{-1} = 513 \text{ W/m}^2\text{K}$$

(Overall heat transfer coefficient calculated using Equation 19.2)

So for the exchanger to be feasible, we would need to raise the steam temperature to T_s , where:

$$Q_{102} = UA \Delta T_m$$

$$621 \times 10^3 = 513 \times 27.5 \times \left(\frac{(T_s - 160) - (T_s - 118.6)}{\ln \left(\frac{T_s - 160}{T_s - 118.6} \right)} \right)$$

$$T_s = 186.5^\circ \text{C}$$

So the heater remains viable if the steam temperature can be increased by 6.5°C . This might be accomplished by raising the local pressure of the medium pressure steam system, for example, by making adjustments in the steam pressure regulator valve set points. If that was not possible, and if the heater was rated for a sufficiently high pressure, then high-pressure steam could be considered instead of medium-pressure steam.

Turning to the cooler, E103. In the revamp case:

$$Q_{103} = 1.5 \times 10^4 (61.4 - 40) = 321 \text{ kW}$$

To meet this extra duty we need to increase the cooling water flow rate by a factor F_{cw} , which changes the cooling water outlet temperature to T_w , where:

$$Q_{103} = 321 \times 10^3 = F_{cw} \times 20 \times 10^3 (T_w - 25)$$

and

$$321 \times 10^3 = UAF \Delta T_{lm}$$

For the revamp case, the logarithmic mean temperature difference ΔT_{lm} depends on T_w and the overall heat transfer coefficient will be:

$$\frac{1}{U} = \frac{1}{700 \times (1.5)^{0.8}} + \frac{1}{700 \times (F_{cw})^{0.8}}$$

These equations must be solved iteratively. This is easily done using a spreadsheet, giving:

$$F_{cw} = 1.89$$

$$T_w = 33.5^\circ \text{C}$$

The exchanger would be feasible if we could accept this increase in cooling water flow rate; however, such a large increase would give a very large increase in pressure drop. The pressure drop is proportional to flow rate squared and so would increase by a factor $(1.89)^2 = 3.57$, which would probably not be acceptable.

An alternative approach for E103 is to see what the outlet temperature would be if cooling water flow were limited. Suppose the cooling water flow cannot be increased more than 20%, giving a 44% increase in pressure drop. If we allow the cold outlet temperature and hot outlet temperature to vary, the same spreadsheet model can be solved to give:

$$T_w = 36.4^\circ \text{C}$$

$$T_3 = 43.1^\circ \text{C}$$

So in this case the designer would have to consider whether the product stream could be made 3.1°C hotter, which depends on the downstream processing.

If this option was also unacceptable, the designer would have to consider adding another cooler. A simple approach would be to split the hot stream, leaving E101 into two streams in a ratio of 2:1. The larger stream could still be sent to E103, and the smaller one could be sent to a parallel cooler half the size of E103. This option would not push the operation of E103 much beyond the current operating mode.

In summary, one option that would allow a 50% increase in throughput would be:

- **E101:** No change, as long as the factor 2.25 increase in pressure drop is acceptable on hydraulic review.
- **E102:** Increase steam temperature to 186.5°C ; no capital modification is needed.
- **E103:** Consider accepting a 3.1°C warmer product. If this is unacceptable, add a new exchanger E104, half the size of E103, in parallel to E103 and split the hot stream, leaving E101 in a 2:1 ratio between E103 and E104.

Alternatively, if E101 is a gasketed plate heat exchanger (see Section 19.12.1), the designer could also consider adding more plates to increase the area available in E101. We can increase the plates in E101 until we achieve the same channel velocity and pressure drop as the base case then solve the remaining problem for E102 and E103, or, alternatively, increase the plates in E101 until the existing E102 and E103 are able to satisfy all of the remaining heating and cooling load without modifications. This alternative solution is explored in Problem 2.11. The addition of plates to E101 would be more expensive than just altering a few temperatures, but would most likely be less costly than adding E104 and could even be cheaper than making the hydraulic modifications necessary to cope with a large increase in pressure drop.

2.5.4 Revamp of plant hydraulics

Any revamp project that leads to increases in plant throughput will have a significant effect on the plant hydraulic equipment. Because pressure drop is proportional to velocity (and hence flow rate) squared, a 40% increase in flow is sufficient to double the pressure drop. The introduction of parallel equipment in the flowsheet may also create a need for additional control valves to regulate flow in the desired ratios, which adds more pressure drop to the plant. Adding equipment to modify the process also has an impact on the hydraulic design.

Much of the design effort in revamp projects goes into evaluating and redesigning the plant hydraulics. All of the pump-and-line calculations and control valve sizing calculations must be repeated for the revamped design case. It is usually not cost-effective to replace the piping with new pipe of more optimal diameter, so usually the designers will accept a higher pressure drop in the pipes and process equipment and then redesign the pumps and control valves accordingly.

The design of pumps, compressors, piping systems, and control valves is discussed in more detail in [Chapter 20](#). The following sections provide some specific guidelines relevant to revamp flowsheet development.

Compressors

Compressors are the largest and most expensive items in the plant hydraulic equipment. Compressor design is covered in more detail in Section 20.6. Because compressors are expensive to replace, experienced designers usually try to reuse the existing compressors in the new flowsheet.

The relationship between flow rate and pressure delivered depends on the type of compressor (Fig. 2.20), but the pressure delivered will usually decrease if the compressor operates at higher flow. The only exception to this rule is large reciprocating compressors, which are usually designed with a recycle from the product to the feed, known as a *spill-back*. If the spill-back flow is large enough, some increase in flow can be obtained by reducing the spill-back, with no loss in delivered pressure.

When two compressors are used in parallel, the total flow increases, but the delivered pressure cannot be greater than the lower pressure delivered by either compressor; see Fig. 2.21. When two compressors are used in series, the pressure delivered increases, but the flow rate does not; see Fig. 2.22.

In a revamp flowsheet design, the designer seeks to increase the pressure delivered to overcome the increased plant pressure drop, as well as an increase in flow rate. From Figs. 2.21 and 2.22 it can be seen that the only way to increase both the pressure delivered *and* the flow rate while adding only one compressor is somewhat paradoxically to *reduce* the flow rate in the existing compressor so that it delivers a higher pressure and then add a second machine in parallel, as shown in Fig. 2.23. The extent to which the flow rate can be reduced depends on the type of compressor and current operating conditions. In this scenario, the revamped flowsheet would need to show two

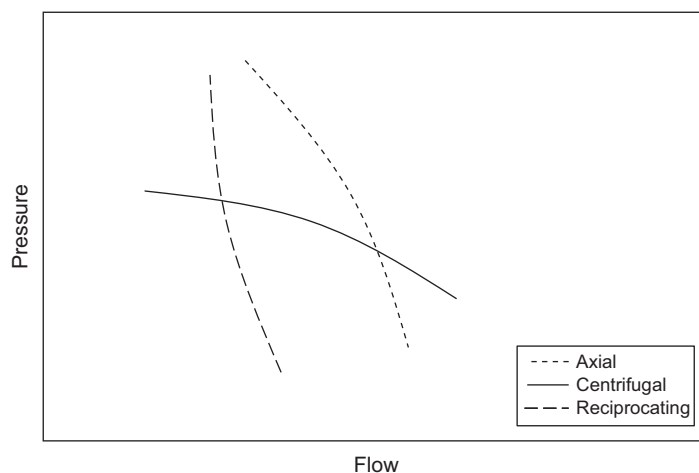


FIG. 2.20 Pressure–flow rate curves for different compressor types.

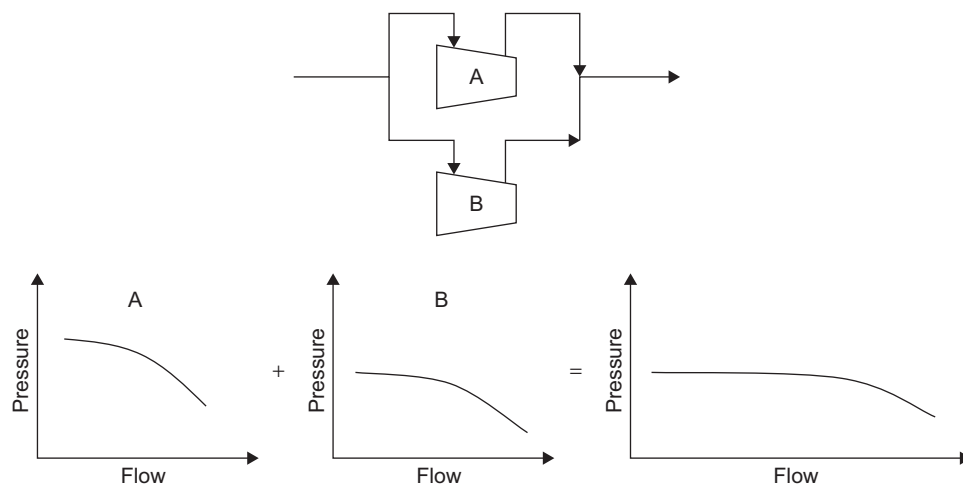


FIG. 2.21 Compressors in parallel.

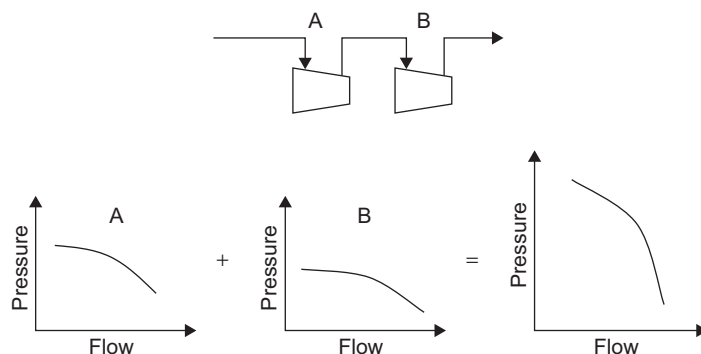


FIG. 2.22 Compressors in series.

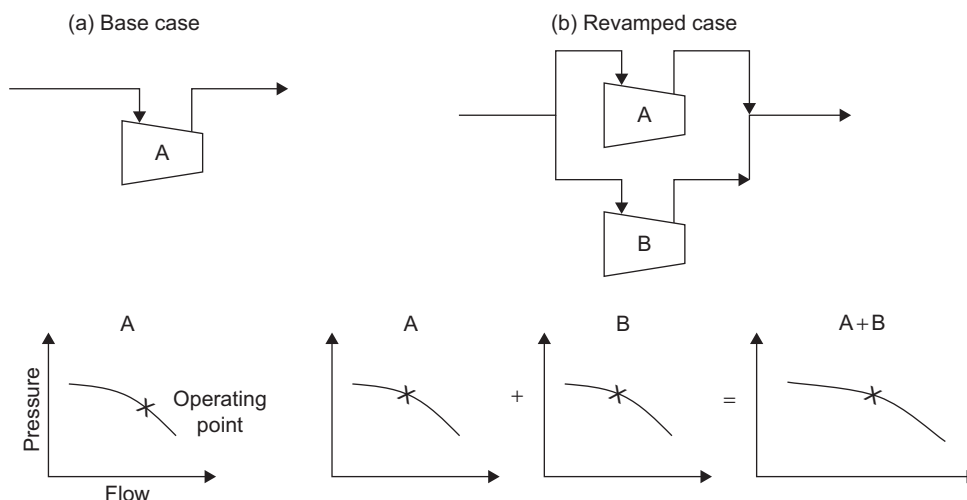


FIG. 2.23 Compressor revamp.

compressors in parallel. A revamp from one machine operating at 100% flow rate to two similar machines in parallel, each at 70% flow rate, would give an overall 40% increase in flow.

If it is not possible to meet the desired flow and pressure with the addition of only one new compressor, it will usually be preferable to replace the existing compressor rather than adding new compressors in series and parallel.

If a compressor cannot be reused in its current location in the revamped flowsheet, it should be evaluated for other process uses. Air compressors and blowers should also always be reevaluated and can be used to assist heaters, boilers, dryers, or even site instrument air systems if there are no suitable process uses.

Pumps

Pumps are relatively inexpensive compared with compressors and will often be replaced entirely in a revamp. Pumps are discussed in more detail in Section 20.7.

The most commonly used pumps are centrifugal pumps. Centrifugal pumps exhibit pressure-flow behavior similar to that shown for centrifugal compressors in Fig. 2.20. A typical pump curve is given in Fig. 20.15. The same pump can deliver a set of different performance curves depending on the impeller diameter and motor speed. The designer can therefore sometimes obtain the required performance just by selecting new impellers for the existing pumps.

Control valves

All control valves must always be rated to confirm that they are correctly sized for the revamped design case and will give the desired controllability, turndown, and ability to meet different operating scenarios. Control valve design and sizing are discussed in Section 20.11.

When parallel equipment is introduced into the revamped flowsheet, the design engineers must determine how the flow is to be split between the existing and new equipment. A simple T-junction or branch will often not be effective, as the rate of fouling or pressure-drop accumulation of the new and old equipment will not be the same, even if they have the same capacity and design, and flow would then preferentially go through one piece of equipment, leading to poor performance of both. The desired split ratio is rarely 1:1 and may need to be adjusted once the plant is put into operation. The most common approach is to put a new control valve on the branch that leads to the new equipment. The existing equipment is likely to be limited by pressure drop, so the new equipment can be designed with lower pressure drop to allow for the pressure drop associated with the control valve. A less costly, but less effective, alternative is to use a manual valve or even a restriction orifice in the bypass line and make manual adjustments until the desired flow ratio is obtained.

Control valves that act discontinuously and handle low flow rates will sometimes be suitable for reuse in the revamped plant as long as the new actuation rate is acceptable. Control valves on main plant flows will usually need to be replaced, as will control valves on gas or vapor streams. The revamped flowsheet does not need to indicate which existing valves are reused and which are replaced, but should show all new control valves that are added.

2.6 Synthesis of novel flowsheets

The terms *process synthesis* and *conceptual process design* are used for the invention of completely new process flowsheets. As stated previously, very few entirely new designs are developed commercially because of the high financial risks inherent in using unproven technology. The primary goals of process synthesis are therefore to reduce commercialization risk and to maximize economic attractiveness so as to generate sufficient financial reward to balance the risk.

Process synthesis has been the subject of a great deal of academic and industrial research over the past 40 years. Many problems that were previously solved using inspired guesswork can now be formally posed and optimized. The use of process simulation programs has also made it much easier to evaluate and optimize alternative flowsheets; see [Chapter 4](#) for more on process simulation. Several prominent researchers in the field of process synthesis have written textbooks on process design that strongly emphasize process synthesis. These are listed in the bibliography at the end of this chapter. Several excellent books have been written on process synthesis in its own right: Rudd et al. (1973); Douglas (1988), and El-Halwagi (2006), as well as on aspects of process synthesis, such as distillation sequencing (Doherty and Malone, 2001), mass integration (El-Halwagi, 2017), and heat integration (Kemp, 2007; Shenoy, 1995). Although it is beyond the scope of this book to cover all aspects of process synthesis, this section sets out an overall framework for flowsheet synthesis that addresses the key issues encountered in developing and commercializing new processes. The reader is encouraged to read the books listed here and in the bibliography for more insights into the subject.

2.6.1 Overall procedure for flowsheet synthesis

Most efforts to systematize process synthesis begin by setting out a sequence or hierarchy of steps for the designer to follow. Design hierarchies recognize that some steps need to come before others and should guide the designer to eliminate unattractive options and focus effort on designs that are most likely to be successful.

The most intuitively obvious design hierarchy is the so-called *onion diagram*. [Fig. 2.24](#) shows a version of the onion diagram given by [Smith \(2016\)](#). The onion diagram represents a design starting with the reactors; adding separation and recycle systems; then proceeding to add heat recovery, utility systems, and environmental systems. [Rudd et al. \(1973\)](#) proposed a more theoretically abstract synthesis hierarchy that essentially follows the same steps but included an additional step of integrating reaction, mixing, separation, or change of state tasks into unit processes or operations.

[Douglas \(1988\)](#) set out a somewhat different approach, summarized in [Table 2.1](#). Douglas emphasized early introduction of process economics to guide the elimination of weaker alternatives, with continuous refinement of the economic model as more detail is added to the flowsheet. This is a useful approach, as it can highlight deficiencies in the design at an early stage.

The reality of process development in industry is usually a lot less systematic than the idealized picture painted by academic researchers. In industrial practice, process development is more interdisciplinary, more iterative, and much less linear than the simple synthesis models suggest. The flowsheet synthesis step is usually part of a larger

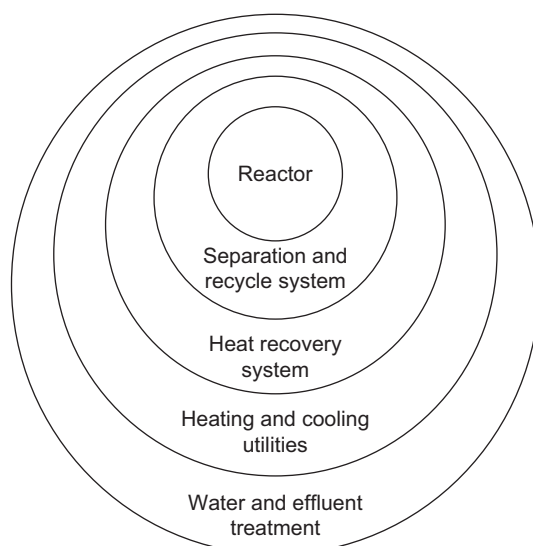


FIG. 2.24 Onion diagram. From Smith, R. [2016]. Chemical process design and integration [2nd ed.] Wiley.

TABLE 2.1 Hierarchy of process synthesis decisions

- | |
|---|
| 1. Batch vs. continuous |
| 2. Input–output structure of the flowsheet |
| 3. Recycle structure of the flowsheet |
| 4. General structure of the separation system |
| a. Vapor recovery system |
| b. Liquid recovery system |
| 5. Heat exchanger network |

Data from Douglas, J. M. (1988). Conceptual design of chemical processes. McGraw-Hill.

effort that involves chemists, biologists, and other engineers and includes laboratory and pilot plant experiments to determine reactor performance and establish yields and product recoveries. The engineers working on synthesis seldom have all the data that are needed to properly optimize the design and often must guide the research members of the team to collect additional data under conditions that will be more favorable to process economics. Many industrial processes involve multiple reaction steps carried out in sequence with intermediate separation steps, and it may be difficult to assess the performance of later steps without good information on the nature and quantity of by-products that are carried over from earlier steps in the sequence. The design team therefore needs to form a rough impression of the PFD and economics, using minimal information on process chemistry, so as to develop an understanding that can guide the efforts of the research team.

Fig. 2.25 sets out an approach to flowsheet synthesis that sets flowsheet development in the context of working with a research team to establish yields and reactor performance. The Douglas hierarchy and onion diagram form substeps in this procedure, as described next.

Step 1. Initial economics

The very first step should be to collect prices for feeds and products and confirm that the cost of production will be attractive if a stoichiometric yield is obtained. If the cost of feed is more than the product value, then there is no hope of developing an economically attractive process and the work should be stopped unless the team has strong evidence that prices will change in the future. This step is particularly important when assessing nontraditional feeds; for example, when looking at processes for converting food-based renewable feeds into chemicals.

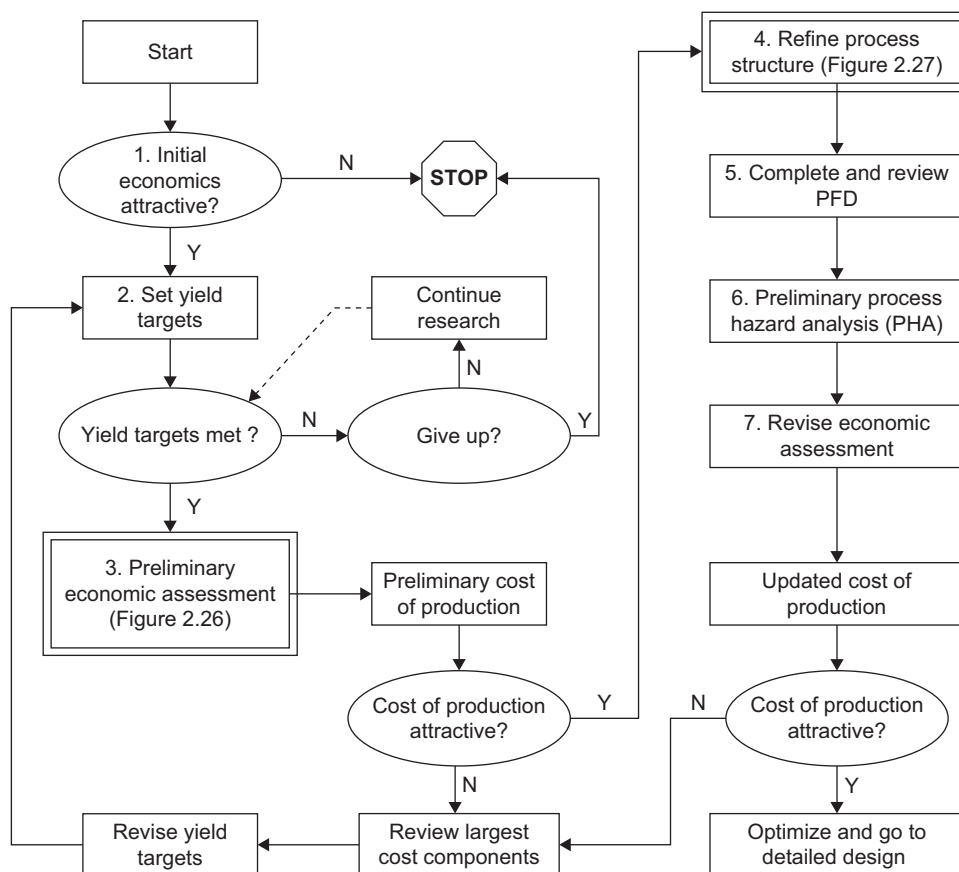


FIG. 2.25 Procedure for process synthesis.

Step 2. Set yield targets

The research team needs to set realistic targets that will lead to an attractive process. The term *yield targets* includes targets for by-product selectivity as well as main product selectivity and conversion. Methods for setting and revising yield targets are given in Section 2.6.3. The researchers will generally need to carry out process development experiments to optimize reactor conditions and catalyst, enzyme, or organism performance to meet the yield targets.

On the first pass through the procedure, the designers may choose to just accept whatever yields and selectivities the research chemists or biologists have already established. When more information on process economics has been generated, the targets can be revised and improved.

If the yield targets are not met, the company must make a strategic decision on whether to continue or abandon the research. Research discoveries are often serendipitous and can be hard to plan. Companies often choose to allow a low level of research activity to continue over a long period once clear success criteria have been established.

Step 3. Preliminary economic assessment

The goal of a preliminary economic assessment is to arrive at a preliminary estimate of the cost of production once the yield targets have been met. The substeps in carrying out a preliminary economic assessment are illustrated in Fig. 2.26. This procedure is similar to that of Douglas (1988), but it should be emphasized that less detail is put into the design and the goal is not to arrive at a PFD or even a detailed block flow diagram at this point.

The components of cost of production and methods for calculating each component are discussed in detail in Chapter 8. For most processes, 80% or more of the cost of production will be feedstock cost minus credits for any economically viable by-products. The rest of the cost of production is chiefly made up of utility costs (mostly energy), fixed costs (mostly labor), and annualized payments to generate an expected return on the capital investment. The split between these depends on the type and scale of the process. Small-scale batch processes will have a higher proportion of fixed costs, whereas large-scale petrochemicals or solids-handling plants will have a higher proportion

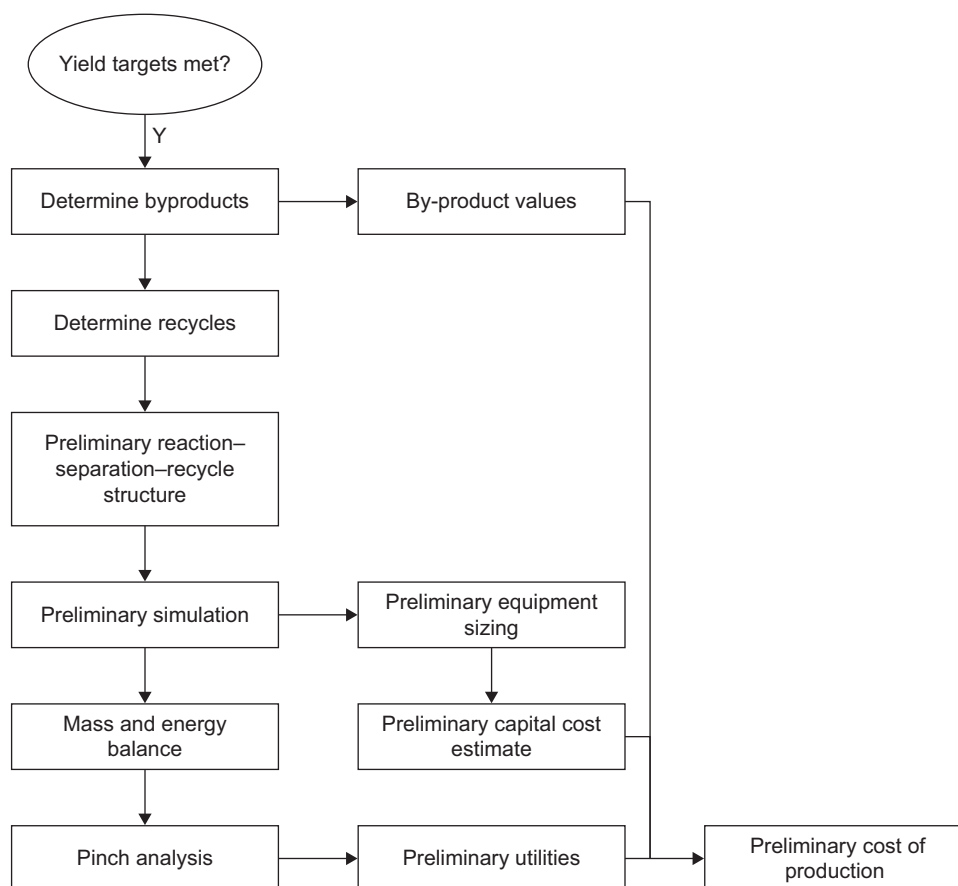


FIG. 2.26 Preliminary economic assessment.

of utility costs. In a preliminary economic assessment, the designer seeks to make a quick estimate of these major components of cost, and hence determine whether the process will be able to make product at an attractive price.

The first step is to look at the yields and identify any significant by-products. By-products can be formed by the reaction stoichiometry, by side reactions, or from extraneous components in the feeds. By-products must be refined and sold, treated as waste streams, or recycled to extinction in the process. A quick examination of the yields and prices of the by-products is usually sufficient to establish which are suitable for recovery. The economic assessment of by-product recovery is discussed in more detail in Section 8.2.3.

Once the designer has a notion of which by-products are worth recovering and which by-products must be recycled, a preliminary reaction-separation-recycle structure can be sketched. It is not important to have the best or optimal flowsheet at this point, and the design team may want to propose a few alternatives to see which is least costly. A process simulation model can then be built and used to generate a mass and energy balance and obtain rough sizing of the major process equipment. The preliminary simulation should include all recycles, reactors, and separation equipment and should capture all changes in temperature and pressure. It does not need to include a heat recovery design and should use heaters and coolers instead of heat exchangers whenever there is a change in temperature. The use of commercial programs for process simulation is discussed in Chapter 4. The preliminary simulation model does not have to be built using a commercial simulation program, but it will usually be convenient to do so, so that the model can be extended as the flowsheet is subsequently refined.

The preliminary equipment sizes from the simulation model (or hand calculations) can be used to develop an initial estimate of the capital cost of building a plant. Estimation of capital costs is covered in Chapter 7. The capital cost is then annualized so that it can be added to the other costs of production. Annualization of capital costs is covered in Section 9.7. In a preliminary economic assessment, the designers usually annualize capital costs by dividing by 3. The basis for this rule of thumb is set out in Section 9.7.2.

The process energy consumption and utility costs are difficult to assess without completing a process simulation and energy balance. Most processes have significant opportunity to reduce energy costs by heat recovery, so simply

adding up all the heating and cooling duties would be a gross overestimate. Instead, a first estimate of energy consumption can be made by carrying out the targeting step of pinch analysis to get hot and cold utility targets. Pinch analysis and other heat recovery methods are described in [Chapter 3](#). At this point in the design it is not necessary to design the heat recovery system, as the targets are adequate for the preliminary economic assessment.

The preliminary estimates of main product and by-product production rates, feed and energy consumption, and capital cost can be used to make a preliminary estimate of the cost of production, as described in [Chapter 8](#). If the cost of production appears attractive, the design team proceeds to the next step. If not, the economic assessment can be used to highlight the major components of cost that must be reduced to make the process economically interesting. Having identified the cost components that must be addressed, the design team can either look at alternative flowsheets that reduce these costs or else set more aggressive yield targets and go back to the research stage.

Step 4. Refine process structure

If the preliminary economic assessment indicates that the process is potentially economically attractive, then it is important to develop a complete PFD and make sure that no costs have been overlooked. The steps in completing a more rigorous design are shown in [Fig. 2.27](#). It can be seen that these follow roughly the same sequence as the onion diagram of [Fig. 2.24](#).

The first step is to optimize the reaction–separation–recycle structure of the flowsheet and confirm the yields under the optimal conditions. The preferred conditions can be estimated by optimization of the preliminary simulation model and economic model. Additional experimental data may be needed if the optimal conditions are different from the conditions originally proposed. The reactor designs must be tested and yields confirmed in the presence of recycle streams, which may require construction of a pilot plant that can operate in recycle mode. The design of reactors is described in [Chapter 15](#).

The design of the separation systems encompasses not only those separations associated with product recovery and recycles but also feed purification, product purification, and by-product recovery. The design of separation processes is covered in detail in [Chapters 16, 17, and 18](#). In some cases, product purification or by-product recovery will require additional reaction steps. For example, in the recovery of ethylene produced by steam cracking of light hydrocarbons, it is easier to hydrogenate by-product acetylene than to separate it by distillation ([Fig. 2.28](#)).

When the by-product separation and recovery sections have been designed in more detail, the attractiveness of recovering the by-products can be revisited. If the costs of producing the by-products are excessive, the designer should revisit the reaction–separation–recycle structure or return to the preliminary economic analysis step.

When designing the reaction and separation steps, as much use as possible should be made of proven process subsections. If a particular reaction, separation, recovery, or purification step is already practiced commercially, then the same method will probably be least costly and will most likely have the least technical risk for the new design. Borrowing proven concepts from established technology is one of the most effective strategies for reducing commercialization risk. Some caution is needed when applying this strategy if the new case is not identical to the commercially proven design, and the designer should take particular care to ensure that slight differences do not lead to potential safety or operability issues. The designer must also ensure that there are no active patents on the features that are borrowed so that there is freedom to practice them.

Once the major process equipment has been specified, the design team should have a good idea of the stream temperature and pressure requirements, and the heat recovery system can be designed. Process heat recovery is described in [Chapter 3](#), and heat transfer equipment is covered in [Chapter 19](#).

The designs of the plant hydraulics and control system are interlinked, as control valves introduce additional pressure drop into the process and can create a requirement for additional pumps. Once the major equipment, including heat exchangers, has been specified, a preliminary PFD can be drafted, which can be developed into a full PFD by adding the location of control valves, pumps, and compressors. The design of plant control systems and location of control valves is discussed in [Chapter 5](#). The design of hydraulic equipment is covered in [Chapter 20](#), and solids handling systems are described in [Chapter 18](#). When all the equipment has been added to the PFD, the process simulation can be updated to produce mass and energy balances to complete the flowsheet. The PFD is then ready for review.

Step 5. PFD review

Review of a PFD is one of the most important steps in flowsheet development. A full PFD review is always carried out in design, regardless of whether the process is a revamp or new unit, or whether it uses novel or proven technology. This vital step is discussed in more detail in [Section 2.7](#).

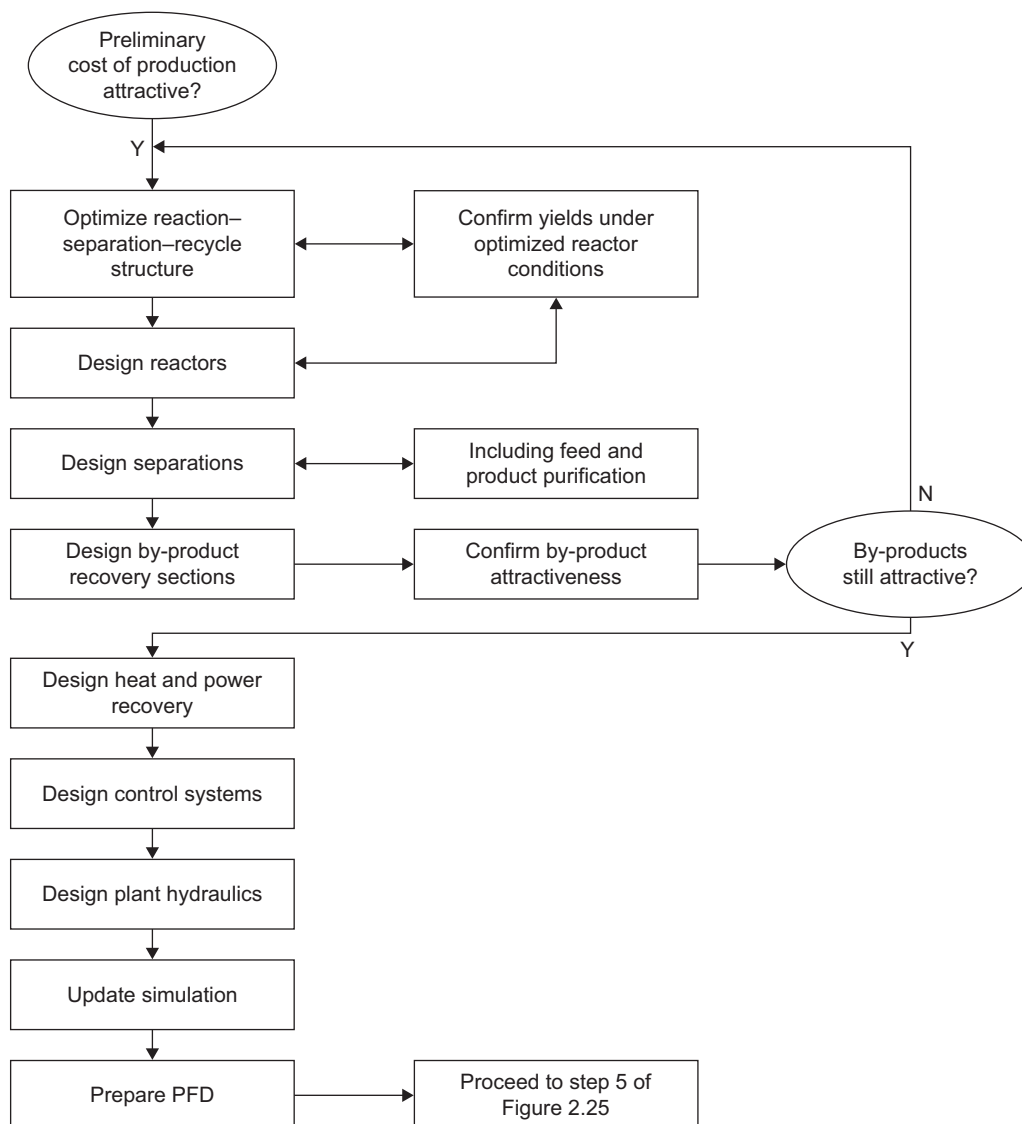


FIG. 2.27 Refining the process flowsheet.

Step 6. Preliminary process hazard analysis

When a completed PFD and mass and energy balance are available, a preliminary process hazard analysis (PHA) can be carried out. A PHA will identify major hazards inherent in the process and may indicate a need to alter some process conditions, substitute different equipment, or completely redesign sections of the process. If the preliminary PHA identifies major modifications to the PFD, the design team should go back to the relevant stage of the procedure and generate a safer alternative design. The role of safety in design is discussed in detail in [Chapter 10](#).

Step 7. Revise economic assessment

The completed PFD and mass and energy balance allow the design team to make more accurate designs of the process equipment, and hence arrive at more accurate estimates of the capital cost and cost of production. If the process still appears to be attractive, then it may be worth developing as an investment, and other methods of economic assessment will be used to determine a viable project for implementing the technology; see [Chapter 9](#). The models that have been developed now have sufficient accuracy to enable more rigorous optimization and can be used as a starting point for detailed design of the plant and equipment. Optimization methods in process design are discussed in [Chapter 12](#).

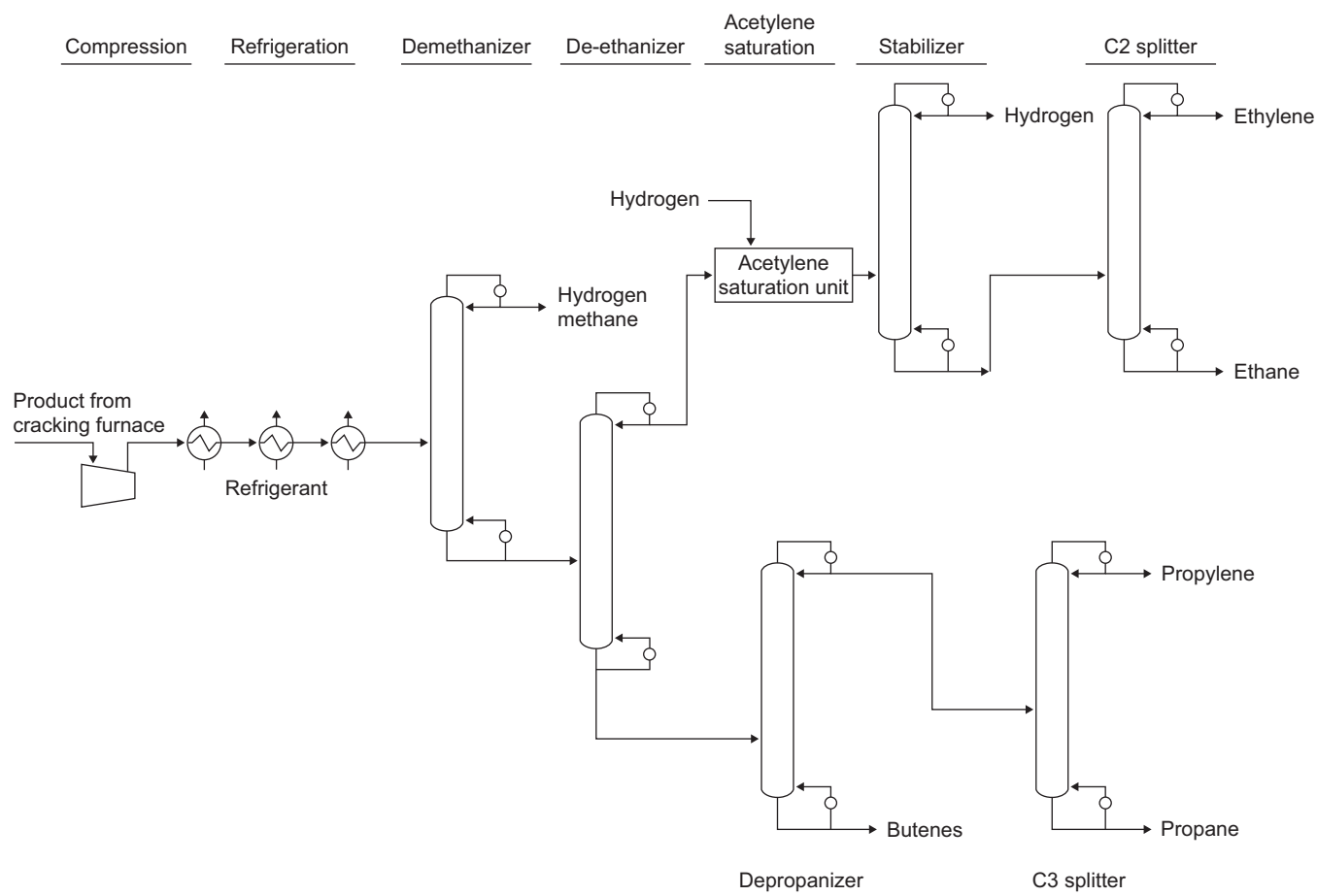


FIG. 2.28 Ethylene recovery from steam cracking.

If the updated cost of production is no longer attractive, the added detail can be used to further define the major cost components and identify areas for process improvement. This will often involve reducing by-product and recycle handling, and hence translate into setting tougher yield targets.

2.6.2 Economic analysis in process synthesis

In process synthesis it is important to use economic analysis to inform decision-making at every step. The procedure outlined in Fig. 2.25 begins with a rough economic analysis and then adds detail to this analysis as information is accumulated. Like an artist beginning with a rough pencil sketch and then filling in details and adding colors, the process design engineer needs to have an overall sense of the composition before getting into the details.

The procedure set out in Fig. 2.25 has three economic check points at steps 1, 3, and 7, corresponding to the initial, preliminary, and updated estimates of the cost of production. At these check points the estimated cost of production should be compared with the product sales price that the marketing organization has forecasted. Usually, the criterion for success will be that the cost of production must be low enough to ensure an acceptable return on the capital deployed. Forecasting of prices and calculation of cost of production are covered in Chapter 8, and methods of economic analysis are described in Chapter 9. Although these three steps in the procedure are formal checks, experienced designers do not wait until the check point to calculate process costs. As soon as information is developed, its impact on the cost of production should be determined. In general, costs accumulate as detail is added to the design, so the design team wants to be aware of large costs as early as possible so that they can start considering alternative design features.

Companies usually use other economic analysis methods, such as net present value (NPV) and internal rate of return (IRR), instead of cost of production when assessing capital investment projects; however, very few companies launch a capital project without having already completed the process synthesis. At the synthesis stage, cost of production is the most useful economic measure, as it is very easily factored into components such as raw material

costs, by-product values, energy costs, etc. Understanding the components of cost of production can help the design team focus on areas of high cost if the target cost of production is not achieved.

2.6.3 Use of targets in process synthesis

Design engineers use targets as a means of setting bounds on design performance that can quickly eliminate unattractive options. Targets also help designers and researchers to focus their efforts on areas that will most effectively improve economic performance. Several different types of targets are used in the synthesis procedure set out in Figs. 2.25, 2.26, and 2.27:

- The cost of production is compared with a price target set by the marketing organization at steps 1, 3, and 7.
- The design team sets targets for the research team at step 2.
- Hot and cold utility targets calculated by pinch analysis are used for initial estimates of process energy consumption.
- The preliminary economic analysis sets targets for capital cost and the components of cost of production that the design team must confirm as they refine the process structure and fill in the PFD.

The basis for targets should always be clearly stated. Whenever possible, targets should be calculated from economic criteria, and the assumptions in the calculations should be made explicit. For example, a badly stated yield target would be:

“Find me a catalyst that doesn’t make by-product X.”

A better way of stating this target might be:

“If reactor selectivity to by-product X is $<0.5\%$ of selectivity to the main product, we can eliminate a separation and purification section of the process, with expected 15% savings in capital cost and 20% savings in energy cost.”

Targets must not be unrealistically tough, or they will never be achieved and will not be taken seriously. Lenient yield targets are usually not as problematic, as they tend to lead to failure at the preliminary economic assessment stage and are then revised to something more realistic. Lenient targets for cost of production are dangerous, as they allow the design to go forward, and much time and effort can be wasted before harsh economic reality kills the project. Good price forecasting and market analysis are critically important in setting the cost of production targets. These topics are discussed in Chapter 8.

It is important to understand whether a target should be treated as a hard constraint or a soft constraint. Companies sometimes address this question by providing *must-have* targets that are hard constraints and *should-have* targets that are soft constraints. The design team can then reject designs that do not meet the hard constraints, but keep concepts that come close to the soft targets. Soft and hard targets are explored in Example 2.8.

Example 2.8

The marketing group is planning to launch a new product and has forecasted that the mean price for the product will be 5 \$/kg, normally distributed with standard deviation 40 ¢/kg. The cost of the stoichiometric amount of feed required to make the product is 3 \$/kg. Propose preliminary targets for cost of production and yields.

Solution

If the forecast is accurate and the mean product price is 5 \$/kg, then there is a 50% probability the project will have economic success if the cost of production (including capital recovery) is 5 \$/kg. Using the standard deviation given in the forecast, we can form the following table:

Cost of production (COP), \$/kg	Probability of success
3.80 ($= 5 - (3 \times 0.4)$)	99.9%
4.20 ($= 5 - (2 \times 0.4)$)	97.7%
4.60 ($= 5 - (1 \times 0.4)$)	84%
5.00	50%
5.40 ($= 5 + (1 \times 0.4)$)	16%

We can note immediately that this process would pass the initial economic assessment even if we chose a target cost of production with 99.9% chance of success.

The probability of success that we require depends on how risk averse or aggressive the company is. A 98% chance of success might be too conservative and would give a high chance that the project would not meet the targets. A 50% chance of success would probably be too aggressive and would allow the project to go forward and spend money with low likelihood of financial success. As a compromise, management might set a *must-have* target for COP of 4.60 \$/kg, with a *should-have* target of 4.40 \$/kg. Note that we are not constrained to using integral multiples of the standard deviation in setting the targets; for example, a price target of 4.34 \$/kg ($= 5 - (1.65 \times 0.4)$) corresponds to 95% probability of success and would be equally easy to justify as a *must-have* or *should-have* target.

The COP target can now be translated into preliminary yield targets. In Section 2.6.1 it was stated that feed costs are typically at least 80% of COP. Using this rule of thumb, we can state:

$$\text{Target feed cost} = 0.8 \times \text{target COP}$$

The yield targets obviously depend on the number of feeds, the relative costs of individual feeds, and the number of reaction and product recovery steps. From Equations 2.4 and 2.6 we know:

$$\text{Yield} = \frac{\text{product formed}}{\text{stoichiometric product formed}}$$

so:

$$\text{Target plant yield} = \frac{\text{stoichiometric cost /kg}}{\text{target feed cost /kg}}$$

So if the target COP is 4.60 \$/kg:

$$\text{Target plant yield} = \frac{3}{0.8 \times 4.60} = 0.815$$

Note that this is the target yield over all the steps in the process. If we assume that we lose roughly 5% of the product during all the steps of product recovery and purification, then:

$$\text{Target plant yield from reactors} = \frac{0.815}{0.95} = 0.858$$

If we have two reaction steps and one feed is more expensive than the others, we could further decompose the yield target into targets for each step:

$$\text{Target plant yield from reactors} = Y_1 Y_2$$

where Y_1 and Y_2 are yields of reaction steps 1 and 2. We could set equal targets for each step:

$$Y_1 = Y_2 = \sqrt{(0.858)} = 0.926$$

or we could use our knowledge of process chemistry or biology to define a suitable allocation between the steps.

Note that the targets calculated are plant yields, not reactor yields. If costly unconverted feeds can be recycled, the plant yields translate into reactor selectivity targets, not reactor yield targets (see Section 2.3.3).

Note also that even in this simplistic example, the apparently very favorable economics quickly translated into rather tough targets for reactor performance.

2.6.4 Use of heuristic rules in process synthesis

Heuristic is an adjective meaning “of, or pertaining to, or based on, experimentation, evaluation or trial and error methods,” which pretty much sums up most engineering knowledge. The terms *heuristic rules* and *design heuristics* are commonly used to describe rules of thumb and design guidelines that have been developed based on experience. Experience is good, but it is rarely acquired instantly or bought cheaply. Design guidelines based on generalizations are only useful if the designer has sufficient experience to understand when to apply the guideline and when to make an exception.

There is often confusion about what constitutes a design heuristic. Consider the following statements, most of which have been made in this chapter:

1. *"The heat of condensation of steam is about 2000 kJ/kg."*
2. *"Pressure drop is usually proportional to velocity squared."*
3. *"Feedstock costs are typically at least 80% of the total cost of production."*
4. *"Capital costs can be annualized by dividing by 3."*
5. *"When designing reaction and separation steps, use proven process subsections as much as possible."*

Statement 1 is a convenient data approximation. It is accurate within $\pm 10\%$ for saturated steam over a temperature range from 100 °C to 240 °C, which covers most temperatures at which steam is used for process heating. Remembering this fact may save an engineer some time when carrying out hand calculations, but it does not provide any guidance for design.

Statement 2 is a convenient summary of several correlations for pressure drop. It may be very useful for making quick calculations in revamp designs, but again provides no guidance for design.

Statement 3 is a generalization that can be useful as a rough check on cost of production calculations. As illustrated in Example 2.8, it can also be used as a basis for setting initial targets in process synthesis. It is, however, rather too general to provide guidance in design.

Statement 4 is one way of annualizing capital costs. It would be equally valid to state *"Capital costs can be annualized by dividing by 2"* or *"...by dividing by 5,"* depending on the assumptions made. The basis for deriving these numbers is given in Section 9.7.2.

Statement 5 is clearly a design guideline based on a general desire to minimize the number of unproven concepts in the flowsheet.

Although all of these statements are useful as rules of thumb and can help make quick calculations to assess a design, only statement 5 actually provides guidance on how to design a process. It might be useful for an engineer to recall approximate data and generalizations, and several compilations of such rules of thumb have been written (Fisher, 1991; Hall, 2017; Hicks & Chopey, 2012), but these are only helpful in process design when making quick calculations in meetings.

Some design texts provide extensive lists of rules of thumb and selection guidelines. That approach is not adopted here, as heuristics easily lose their meaning when taken out of context. For example, one design text gives the rule for vessel design: *"optimal length to diameter ratio = 3."* This is questionable even for horizontal and vertical flash drums (see Section 16.3), but is clearly nonsense when wrongly applied to reactors and distillation columns. Inexperienced engineers often have difficulty determining when to apply such heuristic rules, so in this text all shortcut calculations, convenient approximations, and design guidelines will be presented and explained under the relevant design topic.

The most important heuristic rule, which should always be followed, is:

"Never use a heuristic rule unless you understand where it came from and how it was derived."

2.6.5 Role of optimization in process synthesis

Optimization methods are used in process synthesis to select the best flowsheet options, process conditions, and equipment sizes. The designer must be reasonably sure that design alternatives have been optimized before selecting between them. Optimization underpins all design decisions, and the subject is addressed in more detail in Chapter 12.

The problem that is often encountered when applying optimization methods in process synthesis is that insufficient data are available to properly formulate an optimization problem. For example, it would be good to optimize reactor performance as soon as possible, but the research team may not yet have collected data in the presence of recycle streams or at conditions close to the optimal reactor conditions. Under the circumstances, the designer must optimize the design with the data available and then revisit the optimization later when more data have been collected and the reaction kinetics model has been updated.

Most processes are too complex to formulate a single optimization problem that includes all possible structural flowsheet variations, as well as all continuous process variables. Instead, different aspects of flowsheet synthesis are usually treated as separate optimization tasks. It is important, however, to have an overall optimization model that

captures the major design trade-offs. The cost of production model developed in the preliminary economic assessment can serve as an initial model for optimization. The optimization of subproblems is discussed in Section 12.5.

2.7 PFD review

The most important step in developing a process flowsheet is for the PFD to be rigorously reviewed. This is true regardless of whether the flowsheet is for an established design, a revamp, or a newly invented process. The purpose of a PFD review is to review the design decisions and assumptions and ensure that the flowsheet is complete and shows *all* the equipment needed to really operate the process.

A PFD review is usually attended by the design team and a few outside experts. These may include:

- Senior managers
- Technical experts on process design
- Technical experts on process chemistry, catalysis, or biology
- Equipment or plant design experts
- Process safety experts
- Metallurgists
- Plant operations staff
- Plant mechanical engineers
- Process control engineers

Some companies have rules and procedures governing who must be present at a PFD review, but the review can be effective with only a few people as long as they engage actively with the team.

2.7.1 PFD review procedure

A PFD review is usually carried out as a group exercise. A large printout of the PFD is typically taped or pinned to a wall so that the group can mark up corrections, notes, and other revisions as the review proceeds. The PFD is usually drawn well spaced out to leave room for additions, and may run to several sheets of drawings. One member of the group takes notes and documents any actions that are agreed to or concerns that are raised. If the heat and material balances and stream conditions are not shown on the PFD, printouts are usually provided for the reviewers.

For complex processes, a PFD review may begin with a brief overview of the process chemistry and block flow diagram to establish the context for the reviewers. In some cases the design basis assumptions are also reviewed at the start.

The main part of a PFD review is a “walk through” of the process by the process design engineer. Starting with one feed stream, the designer follows the stream from storage through all the process operations that it encounters. At each process operation, the designer explains the purpose of the operation, the design criteria, and the resulting condition of the stream at the exit. For example, the feed section shown in [Fig. 2.29](#) would be described as follows:

Feed of 99% pure technical grade A leaves floating roof storage tank T101 through line 101 at ambient conditions. The governing ambient temperature for heater design is winter low temperature of -5°C and for pump design is summer high temperature of 30°C . Stream 101 is pumped by centrifugal pump 101 to a pressure of 10 bar gauge, forming stream 102. Pump 101 has a standby spare, shown. Flow of feed A is regulated by flow control valve FCV100, with design pressure drop 1.3 bar. Stream 103 exits FCV100 and is sent to steam heater E101. The purpose of heater E101 is to heat feed to the reaction temperature of 180°C . High-pressure steam at 240°C is used as heat source. The steam rate is controlled by temperature controller TC101, which receives input from temperature indicator on the process stream, leaving E101 in Stream 104. The steam has been placed tube side in E101 because the process stream is nonfouling and steam is at high pressure, so this is expected to lead to the lowest-cost design. A pressure drop allowance of 0.7 bar has been assigned to E101. Stream 104 leaves E101 at the desired reactor feed temperature of 180°C and at reactor pressure 8 bar gauge and enters reactor R1.

As the designer steps through the PFD, the review group asks questions to challenge the design assumptions and identify potential missing equipment. In the previous example, some relevant questions could have been:

- Is it necessary to pass the feed through a filter before FCV100 to remove any crud that accumulated in the tank or came in with the feed?
- Why use high-pressure steam to heat all the way from ambient temperature? Couldn't process heat recovery be used for at least part of the heating?

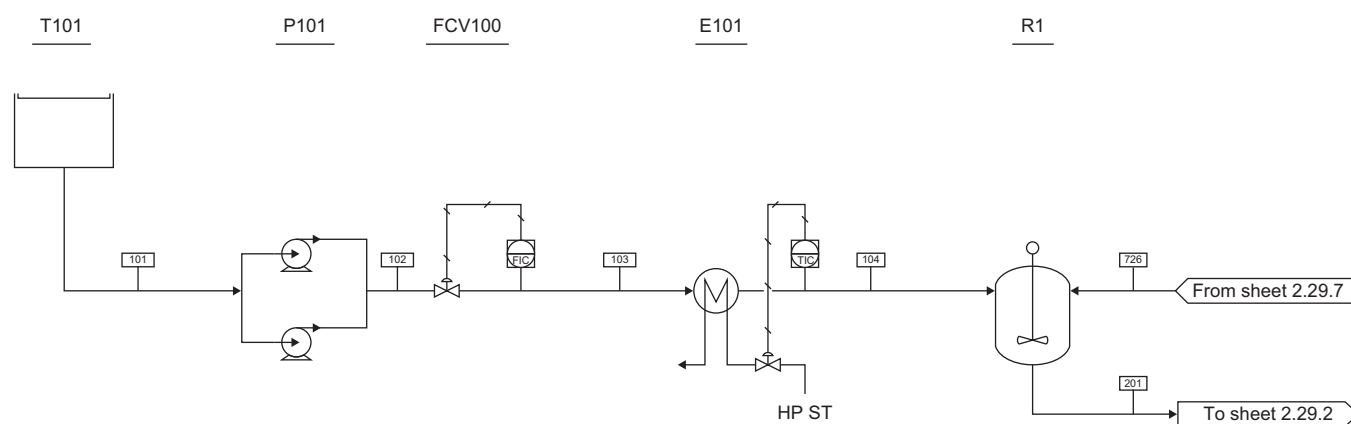


FIG. 2.29 Feed to a reactor.

- Why not use low-pressure steam to heat to 110 °C then high-pressure steam to final temperature?
- Should FCV100 be a separate control loop? Shouldn't it be in ratio to other reactor feeds?
- Did the team look at using a variable-speed drive on the pump to regulate flow instead of a pump and control valve? This would give lower energy consumption.
- What was the basis for choosing 180 °C and 8 bar gauge as the reactor conditions?
- Does the feed need to enter the reactor at reactor conditions? Would a colder or hotter feed reduce the reactor cooling or heating duty?

Some additional typical PFD review questions are given in [Table 2.2](#).

If the questions lead to modifications to the flowsheet that are immediately obvious and agreed to by all the reviewers, then these are marked up as corrections. If further analysis is required before deciding on a modification, it is noted as a follow-up action for the team.

The same procedure is followed for every stream in the PFD. Because it is often necessary to jump from one drawing or section of the flowsheet to another and back again, it is a good idea to mark streams with a highlighter when they have been completed so that the group does not overlook any streams.

Sufficient time must be allowed to complete the PFD review to the satisfaction of all the reviewers. The amount of time needed depends on the complexity and novelty of the design and the familiarity of the reviewers with the technology. For complex designs, a full PFD review can take several days to complete.

2.7.2 PFD review documentation and issue resolution

The notes taken at a PFD review usually include a long list of items that require follow-up by the design team. It is a good idea to include these notes and a description of how any issues and concerns were resolved in the design documentation. The notes should be circulated to meeting attendees immediately after the PFD review meeting to ensure that all issues were correctly captured.

If a PFD review indicated a need for substantial modification of the flowsheet, the group should reconvene after the modifications have been made to review the modified PFD. In process synthesis projects, several rounds of PFD review may be necessary.

2.8 Overall procedure for flowsheet development

[Fig. 2.30](#) shows an overall strategy for flowsheet selection and development. The chart in [Fig. 2.30](#) will generally lead to the selection of a commercially proven process or a modification of such a process when one exists. This reflects the commercial reality that very few business leaders are willing to risk a large sum of money (and their career and reputation) on unproven technology unless the financial return is very good and *no good alternatives are available*. Although process synthesis is enjoyable as a creative activity, the industrial practice of process design is usually more concerned with delivering designs that will work reliably and quickly make money to generate a return on investment. Successful companies are usually good at focusing the creativity of their employees on critical areas where innovation can provide a competitive advantage, without turning every project into an open-ended research problem.

TABLE 2.2 Sample questions that can be asked in a PFD review

Process Section	Questions
Feed preparation	How is (each) feed delivered? How is (each) feed stored? How much inventory of (each) feed is required? How is feed transferred from storage to the process? How is the rate of feed supply controlled? Is any feed pretreatment necessary before the feed is sent to the process? For solid feeds, is any feed size adjustment necessary? Is any heating or cooling needed before the feed is sent to the process?
Reaction	What are the reacting species? What side reactions occur? What are the reaction conditions and why were they chosen? How are the reaction conditions maintained or controlled? How are inventories of solids, liquid, or vapor in the reactor controlled? What are the reactor design specifications (e.g., residence time, interfacial area)? What are the estimated reactor yield and selectivity? What reactor type was chosen and why was it selected? Is a catalyst used? If so, is the catalyst stable or does it require periodic regeneration? Is heat addition or removal necessary? Are there specific safety issues that should be considered? (Additional information on reactor design is given in Chapter 15, which may prompt more questions)
Product Recovery	What is the purpose of each separation? What are the process conditions (temperature, pressure, etc.)? What are the equipment specifications (recovery, purity, etc.)? Why was a particular separation selected? Is heat removal or addition necessary? Can heat addition or removal be accomplished by process-to-process heat transfer? How are inventories of vapor, liquid, or solid controlled in each operation? How is the operation controlled to achieve the desired specifications? Are there specific safety issues that should be considered?
Purification	What are the final product specifications? What are the specifications for any by-products? What are the specifications for any effluents discharged to the environment? What are the specifications for any recycle streams returned to the process? What are the specifications for any waste streams sent to disposal? How are final purity specifications on any stream leaving the process achieved? How is the process controlled to ensure that purity specifications can be achieved?

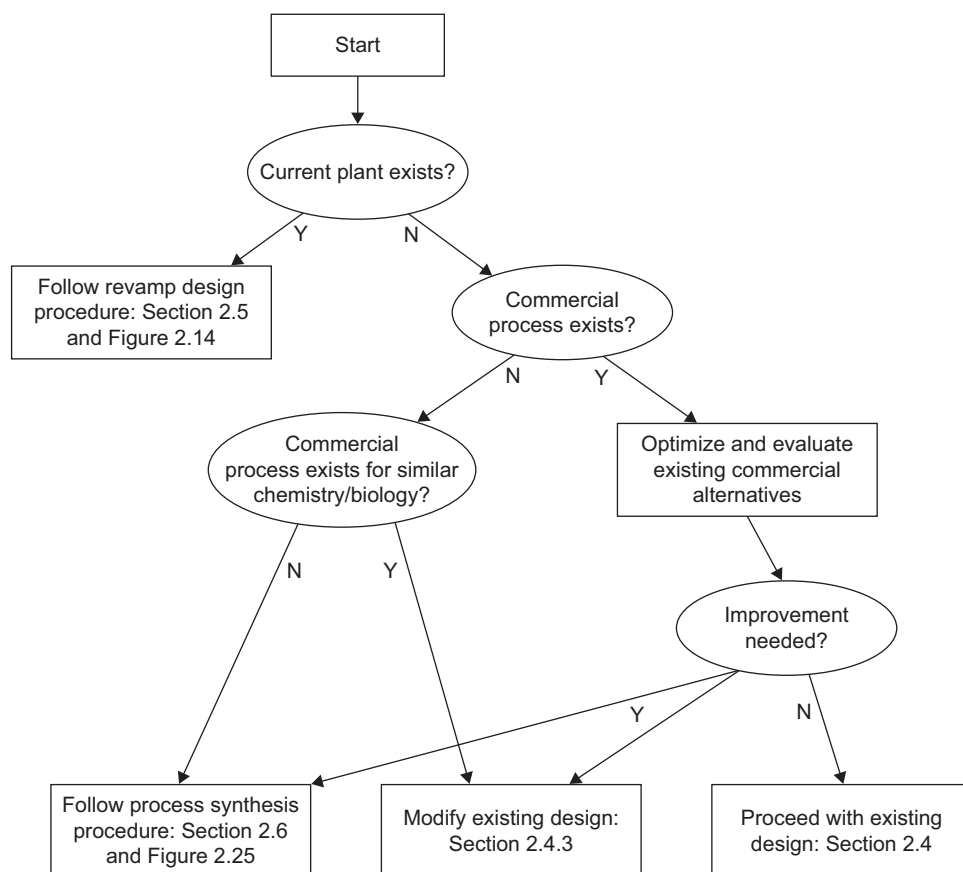


FIG. 2.30 Overall procedure for flowsheet development.

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2.10 Nomenclature

		Dimensions in MLT
A	Area for heat exchange	L^2
$C_{p,f}$	Specific heat capacity of feed	$L^2 T^{-2} \theta^{-1}$
$C_{p,i}$	Specific heat capacity of stream i	$L^2 T^{-2} \theta^{-1}$
$C_{p,p}$	Specific heat capacity of product	$L^2 T^{-2} \theta^{-1}$
F	Unknown flow, Example 2.6	MT^{-1}
F_{cw}	Flow rate of cooling water	MT^{-1}
F_t	Log mean temperature difference correction factor (see Chapter 19)	—
m_f	Mass flow rate of feed	MT^{-1}
m_i	Mass flow rate of stream i	MT^{-1}
m_p	Mass flow rate of product	MT^{-1}
Q	Rate of heat transfer	$ML^2 T^{-3}$
SV	Space velocity	T^{-1}
T_1	Unknown temperature, Example 2.7	θ
T_2	Unknown temperature, Example 2.7	θ
T_3	Unknown temperature, Example 2.7	θ
T_s	Steam temperature	θ
T_w	Cooling water outlet temperature	θ
ΔT_i	Change in temperature of stream i for a stream undergoing sensible heat changes	θ
ΔT_{lm}	Logarithmic mean temperature difference	θ
ΔT_m	Mean temperature difference for heat transfer	θ
U	Overall heat transfer coefficient	$MT^{-3} \theta^{-1}$
V	Reactor volume	L^3
V_{cat}	Catalyst fixed bed volume	L^3
v	Volumetric flow rate	$L^3 T^{-1}$
X	Unknown flow, Examples 2.1, 2.5	MT^{-1}
Y	Unknown flow, Example 2.5	MT^{-1}
Z	Unknown flow, Example 2.5	MT^{-1}
τ	Residence time	T

Common PFD abbreviations

BFW Boiler feed water

CW Cooling water

CTW Cooling tower water

DBO Design by others (used to refer to a section of a flowsheet that is purchased from a vendor as a packaged plant)

DIW Deionized water

HPS High-pressure steam

LPS Low-pressure steam

MPS Medium-pressure steam

NNF Normally no flow (used to indicate discontinuous operations in a continuous plant flowsheet)

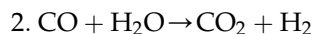
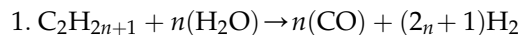
ROW Reverse-osmosis water

Abbreviations for designating process control instruments are given in [Chapter 5](#).

2.11 Problems

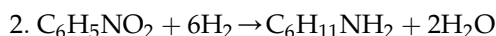
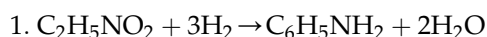
- 2.1. Cinnamic aldehyde (a fragrance compound) can be made by base-catalyzed aldol condensation reaction between benzaldehyde and acetaldehyde. The feeds are contacted with sodium hydroxide in a stirred tank reactor. The product is neutralized and washed with water to remove salts. The washed product is usually separated by batch distillation, in which unreacted feeds are recovered first, followed by product, and a polymeric waste is left as residue. The batch distillation product can be further purified by vacuum distillation. Sketch a block flow diagram of the process.
- 2.2. Cumene is made by alkylation of benzene with propylene over zeolitic catalyst. To maximize selectivity to desired products, several beds of catalyst are used inside the same reactor. A mixture of feed and recycle benzene enters the top of the reactor and the feed propylene is split so that a portion of the propylene is fed to each catalyst bed. A large excess of benzene is used to minimize propylene oligomerization and ensure complete reaction of propylene. The reactor product is cooled and sent to a stabilizer column that removes any light hydrocarbons. The bottoms from this column are sent to a benzene column that recovers benzene overhead for recycle to the alkylation and transalkylation reactors. The bottoms from the benzene column are distilled to give cumene product and a heavy stream. The heavy stream is further distilled in a heavies column to give a mixture of dipropyl- and tripropyl-benzene overhead and a heavy waste stream as bottoms. The distillate from the heavies column is sent to a transalkylation reactor and reacted with excess benzene. The product from the transalkylation reactor is returned to the benzene column. Sketch a block flow diagram of the process.
- 2.3. Cyclosporin A is produced by fermentation using either *Cylindrocarpum lucidum* Booth or *Tolypocladium inflatum* Gams. The fermentation is carried out in batch reactors, which are filled with a feed medium, inoculated with the fungi, and aerated for a period of 13 days. The reactor product is milled and extracted with 90% methanol. The methanol is evaporated off to give an aqueous solution that is then extracted with ethylene chloride. The organic solution is evaporated to dryness, and then the product is purified by chromatography in methanol over aluminum oxide or silica gel. Sketch a block flow diagram of the process.
- 2.4. The composition of a gas derived by the gasification of coal is volume percentage: carbon dioxide 4, carbon monoxide 16, hydrogen 50, methane 15, ethane 3, benzene 2, balance nitrogen. If the gas is burnt in a furnace with 20% excess air, calculate:
- The amount of air required per 100 kmol of gas
 - The amount of flue gas produced per 100 kmol of gas
 - The composition of the flue gases, on a dry basis
- Assume complete combustion.
- 2.5. The off-gases from a gasoline stabilizer are fed to a steam reforming plant to produce hydrogen. The composition of the off-gas, molar%, is CH_4 77.5, C_2H_6 9.5, C_3H_8 8.5, C_4H_{10} 4.5. The gases entering the reformer are at a pressure of 2 bara and 35 °C and the feed rate is 2000 m³/h.

The reactions in the reformer are:



The molar conversion of $\text{C}_2\text{H}_{2n+2}$ in reaction (1) is 96% and of CO in reaction (2) is 92%. Calculate:

- The average molecular mass of the off-gas
 - The mass of gas fed to the reformer, kg/h
 - The mass of hydrogen produced, kg/h
- 2.6. Allyl alcohol can be produced by the hydrolysis of allyl chloride. Together with the main product, allyl alcohol, di-allyl ether is produced as a by-product. The conversion of allyl chloride is typically 97% and the selectivity to alcohol is 90%, both on a molar basis. Assuming that there are no other significant side reactions, calculate masses of alcohol and ether produced per 1000 kg of allyl chloride fed to the reactor.
- 2.7. Aniline is produced by the hydrogenation of nitrobenzene. A small amount of cyclo-hexylamine is produced as a by-product. The reactions are:



Nitrobenzene is fed to the reactor as a vapor, with three times the stoichiometric quantity of hydrogen. The conversion of the nitrobenzene to all products is 96%, and the selectivity for aniline is 95%.

The unreacted hydrogen is separated from the reactor products and recycled to the reactor. A purge is taken from the recycle stream to maintain the inerts in the recycle stream below 5%. The fresh hydrogen feed is 99.5% pure, with the remainder being inerts. All percentages are molar.

For a feed rate of 100 kmol/h of nitrobenzene, calculate:

- The fresh hydrogen feed
 - The purge rate required
 - The composition of the reactor outlet stream
- 2.8. Guaifenesin [guaiacol glyceryl ether, 3-(2-methoxyphenoxy)-1,2-propanediol, $\text{C}_{10}\text{H}_{14}\text{O}_4$] is an expectorant that is found in cough medicines such as Actifed and Robitussin. U.S. Patent 4,390,732 (to Degussa) describes a preparation of the active pharmaceutical ingredient (API) from guaicol (2-methoxyphenol, $\text{C}_7\text{H}_8\text{O}_2$) and glycidol (3-hydroxy propylene oxide, $\text{C}_3\text{H}_6\text{O}_2$). When the reaction is catalyzed by NaOH, the reaction yield is 93.8%. The product is purified in a thin-film evaporator, giving an overall plant yield of 87%.
- Estimate the feed flow rates of glycidol and guaicol that would be needed to produce 100 kg/day of the API.
 - Estimate how much product is lost in the thin-film evaporator.
 - How would you recover the product lost in the evaporator?
- 2.9. 11-[N-ethoxycarbonyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine ($\text{C}_{22}\text{H}_{23}\text{ClN}_2\text{O}_2$) is a non-sedative antihistamine known as loratadine and marketed as Claritin. The preparation of the API is described in U.S. Patent 4,282,233 (to Schering). The patent describes reacting 16.2g of 11-[N-methyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine ($\text{C}_{20}\text{H}_{21}\text{ClN}_2$) in 200 mL of benzene with 10.9 g of ethylchloroformate ($\text{C}_3\text{H}_5\text{ClO}_2$) for 18 hours. The mixture is cooled, poured into ice water, and separated into aqueous and organic phases. The organic layer is washed with water and evaporated to dryness. The residue is triturated (ground to a fine powder) with petroleum ether and recrystallized from isopropyl ether.
- What is the reaction by-product?
 - The reaction appears to be carried out under conditions that maximize both selectivity and conversion (long time at low temperature), as might be expected given the cost of the raw material. If the conversion is 99.9% and the selectivity for the desired ethoxycarbonyl substituted compound is 100%, how much excess ethylchloroformate remains at the end of the reaction?

- c. What fraction of the ethylchloroformate feed is lost to waste products?
 - d. Assuming that the volumes of water and isopropyl ether used in the quenching, washing, and recrystallization steps are the same as the initial solvent volume and that none of these materials are reused in the process, estimate the total mass of waste material produced per kg of the API.
 - e. If the recovery (plant yield) of the API from the washing and recrystallization steps is 92%, estimate the feed flow rates of 11-[N-methyl-4-piperidylidene]-8-chloro-6,11-dihydro-5H-benzo-[5,6]-cyclohepta-[1,2-b]-pyridine and ethylchloroformate required to produce a batch of 10 kg of the API.
 - f. How much API could be produced per batch in a 3.8 m³ (1000 U.S. gal) reactor?
 - g. What would be the advantages and disadvantages of carrying out the other process steps in the same vessel?
 - h. Sketch a block flow diagram of the process.
- 2.10. Describe the main commercial process used to make each of the following compounds. Include a block flow diagram.
- a. Phosphoric acid
 - b. Adipic acid
 - c. Polyethylene terephthalate
 - d. Insulin
 - e. Sorbitol
- 2.11. Example 2.7 introduced a heat-exchange system revamp problem. If the plate heat exchanger E101 in Fig. 2.19 is the gasketed-plate type, then the exchanger area can be increased by adding plates to the exchanger. How much additional area must be added to E101 to allow the system to operate at 50% above base case flow rate with no changes in temperature of steam or cooling water feed temperatures and no modification to exchangers E102 and E103?
- 2.12. Styrene is made by dehydrogenation of ethylbenzene, which is formed by alkylation of benzene with ethylene. Propose target yields for the alkylation and dehydrogenation steps if the forecasted prices per metric ton are styrene \$800, ethylene \$800, and benzene \$500.
- Note: Structures for the compounds in Problems 2.3, 2.8, and 2.9 can be found in the Merck Index, but are not required to solve the problems.

Utilities and energy-efficient design

KEY LEARNING OBJECTIVES

- How processes are heated and cooled
- The systems used for delivering steam, cooling water, and other site utilities
- Methods used for recovering process waste heat
- How to use the pinch design method to optimize process heat recovery
- How to design a heat-exchanger network
- How energy is managed in batch processes

3.1 Introduction

Very few chemical processes are carried out entirely at ambient temperature. Most require process streams to be heated or cooled to reach the desired operation temperature; to add or remove heats of reaction, mixing, adsorption, etc.; to sterilize feed streams; or to cause vaporization or condensation. Gas and liquid streams are usually heated or cooled by indirect heat exchange with another fluid: either another process stream or a utility stream such as steam, hot oil, cooling water, or refrigerant. The design of heat exchange equipment for fluids is addressed in [Chapter 19](#). Solids are usually heated and cooled by direct heat transfer, as described in [Chapter 18](#). This chapter begins with a discussion of the different utilities that are used for heating, cooling, and supplying other needs such as power, water, and air to a process.

The consumption of energy is a significant cost in many processes. Energy costs can be reduced by recovering waste heat from hot process streams and by making use of the fuel value of waste streams. Section 3.4 discusses how to evaluate waste stream combustion as a source of process heat. Section 3.3 introduces other heat recovery approaches.

When it is economically attractive, heating and cooling are accomplished by heat recovery between process streams. The design of a network of heat exchangers for heat recovery can be a complex task if there are many hot and cold streams in a process. Pinch analysis, introduced in Section 3.5, is a systematic method for simplifying this problem.

The efficient use of energy in batch and cyclic processes is made more complicated by the sequential nature of process operations. Some approaches to energy-efficient design of batch and cyclic processes are discussed in Section 3.6.

3.2 Utilities

The word *utilities* is used for the ancillary services needed in the operation of any production process. These services are normally supplied from a central site facility and include:

1. Electricity
2. Fuel for fired heaters
3. Fluids for process heating
 - a. Steam
 - b. Hot oil or specialized heat transfer fluids
4. Fluids for process cooling
 - a. Cooling water
 - b. Chilled water
 - c. Refrigeration systems
5. Process water
 - a. Water for general use
 - b. Demineralized water
6. Compressed air
7. Inert-gas supplies (usually nitrogen)

Most plants are located on sites where the utilities are provided by the site infrastructure. The price charged for a utility is mainly determined by the operating cost of generating and transmitting the utility stream. Some companies also include a capital recovery charge in the utility cost, but if this is done, then the offsite (OSBL) capital cost of projects must be reduced to avoid double counting and biasing the project capital–energy trade-off, leading to poor use of capital.

Some smaller plants purchase utilities “over the fence” from a supplier, such as a larger site or a utility company, in which case the utility prices are set by contract and are typically pegged to the price of natural gas, fuel oil, or electricity.

The utility consumption of a process cannot be estimated accurately without completing the material and energy balances and carrying out a pinch analysis, as described in Section 3.5.6. The pinch analysis gives targets for process heat recovery and hence for the minimum requirements of hot and cold utilities. More detailed optimization then translates these targets into expected demands for fired heat, steam, electricity, cooling water, and refrigeration. In addition to the utilities required for heating and cooling, the process may need process water and air for applications such as washing, stripping, and instrument air supply. Good overviews of methods for design and optimization of utility systems are given by [Smith \(2016\)](#), [Knopf \(2012\)](#), [Zhu \(2014\)](#), and [Kemp \(2007\)](#).

3.2.1 Electricity

The electricity demand of the process is mainly determined by the work required for pumping, compression, air coolers, and solids-handling operations, but also includes the power needed for instruments, lights, and other small users. The power required may be generated on site, but will more usually be purchased from the local supply company. Some plants generate their own electricity using a gas-turbine cogeneration plant with a heat recovery steam generator (waste heat boiler) to raise steam (Fig. 3.1). The overall thermal efficiency of such systems can be

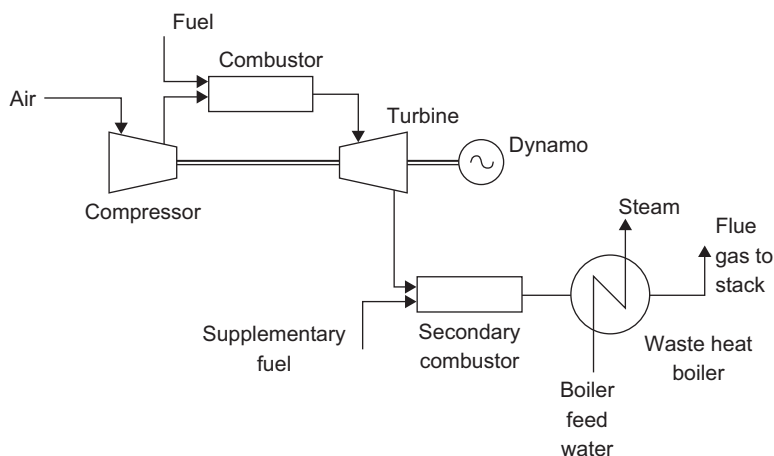


Fig. 3.1 Gas turbine–based cogeneration plant.

in the range of 70% to 80%, compared with the 30% to 40% obtained from a conventional power station, where the heat in the exhaust steam is wasted in the condenser. The cogeneration plant can be sized to meet or exceed the plant electricity requirement, depending on whether the export of electricity is an attractive use of capital. This “make or buy” scenario gives chemical producers strong leverage when negotiating electric power contracts, and they are usually able to purchase electricity at or close to wholesale prices. Wholesale electricity prices vary regionally (see www.eia.gov for details), but are typically about \$0.06/kWh in North America at the time of writing.

The voltage at which the supply is taken or generated will depend on the demand. In the United States, power is 135, 220, 550, or 750 kV. Local substations step the power down to 35 to 69 kV for medium-voltage transmission and then to 4 to 15 kV local distribution lines. Transformers at the plant are used to step down the power to the supply voltages used on site. Most motors and other process equipment run on 208 V three-phase power, whereas 120/240 V single-phase power is used for offices, labs, and control rooms.

On any site it is always worth considering driving large compressors and pumps with steam turbines instead of electric motors and using the exhaust steam for local process heating.

Electric power is rarely used for heating in large-scale chemical plants, although it is often used in smaller batch processes that handle nonflammable materials, such as biological processes. The main disadvantages of electrical heating for large-scale processes are:

- Heat from electricity is typically two to three times more expensive than heat from fuels because of the thermodynamic inefficiency of power generation.
- Electric heating requires very high power draws that would substantially increase the electrical infrastructure costs of the site.
- Electric heating apparatus is expensive, requires high maintenance, and must comply with stringent safety requirements when used in areas where flammable materials may be present.
- Electric heaters are intrinsically less safe than steam systems. The maximum temperature that a steam heater can reach is the temperature of the steam. The maximum temperature of an electric heater is determined by the temperature controller (which could fail) or by burnout of the heating element. Electric heaters therefore have a higher probability of overheating.

Electric heating is more likely to be attractive in small-scale batch or cyclic processes, particularly when the cost of heating is a small fraction of overall process costs and when the process calls for rapid on-off heating.

A detailed account of the factors to be considered when designing electrical distribution systems for chemical process plants and the equipment used (transformers, switch gear, and cables) is given by [Silverman \(1964\)](#). Requirements for electrical equipment used in hazardous (classified) locations are given in the National Electrical Code (NFPA 70), as described in Section 10.3.5.

3.2.2 Fired heat

Fired heaters are used for process heating duties above the highest temperatures that can be reached using high-pressure steam, typically about 250 °C (482 °F). Process streams may be heated directly in the furnace tubes or indirectly using a hot oil circuit or heat transfer fluid, as described in Section 3.2.4. The design of fired heaters is described in Section 19.17. The cost of fired heat can be calculated from the price of the fuel fired. Most fired process heaters use natural gas as fuel, as it is cleaner burning than fuel oil and therefore easier to fit NO_x control systems and obtain permits. Natural gas also requires less maintenance of burners and fuel lines, and natural gas burners can often cofire process waste streams, such as hydrogen, light organic compounds, or air saturated with organic compounds.

Natural gas and heating oil are traded as commodities, and prices can be found at any online trading site or business news site (e.g., www.cnn.com/business). Historic prices for forecasting can be found in the *Oil and Gas Journal* or from the U.S. Energy Information Agency (www.eia.gov).

The fuel consumed in a fired heater can be estimated from the fired heater duty divided by the furnace efficiency. The furnace efficiency will typically be about 0.85 if both the radiant and convective sections are used (see [Chapter 19](#)) and about 0.6 if the process heating is in the radiant section only.

Example 3.1

Estimate the annual cost of providing heat to a process from a fired heater using natural gas as fuel if the process duty is 4 MW and the price of natural gas is \$3.20/MMBtu (million Btu).

Solution

If we assume that the fired heater uses both the radiant and convective sections, then we can start by assuming a heater efficiency of 0.85, so:

$$\text{Fuel required} = \text{Heater duty} / \text{heater efficiency} = 4 / 0.85 = 4.71 \text{ MW}$$

$$1 \text{ Btu/h} = 0.29307 \text{ W, so } 4.71 \text{ MW} = 4.71 / 0.29307 = 16.07 \text{ MMBtu/h}$$

Assuming 8000 operating hours per year, the total annual fuel consumption would be:

$$\text{Annual fuel use} = 16.07 \times 8000 = 128.6 \times 10^3 \text{ MMBtu}$$

$$\text{Annual cost of fired heat} = 128.6 \times 10^3 \times 3.20 = \$411,400$$

Note that if we had decided to carry out all of the heating in the radiant section only, then the fuel required would have been $4 / 0.6 = 6.67 \text{ MW}$ and the annual cost of heating would increase to \$582,600 unless we could find some other process use for the heat available in the convective section of the heater.

3.2.3 Steam

Steam is the most widely used heat source in most chemical plants. Steam has a number of advantages as a hot utility:

- The heat of condensation of steam is high, giving a high heat output per pound of utility at constant temperature (compared with other utilities such as hot oil and flue gas that release sensible heat over a broad temperature range).
- The temperature at which heat is released can be precisely controlled by controlling the pressure of the steam. This enables tight temperature control, which is important in many processes.
- Condensing steam has very high heat transfer coefficients, leading to cheaper heat exchangers.
- Steam is nontoxic, nonflammable, visible if it leaks externally and inert to many (but not all) process fluids.

Most sites have a pipe network supplying steam at three or more pressure levels for different process uses. A typical steam system is illustrated in Fig. 3.2. Boiler feed water (BFW) at high pressure is preheated and fed to boilers, where high-pressure (HP) steam is raised and superheated above the dew point to allow for heat losses in the piping. Boiler feed water preheat can be accomplished using process waste heat or convective section heating in the boiler plant. HP steam is typically at about 40 bar, corresponding to a condensing temperature of 250 °C, but every site is different. Some of the HP steam is used for process heating at high temperatures. The remainder of the HP steam is expanded either through let-down valves or steam turbines known as *back-pressure turbines* to form

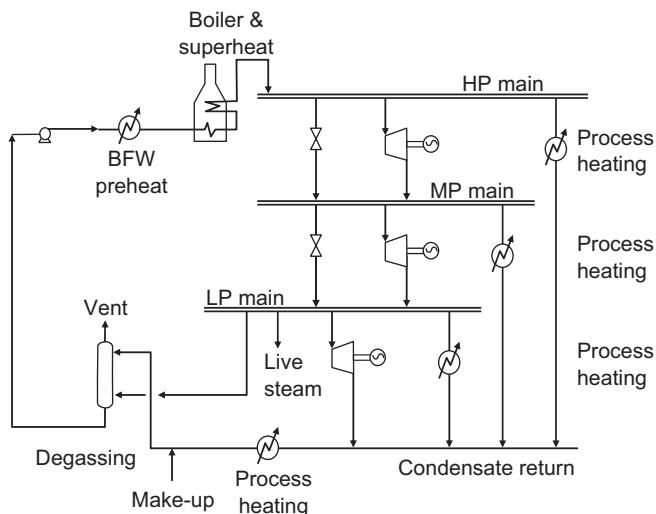


Fig. 3.2 Steam system.

medium-pressure (MP) steam. The pressure of the MP steam mains varies widely from site to site, but is typically about 20 bar, corresponding to a condensing temperature of 212 °C. MP steam is used for intermediate-temperature heating or expanded to form low-pressure (LP) steam, typically at about 3 bar, condensing at 134 °C. Some of the LP steam may be used for process heating if there are low-temperature heat requirements. LP (or MP or HP) steam can also be expanded in condensing turbines to generate shaft work for process drives or electricity production. A small amount of LP steam is used to strip dissolved noncondensable gases such as air from the condensate and make-up water. LP steam is also often used as “live steam” in the process, for example, as stripping vapor or for cleaning, purging, or sterilizing equipment.

When steam is condensed without coming into contact with process fluids, the hot condensate can be collected and returned to the boiler feed water system. Condensate can also sometimes be used as a low-temperature heat source if the process requires low-temperature heat.

The price of HP steam can be estimated from the cost of boiler feed water treatment, the price of fuel, and the boiler efficiency:

$$P_{HPS} = P_F \times \frac{dH_b}{\eta_B} + P_{BFW} \quad (3.1)$$

where: P_{HPS} = price of HP steam (\$/1000 lb, commonly written \$/Mlb)

P_F = price of fuel (\$/MMBtu)

dH_b = heating rate (MMBtu/Mlb steam)

η_B = boiler efficiency

P_{BFW} = price or cost of boiler feed water (\$/Mlb)

Package boilers typically have efficiencies similar to fired heaters, in the range of 0.8 to 0.9. The heating rate should include boiler feed water preheat, the latent heat of vaporization, and the superheat specified. The steam for process heating is usually generated in water-tube boilers, using the most economical fuel available.

The cost of boiler feed water includes allowances for water make-up, chemical treatment, and degassing, and is typically about twice the cost of raw water (see later). If no information on the price of water is available, then 0.50 \$/1000 lb can be used as an initial estimate. If the steam is condensed and the condensate is returned to the boiler feed water (which will normally be the case), then the price of steam should include a credit for the condensate. The condensate credit will often be close enough to the boiler feed water cost that the two terms cancel each other out and can be neglected.

The prices of MP and LP steam are usually discounted from the HP steam price to allow for the shaft work credit that can be gained by expanding the steam through a turbine and also to encourage process heat recovery by raising steam at intermediate levels and using low-grade heat when possible. Several methods of discounting are used. The most rational of these is to calculate the shaft work generated by expanding the steam between levels and price this as equivalent to electricity (which could be generated by attaching the turbine to a dynamo or else would be needed to run a motor to replace the turbine if it is used as a driver). The value of the shaft work then sets the discount between steam at different levels. This is illustrated in the following example.

Example 3.2

A site has steam levels at 40 bar, 20 bar, and 6 bar. The price of fuel is \$6/MMBtu, and electricity costs \$0.05/kWh. If the boiler efficiency is 0.8 and the steam turbine efficiency is 0.85, suggest prices for HP, MP, and LP steam.

Solution

The first step is to look up the steam conditions, enthalpies, and entropies in steam tables:

Steam level	HP	MP	LP
Pressure (bar)	40	20	6
Saturation temperature (°C)	250	212	159

The steam will be superheated above the saturation temperature to allow for heat losses in the pipe network. The following superheat temperatures were set to give an adequate margin above the saturation temperature for HP steam and also to give (roughly) the same specific entropy for each steam level. The actual superheat temperatures of MP and LP steam will be higher due to the nonisentropic nature of the expansion.

Superheat temperature (°C)	400	300	160
Specific entropy, sg, (kJ/kg.K)	6.769	6.768	6.761
Specific enthalpy, hg, (kJ/kg)	3214	3025	2757

We can then calculate the difference in enthalpy between levels for isentropic expansion:

Isentropic delta enthalpy (kJ/kg)	189	268
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Multiplying by the turbine efficiency gives the nonisentropic enthalpy of expansion:

Actual delta enthalpy (kJ/kg)	161	228
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This can be converted to give the shaft work in customary units:

Shaft work (kWh/Mlb)	20.2	28.7
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Multiplying by the price of electricity converts this into a shaft work credit:

Shaft work credit (\$/Mlb)	1.01	1.44
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The price of HP steam can be found from Equation 3.1, assuming that the boiler feed water cost is cancelled out by a condensate credit. The other prices can then be estimated by subtracting the shaft work credits.

Steam price (\$/Mlb)	<u>6.48</u>	<u>5.47</u>	<u>4.03</u>
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For quick estimates, this example can easily be coded into a spreadsheet and updated with the current prices of fuel and power. A sample steam-costing spreadsheet is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

3.2.4 Hot oil and heat transfer fluids

Circulating systems of hot oil or specialized heat transfer fluids are often used as heat sources in situations where fired heat or steam is not suitable. Heat transfer fluids and mineral oils can be used over a temperature range from 50 °C to 400 °C. The upper temperature limit on use of hot oils is usually set by thermal decomposition of the oil, fouling, or coking of heat-exchange tubes. Some heat transfer fluids are designed to be vaporized and condensed in a similar manner to the steam system, though at lower pressures; however, vaporization of mineral oils is usually avoided, as less volatile components in the oil could accumulate and decompose, causing accelerated fouling.

The most common situation where a hot oil system is used is in plants that have many relatively small high-temperature heating requirements. Instead of building several small fired heaters, it can be more economical to supply heat to the process from circulating hot oil streams and build a single fired heater that heats the hot oil. Use of hot oil also reduces the risk of process streams being exposed to high tube-wall temperatures that might be experienced in a fired heater. Hot oil systems are often attractive when there is an HP differential between the process stream and HP steam and use of steam would entail using thicker tubes. Hot oil systems can sometimes be justified on safety grounds if the possibility of steam leakage into the process is hazardous.

The most commonly used heat transfer fluids are mineral oils and Dowtherm A. Mineral oil systems usually require large flow rates of circulating liquid oil. When the oil is taken from a process stream, as is common in oil refining processes, then it is sometimes called a *pump-around system*. Dowtherm A is a mixture of 26.5 wt% diphenyl in diphenyl oxide. Dowtherm A is thermally stable and is usually operated in a vaporization–condensation cycle similar to the steam system, although condensed liquid Dowtherm is sometimes used for intermediate-temperature heating requirements. The design of Dowtherm systems and other proprietary heat transfer fluids is discussed in detail in Singh (1985) and Green and Southard (2018).

The cost of the initial charge of heat transfer fluid usually makes a negligible contribution to the overall cost of running a hot oil system. The main operating cost is the cost of providing heat to the hot oil in the fired heater or

vaporizer. If a pumped liquid system is used, then the pumping costs should also be evaluated. The costs of providing fired heat are discussed in Section 3.2.2. Hot oil heaters or vaporizers usually use both the radiant and convective sections of the heater and have heater efficiencies in the range 80% to 85%.

3.2.5 Cooling water

When a process stream requires cooling at high temperature, various heat recovery techniques should be considered. These include transferring heat to a cooler process stream, raising steam, preheating boiler feed water, etc., as discussed in Section 3.3.

When heat must be rejected at lower temperatures, below about 120 °C (248 °F) (more strictly, below the pinch temperature), then a cold utility stream is needed. Cooling water is the most commonly used cold utility in the temperature range of 120 °C to 40 °C, although air cooling is preferred in regions where water is expensive or the ambient humidity is too high for cooling water systems to operate effectively. The selection and design of air coolers are discussed in Section 19.16. If a process stream must be cooled to a temperature below 40 °C, cooling water or air cooling would be used down to a temperature in the range 40° to 50 °C, followed by chilled water or refrigeration down to the target temperature.

Natural and forced-draft cooling towers are generally used to provide the cooling water required on a site unless water can be drawn from a convenient river or lake in sufficient quantity. Sea water or brackish water can be used at coastal sites and for offshore operations, but if used directly will require the use of more expensive materials of construction for heat exchangers (see Chapter 6). The minimum temperature that can be reached with cooling water depends on the local climate. Cooling towers work by evaporating part of the circulating water to ambient air, causing the remaining water to be chilled. If the ambient temperature and humidity are high, then a cooling water system will be less effective, and air coolers or refrigeration would be used instead.

A schematic of a cooling water system is shown in Fig. 3.3. Cooling water is pumped from the cooling tower to provide coolant for the various process cooling duties. Each process cooler is served in parallel, and cooling water almost never flows to two coolers in series. The warmed water is returned to the cooling tower and cooled by partial evaporation. A purge stream known as a *blowdown* is removed upstream of the cooling tower to prevent the accumulation of dissolved solids as water evaporates from the system. A make-up stream is added to compensate for evaporative losses, blowdown losses, and any other losses from the system. Small amounts of chemical additives are also usually fed into the cooling water to act as biocides and corrosion and fouling inhibitors.

The cooling tower consists of a means of providing high surface area for heat and mass transfer between the warm water and ambient air and a means of causing air to flow counter-current to the water. The surface area for contact is usually provided by flowing the water over wooden slats or high-voidage packing. The cooled water is then collected at the bottom of the tower. In most modern cooling towers the air flow is induced by fans that are placed above the packed section of the tower. For very large cooling loads natural-draft cooling towers are used, in which a large hyperbolic chimney is placed above the packed section to induce air flow. Some older plants use spray ponds instead of cooling towers.

Cooling water systems can be designed using a psychrometric chart if the ambient conditions are known. A psychrometric chart is given in Fig. 3.4. The cooling tower is usually designed so that it will operate effectively except under the hottest (or most humid) conditions that can be expected to occur no more than a few days each year.

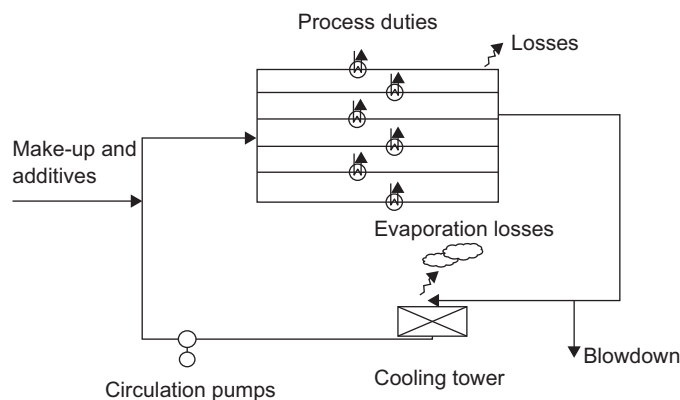


Fig. 3.3 Schematic of cooling water system.

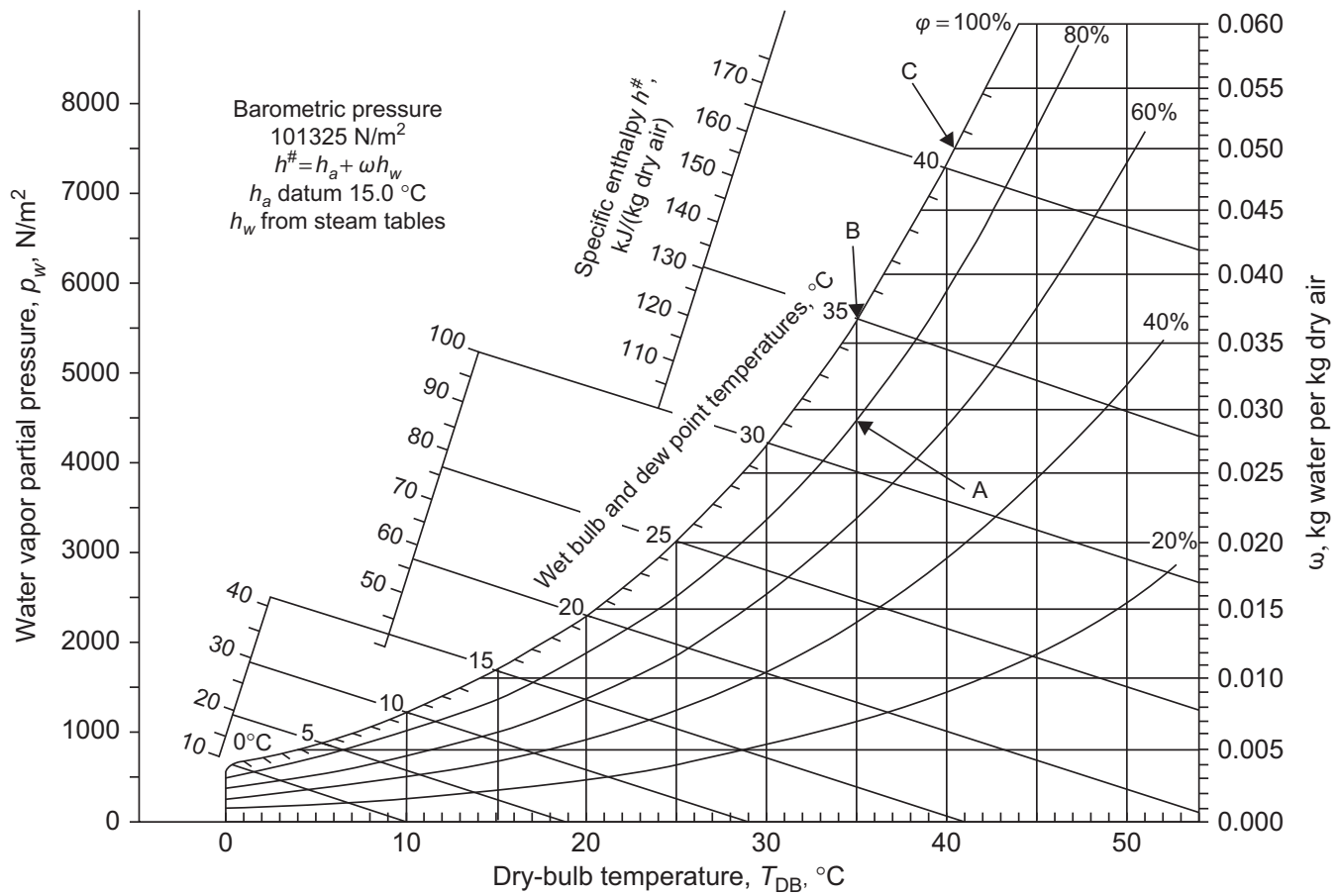


Fig. 3.4 Psychrometric chart. Adapted with permission from Balmer, R. [2010]. *Thermodynamic tables to accompany modern engineering thermodynamics* [Academic Press].

The ambient temperature and humidity can be plotted on the psychrometric chart, allowing the inlet air wet bulb temperature to be determined. This is the coldest temperature that the cooling water could theoretically reach; however, in practice most cooling towers operate with a temperature approach to the air wet bulb temperature of at least 2.8 °C (5 °F). Adding the approach temperature to the inlet air wet bulb temperature, we can then mark the outlet water condition on the saturation curve. For example, if the hottest ambient condition for design purposes is a dry bulb temperature of 35 °C (95 °F) with 80% humidity, then we can mark this point on the psychrometric chart (point A) and read the wet bulb temperature as roughly 32 °C (89.6 °F). Adding a 2.8 °C temperature approach would give a cold water temperature of about 35 °C (95 °F), which can then be marked on the saturation line (point B).

The inlet water condition, or cooling water return temperature, is found by optimizing the trade-off between water circulation costs and cooling tower cost. The difference between the cooling water supply (coldest) and return (hottest) temperatures is known as the *range* or *cooling range* of the cooling tower. As the cooling range increases, the cost of the cooling tower increases, but the water flow rate that must be circulated decreases, and hence the pumping cost decreases. Note that because most of the cooling is accomplished by evaporation of water rather than transfer of sensible heat to the air, the evaporative losses do not vary much with the cooling range. Most cooling towers are operated with a cooling range between 2.8° to 11.1 °C (5° and 20 °F). A typical initial design point would be to assume a cooling water return temperature about 5.5 °C (10 °F) hotter than the cold water temperature. In the example earlier, this would give a cooling water return temperature of 40.5 °C (105 °F), which can also be marked on the psychrometric chart (point C). The difference in air humidity across the cooling tower can now be read from the right-hand axis as the difference in moisture loadings between the inlet air (point A) and the outlet air (point C). The cooling tower design can then be completed by determining the cooling load of the water from an energy balance and hence determining the flow rate of air that is needed based on the change in air humidity between ambient air and the air exit condition. The exit air is assumed to have a dry bulb temperature equal to the cooling water return temperature, and the minimum air flow will be obtained when the air leaves saturated with moisture. Examples of the detailed design of cooling towers are given in [Green and Southard \(2018\)](#).

When carrying out the detailed design of a cooling tower, it is important to check that the cooling system has sufficient capacity to meet the site cooling needs over a range of ambient conditions. In practice, multiple cooling water pumps are usually used so that a wide range of cooling water flow rates can be achieved. When new capacity is added to an existing site, the limit on the cooling system is usually the capacity of the cooling tower. If the cooling tower fans cannot be upgraded to meet the increased cooling duty, additional cooling towers must be added. In such cases, it is often cheaper to install air coolers for the new process rather than upgrading the cooling water system.

The cost of providing cooling water is mainly determined by the cost of electric power. Cooling water systems use power for pumping the cooling water through the system and for running fans (if installed) in the cooling towers. They also have costs for water make-up and chemical treatment. The power used in a typical recirculating cooling water system is usually between 1 and 2 kWh/1000 gal of circulating water. The costs of water make-up and chemical treatment usually add about \$0.02/1000 gal.

3.2.6 Refrigeration

Refrigeration is needed for processes that require temperatures below those that can be economically obtained with cooling water (i.e., below about 40 °C). For temperatures down to around 10 °C, chilled water can be used. For lower temperatures, down to −30 °C, salt brines (NaCl and CaCl₂) are sometimes used to distribute the “refrigeration” around the site from a central refrigeration machine. Large refrigeration duties are usually supplied by a stand-alone packaged refrigeration system.

Vapor compression refrigeration machines are normally used, as illustrated in Fig. 3.5. The working fluid (refrigerant) is compressed as a vapor and then cooled and condensed at HP, allowing heat rejection at high temperature in an exchanger known as a *condenser*. Heat is usually rejected to a coolant such as cooling water or ambient air. The liquid refrigerant is then expanded across a valve to a lower pressure, where it is vaporized in an exchanger known as an *evaporator*, taking up heat at low temperature. The vapor is then returned to the compressor, completing the cycle.

The working fluid for a refrigeration system must satisfy a broad range of requirements. It should have a boiling point that is colder than the temperature that must be reached in the process at a pressure that is above atmospheric pressure (to prevent leaks into the system). It should have a high latent heat of evaporation to reduce refrigerant flow rate. The system should operate well below the critical temperature and pressure of the refrigerant, and the condenser pressure should not be too high or the cost will be excessive. The freezing temperature of the refrigerant must be well below the minimum operating temperature of the system. The refrigerant should also be nontoxic, nonflammable, and have minimal environmental impact.

A wide range of materials have been developed for use as refrigerants, most of which are halogenated hydrocarbons. In some situations ammonia, nitrogen, and carbon dioxide are used. Cryogenic gas separation processes usually use the process fluids as working fluid; for example, ethylene and propylene refrigeration cycles are used in ethylene plants.

Refrigeration systems use power to compress the refrigerant. The power can be estimated using the cooling duty and the refrigerator coefficient of performance (COP).

$$COP = \frac{\text{Refrigeration produced (Btu/hr or MW)}}{\text{Shaft work used (Btu/hr or MW)}} \quad (3.2)$$

The COP is a strong function of the temperature range over which the refrigeration cycle operates. For an ideal refrigeration cycle (a reverse Carnot cycle), the COP is:

$$COP = \frac{T_e}{(T_c - T_e)} \quad (3.3)$$

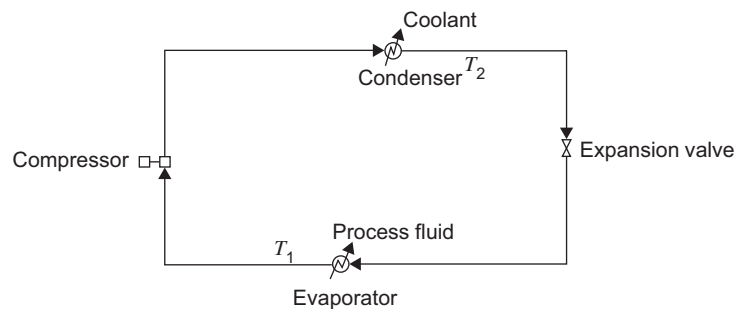


Fig. 3.5 Simple refrigeration cycle.

where: T_e = evaporator absolute temperature (K)

T_c = condenser absolute temperature (K)

The *COP* of real refrigeration cycles is always less than the Carnot efficiency. It is usually about 0.6 times the Carnot efficiency for a simple refrigeration cycle, but can be as high as 0.9 times the Carnot efficiency if complex cycles are used. If the temperature range is too large, then it may be more economical to use a cascaded refrigeration system, in which a low-temperature cycle rejects heat to a higher-temperature cycle that rejects heat to cooling water or ambient air. Good overviews of refrigeration cycle design are given by [Dincer \(2017\)](#), [Stoecker \(1998\)](#), and [Hundy et al. \(2008\)](#).

The operating cost of a refrigeration system can be determined from the power consumption and the price of power. Refrigeration systems are usually purchased as packaged modular plants, and the capital cost can be estimated using commercial cost estimating software, as described in Section 7.10. An approximate correlation for the capital cost of packaged refrigeration systems is also given in Table 7.2.

Example 3.3

Estimate the annual operating cost of providing refrigeration to a condenser with duty 1.2 MW operating at -5°C . The refrigeration cycle rejects heat to cooling water that is available at 40°C and has an efficiency of 80% of the Carnot cycle efficiency. The plant operates for 8000 hours per year, and electricity costs \$0.06/kWh.

Solution

The refrigeration cycle needs to operate with an evaporator temperature below -5°C , say at -10°C or 263 K. The condenser must operate above 40°C , say at 45°C (318 K).

For this temperature range the Carnot cycle efficiency is:

$$COP = \frac{T_e}{(T_c - T_e)} = \frac{263}{318 - 263} = 4.78 \quad (3.3)$$

If the cycle is 80% efficient, then the actual $COP = 4.78 \times 0.8 = 3.83$

The shaft work needed to supply 1.2 MW of cooling is given by:

$$\text{Shaft work required} = \frac{\text{Cooling duty}}{COP} = \frac{1.2}{3.83} = 0.313 \text{ MW}$$

The annual operating cost is then $= 313 \text{ kW} \times 8000 \text{ h/y} \times 0.06 \text{ \$/kWh} = 150,000 \text{ \$/y}$

3.2.7 Water

The water required for general purposes on a site will usually be taken from the local mains supply, unless a cheaper source of suitable quality water is available from a river, lake, or well. Raw water is brought in to make up for losses in the steam and cooling water systems and is also treated to generate demineralized and deionized water for process use. Water is also used for process cleaning operations and to supply fire hydrants.

The price of water varies strongly by location, depending on fresh water availability. Water prices are often set by local government bodies and often include a charge for waste water rejection. This charge is usually applied on the basis of the water consumed by the plant, regardless of whether that water is actually rejected as a liquid (as opposed to being lost as vapor or incorporated into a product by reaction). A very rough estimate of water costs can be made by assuming \$2 per 1000 gal (\$0.5 per metric ton).

Demineralized water

Demineralized water, from which all the minerals have been removed by ion exchange, is used where pure water is needed for process use and as boiler feed-water. Mixed and multiple-bed ion exchange units are used, with one resin converting the cations to hydrogen and the other removing the anions. Water with less than 1 part per million of dissolved solids can be produced. The design of ion exchange units is discussed in Section 16.5.5. Demineralized water typically costs about double the price of raw water, but this obviously varies strongly with the mineral content of the water and the disposal cost of blowdown from the demineralization system. A correlation for the cost of a water ion exchange plant is given in Table 7.2.

3.2.8 Compressed air

Compressed air is needed for general use, oxidation reactions, air strippers, aerobic fermentation processes, and pneumatic control actuators that are used for plant control. Air is normally distributed at a mains pressure of 6 bar (100 psig), but large process air requirements are typically met with stand-alone air blowers or compressors. Rotary and reciprocating single-stage or two-stage compressors are used to supply utility and instrument air. Instrument air must be dry and clean (free from oil). Air is usually dried by passing it over a packed bed of molecular sieve adsorbent. For most applications, the adsorbent is periodically regenerated using a temperature swing cycle. Temperature swing adsorption (TSA) is discussed in more detail in Section 16.2.1.

Air at 1 atmosphere pressure is freely available in most chemical plants. Compressed air can be priced based on the power needed for compression (see Section 20.6). Drying the air—for example, for instrument air—typically adds about \$0.005 per standard m^3 (\$0.14/1000 scf).

Cooling air

Ambient air is used as a coolant in many process operations; for example, air-cooled heat exchangers, cooling towers, solids coolers, and prilling towers. If the air flow is caused by natural draft, then the cooling air is free, but the air velocity will generally be low, leading to high equipment cost. Fans or blowers are commonly used to ensure higher air velocities and reduce equipment costs. The cost of providing cooling air is then the cost of operating the fan, which can be determined from the fan power consumption. Cooling fans typically operate with very high flow rates and very low pressure drop, on the order of a few inches of water. The design of a cooling fan is illustrated in the discussion of air-cooled heat exchangers in Section 19.16.

3.2.9 Nitrogen

Where a large quantity of inert gas is required for the inert blanketing of tanks and for purging (see [Chapter 10](#)), this will usually be supplied from a central facility. Nitrogen is normally used and can be manufactured on site in an air liquefaction plant or purchased as a liquid in tankers.

Nitrogen and oxygen are usually purchased from one of the industrial gas companies via pipeline or a small dedicated “over-the-fence” plant. The price varies depending on local power costs, but is typically in the range of \$0.01 to \$0.03 per lb for large facilities.

3.3 Energy recovery

Process streams at HP or high temperature contain energy that can be usefully recovered. Whether it is economic to recover the energy content of a particular stream depends on the value of the energy that can be usefully extracted and the cost of recovery. The value of the energy is related to the marginal cost of energy at the site. The cost of recovery will be the capital and operating cost of any additional equipment required. If the savings exceed the total annualized cost, including capital charges, then the energy recovery will usually be worthwhile. Maintenance costs should be included in the annualized cost (see [Chapter 9](#)).

Some processes, such as air separation, depend on efficient energy recovery for economic operation, and in all processes the efficient use of energy will reduce product cost.

When setting up process simulation models, the design engineer needs to pay careful attention to operations that have an impact on the energy balance and heat use within the process. Some common problems to watch out for include:

1. Avoid mixing streams at very different temperatures. This usually represents a loss of heat (or cooling) that could be better used in the process.
2. Avoid mixing streams at different pressures. The mixed stream will be at the lowest pressure of the feed streams. The higher-pressure streams will undergo cooling as a result of adiabatic expansion. This may lead to increased heating or cooling requirements or lost potential to recover shaft work during the expansion.
3. Segment heat exchangers to avoid internal pinches. This is particularly necessary for exchangers where there is a phase change. When a liquid is heated, boiled, and superheated, the variation of stream temperature with enthalpy added looks like [Fig. 3.6](#). Liquid is heated to the boiling point (A-B), then the heat of vaporization is added (B-C), and the vapor is superheated (C-D). This is a different temperature–enthalpy profile than a straight line between the initial and final states (A-D). If the stream in [Fig. 3.6](#) were matched against a heat source that had

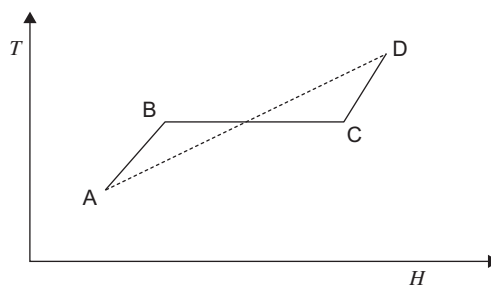


Fig. 3.6 Temperature–enthalpy profile for a stream that is vaporized and superheated.

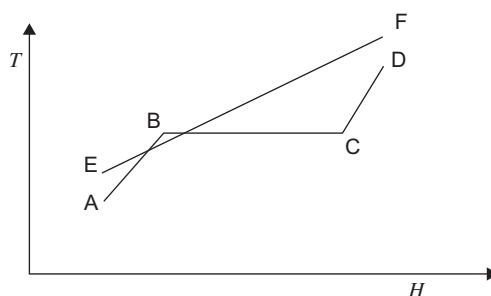


Fig. 3.7 Heat transfer to a stream that is vaporized and superheated.

a temperature profile like line E–F in Fig. 3.7, then the exchanger would appear feasible based on the inlet and outlet temperatures, but would in fact be infeasible because of the cross-over of the temperature profiles at B. A simple way to avoid this problem is to break up the preheat, boiling, and superheat into three exchangers in the simulation model, even if they will be carried out in a single piece of equipment in the final design. The same problem also occurs with condensers that incorporate desuperheat and subcooling.

4. Check for heat of mixing. This is important whenever acids or bases are mixed with water. If the heat of mixing is large, two or more stages of mixing with intercoolers may be needed. If a large heat of mixing is expected but is not predicted by the model, then check that the thermodynamic model includes heat of mixing effects.
5. Remember to allow for process inefficiency and design margins. For example, when sizing a fired heater, if process heating is carried out in the radiant section only, the heating duty calculated in the simulation is only 60% of the total furnace duty (see Sections 3.2.2 and 19.17). The operating duty will then be the process duty divided by 0.6. The design duty must be increased further by a suitable design factor, say 10%. The design duty of the fired heater is then $1.1/0.6 = 1.83$ times the process duty calculated in the simulation.

Some of the techniques used for energy recovery in chemical process plants are described briefly in the following sections. The references cited give fuller details of each technique. Miller (1968) gives a comprehensive review of process energy systems, including heat exchange and power recovery from HP fluid streams. Kenney (1984) reviews the application of thermodynamic principles to energy recovery in the process industries. Zhu (2014) and Knopf (2012) provide comprehensive treatments of energy recovery and process energy optimization. Kemp (2007) gives a detailed description of pinch analysis and several other methods for heat recovery.

3.3.1 Heat exchange

The most common energy-recovery technique is to use the heat in a high-temperature process stream to heat a colder stream. This saves part or all of the cost of heating the cold stream, as well as part or all of the cost of cooling the hot stream. Conventional shell and tube exchangers are normally used. The cost of the heat-exchange surface may be increased relative to using a hot utility as a heat source due to the reduced-temperature driving forces, or it may be decreased due to needing fewer exchangers. The cost of recovery will be reduced if the streams are located conveniently close within the plant.

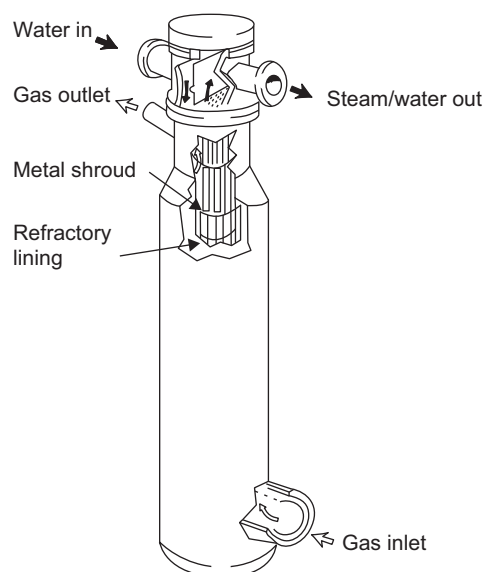


Fig. 3.8 Reformed gas waste-heat boiler arrangement of vertical U-tube water tube boiler. Reprinted by permission of the Council of the Institution of Mechanical Engineers from the *Proceedings of the Conference on Energy Recovery in the Process Industries*, London, 1975.

The amount of energy that can be recovered depends on the temperature, flow, heat capacity, and temperature change possible in each stream. A reasonable temperature-driving force must be maintained to keep the exchanger area to a practical size. The most efficient exchanger will be the one in which the shell and tube flows are truly counter-current. Multiple tube-pass exchangers are usually used for practical reasons. With multiple tube passes, the flow is part counter-current and part co-current, and temperature crosses can occur, which reduce the efficiency of heat recovery (see [Chapter 19](#)). In cryogenic processes, where heat recovery is critical to process efficiency, brazed or welded plate exchangers are used to obtain true counter-current performance, and very low-temperature approaches on the order of a few degrees Celsius are common.

The hot process streams leaving a reactor or a distillation column are frequently used to preheat the feed streams (“feed-effluent” or “feed-bottoms” exchangers).

In an industrial process there will be many hot and cold streams, and there will be an optimum arrangement of the streams for energy recovery by heat exchange. The problem of synthesizing a network of heat exchangers has been the subject of much research and is covered in more detail in [Section 3.5](#).

3.3.2 Waste-heat boilers

If the process streams are at a sufficiently high temperature and there are no attractive options for process-to-process heat transfer, then the heat recovered can be used to generate steam.

Waste-heat boilers are often used to recover heat from furnace flue gases and to process gas streams from high-temperature reactors. The pressure and superheat temperature of the steam generated depend on the temperature of the hot stream and the approach temperature permissible at the boiler exit. As with any heat-transfer equipment, the area required increases as the mean temperature driving force (log mean ΔT) is reduced. The permissible exit temperature may also be limited by process considerations. If the gas stream contains water vapor and soluble corrosive gases, such as HCl or SO₂, the exit gas temperature must be kept above the dew point.

[Hinchley \(1975\)](#) discusses the design and operation of waste heat boilers for chemical plants. Both fire-tube and water-tube boilers are used. A typical arrangement of a water-tube boiler on a reformer furnace is shown in [Fig. 3.8](#) and a fire-tube boiler in [Fig. 3.9](#).

The application of a waste-heat boiler to recover energy from the reactor exit streams in a nitric acid plant is shown in [Fig. 3.10](#). The selection and operation of waste heat boilers for industrial furnaces are discussed by [Dryden \(2013\)](#).

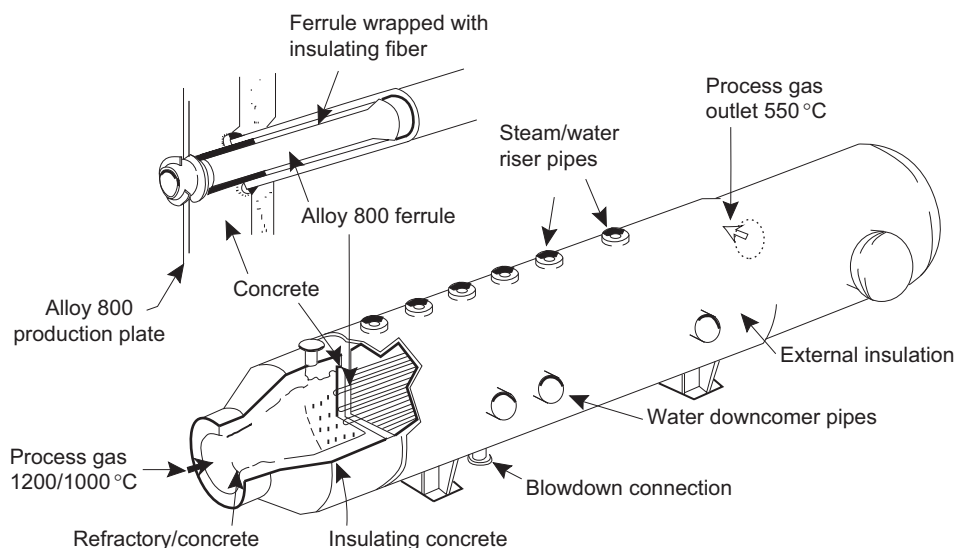
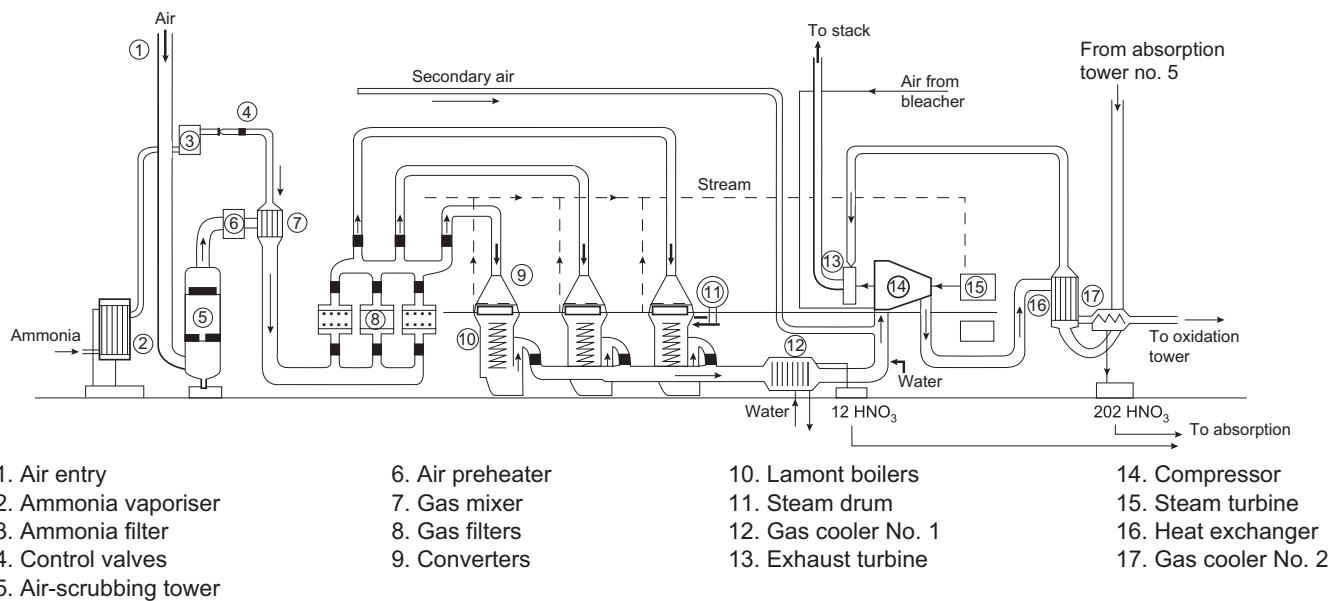


Fig. 3.9 Reformed gas waste-heat boiler, principal features of typical natural circulation fire-tube boilers. Reprinted by permission of the Council of the Institution of Mechanical Engineers from the Proceedings of the Conference on Energy Recovery in the Process Industries, London, 1975.



(From nitric acid manufacture, Miles (1961), with permission)

Fig. 3.10 Connections of a nitric acid plant, intermediate-pressure type.

3.3.3 High-temperature reactors

If a reaction is highly exothermic, cooling will be needed. If the reactor temperature is high enough, the heat removed can be used to generate steam. The lowest steam pressure normally used in the process industries is about 2.7 bar (25 psig), so any reactor with a temperature above 150 °C is a potential steam generator. Steam is preferentially generated at as high a pressure as possible, as HP steam is more valuable than LP steam (see Section 3.2.3). If the steam production exceeds the site steam requirements, some steam can be fed to condensing turbines to produce electricity to meet site power needs.

Three systems are used:

1. Fig. 3.11(a). An arrangement similar to a conventional water-tube boiler. Steam is generated in cooling pipes within the reactor and separated in a steam drum.
2. Fig. 3.11(b). Similar to the first arrangement but with the water kept at high pressure to prevent vaporization. The HP water is flashed to steam at a lower pressure in a flash drum. This system gives more responsive control over the reactor temperature.

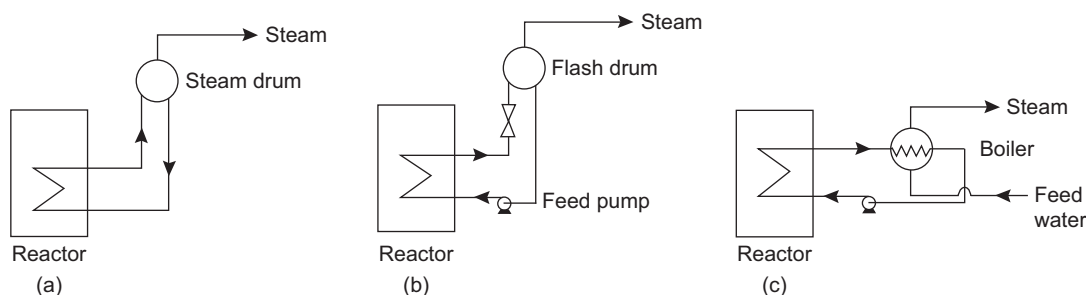


Fig. 3.11 Steam generation.

3. Fig. 3.11(c). In this system a heat-transfer fluid, such as Dowtherm A (see Section 3.2.4 and Singh [1985] for details of heat-transfer fluids) is used to avoid the need for HP tubes. The steam is raised in an external boiler.

3.3.4 High-pressure process streams

Where HP gas or liquid process streams are throttled to lower pressures, energy can be recovered by carrying out the expansion in a suitable turbine.

Gas streams

The economic operation of processes that involve the compression and expansion of large quantities of gases—such as ammonia synthesis, nitric acid production, and air separation—depends on the efficient recovery of the energy of compression. The energy recovered by expansion is often used to drive the compressors directly, as shown in Fig. 3.10. If the gas contains condensable components, it may be advisable to consider heating the gas by heat exchange with a higher-temperature process stream before expansion. The gas can then be expanded to a lower pressure without condensation and the power generated increased.

The process gases do not have to be at a particularly high pressure for expansion to be economical if the gas flow rate is high. For example, Luckenbach (1978) in U.S. patent 4,081,508 describes a process for recovering power from the off-gas from a fluid catalytic cracking process by expansion from about 2 to 3 bar (15 to 25 psig) down to just over atmospheric pressure (1.5 to 2 psig).

The energy recoverable from the expansion of a gas can be estimated by assuming polytropic expansion; see Section 20.6.3 and Example 20.4. The design of turboexpanders for the process industries is discussed by Bloch et al. (1982).

Liquid streams

As liquids are essentially incompressible, less energy is stored in a compressed liquid than a gas; however, it is often worth considering power recovery from HP liquid streams (>15 bar), as the equipment required is relatively simple and inexpensive. Centrifugal pumps are used as expanders and are often coupled directly to other pumps. The design, operation, and cost of energy recovery from HP liquid streams is discussed by Jenett (1968), Chada (1984), and Buse (1981).

3.3.5 Heat pumps

A heat pump is a device for raising low-grade heat to a temperature at which the heat can be used. It pumps the heat from a low-temperature source to the higher-temperature sink, using a small amount of energy relative to the heat energy recovered. A heat pump is essentially the same as a refrigeration cycle (Section 3.2.6 and Fig. 3.5), but the objective is to deliver heat to the process in the condensation step of the cycle, as well as (or instead of) removing heat in the evaporation step.

Heat pumps are increasingly finding applications in the process industries. A typical application is the use of the low-grade heat from the condenser of a distillation column to provide heat for the reboiler; see Barnwell and Morris (1982) and Meili (1990). Heat pumps are also used with dryers, with heat being abstracted from the exhaust air and used to preheat the incoming air.

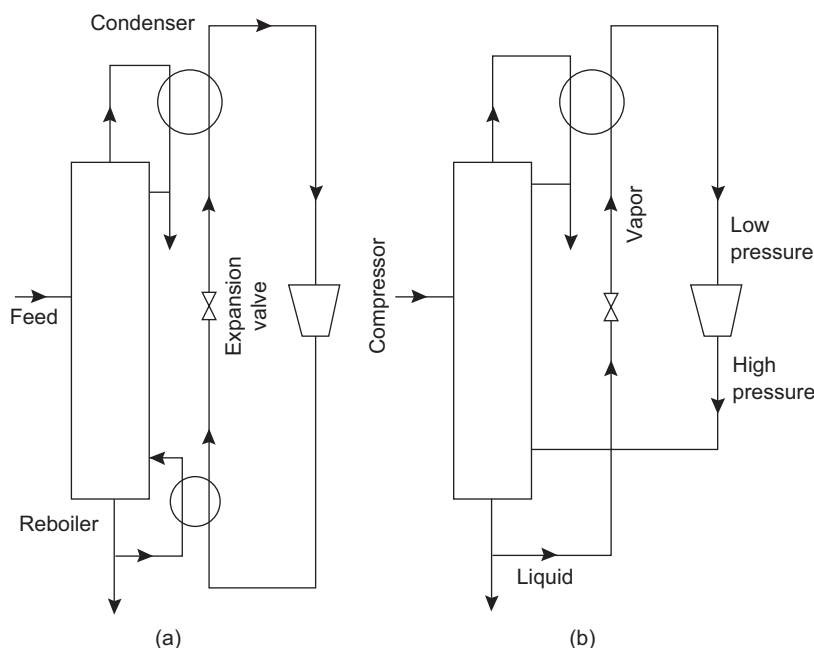


Fig. 3.12 Distillation column with heat pump. (a) Separate refrigerant circuit. (b) Using column fluid as the refrigerant.

Details of the thermodynamic cycles used for heat pumps can be found in most textbooks on engineering thermodynamics, and in [Reay and MacMichael \(1988\)](#). In the process industries, heat pumps operating on the mechanical vapor compression cycle are normally used. A vapor compression heat pump applied to a distillation column is shown in [Fig. 3.12\(a\)](#). The working fluid, usually a commercial refrigerant, is fed to the reboiler as a vapor at HP and condenses, giving up heat to vaporize the process fluid. The liquid refrigerant from the reboiler is then expanded over a throttle valve, and the resulting wet vapor is fed to the column condenser. In the condenser the wet refrigerant is dried, taking heat from the condensing process vapor. The refrigerant vapor is then compressed and recycled to the reboiler, completing the working cycle.

If the conditions are suitable, the process fluid can be used as the working fluid for the heat pump. This arrangement is shown in [Fig. 3.12\(b\)](#). The hot process liquid at HP is expanded over the throttle valve and fed to the condenser to provide cooling to condense the vapor from the column. The vapor from the condenser is compressed and returned to the base of the column. In an alternative arrangement, the process vapor is taken from the top of the column, compressed, and fed to the reboiler to provide heating.

The “efficiency” of a heat pump is measured by the heat pump coefficient of performance, COP_h :

$$COP_h = \frac{\text{energy delivered at higher temperature}}{\text{energy input to the compressor}} \quad (3.4)$$

The COP_h depends principally on the working temperatures. Heat pumps are more efficient (higher COP_h) when operated over a narrow temperature range. They are thus most often encountered on distillation columns that separate close-boiling compounds. Note that the COP_h of a heat pump is not the same as the COP of a refrigeration cycle (Section 3.2.6).

The economics of the application of heat pumps in the process industries is discussed by [Holland and Devotta \(1986\)](#). Details of the application of heat pumps in a wide range of industries are given by [Moser and Schnitzer \(1985\)](#).

3.4 Waste stream combustion

Process waste products that contain significant quantities of combustible material can be used as low-grade fuels for raising steam or direct process heating. Their use will only be economic if the intrinsic value of the fuel justifies

the cost of special burners and other equipment needed to burn the waste. If the combustible content of the waste is too low to support combustion, the waste must be supplemented with higher-calorific-value primary fuels.

3.4.1 Reactor off-gases

Reactor off-gases (vent gas) and recycle stream purges are often of high enough calorific value to be used as fuels. Vent gases will typically be saturated with organic compounds such as solvents and high-volatility feed compounds. The calorific value of a gas can be calculated from the heats of combustion of its constituents; the method is illustrated in Example 3.4.

Other factors that, together with the calorific value, determine the economic value of an off-gas as a fuel are the quantity available and the continuity of supply. Waste gases are best used for steam raising rather than for direct process heating, as this decouples the source from the use and gives greater flexibility.

Example 3.4 Calculation of waste-gas calorific value

The typical vent gas analysis from the recycle stream in an oxyhydrochlorination process for the production of dichloroethane (DCE) (British patent BP 1,524,449) is given here, with percentages on volume basis.

O₂ 7.96, CO₂ + N₂ 87.6, CO 1.79, C₂H₄ 1.99, C₂H₆ 0.1, DCE 0.54

Estimate the vent gas calorific value.

Solution

Component calorific values, from Perry and Chilton (1973)

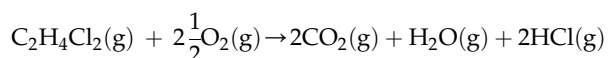
$$\text{CO } 67.6 \text{ kcal/mol} = 283 \text{ kJ/mol}$$

$$\text{C}_2\text{H}_4 \text{ } 372.8 = 1560.9$$

$$\text{C}_2\text{H}_6 \text{ } 337.2 = 1411.9$$

The value for DCE can be estimated from the heats of formation.

Combustion reaction:



The heats of formation ΔH_f° are given in Appendix C, which is available in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793.

$$\text{CO}_2 = -393.8 \text{ kJ/mol}$$

$$\text{H}_2\text{O} = -242.0$$

$$\text{HCl} = -92.4$$

$$\text{DCE} = -130.0$$

$$\begin{aligned} \Delta H_c^\circ &= \sum \Delta H_f^\circ \text{ products} - \sum \Delta H_f^\circ \text{ reactants} \\ &= [2(-393.8) - 242.0 + 2(-92.4)] - [-130.0] \\ &= -1084.4 \text{ kJ} \end{aligned}$$

Estimation of vent gas calorific value, basis 100 mol.

Component	mol/100 mol		Calorific Value		Heating Value (kJ/mol)
CO	1.79	×	283.0	=	506.6
C ₂ H ₄	1.99		1560.9		3106.2
C ₂ H ₆	0.1		1411.9		141.2
DCE	0.54		1084.4		585.7
				Total	4339.7

$$\text{Calorific value of vent gas} = \frac{4339.7}{100} = 43.4 \text{ kJ/mol}$$

$$= \frac{43.4}{22.4} \times 10^3 = 1938 \text{ kJ/m}^3 (52 \text{ Btu/ft}^3) \text{ at 1 bar, } 0^\circ\text{C}$$

This calorific value is very low compared with 37 MJ/m³ (1000 Btu/ft³) for natural gas. The vent gas is barely worth recovery, but if the gas has to be burnt to avoid pollution, it could be used in an incinerator such as that shown in Fig. 3.13, giving a useful steam production to offset the cost of disposal.

3.4.2 Liquid and solid wastes

Combustible liquid and solid waste can be disposed of by burning, which is usually preferred to dumping. Incorporating a steam boiler in the incinerator design will enable an otherwise unproductive, but necessary, operation to save energy. If the combustion products are corrosive, corrosion-resistant materials will be needed, and the flue gases must be scrubbed to reduce air pollution. An incinerator designed to handle chlorinated and other liquid and solid wastes is shown in Fig. 3.13. This incinerator incorporates a steam boiler and a flue-gas scrubber. The disposal of chlorinated wastes is discussed by Santoleri (1973).

Dunn and Tomkins (1975) discuss the design and operation of incinerators for process wastes. They give particular attention to the need to comply with the current clean air legislation and the problem of corrosion and erosion of refractories and heat-exchange surfaces.

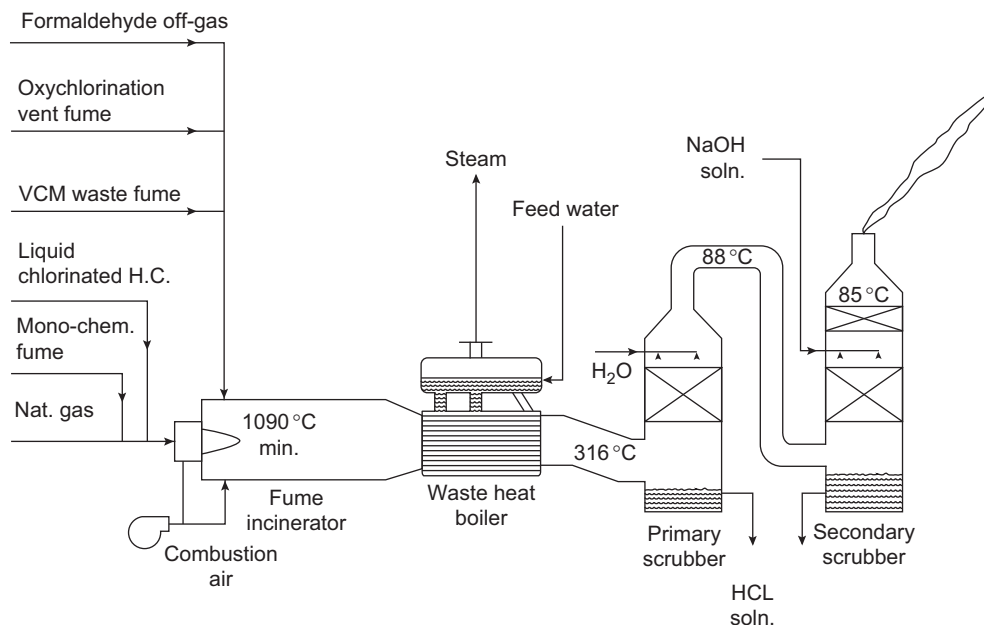


Fig. 3.13 Typical incinerator-heat recovery-scrubber system for vinyl chloride-monomer process waste. Courtesy John Thurley Ltd.

3.5 Heat exchanger networks

The design of a heat exchanger network for a simple process with only one or two streams that need heating and cooling is usually straightforward. When there are multiple hot and cold streams, the design is more complex, and there may be many possible heat-exchange networks. The design engineer must determine the optimum extent of heat recovery while ensuring that the design is flexible to changes in process conditions and can be started up and operated easily and safely.

In the 1980s, there was a great deal of research into design methods for heat exchanger networks; see [Gundersen and Naess \(1988\)](#). One of the most widely applied methods that emerged was a set of techniques termed *pinch technology* that was developed by Bodo Linnhoff and his collaborators at ICI, Union Carbide, and the University of Manchester. The term derives from the fact that in a plot of the system temperatures versus the heat transferred, a *pinch* usually occurs between the hot stream and cold stream curves (see [Fig. 3.19](#)). It has been shown that the pinch represents a distinct thermodynamic break in the system and that for minimum energy requirements, heat should not be transferred across the pinch ([Linnhoff et al. \[1982\]](#)).

In this section the fundamental principles of the pinch technology method for energy integration will be outlined and illustrated with reference to a simple problem. The method and its applications are described fully in a guide published by the Institution of Chemical Engineers, Kemp (2007); see also [Douglas \(1988\)](#), [Knopf \(2012\)](#), [Smith \(2016\)](#), [Zhu \(2014\)](#), and [El-Halwagi \(2006\)](#).

3.5.1 Pinch technology

The development and application of the method can be illustrated by considering the problem of recovering heat between four process streams: two hot streams that require cooling and two cold streams that must be heated. The process data for the streams are set out in [Table 3.1](#). Each stream starts from a source temperature T_s and is to be heated or cooled to a target temperature T_t . The heat capacity flow rate of each stream is shown as CP . For streams where the specific heat capacity can be taken as a constant and there is no phase change, CP will be given by:

$$CP = mC_p \quad (3.5)$$

where m = mass flow rate, kg/s

C_p = average specific heat capacity between T_s and T_t kJ kg⁻¹°C⁻¹

The heat load shown in the table is the total heat required to heat or cool the stream from the source to the target temperature.

There is clearly scope for energy integration between these four streams. Two require heating and two cooling, and the stream temperatures are such that heat can be transferred from the hot to the cold streams. The task is to find the best arrangement of heat exchangers to achieve the target temperatures.

Simple two-stream problem

Before investigating the energy integration of the four streams shown in [Table 3.1](#), the use of a temperature–enthalpy diagram will be illustrated for a simple problem involving only two streams. The general problem of heating and cooling two streams from source to target temperatures is shown in [Fig. 3.14](#). Some heat is exchanged between the streams in the heat exchanger. Additional heat to raise the cold stream to the target temperature is provided by the hot utility (usually steam) in the heater, and additional cooling to bring the hot stream to its target temperature is provided by the cold utility (usually cooling water) in the cooler.

TABLE 3.1 Data for heat integration problem

Stream number	Type	Heat capacity flow rate CP , kW/°C	T_s °C	T_t °C	Heat load kW
1	Hot	3.0	180	60	360
2	Hot	1.0	150	30	120
3	Cold	2.0	20	135	230
4	Cold	4.5	80	140	270

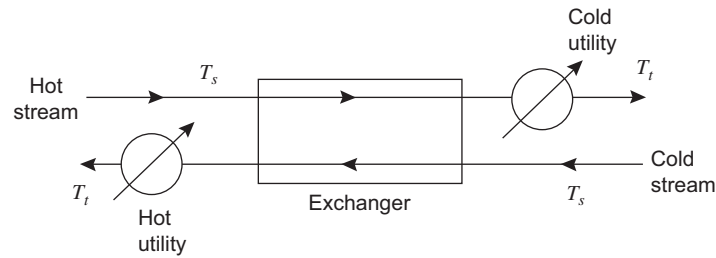
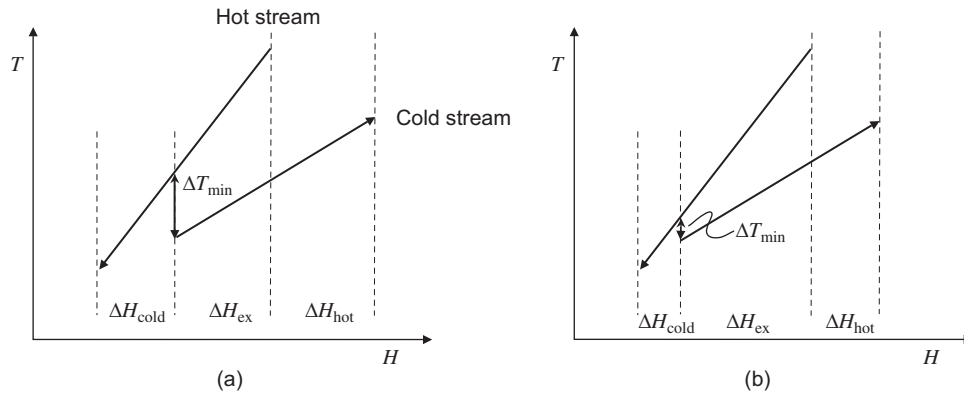


Fig. 3.14 Two-stream exchanger problem.

Fig. 3.15 Temperature–enthalpy (T - H) diagram for two-stream example.

In Fig. 3.15(a) the stream temperatures are plotted on the y -axis and the enthalpy change of each stream on the x -axis. This is known as a *temperature–enthalpy (T - H) diagram*. For heat to be exchanged, a minimum temperature difference must be maintained between the two streams. This is shown as $\Delta T_{\min}$ on the diagram. The practical minimum temperature difference in a heat exchanger will usually be between 5° and 30°C ; see Chapter 19.

The slope of the lines in the T - H plot is proportional to $1/CP$, because $\Delta H = CP \times \Delta T$, so $dT/dH = 1/CP$. Streams with low heat capacity flow rate thus have steep slopes in the T - H plot, and streams with high heat capacity flow rate have shallow slopes.

The heat transferred between the streams is given by the range of enthalpy over which the two curves overlap each other and is shown on the diagram as ΔH_{ex} . The heat transferred from the hot utility, ΔH_{hot} , is given by the part of the cold stream that is not overlapped by the hot stream. The heat transferred to the cold utility, ΔH_{cold} , is similarly given by the part of the hot stream that is not overlapped by the cold stream. The heats can also be calculated as:

$$\Delta H = CP \times (\text{temperature change})$$

Because we are only concerned with changes in enthalpy, we can treat the enthalpy axis as a relative scale and slide either the hot stream or the cold stream horizontally. As we do so, we change the minimum temperature difference between the streams, $\Delta T_{\min}$, and also the amount of heat exchanged and the amounts of hot and cold utilities required.

Fig. 3.15(b) shows the same streams plotted with a lower value of $\Delta T_{\min}$. The amount of heat exchanged is increased, and the utility requirements have been reduced. The temperature-driving force for heat transfer has also been reduced, so the heat exchanger has both a larger duty and a smaller log-mean temperature difference. This leads to an increase in the heat transfer area required and in the capital cost of the exchanger. The capital cost increase is partially offset by capital cost savings in the heater and cooler, which both become smaller, as well as by savings in the costs of hot and cold utility. In general, there will be an optimum value of $\Delta T_{\min}$, as illustrated in Fig. 3.16. This optimum is usually rather flat over the range of 10° to 30°C .

The maximum feasible heat recovery is reached at the point where the hot and cold curves touch each other on the T - H plot, as illustrated in Fig. 3.17. At this point, the temperature-driving force at one end of the heat exchanger is

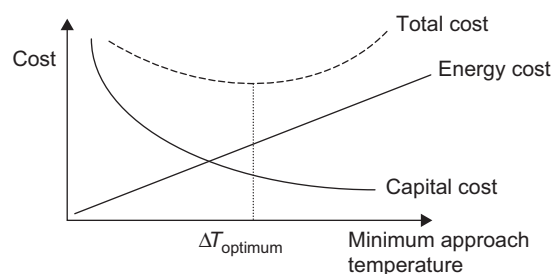


Fig. 3.16 The capital–energy trade-off in process heat recovery.

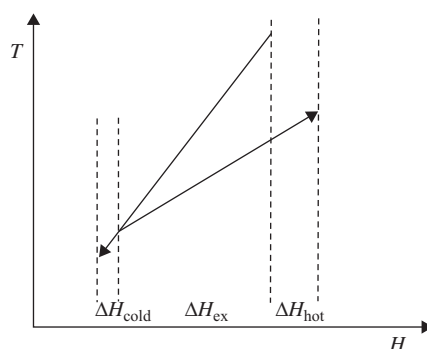


Fig. 3.17 Maximum feasible heat recovery for two-stream example.

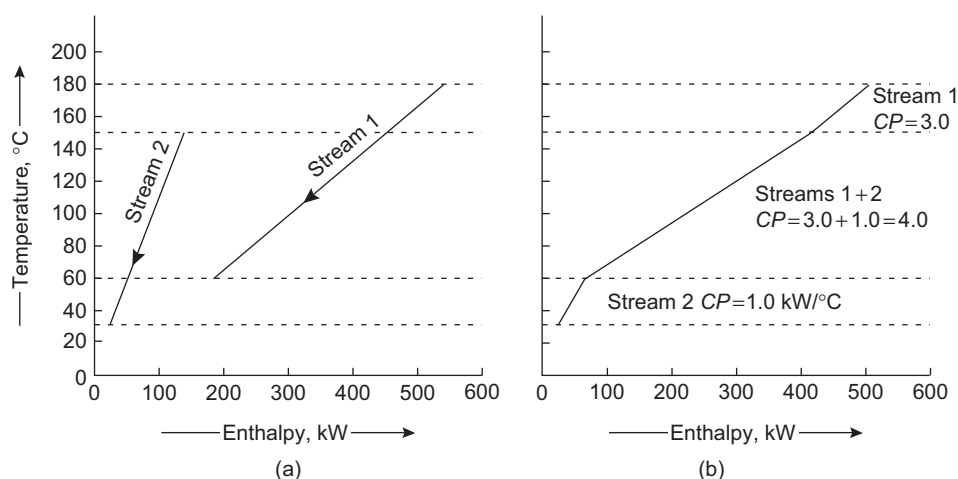


Fig. 3.18 Hot stream temperature vs. enthalpy. (a) Separate hot streams. (b) Composite hot streams. $CP, ^\circ$.

zero, and an infinite heat exchange surface is required, so the design is not practical. The exchanger is said to be *pinched* at the end where the hot and cold curves meet. In Fig. 3.17, the heat exchanger is pinched at the cold end.

It is not possible for the hot and cold streams to cross each other, as this would be a violation of the second law of thermodynamics and would give an infeasible design.

Four-stream problem

In Fig. 3.18(a) the hot streams given in Table 3.1 are shown plotted on a T - E diagram.

As the diagram shows changes in the enthalpy of the streams, it does not matter where a particular curve is plotted on the enthalpy axis as long as the curve runs between the correct temperatures. This means that where more than one stream appears in a temperature interval, the stream heat capacities can be added to form a composite curve, as shown in Fig. 3.18(b).

In Fig. 3.19, the composite curve for the hot streams and the composite curve for the cold streams are drawn with a minimum temperature difference—the displacement between the curves—of 10 °C. This implies that in any of the exchangers to be used in the network, the temperature difference between the streams will not be less than 10 °C.

As for the two-stream problem, the overlap of the composite curves gives a target for heat recovery, and the displacements of the curves at the top and bottom of the diagram give the hot and cold utility requirements. These will be the minimum values needed to satisfy the target temperatures. This is valuable information. It gives the designer target values for the utilities to aim for when designing the exchanger network. Any design can be compared with the minimum utility requirements to check if further improvement is possible.

In most exchanger networks the minimum temperature difference will occur at only one point. This is termed the *pinch*. In the problem being considered, the pinch occurs at between 90 °C on the hot stream curve and 80 °C on the cold stream curve.

For multistream problems, the pinch will usually occur somewhere in the middle of the composite curves, as illustrated in Fig. 3.19. The case when the pinch occurs at the end of one of the composite curves is termed a *threshold problem* and is discussed in Section 3.5.5.

Thermodynamic significance of the pinch

The pinch divides the system into two distinct thermodynamic regions. The region above the pinch can be considered a heat sink, with heat flowing into it, from the hot utility but no heat flow out of it. Below the pinch the converse is true. Heat flows out of the region to the cold utility. No heat flows across the pinch, as shown in Fig. 3.20(a).

If a network is designed in which heat is transferred from any hot stream at a temperature above the pinch (including hot utilities) to any cold stream at a temperature below the pinch (including cold utilities), then heat is transferred across the pinch. If the amount of heat transferred across the pinch is ΔH_{xp} , then to maintain energy balance, the hot utility and cold utility must both be increased by ΔH_{xp} , as shown in Fig. 3.20(b). Cross-pinch heat transfer thus always leads to consumption of both hot and cold utilities that is greater than the minimum values that could be achieved.

The pinch decomposition is useful in heat exchanger network design, as it decomposes the problem into two smaller problems. It also indicates the region where heat transfer matches are most constrained, at or near the pinch. When multiple hot or cold utilities are used, there may be other pinches, termed *utility pinches*, that cause further problem decomposition. Problem decomposition can be exploited in algorithms for automatic heat exchanger network synthesis.

3.5.2 The problem table method

The problem table is a numerical method for determining the pinch temperatures and the minimum utility requirements, introduced by Linnhoff and Flower (1978). It eliminates the sketching of composite curves, which

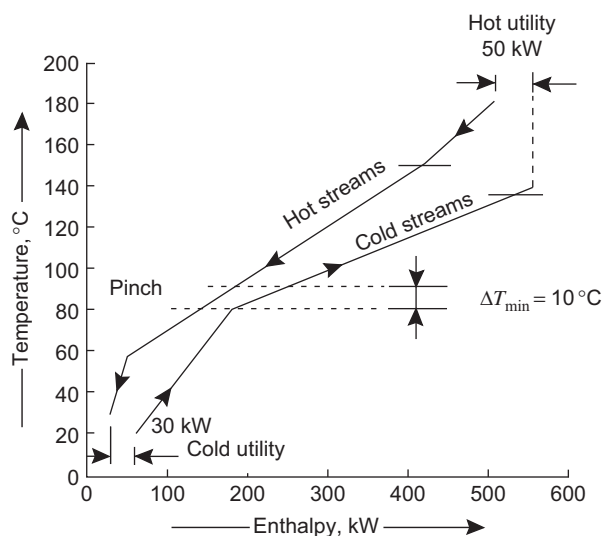


Fig. 3.19 Hot and cold stream composite curves.

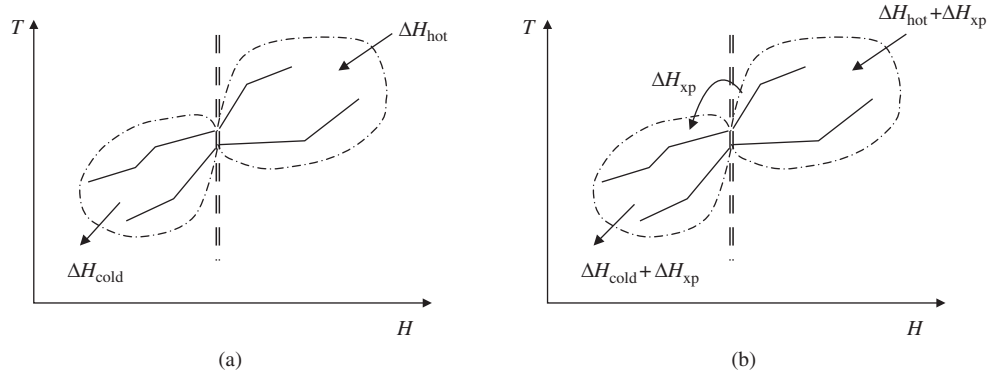


Fig. 3.20 Pinch decomposition.

can be useful if the problem is being solved manually. It is not widely used in industrial practice any more due to the wide availability of computer tools for pinch analysis (see Section 3.5.7).

The procedure is as follows:

1. Convert the actual stream temperatures T_{act} into interval temperatures T_{int} by subtracting half the minimum temperature difference from the hot stream temperatures and by adding half to the cold stream temperatures:

$$\text{hot streams } T_{\text{int}} = T_{\text{act}} - \frac{\Delta T_{\min}}{2}$$

$$\text{cold streams } T_{\text{int}} = T_{\text{act}} + \frac{\Delta T_{\min}}{2}$$

The use of the interval temperature rather than the actual temperatures allows the minimum temperature difference to be taken into account. $\Delta T_{\min} = 10^\circ\text{C}$ for the problem being considered (Table 3.2).

2. Note any duplicated interval temperatures. These are bracketed in Table 3.2.
3. Rank the interval temperatures in order of magnitude, showing the duplicated temperatures only once in the order (Table 3.3).

TABLE 3.2 Interval temperatures for $\Delta T_{\min} = 10^\circ\text{C}$

Stream	Actual	Temperature	Interval	Temperature
1	180	60	175	55
2	150	30	145	25
3	20	135	(25)	140
4	80	140	85	(145)

TABLE 3.3 Ranked order of interval temperatures

Rank	Interval ΔT °C	Streams in interval
175 °C		
145	30	–1
140	5	4 – (2 + 1)
85	55	(3 + 4) – (1 + 2)
55	30	3 – (1 + 2)
25	30	3 – 2

Note: Duplicated temperatures are omitted. The interval ΔT and streams in the intervals are included, as they are needed for Table 3.4.

TABLE 3.4 Problem table

Interval	Interval temp. °C	ΔT_n -C	$\Sigma CP_c - \Sigma CP_h^*$ kW/°C	ΔH kW	Surplus or deficit
	175				
1	145	30	-3.0	-90	s
2	140	5	0.5	2.5	d
3	85	55	2.5	137.5	d
4	55	30	-2.0	-60	s
5	25	30	1.0	30	d

*Note: The streams in each interval are given in Table 3.3.

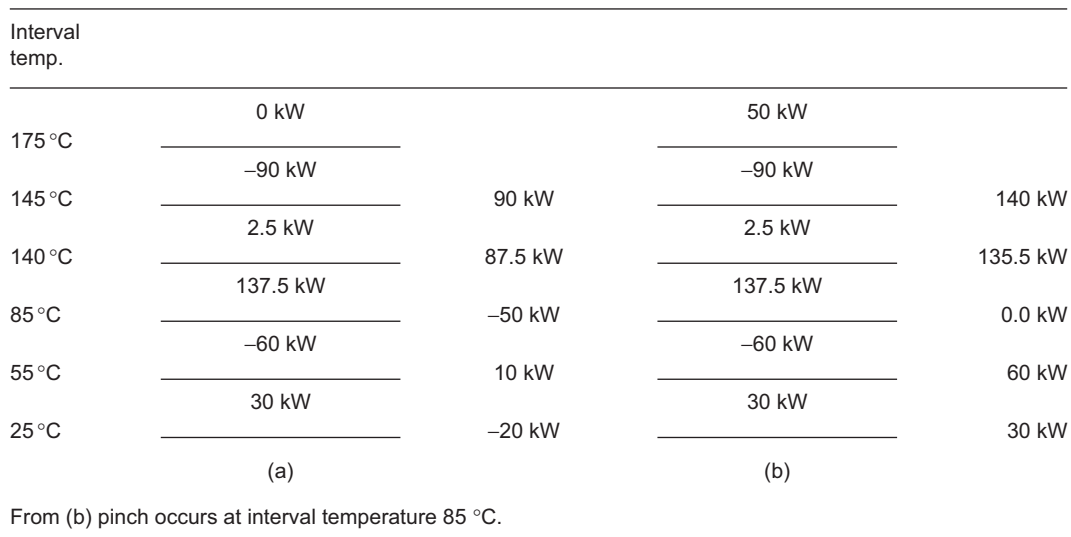


Fig. 3.21 Heat cascade.

4. Carry out a heat balance for the streams falling within each temperature interval:
For the n th interval:

$$\Delta H_n = (\Sigma CP_c - \Sigma CP_h) (\Delta T_n)$$

where ΔH_n = net heat required in the n th interval

ΣCP_c = sum of the heat capacities of all the cold streams in the interval

ΣCP_h = sum of the heat capacities of all the hot streams in the interval

ΔT_n = interval temperature difference = $(T_{n-1} - T_n)$

See Table 3.4.

5. "Cascade" the heat surplus from one interval to the next down the column of interval temperatures, as shown in Fig. 3.21(a).

Cascading the heat from one interval to the next implies that the temperature difference is such that the heat can be transferred between the hot and cold streams. The presence of a negative value in the column indicates that the temperature gradient is in the wrong direction and that the exchange is not thermodynamically possible.

This difficulty can be overcome if heat is introduced into the top of the cascade:

6. Introduce just enough heat to the top of the cascade to eliminate all the negative values; see Fig. 3.21(b).

Comparing the composite curve (see Fig. 3.19) with Fig. 3.21(b) shows that the heat introduced to the cascade is the minimum hot utility requirement and the heat removed at the bottom is the minimum cold utility required. The pinch occurs in Fig. 3.21(b) where the heat flow in the cascade is zero. This is as would be expected from the rule that for minimum utility requirements no heat flows across the pinch. In Fig. 3.21(b) the pinch is at an interval

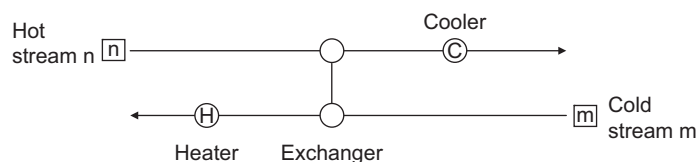


Fig. 3.22 Grid representation.

					CP (kW/°C)
1	180 °C	90 °C	90 °C	60 °C	3.0
2	150 °C	90 °C	90 °C	30 °C	1.0
	135 °C	80 °C	80 °C	20 °C	3
	140 °C	80 °C			4

Fig. 3.23 Grid for four-stream problem.

temperature of 85 °C, corresponding to a cold stream temperature of 80 °C and a hot stream temperature of 90 °C, as was found using the composite curves.

It is not necessary to draw up a separate cascade diagram. This was done in Fig. 3.21 to illustrate the principle. The cascaded values can be added to the problem table as two additional columns; see Example 3.5.

Summary

For maximum heat recovery and minimum use of utilities:

1. Do not transfer heat across the pinch.
2. Do not use hot utilities below the pinch.
3. Do not use cold utilities above the pinch.

3.5.3 Heat exchanger network design

Grid representation

It is convenient to represent a heat exchanger network as a grid (Fig. 3.22). The process streams are drawn as horizontal lines, with the stream numbers shown in square boxes. Hot streams are drawn at the top of the grid and flow from left to right. The cold streams are drawn at the bottom and flow from right to left. The stream heat capacities CP are shown in a column at the end of the stream lines.

Heat exchangers are drawn as two circles connected by a vertical line. The circles connect the two streams between which heat is being exchanged; that is, the streams that would flow through the actual exchanger. Heaters and coolers can be drawn as a single circle, connected to the appropriate utility. If multiple utilities are used, then these can also be shown as streams. Exchanger duties are usually marked under the exchanger, and temperatures are also sometimes indicated on the grid diagram.

Network design for maximum energy recovery

The analysis carried out in Fig. 3.19 and Fig. 3.21 has shown that the minimum utility requirements for the problem set out in Table 3.1 are 50 kW of the hot and 30 kW of the cold utility and that the pinch occurs where the cold streams are at 80 °C and the hot streams are at 90 °C.

The grid representation of the streams is shown in Fig. 3.23. The vertical dotted lines represent the pinch and separate the grid into the regions above and below the pinch. Note that the hot and cold streams are off-set at the pinch because of the difference in pinch temperature.

For maximum energy recovery (minimum utility consumption) the best performance is obtained if no cooling is used above the pinch. This means that the hot streams above the pinch should be brought to the pinch temperature solely by exchange with the cold streams. The network design is therefore started at the pinch, finding feasible matches between streams to fulfill this aim. In making a match adjacent to the pinch the heat capacity CP of the hot stream must be equal to or less than that of the cold stream. This is to ensure that the minimum temperature difference between the curves is maintained. The slope of a line on the T - E diagram is equal to the reciprocal of the heat capacity. So, above the pinch the lines will converge if CP_h exceeds CP_c , and as the streams start with a separation at the pinch equal to ΔT_{min} , the minimum temperature condition would be violated. Every hot stream must be matched with a cold stream immediately above the pinch; otherwise, it will not be able to reach the pinch temperature.

Below the pinch the procedure is the same, with the aim being to bring the cold streams to the pinch temperature by exchange with the hot streams. For streams adjacent to the pinch, the criterion for matching streams is that the heat capacity of the cold stream must be equal to or greater than the hot stream to avoid breaking the minimum temperature difference condition. Every cold stream must be matched with a hot stream immediately below the pinch.

Network design above the pinch

$$CP_h \leq CP_c$$

1. Applying this condition at the pinch, stream 1 can be matched with stream 4 but not with stream 3.

Matching streams 1 and 4 and transferring the full amount of heat required to bring stream 1 to the pinch temperature gives:

$$\Delta H_{ex} = CP(T_s - T_{pinch})$$

$$\Delta H_{ex} = 3.0(180 - 90) = 270 \text{ kW}$$

This will also satisfy the heat load required to bring stream 4 to its target temperature:

$$\Delta H_{ex} = 4.5(140 - 80) = 270 \text{ kW}$$

2. Stream 2 can be matched with stream 3 while satisfying the heat capacity restriction. Transferring the full amount to bring stream 2 to the pinch temperature:

$$\Delta H_{ex} = 1.0(150 - 90) = 60 \text{ kW}$$

3. The heat required to bring stream 3 to its target temperature from the pinch temperature is:

$$\Delta H = 2.0(135 - 80) = 110 \text{ kW}$$

So a heater will have to be included to provide the remaining heat load:

$$\Delta H_{hot} = 110 - 60 = 50 \text{ kW}$$

This checks with the value given by the problem table in Fig. 3.21(b).

The proposed network design above the pinch is shown in Fig. 3.24.

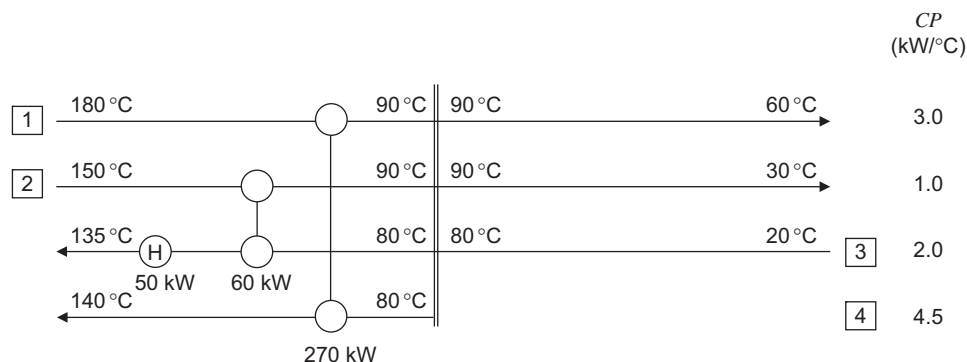


Fig. 3.24 Network design above the pinch.

Network design below the pinch

$$CP_h \geq CP_c$$

4. Stream 4 begins at the pinch temperature, $T_s = 80^\circ\text{C}$ and so is not available for any matches below the pinch.
5. A match between stream 1 and 3 adjacent to the pinch will satisfy the heat capacity restriction but not one between streams 2 and 3. So stream 1 is matched with stream 3, transferring the full amount to bring stream 1 to its target temperature:

$$\Delta H_{\text{ex}} = 3.0(90 - 60) = 90 \text{ kW}$$

6. Stream 3 requires more heat to bring it to the pinch temperature; the amount needed:

$$\Delta H = 2.0(80 - 20) - 90 = 30 \text{ kW}$$

This can be provided from stream 2, as the match is now away from the pinch.

The rise in temperature of stream 3 will be given by:

$$\Delta T = \Delta H / CP$$

So transferring 30 kW will raise the temperature from the source temperature to:

$$20 + 30/2.0 = 35^\circ\text{C}$$

and this gives a stream temperature difference on the outlet side of the exchanger of:

$$90 - 35 = 55^\circ\text{C}$$

So the minimum temperature difference condition, 10°C , will not be violated by this match.

7. Stream 2 needs further cooling to bring it to its target temperature, so a cooler must be included; the cooling required is:

$$\Delta H_{\text{cold}} = 1.0(90 - 30) - 30 = 30 \text{ kW}$$

which is the amount of the cold utility predicted by the problem table.

The proposed network for maximum energy recovery is shown in Fig. 3.25.

Stream splitting

If the heat capacities of streams are such that it is not possible to make a match at the pinch without violating the minimum temperature difference condition, then the heat capacity can be altered by splitting a stream. Dividing the stream will reduce the mass flow rates in each leg and hence the heat capacities. This is illustrated in Example 3.5.

Similarly, if there are not enough streams available to make all of the required matches at the pinch, then streams with large CP can be split to increase the number of streams.

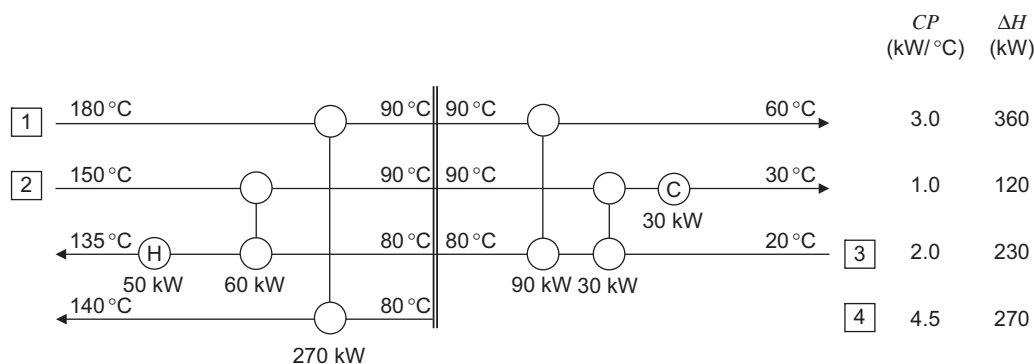


Fig. 3.25 Proposed heat exchanger network for $\Delta T_{\min} = 10^\circ\text{C}$.

Guide rules for stream matching and splitting are given in the *Institution of Chemical Engineers Guide*, [Kemp \(2007\)](#), and [Smith \(2016\)](#).

Summary

The guide rules for devising a network for maximum heat recovery are as follows:

1. Divide the problem at the pinch.
2. Design away from the pinch.
3. Above the pinch, match streams adjacent to the pinch, meeting the restriction:

$$CP_h \leq CP_c$$

4. Below the pinch, match streams adjacent to the pinch, meeting the restriction:

$$CP_h \geq CP_c$$

5. If the stream matching criteria cannot be satisfied, split a stream.
6. Maximize the exchanger heat loads.
7. Supply external heating only above the pinch and external cooling only below the pinch.

3.5.4 Minimum number of exchangers

The network shown in [Fig. 3.25](#) was designed to give the maximum heat recovery and will therefore give the minimum consumption, and cost, of the hot and cold utilities.

This will not necessarily be the optimum design for the network. The optimum design will be that which gives the lowest total annualized cost, taking into account the capital cost of the system, in addition to the utility and other operating costs. The number of exchangers in the network, and their size, will determine the capital cost.

In [Fig. 3.25](#) it is clear that there is scope for reducing the number of exchangers. The 30 kW exchanger between streams 2 and 3 can be deleted and the heat loads of the cooler and heater increased to bring streams 2 and 3 to their target temperatures. Heat would cross the pinch, and the consumption of the utilities would be increased. Whether the revised network would be better, or more economic, depends on the relative cost of capital and utilities and the operability of each design. For any network, there will be an optimum design that gives the least annual cost: capital charges plus utility and other operating costs. The estimation of capital and operating costs is covered in [Chapters 7 and 8](#).

To find the optimum design, it is necessary to cost a number of alternative designs, seeking a compromise between the capital costs, determined by the number and size of the exchangers, and the utility costs, determined by the heat recovery achieved.

For simple networks, [Holmann \(1971\)](#) has shown that the minimum number of exchangers is given by:

$$Z_{\min} = N' - 1 \quad (3.6)$$

where $Z_{\min}$ = minimum number of exchangers needed, including heaters and coolers

N' = the number of streams, including the utilities

For complex networks, a more general expression is needed to determine the minimum number of exchangers:

$$Z_{\min} = N' + L' - S \quad (3.7)$$

where L' = the number of internal loops present in the network

S = the number of independent branches (subsets) that exist in the network

A loop exists where a closed path can be traced through the network. A loop in the network shown in [Fig. 3.25](#) is shown in [Fig. 3.26](#). The presence of a loop indicates that there is scope for reducing the number of exchangers.

For a full discussion of Equation 3.7 and its applications see [Linnhoff et al. \(1979\)](#), [Smith \(2016\)](#), [Zhu \(2014\)](#), or [Kemp \(2007\)](#).

In summary, to seek the optimum design for a network:

1. Start with the design for maximum heat recovery. The number of exchangers needed will be equal to or less than the number for maximum energy recovery.

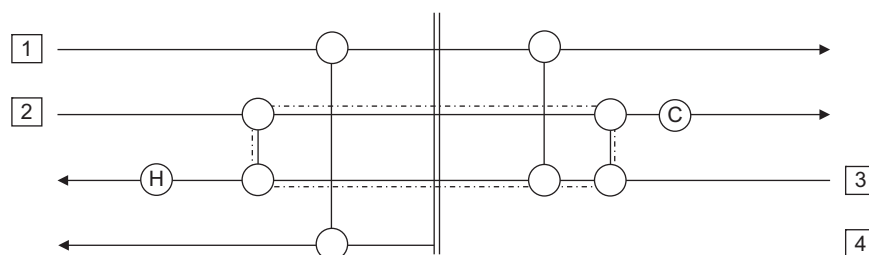


Fig. 3.26 Loop in network.

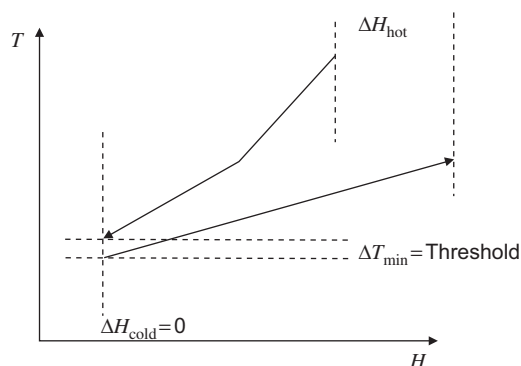


Fig. 3.27 Threshold problem.

2. Identify loops that cross the pinch. The design for maximum heat recovery will usually contain loops.
3. Starting with the loop with the least heat load, break the loops by adding or subtracting heat.
4. Check that the specified minimum temperature difference $\Delta T_{\min}$ has not been violated. If the violation is significant, revise the design as necessary to restore $\Delta T_{\min}$. If the violation is small, it may not have much impact on the total annualized cost and can be ignored.
5. Estimate the capital and operating costs and the total annual cost.
6. Repeat the loop breaking and network revision to find the lowest-cost design.
7. Consider the safety, operability, and maintenance aspects of the proposed design.

3.5.5 Threshold problems

Problems that show the characteristic of requiring only either a hot utility or a cold utility (but not both) over a range of minimum temperature differences, from zero up to a threshold value, are known as *threshold problems*. A threshold problem is illustrated in Fig. 3.27.

To design the heat exchanger network for a threshold problem, it is normal to start at the most constrained point. The problem can often be treated as one-half of a problem exhibiting a pinch.

Threshold problems are often encountered in the process industries. A pinch can be introduced in such problems if multiple utilities are used, as in the recovery of heat to generate steam, or if the chosen value of $\Delta T_{\min}$ is greater than the threshold value.

The procedures to follow in the design of threshold problems are discussed by Smith (2016) and Kemp (2007).

3.5.6 Determining utility consumption

Pinch analysis can be used to determine targets for process utility consumption. Initial targets for total hot and cold utility use can be calculated directly from the problem table algorithm or read from the composite curves. A more detailed breakdown of the utility needs can be determined from the initial heat exchanger network.

The following guidelines should be followed when using the pinch method to determine utility consumption targets:

1. Do not use cold utilities above the pinch temperature. This means that no process stream should be cooled from a temperature above the pinch temperature using a cold utility.
2. Do not use hot utilities below the pinch. This means no process stream should be heated from a temperature below the pinch temperature using a hot utility.
3. On either side of the pinch, maximize use of the cheapest utility first. Above the pinch this means use LP steam wherever possible before considering MP steam, then HP steam, hot oil, etc. Below the pinch, maximize use of cooling water before considering refrigeration.
4. If the process pinch is at a high temperature, consider boiler feed water preheat and steam generation as potential cold utility streams.
5. If the process pinch is at a low temperature, consider steam condensate and spent cooling water as hot streams.
6. If the process requires cooling to a very low temperature, consider using cascaded refrigeration cycles to improve the overall COP.
7. If the process requires heating to a very high temperature and a fired heater is needed, consider using the convective section heat either for process heating or for steam generation. For process control reasons, it may be necessary to operate the heater with process heating in the radiant section only, but the convective section heat is still available for use. In strict pinch terms, this heat can be used at any temperature above the pinch temperature, but in practice, convective section heat recovery is usually limited by the acid–gas dew point of the flue gas or other furnace design considerations (see [Section 19.17](#)).
8. If a process condition leads to the use of a more expensive utility, then consider process modifications that would make this unnecessary. For example, if a product must be cooled and sent to storage at 30 °C, the cooling cannot be carried out using cooling water and refrigeration must be used. The designer should question why 30 °C was specified for the storage. If it was because a vented tank was selected, then choosing a nonvented (floating roof) tank instead might allow the product to be sent to storage at 40 °C, in which case the refrigeration system could be eliminated.

Graphical methods and numerical approaches have been developed to assist in the optimal design of utility systems. For simple problems, these methods are not needed, as the heaters and coolers that have been identified in the heat exchange network can be assigned to the appropriate utility stream using the simple rules provided earlier. When a stream requires heating or cooling over a broad temperature range, the designer should consider whether it is cheaper to break the duty into several exchangers, each served by the appropriate utility for a given temperature range, or whether it makes more economic sense to use a single exchanger served by the hottest or coldest utility. The problem of placing multiple utilities is illustrated in Example 3.6.

3.5.7 Process integration: Integration of other process operations

The pinch technology method can give many other insights into process synthesis beyond the design of heat exchanger networks. The method can also be applied to the integration of other process units, such as separation columns, reactors, compressors and expanders, boilers, and heat pumps. The wider applications of pinch technology are discussed in the *Institution of Chemical Engineers Guide*, [Kemp \(2007\)](#), [El-Halwagi \(2006\)](#), and [Smith \(2016\)](#).

The techniques of process integration have been expanded for use in optimizing mass transfer operations and have been applied in waste reduction, water conservation, and pollution control; see [El-Halwagi \(2017\)](#) and [Dunn and El-Halwagi \(2003\)](#).

3.5.8 Computer tools for heat exchanger network design

Most pinch analysis in industry is carried out using commercial pinch analysis software. Programs such as Aspen HX-Net (Aspen Technology, Inc.), SUPERTARGET (Linnhoff March Ltd.), and UniSim ExchangerNet (Honeywell International, Inc.) allow the design engineer to plot composite curves, optimize $\Delta T_{\min}$, set targets for multiple utilities, and design the heat exchanger network.

Most of these programs are able to automatically extract stream data from process simulation programs, although great care should be taken to check the extracted data. There are many possible pitfalls in data extraction—for example, not recognizing changes in the CP of a stream or partial vaporization or condensation of a stream, any of which could lead to a kink in the stream *T-H* profile. See [Smith \(2016\)](#), [Knopf \(2012\)](#), [Zhu \(2014\)](#), or [Kemp \(2007\)](#) for more information on data extraction.

The commercial pinch technology tools also usually include automatic heat exchanger network synthesis features. The automatic synthesis methods are based on MINLP optimization of superstructures of possible exchanger options (see Chapter 12 for a discussion of MINLP methods). These tools can be used to arrive at a candidate network, but the optimization must be properly constrained so that it does not introduce a large number of stream splits and add a lot of small exchangers. Experienced designers seldom use automatic heat exchanger network synthesis methods, as it usually requires more effort to turn the resulting network into something practical than it would take to design a practical network manually. The NLP optimization capability of the software is widely used, though, for fine-tuning the network temperatures by exploitation of loops and stream split ratios.

Example 3.5

Determine the pinch temperatures and the minimum utility requirements for the streams set out in the following table for a minimum temperature difference between the streams of 20 °C. Devise a heat exchanger network to achieve the maximum energy recovery.

Stream Number	Type	Heat Capacity Flow Rate, kW/°C	Source Temp. °C	Target Temp. °C	Heat Load, kW
1	hot	40.0	180	40	5600
2	hot	30.0	150	60	1800
3	cold	60.0	30	180	9000
4	cold	20.0	80	160	1600

Solution

The problem table to find the minimum utility requirements and the pinch temperature can be built in a spreadsheet. The calculations in each cell are repetitive, and the formula can be copied from cell to cell using the cell copy commands. A spreadsheet template for the problem table algorithm is available in Microsoft Excel format in the online material at www.elsevier.com/books-and-journals/book-companion/9780128211793. The use of the spreadsheet is illustrated in Fig. 3.28 and is described here.

Company Name Address PROBLEM TABLE ALGORITHM Form XXXXX-YY-ZZ		Project Name				Sheet 1 of 1																																																																						
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2. Stream data <table border="1" style="width: 100%;"> <thead> <tr> <th rowspan="2">Stream No.</th> <th colspan="2">Actual temperature (°C)</th> <th colspan="2">Interval temperature (°C)</th> <th rowspan="2">Heat capacity flow rate CP (kW/°C)</th> <th rowspan="2">Heat load (kW)</th> </tr> <tr> <th>Source</th> <th>Target</th> <th>Source</th> <th>Target</th> </tr> </thead> <tbody> <tr><td>1</td><td>180</td><td>40</td><td>170</td><td>30</td><td>40</td><td>5600</td></tr> <tr><td>2</td><td>150</td><td>60</td><td>140</td><td>50</td><td>30</td><td>2700</td></tr> <tr><td>3</td><td>30</td><td>180</td><td>40</td><td>190</td><td>60</td><td>9000</td></tr> <tr><td>4</td><td>80</td><td>160</td><td>90</td><td>170</td><td>20</td><td>1600</td></tr> <tr><td>5</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr> <tr><td>6</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr> <tr><td>7</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr> <tr><td>8</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr> </tbody> </table>										Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)	Source	Target	Source	Target	1	180	40	170	30	40	5600	2	150	60	140	50	30	2700	3	30	180	40	190	60	9000	4	80	160	90	170	20	1600	5						0	6						0	7						0	8						0
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5	90	50	10	500	–2900	0																																																																						
6	50	40	–10	–400	–2500	400																																																																						
7	40	10	20	200	–2700	200																																																																						
8	30	10	–40	–400	–2300	600																																																																						

Fig. 3.28 Problem table algorithm spreadsheet.

First calculate the interval temperatures for $\Delta T_{\min} = 20^\circ\text{C}$

$$\text{hot streams } T_{\text{int}} = T_{\text{act}} - 10^\circ\text{C}$$

$$\text{cold streams } T_{\text{int}} = T_{\text{act}} + 10^\circ\text{C}$$

Stream	Actual temp. $^\circ\text{C}$		Interval temp. $^\circ\text{C}$	
	Source	Target	Source	Target
1	180	40	170	30
2	150	60	140	50
3	30	180	40	190
4	80	160	90	170

In the spreadsheet this can be done by using an IF function to determine whether the source temperature is lower than the target temperature, in which case the stream is a cold stream and should have $\Delta T_{\min} / 2$ added.

Next rank the interval temperatures, ignoring any duplicated values. In the spreadsheet this is done using the LARGE function. Determine which streams occur in each interval. For a stream to be present in a given interval, the largest stream interval temperature must be greater than the lower end of the interval range, and the lowest stream interval temperature must also be greater than or equal to the lower end of the interval range. This can be calculated in the spreadsheet using IF, AND, and OR functions. Once the streams in each interval have been determined, it is possible to calculate the combined stream heat capacities. These calculations are not strictly part of the problem table, so they have been hidden in the spreadsheet in columns to the right of the table.

The sum of CP values for the cold streams minus that for the hot streams can then be multiplied by the interval ΔT to give the interval ΔH , and the interval ΔH values can be cascaded to give the overall heat flow. The amount of heat that must be put in to prevent the heat flow from becoming negative is the lowest value in the column, which can be found using the SMALL function. The final column then gives a cascade showing only positive values, with zero energy cascading at the pinch temperature.

In the last column 2900 kW of heat have been added to eliminate the negative values in the previous column, so the hot utility requirement is 2900 kW and the cold utility requirement, the bottom value in the column, is 600 kW.

The pinch occurs where the heat transferred is zero, that is, at interval number 4, interval temperature 90°C .

So at the pinch hot streams will be at:

$$90 + 10 = 100^\circ\text{C}$$

and the cold streams will be at:

$$90 - 10 = 80^\circ\text{C}$$

Note that in the table both stream 1 and stream 4 had an interval temperature of 170°C , which led to a duplicate line in the list of ranked interval temperatures. Strictly, this line could have been eliminated, but because it gave a zero value for the ΔT , it did not affect the calculation. The programming of the spreadsheet is a lot easier if duplicate temperatures are handled in this manner.

To design the network for maximum energy recovery, start at the pinch and match streams, following the rules on stream heat capacities for matches adjacent to the pinch. Where a match is made, transfer the maximum amount of heat.

The proposed network is shown in Fig. 3.29. The methodology followed in devising this network was as follows.

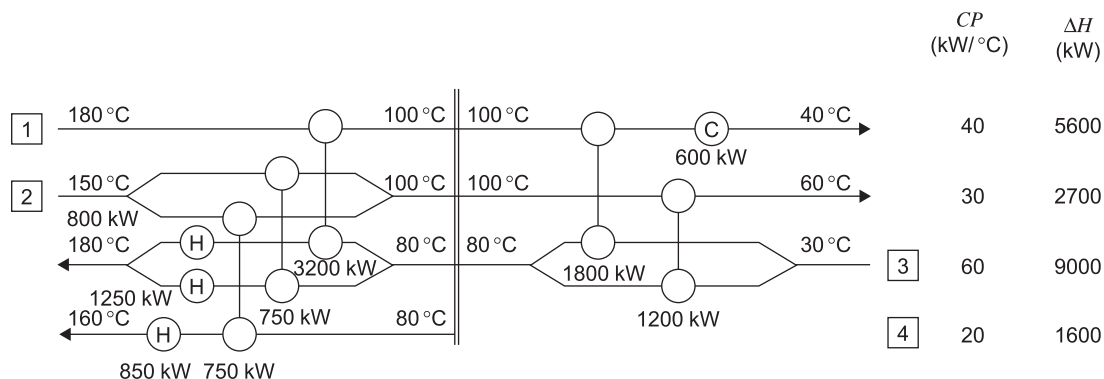


Fig. 3.29 Proposed heat exchanger network for Example 3.5.

Above pinch

1. $CP_h \leq CP_c$
2. Can match stream 1 or 2 with stream 3, but neither stream can match with stream 4. This creates a problem, because if we match stream 1 with 3, then stream 2 will not be able to make a match at the pinch. Likewise, if we match stream 2 with 3, then stream 1 will not be able to make a match at the pinch.
3. Check the heat available in bringing the hot streams to the pinch temperature.

$$\text{stream 1 } \Delta H = 40.0(180 - 100) = 3200 \text{ kW}$$

$$\text{stream 2 } \Delta H = 30.0(150 - 100) = 1500 \text{ kW}$$

4. Check the heat required to bring the cold streams from the pinch temperature to their target temperatures.

$$\text{stream 3 } \Delta H = 60.0(180 - 80) = 6000 \text{ kW}$$

$$\text{stream 4 } \Delta H = 20.0(160 - 80) = 1600 \text{ kW}$$

5. If we split stream 3 into two branches with CP of 40.0 and 20.0, then we can match the larger branch with stream 1 and transfer 3200 kW; that satisfies (ticks off) stream 1.
6. We now have two cold streams, both with CP of 20.0 and one hot stream (2) with CP of 30.0. We need to split stream 2 into two branches. As an initial guess, these can both have CP of 15.0. We can then match one branch of stream 2 with the smaller branch of 4 and transfer 750 kW, and the other branch with stream 3, also for 750 kW, which then ticks off stream 2.
7. Include a heater on the larger branch of stream 3 to bring it to its target temperature:

$$\Delta H_{\text{hot}} = 40(100) - 3200 = 800 \text{ kW}$$

8. Include a heater on the smaller branch of stream 3 to provide the balance of the heat required:

$$\Delta H_{\text{hot}} = 20(100) - 750 = 1250 \text{ kW}$$

9. Include a heater on stream 4 to provide the balance of the heat required:

$$\Delta H_{\text{hot}} = 1600 - 750 = 850 \text{ kW}$$

Check sum of heater duties = $800 + 1250 + 850 = 2900 \text{ kW} = \text{hot utility target}$.

Below pinch

9. $CP_h \geq CP_c$
10. Note that stream 4 starts at the pinch temperature so it cannot provide any cooling below the pinch.
11. Cannot match stream 1 or 2 with stream 3 at the pinch.
12. Split stream 3 to reduce CP . An even split will allow both streams 1 and 2 to be matched with the split streams adjacent to the pinch, so try this initially.
13. Check the heat available from bringing the hot streams from the pinch temperature to their target temperatures.

$$\text{stream 1 } \Delta H = 40.0(100 - 40) = 2400 \text{ kW}$$

$$\text{stream 2 } \Delta H = 30.0(100 - 60) = 1200 \text{ kW}$$

14. Check the heat required to bring the cold streams from their source temperatures to the pinch temperature:

$$\text{stream 3 } \Delta H = 60.0(80 - 30) = 3000 \text{ kW}$$

stream 4 is at the pinch temperature.

15. Note that stream 1 cannot be brought to its target temperature of 40°C by full interchange with stream 3, as the source temperature of stream 3 is 30°C , so ΔT_{min} would be violated. So transfer 1800 kW to one leg of the split stream 3.

16. Check the temperature at the exit of this exchanger:

$$\text{Temp out} = 100 - \frac{1800}{40} = 55^\circ\text{C, satisfactory}$$

17. Provide a cooler on stream 1 to bring it to its target temperature; the cooling needed:

$$\Delta H_{\text{cold}} = 2400 - 1800 = 600 \text{ kW}$$

18. Transfer the full heat load from stream 2 to the second leg of stream 3; this satisfies both streams.

Note that the heating and cooling loads, 2900 kW and 600 kW, respectively, match those predicted from the problem table.

Also note that to satisfy the pinch decomposition and the stream matching rules we ended up introducing a large number of stream splits. This is quite common in heat exchanger network design. None of the three split fractions was optimized, so substantial savings, as well as simplification of the network, could be possible. For example, loops exist between the branches of stream 3 and stream 1 and between the branches of stream 3 and stream 2. With the current split ratios, these loops cannot be eliminated, but with other ratios, it might be possible to eliminate one or two exchangers.

The introduction of multiple stream splits is often cited as a drawback of the pinch method. Stream splits can be problematic in process operation. For example, when an oil or other multicomponent stream is heated and partially vaporized, then the stream is a two-phase mixture. It is difficult to control the splitting of such streams to give the required flow rate in each branch. Experienced designers usually constrain the network to avoid multiple stream splits whenever possible, even if this leads to designs that have higher-than-minimum utility consumption.

Example 3.6

Determine the mix of utilities to use for the process introduced in Example 3.5 if the following utility streams are available:

Utility Stream	$T_{\text{supply}} (^\circ\text{C})$	$T_{\text{return}} (^\circ\text{C})$	Cost
MP steam (20 bar)	212	212	\$5.47/1000 lb
LP steam (6 bar)	159	159	\$4.03/1000 lb
Cooling water	30	40	\$0.10/1000 gal
Chilled water	10	20	\$4.50/GJ

Solution

From the solution to Example 3.5, we have the following heating and cooling duties that require utilities:

- Cooler on stream 1, duty 600 kW, to cool stream 1 from 55 °C to 40 °C
- Heater on large branch of stream 3, duty 800 kW, to heat from 160 °C to 180 °C
- Heater on small branch of stream 3, duty 1250 kW, to heat from 117.5 °C to 180 °C
- Heater on stream 4, duty 750 kW, to heat from 117.5 °C to 160 °C

It is obvious by inspection that if we are to maintain an approach temperature of 20 °C, then we will need to use MP steam and chilled water in at least some of the utility exchangers.

We can start by converting the utility costs into annual costs to provide a kW of heating or cooling based on an assumed 8000 hours per year of operation.

For MP steam at 20 bar:

- Heat of condensation (by interpolation in steam tables) $\approx 1889 \text{ kJ/kg}$
- $1 \text{ kW} = 3600 \times 8000 \text{ kJ/yr}$, therefore requires $3600 \times 8000 / 1889 = 15.2 \times 10^3 \text{ kg/y}$
- Annual cost per kW = $15.2 \times 10^3 \times 2.205 \text{ (lb/kg)} \times 5.47 \text{ (\$/1000 lb)} / 1000 = \$183 / \text{y}$

Similarly for LP steam at 6 bar:

- Heat of condensation (by interpolation in steam tables) $\approx 2085 \text{ kJ/kg}$
- $1 \text{ kW} = 3600 \times 8000 \text{ kJ/yr}$, therefore requires $3600 \times 8000 / 2085 = 13.8 \times 10^3 \text{ kg/y}$
- Annual cost per kW = $13.8 \times 10^3 \times 2.205 \text{ (lb/kg)} \times 4.03 \text{ (\$/1000 lb)} / 1000 = \$123 / \text{y}$

For cooling water with a cooling range of 10 °C

- 1 kW of cooling requires $CP = 1/10 = 0.1 \text{ kW/}^\circ\text{C}$
- Heat capacity of water $\approx 4.2 \text{ kJ/kg}^\circ\text{C}$, so:
- Annual flow rate of cooling water per kW = $0.1 \times 3600 \times 8000 / 4.2 = 686 \times 10^3 \text{ kg/y}$
- 1000 gal of water = 3785 liters and has mass roughly 3785 kg, so:
- water flow rate = $686 \times 10^3 / 3785 = 181.2$ thousand gallons per year, which has annual cost $0.1 \times 181.2 = \$18.1 / \text{y}$

For chilled water:

1 kW of cooling = $3600 \times 8000 = 28.8 \times 10^6 \text{ kJ/y} = 28.8 \text{ GJ/y}$

So, annual cost = $28.8 \times 4.50 = \$129.6 / \text{y}$

It is clearly cheaper to use LP steam rather than MP steam and to use cooling water instead of chilled water whenever it is feasible to do so.

Beginning with the design below the pinch, if we are to maintain a minimum temperature difference of 20 °C, then we cannot use cooling water below $30 + 20 = 50^\circ\text{C}$. The lowest utility cost design would therefore use cooling water to cool stream 1 from 55 °C to 50 °C (duty 200 kW). A second cooler would then be needed to cool stream 1 from 50 °C to 40 °C using chilled water (duty 400 kW). The annual utility cost of this design would be $200(18.1) + 400(129.6) = \$55,460$.

It might reasonably be argued that the utility savings from using the minimum cost of coolant do not justify the capital cost of an extra exchanger. Two possible alternatives can be considered. If all of the cooling is carried out using chilled water, then the minimum temperature difference constraint is not violated and a single cooler of duty 600 kW can be used. The annual utility cost would be $600(129.6) = \$77,760$. The use of chilled water gives larger log-mean temperature difference in the cooler, so the total surface area required in this design is less than the sum of the areas needed for the two exchangers proposed earlier. The incremental operating cost would have to be traded against the capital cost savings. Alternatively, if we jettison the 20 °C minimum temperature difference and allow a 10 °C minimum temperature difference in the cooler, then we can cool stream 1 using only cooling water in a single cooler of duty 600 kW. The annual utility cost would be $600(18.1) = \$10,860$. The savings in operating cost would have to be traded against the increased capital cost that would result from having a lower log-mean temperature difference for this exchanger.

Turning now to the design above the pinch, LP steam cannot be used for heating any stream that is above a temperature of $159 - 20 = 139^\circ\text{C}$. The minimum utility cost design would therefore use the following heaters:

- LP steam to heat stream 4 from 117.5 °C to 139 °C
- LP steam to heat the small branch of stream 3 from 117.5 °C to 139 °C
- MP steam to heat the small branch of stream 3 from 139 °C to 180 °C
- MP steam to heat the large branch of stream 3 from 160 °C to 180 °C
- MP steam to heat stream 4 from 139 °C to 160 °C

Again, although this design has the minimum utility cost, other designs may be more optimal when capital costs are also considered. For example, there is no reason why the two branches of stream 3 must be sent to separate MP steam heaters. These two heaters could be combined, even though that violates the rule of thumb about not mixing streams at different temperatures, as we are well away from the pinch and have already ensured maximum use of LP steam. This modification would reduce capital cost with no increase in operating cost, so would almost certainly be adopted. Another modification to consider would be to examine allowing a smaller minimum temperature difference for the heaters that use LP steam. This would increase LP steam use at the expense of more capital (reduced temperature difference in the exchangers) and so would require a trade-off between the additional capital and the energy cost savings.

Note that by introducing the lowest-cost utilities into the design we went from needing three heaters and one cooler in Fig. 3.29 to using two coolers and five heaters in the lowest-cost utility design. The introduction of multiple utilities almost always leads to an increase in the number of heat exchangers needed in a design, as well as the surface area requirements, and the energy cost savings must justify the resulting increase in capital cost.

3.6 Energy management in unsteady processes

The energy recovery approaches described earlier have been for steady-state processes, where the rate of energy generation or consumption did not vary with time. Batch and cyclic processes present multiple challenges for energy management. The designer must not only consider the amount of heat that must be added to or removed from the process but also the dynamics of heat transfer. Limitations on the rate of heat transfer often cause heating and cooling steps to become the rate-limiting steps in determining the overall cycle time. The sequential nature of batch operations can also reduce the possibilities for heat recovery by heat exchange unless multiple batches are processed in parallel and sequenced such that heat can be transferred from one batch to the next.

3.6.1 Differential energy balances

If a batch process is considered, or if the rate of energy generation or removal varies with time, it is necessary to set up a differential energy balance. For batch processes, the total energy requirements can usually be estimated by taking a single batch as the time basis for the calculation, but the maximum rate of heat generation must also be estimated to size any heat transfer equipment needed.

A generalized differential energy balance can be written as:

$$\text{Energy out} = \text{Energy in} + \text{generation} - \text{consumption} - \text{accumulation} \quad (3.8)$$

The energy in and energy out terms should include both heat transfer and convective heat flows, and the generation and consumption terms include heat of mixing, heat of reaction, etc. An unsteady-state mass balance must usually be solved simultaneously with the differential energy balance.

Most batch processing operations are carried out in the liquid phase in stirred tanks. In the simplest case, heat is only added or removed when the vessel is full, and the convective heat flows can be neglected. If there is no heat of reaction or mixing, then Equation 3.8 simplifies to:

$$\text{Rate of heat accumulation} = \text{rate of heat transfer into vessel} \quad (3.9)$$

$$MC_p \frac{dT}{dt} = UA\Delta T_m \quad (3.10)$$

where M = the mass contained in the vessel, kg

C_p = the specific heat capacity of the vessel contents, J/kg°C

T = temperature of the vessel contents, °C

t = time, s

U = the overall heat transfer coefficient, W/m²°C

A = heat transfer area, m²

ΔT_m = the mean temperature difference, the temperature driving force, °C

The mean temperature difference for heat transfer, ΔT_m , will generally be a function of the temperature of the vessel contents, T , as well as depending on the nature of the heating or cooling medium (isothermal or nonisothermal) and the type of heat transfer surface used. Batch tanks are usually heated or cooled using internal coils, jacketed vessels, or external heat exchangers. Heat transfer to vessels is discussed in more detail in Section 19.18.

In more complex cases, it is usually a good idea to set up a dynamic simulation model of the process. The use of dynamic simulation allows the designer to include additional heat source and sink terms such as losses to the environment. The designer can also use the dynamic model to investigate the interaction between the process, the heat transfer equipment, and the process control system, and hence to develop control algorithms that ensure rapid heating or cooling but do not cause excessive overshoot of the target temperature.

The application of differential energy balances to simple problems is illustrated in Examples 3.7 and 15.6.

3.6.2 Energy recovery in batch and cyclic processes

Most batch processes operate at relatively low temperatures, below 200 °C, where use of steam or hot oil will give high heat transfer rates for process heating. High heat transfer rates allow shorter heating times and enable use of internal coils and jacketed vessels, reducing the number of pieces of equipment in the plant. If the energy cost is a very small fraction of the total cost of production, then recovering heat from the process may not be economically attractive, as the resulting increase in capital cost will not be justified.

Many batch processes need cooling to temperatures that require some degree of refrigeration. Fermentation processes are often operated at temperatures below 40 °C, where use of cooling water can be problematic and chilled water or other refrigerants are used instead. Food processes often require refrigeration or freezing of the product. Recovery of “cooling” from chilled streams is not possible when the product must be delivered in chilled form.

Three of the most commonly used methods for recovering heat in batch and cyclic processes are described next. Energy optimization in batch plants has been the subject of much research and is discussed in more detail in the papers by Vaselenak et al. (1986), Kemp and Deakin (1989), and Lee and Reklaitis (1995) and the books by Smith (2016), Kemp (2007), and Majozi (2010).

Semicontinuous operation

The simplest approach to allow some degree of heat recovery in a batch process is to operate part of the plant in a continuous mode. The use of intermediate accumulation tanks can allow sections of the plant to be fed continuously or to accumulate product for batching into other operations.

Semicontinuous operation is often deployed for feed sterilizers and pasteurizers in food processing and fermentation plants. In a pasteurization operation, the feed must be heated to a target temperature, held at that temperature for long enough to kill unwanted species that may be present in the feed, and then cooled to the process temperature. The high-temperature residence time is usually obtained by passing the process fluid through a steam-traced or well-insulated pipe coil. The initial heating of the feed can be accomplished by heat exchange, with the hot fluid leaving the coil, allowing the use of a smaller steam heater to reach the target temperature, as shown in Fig. 3.30. This design is common in food processing plants, but care must be taken to ensure that there is no leakage across the heat exchanger, which could potentially lead to contamination of the “sterile” feed with components from the raw feed.

Another situation where semicontinuous operation is often adopted is in the separation section of a batch plant. Some energy-intensive separations such as distillation and crystallization are easier to control to high recovery and tight product specifications when operated in continuous mode. In these cases a surge tank can feed the continuous section of the plant, and typical heat recovery schemes such as feed-bottoms heat exchange can be considered.

If a batch plant is designed so that batches are transferred from one vessel to another (as opposed to undergoing successive steps in the same vessel), then heat can be transferred between streams as they are pumped from one vessel to the next. During the pumping operation the flow is at a pseudo-steady state, and a heat exchanger between two streams behaves the same as a heat exchanger in a continuous plant. Fig. 3.31 shows such an arrangement in which a hot stream flows from vessel R1 to vessel R2, while a cold stream flows from vessel R3 to vessel R4. The flowing streams exchange heat in a heat exchanger that is shown as being counter-current, but could equally well be cross-flow or co-current if the temperatures were suitable. This arrangement is sometimes referred to as a “counter-current” heat integration, although it should be stressed that the exchanger can be co-current or cross-flow.

When stream-to-stream heat transfer is used, a high degree of heat recovery can be obtained. The exchanger will perform well and maintain roughly constant stream outlet temperatures during the period when the vessels are being pumped out. When the liquid level in the vessels becomes too low for pump operation, the flow rates in the exchanger become too low for the exchanger to function effectively. If batch-to-batch contamination is not important and there are no safety hazards, product quality issues, or fouling concerns, then the exchanger can be isolated (“blocked in”) while the remaining tank contents are drained through bypass lines, and the exchanger is then ready to be reused when tanks R1 and R3 are again ready to be drained. In the case where batch-to-batch

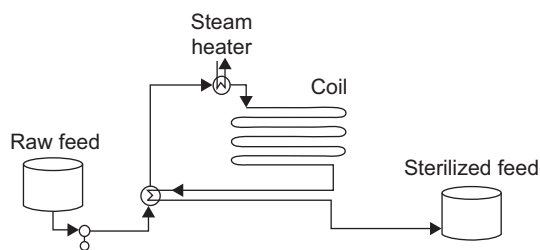


Fig. 3.30 Heat integration of feed sterilization system.

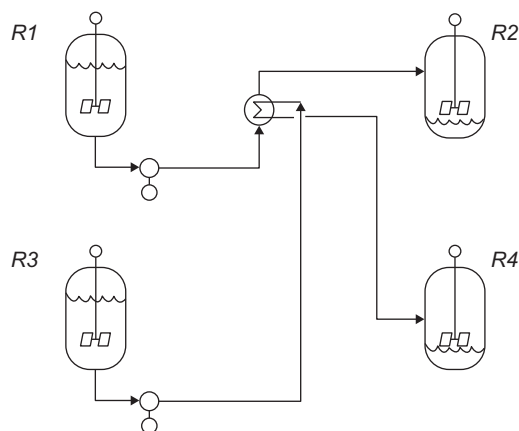


Fig. 3.31 Stream-to-stream ("counter-current") heat integration of batch vessels.

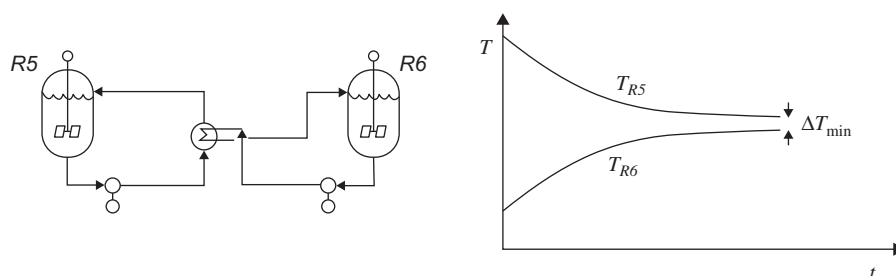


Fig. 3.32 Tank-to-tank ("co-current") heat integration of batch vessels.

mixing is not desired or where there are other reasons why the exchanger cannot be left full of process fluid, provision must be made to flush, drain, and clean the exchanger once the upstream tanks are empty.

Sequencing multiple batches

If a plant contains several batches that are undergoing different steps of a process at the same time, or if several different batch plants are grouped close to each other, then the batches can sometimes be sequenced so that heat can be transferred from one batch to another.

Suppose a batch process contains the steps of heating reagents, reacting them at a desired temperature and then cooling the products before sending them for further processing. If two reactors are used, a heat exchanger can be employed to exchange heat from the reactor that is being cooled to the reactor that is being heated. For example, in Fig. 3.32, hot fluid from vessel R5 is pumped through an exchanger, where it transfers heat to cold fluid that is pumped from vessel R6. The fluid from each vessel is returned to the vessel that it came from. The heat exchanger in Fig. 3.31 is shown as being counter-current, but co-current or cross-flow heat exchange could be used if the temperatures were appropriate.

The graph on the right of Fig. 3.32 is a schematic of the temperature–time profile for both vessels. As time progresses, they become closer in temperature and would eventually reach thermal equilibrium. In practice, it is usually not economic to run the exchanger for very long times, and heat transfer is stopped when an acceptable minimum temperature difference between the vessels is reached, shown as ΔT_{min} in the figure. Tank-to-tank heat transfer does not allow as efficient heat recovery as stream-to-stream, as the hottest temperatures in the hot tank are matched with the coldest temperatures in the cold tank, as they would be in a co-current heat exchanger; hence Vaselenak et al. (1986) named this type of batch heat integration "co-current" heat integration. It should again be stressed that the heat exchanger is usually designed to be counter-current or cross-flow.

An improvement on this scheme is to use stream-to-tank heat transfer, shown in Fig. 3.33, in which a stream that is transferred from one vessel to another exchanges heat with a stream that is returned to the tank from which it originated. In Fig. 3.33, hot fluid flows from R7 to R8 and transfers heat to a cold stream that is pumped from R9 and returned to R9. The graph on the right of Fig. 3.33 is a schematic of the temperature behavior of R9, R8, and the location marked as A on the line entering R8. The temperature of the cold fluid in R9 increases over

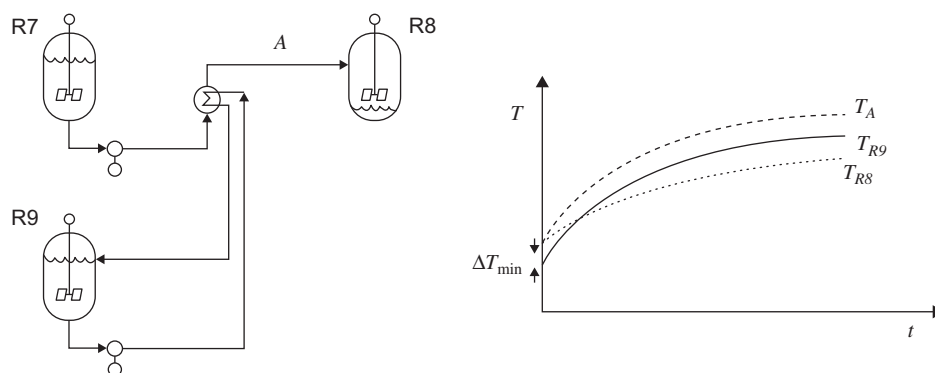


Fig. 3.33 Stream-to-tank ("co-current/counter-current") heat integration of batch vessels.

time as heat is transferred to it. The temperature at A is the temperature of the hot fluid at the exit of the heat exchanger. The heat exchanger will usually be designed to pinch at the cold end, because the recirculating flow from $R9$ can be much greater than the pump-out flow from $R7$. Consequently, the temperature at A will be equal to the temperature in $R9$ plus the temperature approach of the heat exchanger, and so the temperature at A has a profile offset above the $R9$ temperature profile. The temperature in $R8$ is the time-averaged integral of the temperature of the feed to the vessel (i.e., the time-averaged integral of the temperature at A). Although the fluid entering $R8$ becomes hotter with time, it is mixed with an accumulating volume of colder fluid, so the temperature in $R8$ does not increase so rapidly as the temperature in $R9$, and $R8$ can even be colder than $R9$ when the heat transfer is complete. This process is therefore intermediate in terms of thermal efficiency between tank-to-tank heat transfer and stream-to-stream heat transfer. It is sometimes known as "co-current/counter-current" heat integration. The derivation of the equations needed to accurately describe the temperature profiles for this arrangement is given by Vaselenak et al. (1986).

When tank-to-tank or tank-to-stream heat transfer is selected, care must be taken to ensure that the heat exchanger doesn't cause problems when not in use. If the designer anticipates that there could be problems with fouling, corrosion, batch-to-batch contamination, product degradation, safety issues, or any other issue with leaving the exchanger filled, then the design must include means to drain, flush, and clean the exchanger between batches.

When considering the use of stream-to-stream, stream-to-tank, or tank-to-tank heat transfer in a batch process, the designer must ensure that the batch schedules allow both streams to be available at the same time and for a sufficient time to accomplish the desired heat recovery. When draining, flushing, and cleaning of the heat exchanger are necessary, these steps must also be taken into account. For a process that handles multiple batches simultaneously or a site with multiple batch plants, the resulting scheduling problem becomes too large to optimize by hand and numerical methods must be used. See Vaselenak et al. (1986), Kemp and Deakin (1989), and Lee and Reklaitis (1995) for approaches to solving such problems.

Indirect heat recovery

An alternative method of heat recovery that can be used in batch processing is to recover heat indirectly through the utility system or using a heat storage system. Although less thermally efficient than process-to-process heat recovery, this method eliminates problems from sequencing of operations.

In indirect heat recovery, heat from a hot process stream is transferred to a utility stream, such as a reservoir of heat transfer fluid. The heat transfer fluid can then be used for heating elsewhere in the process. Indirect heat recovery can be used in any of the flow schemes described earlier, but in all cases the use of an intermediate stream will reduce the thermal efficiency and the amount of heat that can be recovered. Heat storage systems can only be used when there is a large enough temperature difference between the process heat source and process heat sink to allow for the thermal inefficiency of transfer of heat to the storage medium, cooling losses during storage, and transfer of heat to the process heat sink.

Example 3.7 Differential energy balance

In the batch preparation of an aqueous solution, the water is first heated to 80 °C in a jacketed, agitated vessel; 1000 Imp. gal. (4545 kg) is heated from 15 °C. If the jacket area is 300 ft² (27.9 m²) and the overall heat transfer coefficient can be taken as 50 Btu ft⁻² h⁻¹ °F⁻¹ (285 W m⁻² K⁻¹), estimate the heating time. Steam is supplied at 25 psig (2.7 bar).

Solution

The rate of heat transfer from the jacket to the water will be given by Equation 3.10:

$$MC_p \frac{dT}{dt} = UA\Delta T_m \quad (3.10)$$

Because steam is used as the heating medium, the hot side is isothermal and we can write:

$$\Delta T_m = T_s - T$$

where T_s = the steam saturation temperature.

Integrating:

$$\int_0^{t_B} dt = \frac{MC_p}{UA} \int_{T_1}^{T_2} \frac{dT}{(T_s - T)}$$

Batch heating time, t_B :

$$t_B = -\frac{MC_p}{UA} \ln \frac{T_s - T_2}{T_s - T_1}$$

For this example

$$MC_p = 4.18 \times 4545 \times 10^3 \text{ JK}^{-1}$$

$$UA = 285 \times 27 \text{ WK}^{-1}$$

$$T_1 = 15^\circ\text{C}, T_2 = 80^\circ\text{C}, T_s = 130^\circ\text{C}$$

$$\begin{aligned} t_B &= -\frac{4.18 \times 4545 \times 10^3}{285 \times 27.9} \ln \frac{130 - 80}{130 - 15} \\ &= 1990 \text{ s} = 33.2 \text{ min} \end{aligned}$$

In this example the heat capacity of the vessel and the heat losses have been neglected for simplicity. They would increase the heating time by 10% to 20%.

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3.8 Nomenclature

		Dimensions in \$MLT\theta\$
A	Area	L^2
CP	Stream heat capacity flow rate	$ML^2T^{-2}\theta^{-1}$
CP_c	Stream heat capacity flow rate, cold stream	$ML^2T^{-2}\theta^{-1}$
CP_h	Stream heat capacity flow rate, hot stream	$ML^2T^{-2}\theta^{-1}$
C_p	Specific heat at constant pressure	$L^2T^{-2}\theta^{-1}$
ΣCP_c	Sum of heat capacity flow rates of cold streams	$ML^2T^{-2}\theta^{-1}$
ΣCP_h	Sum of heat capacity flow rates of hot streams	$ML^2T^{-2}\theta^{-1}$
COP	Coefficient of performance for a refrigeration cycle	—
COP_h	Coefficient of performance for a heat pump	—
dH_b	Boiler heating rate	$L^{-2}T^2$
H	Enthalpy	ML^2T^{-2}
ΔH	Change in enthalpy	ML^2T^{-2}
ΔH_{cold}	Heat transfer from cold utility	ML^2T^{-3}
ΔH_{ex}	Heat transfer in exchanger	ML^2T^{-3}
ΔH_{hot}	Heat transfer from hot utility	ML^2T^{-3}
ΔH_n	Net heat required in n th interval	ML^2T^{-3}

(Continued)

Dimensions in \$MLT θ

ΔH_{xp}	Cross-pinch heat transfer	ML^2T^{-3}
$-\Delta H^\circ_c$	Standard heat of combustion	L^2T^{-2}
ΔH°_f	Standard enthalpy of formation	L^2T^{-2}
h_g	Specific enthalpy of steam	L^2T^{-2}
L'	Number of internal loops in network	—
M	Mass	M
m	Mass flow rate	MT^{-1}
N	Number of cold streams, heat exchanger networks	—
N'	Number of streams	—
P_{BFW}	Price of boiler feed water	$\$M^{-1}$
P_F	Price of fuel	$\$M^{-1}L^{-2}T^2$
P_{HPS}	Price of high-pressure steam	$\$M^{-1}$
S	Number of independent branches	—
s_g	Specific entropy	$L^2T^{-2}\theta^{-1}$
T	Temperature, absolute	θ
T_1	Initial temperature	θ
T_2	Final temperature	θ
T_{act}	Actual stream temperature	θ
T_c	Condenser temperature	θ
T_e	Evaporator temperature	θ
T_{int}	Interval temperature	θ
T_n	Temperature in n th interval	θ
T_{pinch}	Pinch temperature	θ
T_{return}	Return temperature for utility	θ
T_s	Source temperature	θ
T_{supply}	Supply temperature for utility	θ
T_s	Steam saturation temperature	θ
T_t	Target temperature	θ
ΔT_m	Mean temperature difference	θ
ΔT_{min}	Minimum temperature difference (minimum approach) in heat exchanger	θ
ΔT_n	Interval temperature difference	θ
t	Time	T
t_B	Batch heating time	T
U	Overall heat transfer coefficient	$MT^{-3}\theta^{-1}$
Z_{min}	Minimum number of heat exchangers in network	—
η_B	Boiler efficiency	—

3.9 Problems

- 3.1. A process heater uses Dowtherm A heat transfer fluid to provide 850 kW of heat. Estimate the annual operating cost of the heater if the Dowtherm evaporator is 80% efficient and the price of natural gas is \$4.60/MMBtu. Assume 8000 operating hours per year.
- 3.2. A site steam system consists of HP steam at 40 bar, MP steam at 18 bar, and LP steam at 3 bar. If natural gas costs \$3.50/MMBtu and electricity is worth \$0.07/kWh, estimate the cost of steam at each level in \$/metric ton.
- 3.3. Make a rough estimate of the cost of steam per ton produced from a packaged boiler with 10,000 kg per hour of steam required at 15 bar. Natural gas will be used as the fuel, with calorific value 39 MJ/m³ (roughly 1 MMBtu/1000 scf). Take the boiler efficiency as 80%. No condensate will be returned to the boiler.
- 3.4. A crystallization process requires operation at -5°C . The refrigeration system can reject heat to cooling water that is available at 35°C . If a single refrigeration cycle has an efficiency of 60% of Carnot cycle performance, estimate the cost of providing 1 kW of cooling to this process using a single-stage cycle and using a cascaded-two stage cycle (in which the colder cycle rejects heat to the warmer cycle). Electricity costs \$0.07/kWh, and the cost of cooling water can be neglected.
- 3.5. A gas produced as a by-product from the carbonization of coal has the following composition, mole percent: carbon dioxide 4, carbon monoxide 15, hydrogen 50, methane 12, ethane 2, ethylene 4, benzene 2, balance

nitrogen. Using the data given in Appendix C (available online at www.elsevier.com/books-and-journals/book-companion/9780128211793), calculate the gross and net calorific values of the gas. Give your answer in MJ/m³ at standard temperature and pressure.

- 3.6. Determine the pinch temperature and the minimum utility requirements for the process set out here. Take the minimum approach temperature as 15 °C. Devise a heat exchanger network to achieve maximum energy recovery.

Stream Number	Type	Heat capacity kW/°C	Source Temp. °C	Target Temp. °C
1	Hot	13.5	180	80
2	Hot	27.0	135	45
3	Cold	53.5	60	100
4	Cold	23.5	35	120

- 3.7. Determine the pinch temperature and the minimum utility requirements for the process set out here. Take the minimum approach temperature as 15 °C. Devise a heat exchanger network to achieve maximum energy recovery.

Stream Number	Type	Heat capacity kW/°C	Source Temp. °C	Target Temp. °C
1	Hot	10.0	200	80
2	Hot	20.0	155	50
3	Hot	40.0	90	35
4	Cold	30.0	60	100
5	Cold	8.0	35	90

- 3.8. To produce a high-purity product two distillation columns are operated in series. The overhead stream from the first column is the feed to the second column. The overhead from the second column is the purified product. Both columns are conventional distillation columns fitted with reboilers and total condensers. The bottom products are passed to other processing units, which do not form part of this problem. The feed to the first column passes through a preheater. The condensate from the second column is passed through a product cooler. The duty for each stream is summarized here:

No.	Stream	Type	Source Temp. °C	Target Temp. °C	Duty, kW
1	Feed preheater	cold	20	50	900
2	First condenser	hot	70	60	1350
3	Second condenser	hot	65	55	1100
4	First reboiler	cold	85	87	1400
5	Second reboiler	cold	75	77	900
6	Product cooler	hot	55	25	30

Find the minimum utility requirements for this process, for a minimum approach temperature of 10 °C.

Note: The stream heat capacity is given by dividing the exchanger duty by the temperature change.

- 3.9. At what value of the minimum approach temperature does the problem in Example 3.5 become a threshold problem? Design a heat exchanger network for the resulting threshold problem. What insights does this give into the design proposed in Example 3.5?

Process simulation

KEY LEARNING OBJECTIVES

- How to use commercial process simulation software to build a process heat and material balance model
- How to select thermodynamic models for prediction of phase equilibrium and stream properties
- How to use user-specified models and components when the simulator does not have what you need
- How to converge flowsheets that include recycles and overcome convergence problems
- How to optimize converged flowsheets

4.1 Introduction

This chapter addresses the use of process simulation tools in developing an overall process mass and energy balance. A process flow diagram or flowsheet typically includes material balances made over the complete process and each individual unit. Energy balances are also made to determine the energy flows and the utility requirements.

Most flowsheet calculations are carried out using commercial process simulation programs. The process simulation programs contain models for most unit operations as well as thermodynamic and physical property models. All the commercial programs feature some level of custom modeling capability that allows the designer to add models for nonstandard operations.

Many companies developed proprietary flowsheeting programs between 1960 and 1980. The cost of maintaining and updating proprietary software is high; consequently, very few of the proprietary flowsheeting programs are still in use, and most companies now rely entirely on commercially available software. Each of the commercial process simulation programs has its own unique idiosyncrasies, but they share many common features. The discussion in this chapter addresses general problems of process simulation and flowsheeting rather than software-specific issues. The latter are usually thoroughly documented in the user manuals and online help that come with the software. Examples have been provided in this chapter using both Aspen Plus (Aspen Technology, Inc.) and UniSim Design Suite (Honeywell International, Inc.). UniSim Design is based on the Hysys software that was originally developed by Hyprotech Ltd. and is now owned and licensed by Honeywell.

Cyclic and batch process operations that do not run at steady state require dynamic simulation models. Several of the steady-state process simulation programs can be modified to run as dynamic simulators. Specialized programs have also been developed for batch processing, such as SuperPro Batch Designer, which has many features tailored to the simulation of biological processes. Simulation of batch processes is discussed in Section 4.9.

Because flowsheeting is usually carried out using computer programs, it is necessary for the design engineer to have a good understanding of how to set up and solve computer models. The flowsheet model that is solved on the computer to generate a mass and energy balance is often not an exact representation of the process flow diagram. The designer may need to use combinations of simulation library models and user models to capture the performance of process equipment. Spreadsheet or hand calculations are also often helpful in setting up process simulation models and providing good initial estimates so as to accelerate convergence.

4.2 Process simulation programs

The most commonly used commercial process simulation programs are listed in Table 4.1. Most of these programs can be licensed by universities for educational purposes at nominal cost.

Detailed discussion of the features of each of these programs is beyond the scope of this book. For a general review of the requirements, methodology, and application of process simulation programs, the reader is referred to the books by Husain (1986), Wells and Rose (1986), Leesley (1982), Benedek (1980), and Westerberg et al. (1979). The features of the individual programs are described in their user manuals and online help. Two of these simulators have been used to generate the examples in this chapter: Aspen Plus (v.11.1) and UniSim Design (R360.1). More recent versions of these programs are now available with additional features, but the screen appearance has not changed significantly since the examples were developed.

Process simulation programs can be divided into two basic types:

Sequential-modular programs, in which the equations describing each process unit (module) are solved module by module in a stepwise manner. Iterative techniques are then used to solve the problems arising from the recycle of information.

Simultaneous (also known as *equation-oriented*) programs, in which the entire process is described by a set of equations and the equations are solved simultaneously, not stepwise as in the sequential approach. Simultaneous programs can simulate the unsteady-state operation of processes and equipment and can give faster convergence when multiple recycles are present.

TABLE 4.1 Simulation packages

Name	Type	Source	Internet address www.—
Aspen Plus	steady-state	Aspen Technology Inc. Ten Canal Park Cambridge, MA 02141-2201, USA	Aspentech.com
CHEMCAD	steady-state	2901 Wilcrest, Suite 305 Houston, TX 77042 USA	Chemstations.net
DESIGN II	steady-state	WinSim Inc. P.O. Box 1885 Houston, TX 77251-1885, USA	Winsim.com
HYSYS	steady-state and dynamic	Aspen Technology Inc. Ten Canal Park Cambridge, MA 02141-2201, USA	Aspentech.com
PRO/II and DYNsIM	steady-state and dynamic	SimSci-Esscor 5760 Fleet Street Suite 100, Carlsbad, CA 92009, USA	Simsci.com
UniSim Design	steady-state and dynamic	Honeywell 300-250 York Street London, Ontario N6A 6K2, Canada	Honeywell.com

Note: Contact the website to check the full features of the most recent versions of the programs.

In the past, most simulation programs available to designers were of the sequential-modular type. They were simpler to develop than the equation-oriented programs and required only moderate computing power. The modules are processed sequentially, so essentially only the equations for a particular unit are in the computer memory at one time. Also, the process conditions, temperature, pressure, flow rate, etc., are fixed in time. With the sequential modular approach, computational difficulties can arise due to the iterative methods used to solve recycle problems and obtain convergence. A major limitation of sequential modular simulators is the inability to simulate the dynamic, time-dependent behavior of a process.

Simultaneous dynamic simulators require appreciably more computing power than steady-state simulators to solve the thousands of differential equations needed to describe a process, or even a single item of equipment. With the development of fast, powerful computers, this is no longer a restriction. By their nature, simultaneous programs do not experience the problems of recycle convergence inherent in sequential simulators; however, as temperature, pressure, and flow rate are not fixed and the input of one unit is not determined by the calculated output from the previous unit in the sequence, simultaneous programs demand more computer time. This has led to the development of hybrid programs in which the steady-state simulator is used to generate the initial conditions for the equation-oriented or dynamic simulation.

The principal advantage of simultaneous, dynamic simulators is their ability to model the unsteady-state conditions that occur at start-up and during fault conditions. Dynamic simulators are being increasingly used for safety studies and in the design of control systems, as discussed in Section 4.9.

The structure of a typical simulation program is shown in Fig. 4.1.

The program consists of:

1. A main executive program that controls and keeps track of the flowsheet calculations and the flow of information to and from the subroutines.
2. A library of equipment performance subroutines (modules) that simulate the equipment and enable the output streams to be calculated from information on the inlet streams.
3. A data bank of physical properties. To a large extent, the utility of a sophisticated flowsheeting program depends on the comprehensiveness of the physical property data bank. The collection of the physical property data required for the design of a particular process and its transformation into a form suitable for a particular flowsheeting program can be very time consuming.
4. Subroutines for thermodynamics, such as the calculation of vapor–liquid equilibrium and stream enthalpies.
5. Subprograms and data banks for equipment sizing and costing. Process simulation programs enable the designer to consider alternative processing schemes, and the cost routines allow quick economic comparisons to be made. Some programs include optimization routines. To make use of a costing routine, the program must be capable of producing at least approximate equipment designs.

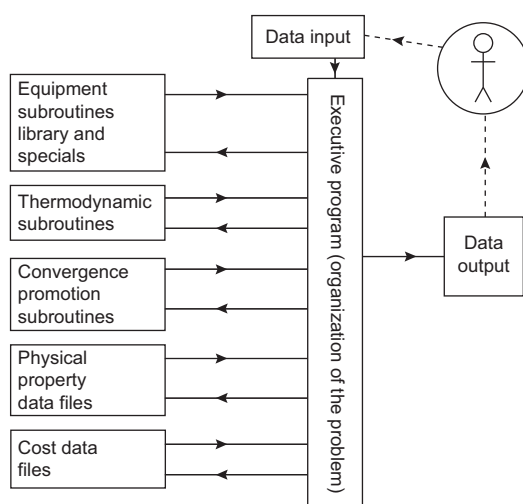


FIG. 4.1 A typical simulation program.

In a sequential-modular program, the executive program sets up the flowsheet sequence, identifies the recycle loops, and controls the unit operation calculations, while interacting with the unit operations library, physical property data bank, and the other subroutines. The executive program also contains procedures for the optimum ordering of the calculations and routines to promote convergence.

In an equation-oriented simulator, the executive program sets up the flowsheet and the set of equations that describe the unit operations and then solves the equations using data from the unit operations library and the physical property data bank and calling on the file of thermodynamics subroutines.

All process simulators use graphical user interfaces to display the flowsheet and facilitate the input of information to the package. The entry of data is usually intuitive to anyone familiar with the Microsoft Windows operating systems.

4.3 Specification of components

The first step in building a process simulation is usually establishing the chemical basis for the model. This consists of choosing the components that will be included in the mass balance and deciding which models to use for the prediction of physical properties and phase equilibrium. This section focuses on the selection of suitable components; the selection of physical property models is discussed in Section 4.4.

4.3.1 Pure components

Each of the commercial process simulation programs contains a large data bank of pure component compounds. Most of the pure components are organic compounds, but inorganic compounds and electrolytes are also included.

The fact that a pure component is listed in a simulator data bank does not guarantee that any of the properties given for that component are based on measured data. If the properties of a compound are critical to process performance, then the scientific literature should be consulted to confirm that the values used in the simulation are realistic.

The most important decision when building a pure component model is choosing the right number of components. The design engineer needs to consider carefully which components will have a significant impact on process design, operation, and economics. If too few components are used, the model will be inadequate for process design, as it will not correctly predict the performance of reactors and separation equipment. Conversely, if too many components are used, the model can become difficult to converge, particularly if there are multiple recycles in the design.

Some guidelines to keep in mind when building a component list include:

1. Always include any component that has a specified limit in any of the products if that component is present in any of the feeds or could be formed in the process. This is critical to determining whether the separations are meeting product specifications.
2. Always include any component that has a specified limit in any of the feeds. These components can be a source of by-products or can act as catalyst or enzyme inhibitors. They must be tracked to ensure that they do not accumulate in the process or make it difficult to meet product specifications. In some cases, an additional separation may be needed to remove a feed contaminant.
3. Always include components that are expected to be formed in side reactions or consecutive reactions. It is important to understand where these components will accumulate or leave the process, even if their yield is not yet known.
4. Always include any compounds that are expected to be present and are known to have significant health, safety, or environmental concerns, such as compounds with high toxicity or explosivity, known carcinogens, or listed hazardous air pollutants (see [Chapter 10](#)). These compounds must be tracked to make sure that they do not reach unsafe levels in any stream and to understand where they might be released to the environment.
5. Usually include any compound that might be present at a mass or mole fraction greater than 2% in any stream in the process.
6. Do not include isomers unless the process specifically requires distinction between isomers (for example, if the process is selective for one isomer, gives different products for different isomers, or is designed to separate isomers). Considering all of the possible isomers of organic compounds becomes combinatorically explosive at high carbon numbers. For fuels and bulk petrochemical processes that are carried out at relatively high temperatures, it is often reasonable to assume an equilibrium distribution of isomers. For fine chemical and

pharmaceutical processes, it is usually important to track isomers separately, particularly enantiomers, as the desired product is often only one of the isomers.

In general, pure component models solve more efficiently with fewer than about 40 components. If the number of components becomes too large and there are many recycles, it may be necessary to build two models. The first is a high-level model that contains only the main bulk components. This model is then used to initialize a second, more detailed model that has the full component list.

4.3.2 Pseudocomponents

Pseudocomponents (hypocomponents) are components created by the simulator to match the boiling curves of petroleum mixtures.

Crude oil; fuels such as gasoline, kerosene, and diesel; and most intermediate streams in an oil refinery consist of many different hydrocarbon compounds. The number of possible hydrocarbon isomers present depends on the carbon number, and both increase with boiling range. For diesel, crude oil, and heavy fuel oils, the number of possible compounds can be from 10^4 to $>10^6$. At the time of writing, there is no analytical method that can uniquely identify all of these compounds, so it would be impossible to include them all in a model even if the resulting model could be solved. Instead, a large number of possible compounds with boiling points in a given range are “lumped” together and represented by a single pseudocomponent with a boiling point in the middle of that range. A set of 10 to 30 pseudocomponents can then be fitted to any petroleum assay and used to model that oil.

Pseudocomponent models are useful for oil fractionation and blending problems. They can also be used to characterize heavy products in some chemical processes, such as ethane cracking. Pseudocomponents are treated as inert in most of the reactor models, but they can be converted or produced in yield shift reactors (see Section 4.5.1).

Some of the commercial simulation programs use a standard default set of pseudocomponents and fit the composition of each to match a boiling curve of the oil that is entered by the user. This can sometimes lead to errors when predicting ASTM D86 or D2887 curves for products from a feed that has been defined based on a true boiling point (TBP) curve or when making many subcuts or cuts with tight distillation specifications. It is often better to work back from the product distillation curves and add extra pseudocomponents around the cut points to make sure that the recoveries and 5% and 95% points on the product distillation curves are predicted properly. All of the simulators have the option to add pseudocomponents to the default set or use a user-generated curve.

4.3.3 Solids and salts

Most chemical and pharmaceutical processes involve some degree of solids handling. Examples of solids that must be modeled include:

- Components that are crystallized for separation, recovery, or purification
- Pharmaceutical products that are manufactured as powders or tablets
- Insoluble salts formed by the reaction of acids and bases or other electrolytes
- Hydrates, ice, and solid carbon dioxide that can form in cryogenic processes
- Cells, bacteria, and immobilized enzymes in biological processes
- Pellets or crystals of polymer formed in polymerization processes
- Coal and ash particles in power generation
- Catalyst pellets in processes in which the catalyst is fluidized or transported as a slurry
- Mineral salts and ores that are used as process feeds
- Fertilizer products
- Fibers in paper processing

Some solid phase components can be characterized as pure components and can interact with other components in the model through phase and reaction equilibrium. Others, such as cells and catalysts, are unlikely to equilibrate with other components, although they can play a vital role in the process.

In Aspen Plus, solid components are identified as different types. Pure materials with measurable properties such as molecular weight, vapor pressure, and critical temperature and pressure are known as *conventional solids* and are present in the MIXED substream with other pure components. They can participate in any of the phase or reaction equilibria specified in any unit operation. If the solid phase participates only in reaction equilibrium but not in phase equilibrium (for example, when the solubility in the fluid phase is known to be very low), then it is called a

conventional inert solid and is listed in a substream CISOLID. If a solid is not involved in either phase or reaction equilibrium, then it is a nonconventional solid and is assigned to substream NC. Nonconventional solids are defined by attributes rather than molecular properties and can be used for coal, cells, catalysts, bacteria, wood pulp, and other multicomponent solid materials.

In UniSim Design, nonconventional solids can be defined as hypothetical components (see Section 4.3.4). The solid phases of pure components are predicted in the phase and reaction equilibrium calculations and do not need to be identified separately.

Many solids-handling operations have an effect on the particle size distribution (PSD) of the solid phase. The PSD can also be an important product property. Aspen Plus allows the user to enter a PSD as an attribute of a solid substream. In UniSim Design, the PSD is entered on the “PSD Property” tab, which appears under “worksheet” on the stream editor window for any stream that contains a pure or hypothetical solid component. Unit operations such as yield shift reactor, crusher, screen, cyclone, electrostatic precipitator, and crystallizer can then be set up to modify the PSD, typically by using a conversion function or a particle capture efficiency in each size range.

When inorganic solids and water are present, an electrolyte phase equilibrium model must be selected for the aqueous phase to properly account for the dissolution of the solid and formation of ions in solution.

4.3.4 User components

The process simulators were originally developed for petrochemical and fuels applications; consequently, many molecules that are made in specialty chemical and pharmaceutical processes are not listed in the component data banks. All of the simulators allow the designer to overcome this drawback by adding new molecules to customize the data bank.

In UniSim Design, new molecules are added as hypothetical components. The minimum information needed to create a new hypothetical pure component is the normal boiling point, although the user is encouraged to provide as much information as is available. If the boiling point is unknown, then the molecular weight and density are used instead. The input information is used to tune the UNIFAC correlation to predict the physical and phase equilibrium properties of the molecule, as described in Section 4.4.

User-defined components are created in Aspen Plus using a “user-defined component wizard.” The minimum required information is the molecular weight and normal boiling point. The program also allows the designer to enter molecular structure, specific gravity, enthalpy and Gibbs energy of formation, ideal gas heat capacity, and Antoine vapor pressure coefficients, but for complex molecules usually only the molecular structure is known.

It is often necessary to add user components to complete a simulation model. The design engineer should always be cautious when interpreting simulation results for models that include user components. Phase equilibrium predictions for flashes, decanters, extraction, distillation, and crystallization operations should be carefully checked against laboratory data to ensure that the model is correctly predicting the component distribution between the phases. If the fit is poor, the binary interaction parameters in the phase equilibrium model can be tuned to improve the prediction.

4.4 Selection of physical property models

The process simulation programs all contain subroutines for calculating component and stream physical properties and for determining phase equilibrium in process operations. The user must select a thermodynamic model that provides a sufficiently accurate representation of the system for design purposes. When the design is sensitive to the choice of thermodynamic model, the models should be checked against measured data and the most accurate model selected. In some cases, it may be necessary to tune the library models in the simulator by adjusting some of the parameters to provide a better fit to the data.

4.4.1 Sources of physical property data

It is always a good practice to benchmark the physical properties predicted using a process simulation program against measured data. There are many good literature sources of data for properties of individual compounds, but much less data are available for mixtures. Caution should be exercised when taking data from the literature, as

typographical errors often occur. If a value looks suspicious, it should be cross-checked in an independent reference or by estimation.

The values of some properties are dependent on the method of measurement—for example, surface tension and flash point—and the method used should be checked by reference to the original paper, if necessary, if an accurate value is required.

International Critical Tables (ICT) (Washburn, 1933) is still probably the most comprehensive compilation of physical properties and is available in most reference libraries. Though it was first published in 1933, physical properties do not change, except in as much as experimental techniques improve, and ICT is still a useful source of engineering data. ICT is now available as an e-book and can be referenced on the Internet through Knovel (2003).

Tables and graphs of physical properties are given in many handbooks and textbooks on chemical engineering and related subjects. Many of the data given are duplicated from book to book, but the various handbooks do provide quick, easy access to data on the more commonly used substances.

An extensive compilation of thermophysical data has been published by Plenum Press (Touloukian, 1970–1977). This multiple-volume work covers conductivity, specific heat, thermal expansion, viscosity, and radiative properties (emittance, reflectance, absorptance, and transmittance).

The Engineering Sciences Data Unit (ESDU, www.ihsesdu.com) was set up to provide validated data for engineering design, developed under the guidance and approval of engineers from industry, the universities, and research laboratories. ESDU data include equipment design data and extensive high-quality physical property data—mostly for pure fluids that are in use in the oil and process industries.

The results of research work on physical properties are reported in the general engineering and scientific literature. The *Journal of Chemical Engineering Data* specializes in publishing physical property data for use in chemical engineering design. A quick search of the literature for data can be made by using the abstracting journals, such as *Chemical Abstracts* (American Chemical Society) and *Engineering Index* (Engineering Index, Inc.). *Engineering Index* is now called Engineering Information (Ei) and is a web-based reference source owned by Elsevier information (www.ei.org). *Chemical Abstracts* can be searched using the ACS SciFinder service.

Experimental phase equilibrium data have been published for several thousand binary and many multicomponent systems. Virtually all the published experimental data have been collected together in the volumes comprising the DECHEMA vapor–liquid and liquid–liquid data collection (DECHEMA, 1977). The books by Chu et al. (1956), Hala et al. (1968, 1973), Hirata et al. (1975), and Ohe (1989, 1990) are also useful sources.

Computerized physical property data banks have been set up by various organizations to provide a service to the design engineer. They can be incorporated into computer-aided design programs and are increasingly being used to provide reliable, authenticated design data. Examples of such programs are the Physical Property Data Service (PPDS) and the The Design Institute for Physical Properties (DIPPR) databases.

PPDS was originally developed in the United Kingdom by the Institution of Chemical Engineers and the National Physical Laboratory. It is now available as a Microsoft Windows version from NEL, a division of the TUV Sudddeutschland Group (www.tuvnel.com/content/ppds.aspx). PPDS is made available to universities at a discount.

The DIPPR databases were developed by the Design Institute for Physical Properties of the AIChE. The DIPPR projects are aimed at providing evaluated process design data for the design of chemical processes and equipment (www.aiche.org/TechnicalSocieties/DIPPR/index.aspx). The DIPPR Project 801 has been made available to university departments; see Rowley et al. (2004).

Many of the important sources of engineering information are subscription services. The American Chemical Society's Chemical Abstracts Service is the best source for chemical properties and reaction kinetics data. Chemical abstracts can be searched online through the SciFinder subscription service (www.cas.org). This is available in most university libraries.

Another important source of information is Knovel. Knovel provides online access to most standard reference books. It is a subscription service but can be accessed through many libraries, including those of the professional engineering institutions and most universities. At the time of writing, Knovel is available for free to members of the AIChE. In addition to having many reference books in .pdf format, Knovel has interactive graphs and look-up tables for books such as *Perry's Chemical Engineers Handbook* and the *International Critical Tables*.

4.4.2 Prediction of physical properties

The process simulation programs contain subroutines that predict the physical properties of pure compounds and mixtures as functions of temperature, pressure, and composition. The algorithms used have been developed based on decades of research in thermodynamics and property estimation. Techniques are available for the prediction of

most physical properties, with sufficient accuracy for use in process and equipment design; however, the accuracy of the predictions should always be assessed by comparing the model output with data from experiments, pilot plants, or operating units. A detailed review of all the different methods available is beyond the scope of this book. If accurate values are required, then specialized texts on physical property estimation should be consulted, such as those by Reid et al. (1987), Poling et al. (2000), Bretsznajder (1971), Sterbacek et al. (1979), and AIChE (1983, 1985), and the data should be confirmed experimentally.

The techniques used for prediction are also useful for the correlation, and extrapolation and interpolation, of experimental values. The two most common approaches used in predicting properties are group contribution methods and the use of reduced properties.

Group contribution methods

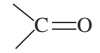
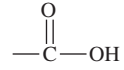
Group contribution techniques are based on the concept that a particular physical property of a compound can be considered to be made up of contributions from the constituent atoms, groups, and bonds, with the contributions being determined from experimental data. They provide the designer with simple, convenient methods for physical property estimation, requiring only a knowledge of the structural formula of the compound.

Group contribution methods are used to predict a wide range of physical properties when no data are available for regression. For example, the group contribution method proposed by Chueh and Swanson (1973a, 1973b) gives reasonably accurate predictions of specific heat capacity for organic liquids. The contributions to be assigned to each molecular group are given in Table 4.2, and the method is illustrated in Example 4.1. The most widely used group contribution model is the UNIFAC method for predicting the parameters for phase equilibrium models.

TABLE 4.2 Group contributions for liquid heat capacities at 20 °C, kJ/kmol°C

Group	Value	Group	Value
Alkane		$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{O}- \end{array}$	60.71
$-\text{CH}_3$	36.84	$-\text{CH}_2\text{OH}$	73.27
$-\text{CH}_2-$	30.40	$\begin{array}{c} \\ -\text{CH}- \end{array}$	76.20
$\begin{array}{c} \\ -\text{CHOH} \end{array}$	20.93	$\begin{array}{c} \\ -\text{C}- \\ \end{array}$	111.37
$\begin{array}{c} \\ -\text{COH} \\ \end{array}$	7.37	$-\text{OH}$	44.80
Olefin		$-\text{ONO}_2$	119.32
$=\text{CH}_2$	21.77	Halogen	36.01
$\begin{array}{c} \\ =\text{C}-\text{H} \end{array}$	21.35	$-\text{Cl}$ (first or second on a carbon)	25.12
$\begin{array}{c} \\ =\text{C}- \end{array}$	15.91	$-\text{Cl}$ (third or fourth on a carbon)	
Alkyne		$-\text{Br}$	37.68
$-\text{C}\equiv\text{H}$	24.70	$-\text{F}$	16.75
$-\text{C}\equiv$	24.70	$-\text{I}$	36.01
In a ring		Nitrogen	
$\begin{array}{c} \\ -\text{CH}= \end{array}$	18.42	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{N}- \end{array}$	58.62
$\begin{array}{c} \\ -\text{C}= \text{ or } -\text{C}- \\ \end{array}$	12.14	$\begin{array}{c} \text{H} \\ \\ -\text{N}- \end{array}$	43.96
$-\text{C}=$	22.19	$\begin{array}{c} \\ -\text{N}- \end{array}$	31.40
$-\text{CH}_2-$	25.96	$-\text{N}=(\text{in a ring})$	18.84

TABLE 4.2 Group contributions for liquid heat capacities at 20 °C, kJ/kmol°C—cont'd

Group	Value	Group	Value
Oxygen		—C≡N	58.70
—O—	35.17	Sulphur	
	53.00	—SH	44.80
—C—O H	53.00	—S—	33.49
	79.97	Hydrogen	
		H— (for formic acid, formates, hydrogen cyanide, etc.)	14.65

Add 18.84 for any carbon group that fulfils the following criterion: a carbon group that is joined by a single bond to a carbon group connected by a double or triple bond with a third carbon group. In some cases a carbon group fulfils this criterion in more ways than one; 18.84 should be added each time the group fulfils the criterion. Exceptions to the 18.84 rule:

1. No such extra 18.84 additions for —CH₃ groups.
2. For a —CH₂— group fulfilling the 18.84 addition criterion, add 10.47 instead of 18.84. However, when the —CH₂— group fulfils the addition criterion in more ways than one, the addition should be 10.47 the first time and 18.84 for each subsequent addition.
3. No such extra addition for any carbon group in a ring.

Source: Chueh and Swanson (1973a, 1973b).

Example 4.1

Using Chueh and Swanson's method, estimate the specific heat capacity of ethyl bromide at 20 °C.

Solution

Ethyl bromide CH₃CH₂Br

Group	Contribution	No. of	
—CH ₃	36.84	1	= 36.84
—CH ₂ —	30.40	1	= 30.40
—Br	37.68	1	= 37.68
		Total	104.92 kJ/kmol°C

mol. wt. = 109

$$\text{Specific heat capacity} = \frac{104.92}{109} = \underline{\underline{0.96 \text{ kJ/kg}^\circ\text{C}}}$$

Experimental value 0.90 kJ/kg°C

Reduced properties

Reduced property models (also known as *method of corresponding states models*) predict properties based on knowledge of the critical conditions of a compound. They are useful if values for the critical properties are available or can be estimated with sufficient accuracy; see Sterbacek et al. (1979). An example of a reduced property model is the method for estimating latent heat of vaporization developed by Haggenmacher (1946), which is derived from the Antoine vapor pressure equation:

$$L_v = \frac{8.32 BT^2 \Delta z}{+(TC)^2} \quad (4.1)$$

where L_v = latent heat at the required temperature, kJ/kmol

T = temperature, K

B, C = coefficients in the Antoine equation (Antoine, 1888):

$$\ln P = A - \frac{B}{T + C} \quad (4.2)$$

where P = vapor pressure, mmHg

A, B, C = the Antoine coefficients

T = temperature, K

and

$\Delta z = z_{\text{gas}} - z_{\text{liquid}}$ (where z is the compressibility constant), calculated from the equation:

$$\Delta z = \left[1 - \frac{P_r}{T_r^3} \right]^{0.5} \quad (4.3)$$

P_r = reduced pressure = P/P_c

T_r = reduced temperature = T/T_c

Example 4.2

Estimate the latent heat of vaporization of acetic anhydride, $\text{C}_4\text{H}_6\text{O}_3$, at its boiling point, 139.6°C (412.7 K) and at 200°C (473 K).

Solution

For acetic anhydride $T_c = 569.1\text{ K}$, $P_c = 46\text{ bar}$

Antoine constants $A = 16.3982$

$B = 3287.56$

$C = -75.11$

Experimental value at the boiling point 41,242 kJ/kmol.

From the Haggemacher equation:

$$\text{at the b.p. } P_r = \frac{1}{46} = 0.02124$$

$$T_r = \frac{412.7}{569.1} = 0.7252$$

$$\Delta z = \left[1 - \frac{0.02124}{0.7252^3} \right]^{0.5} = 0.972$$

$$L_{v,b} = \frac{8.32 \times 3287.6 \times (412.7)^2 \times 0.972}{(412.7 - 75.11)^2} = \underline{\underline{39,733\text{ kJ/mol}}}$$

At 200°C , the vapor pressure must first be estimated from the Antoine equation:

$$\ln P = A - \frac{B}{T + C}$$

$$\ln P = 16.3982 - \frac{3287.56}{473 - 75.11} = 8.14$$

$$P = 3421.35\text{ mmHg} = 4.5\text{ bar}$$

$$P_r = \frac{4.5}{46} = 0.098$$

$$T_r = \frac{473}{569.1} = 0.831$$

$$\Delta z = \left[1 - \frac{0.098}{0.831^3} \right]^{0.5} = 0.911$$

$$L_v = \frac{8.32 \times 3287.6 \times (473)^2 \times 0.911}{(473 - 75.11)^2} = \underline{\underline{35,211\text{ kJ/kmol}}}$$

If reliable experimental values of the critical constants cannot be found, techniques are available for estimating the critical constants with sufficient accuracy for most design purposes. For organic compounds Lydersen's method is normally used (Lydersen, 1955):

$$T_c = \frac{T_b}{\left[0.567 + \Sigma \Delta T - (\Sigma \Delta T)^2\right]} \quad (4.4)$$

$$P_c = \frac{M}{(0.34 + \Sigma \Delta P)^2} \quad (4.5)$$

$$V_c = 0.04 + \Sigma \Delta V \quad (4.6)$$

where T_c = critical temperature, K

P_c = critical pressure, atm (1.0133 bar)

V_c = molar volume at the critical conditions, m³/kmol

T_b = normal boiling point, K

M = relative molecular mass

ΔT = critical temperature increments, Table 4.3

ΔP = critical pressure increments, Table 4.3

ΔV = molar volume increments, Table 4.3

TABLE 4.3 Critical constant increments

	ΔT	ΔP	ΔV		ΔT	ΔP	ΔV
<i>Nonring increments</i>							
—CH ₃	0.020	0.227	0.055	$\begin{array}{c} \\ \text{—C—} \end{array}$	0.0	0.198	0.036
$\begin{array}{c} \\ \text{—CH}_2 \end{array}$	0.020	0.227	0.055	=C=	0.0	0.198	0.036
$\begin{array}{c} \\ \text{—CH} \\ \end{array}$	0.012	0.210	0.051	$\equiv \text{CH}$	0.005	0.153	0.036*
$\begin{array}{c} \\ \text{—C—} \\ \end{array}$	0.00	0.210	0.041	$\equiv \text{C—}$	0.005	0.153	0.036*
=CH_2	0.018	0.198	0.045	H	0	0	0
$\begin{array}{c} \\ \text{=CH} \end{array}$	0.018	0.198	0.045				
<i>Ring increments</i>							
—CH ₂ —	0.013	0.184	0.0445	$\begin{array}{c} \\ \text{=CH} \end{array}$	0.011	0.154	0.037
$\begin{array}{c} \\ \text{—CH} \\ \end{array}$	0.012	0.192	0.046	$\begin{array}{c} \\ \text{=C—} \end{array}$	0.011	0.154	0.036
$\begin{array}{c} \\ \text{—C—} \\ \end{array}$	−0.007*	0.154*	0.031*	=C=	0.011	0.154	0.036
<i>Halogen increments</i>							
—F	0.018	0.224	0.018	—Br	0.010	0.50*	0.070*
—Cl	0.017	0.320	0.049	—I	0.012	0.83*	0.095*

Continued

TABLE 4.3 Critical constant increments—cont'd

	ΔT	ΔP	ΔV		ΔT	ΔP	ΔV
Oxygen increments							
—OH (alcohols)	0.082	0.06	0.018*	$\begin{array}{c} \\ \text{—CO—} \end{array}$ (ring)	0.033*	0.2*	0.050*
—OH (phenols)	0.031	−0.02*	0.030*	$\begin{array}{c} \\ \text{HC=O} \end{array}$ (aldehyde)	0.048	0.33	0.073
—O— (nonring)	0.021	0.16	0.020	—COOH (acid)	0.085	0.4*	0.080
—O— (ring)	0.014*	0.12*	0.080*	—COO— (ester)	0.047	0.47	0.080
$\begin{array}{c} \\ \text{—C=O} \end{array}$ (nonring)	0.040	0.29	0.060	=O (except for combinations noted earlier)	0.02*	0.12*	0.011*
Nitrogen increments							
—NH ₂	0.031	0.095	0.028	$\begin{array}{c} \\ \text{—N—} \end{array}$ (ring)	0.007*	0.013*	0.032*
$\begin{array}{c} \\ \text{—NH} \end{array}$ (nonring)	0.031	0.135	0.037*	—CN	0.060*	0.36*	0.080*
$\begin{array}{c} \\ \text{—NH} \end{array}$ (ring)	0.024*	0.09*	0.027*	—NO ₂	0.055*	0.42*	0.078*
$\begin{array}{c} \\ \text{—N—} \end{array}$ (nonring)	0.014	0.17	0.042*				
Sulfur increments							
—SH	0.015	0.27	0.055	—S— (ring)	0.008*	0.24*	0.045*
—S— (nonring)	0.015	0.27	0.055	S	0.003*	0.24*	0.047*
Miscellaneous							
$\begin{array}{c} \\ \text{—Si—} \\ \end{array}$	0.03	0.54*		$\begin{array}{c} \\ \text{—B—} \\ \end{array}$	0.03*		

Dashes represent bonds with atoms other than hydrogen.

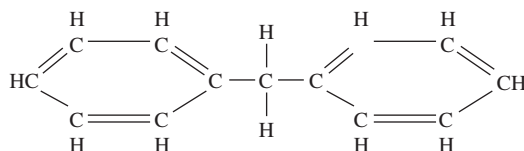
Values marked with an asterisk are based on too few experimental points to be reliable.

Source: [Lydersen \(1955\)](#).

Lydersen's method illustrates how process simulation programs can predict the properties of user-specified components using just a molecular structure and a boiling point. Application of Lydersen's method generates critical constants, which can then be used in reduced parameter models to generate other properties. Although the final values that are predicted may still be suitable for preliminary design purposes, inaccuracy is introduced and propagated at each stage of such calculations, and the predictions should be confirmed against experimental values before detailed design.

Example 4.3

Estimate the critical constants for diphenylmethane using Lydersen's method: normal boiling point 537.5 K, molecular mass 168.2, structural formula:



Solution

Group	No. of	Total contribution		
		ΔT	ΔP	ΔV
$\begin{array}{c} \parallel \\ \text{H}-\text{C}-(\text{ring}) \end{array}$	10	0.11	1.54	0.37
$\begin{array}{c} \\ =\text{C}-(\text{ring}) \end{array}$	2	0.022	0.308	0.072
$-\text{CH}_2-$	1	0.02	0.227	0.055
		$\Sigma 0.152$	2.075	0.497

$$T_c = \frac{537.5}{(0.567 + 0.152 - 0.152^2)} = \underline{\underline{772 \text{ K}}}$$

experimental value 767 K,

$$P_c = \frac{168.2}{(0.34 + 2.075)^2} = \underline{\underline{28.8 \text{ atm}}}$$

experimental value 28.2 atm,

$$V_c = 0.04 + 0.497 = \underline{\underline{0.537 \text{ m}^3/\text{kmol}}}$$

4.4.3 Phase equilibrium models

The choice of the best method for deducing vapor–liquid and liquid–liquid equilibria for a given system will depend on three factors:

1. The composition of the mixture (the system chemistry)
2. The operating pressure (low, medium, or high)
3. The experimental data available

The criterion for thermodynamic equilibrium between two phases of a multicomponent mixture is that for every component, i :

$$f_i^v = f_i^L \quad (4.7)$$

where f_i^v is the vapor phase fugacity and f_i^L the liquid phase fugacity of component i :

$$f_i^v = P\phi_i y_i \quad (4.8)$$

and

$$f_i^L = f_i^{OL} \gamma_i x_i \quad (4.9)$$

where P = total system pressure

ϕ_i = vapor fugacity coefficient

y_i = concentration of component i in the vapor phase

f_i^{OL} = standard-state fugacity of the pure liquid

γ_i = liquid phase activity coefficient

x_i = concentration of component i in the liquid phase

Substitution from Equations 4.8 and 4.9 into Equation 4.7 and rearranging gives:

$$K_i = \frac{y_i}{x_i} = \frac{\gamma_i f_i^{OL}}{P\phi_i} \quad (4.10)$$

where K_i is the distribution coefficient (the K value)

ϕ_i can be calculated from an appropriate equation of state

f_i^{OL} can be computed from the following expression:

$$f_i^{OL} = P_i^o \phi_i^s \left\{ \exp \left[\frac{(P - P_i^o)}{RT} v_i^L \right] \right\} \quad (4.11)$$

where P_i^o = the pure component vapor pressure (which can be calculated from the Antoine equation, Equation 4.2), N/m²

ϕ_i^s = the fugacity coefficient of the pure component i at saturation

v_i^L = the liquid molar volume, m³/mol

The exponential term in Equation 4.11 is known as the *Poynting correction* and corrects for the effect of pressure on the liquid phase fugacity.

ϕ_i^s is calculated using the same equation of state used to calculate ϕ_i .

For systems in which the vapor phase imperfections are not significant, Equation 4.10 reduces to the familiar Raoult's law equation:

$$K_i = \frac{\gamma_i P_i^o}{P} \quad (4.12)$$

Vapor phase nonideality is usually modeled using an equation of state. An equation of state is a model for the molar volume of a real gas or liquid as a function of temperature and pressure. The features and limitations of the most commonly used equations of state are given in Table 4.4. For low-pressure systems with no known chemical interactions in the vapor phase, it is often acceptable to assume ideal gas behavior. For details of the equations, the reader should consult the reference cited or the books by Reid et al. (1987), Prausnitz et al. (1998), and Walas (1985). To select the best equation to use for a particular process design, refer to Fig. 4.2.

Liquid phase nonideality is encountered much more often than vapor phase nonideality and is modeled using activity coefficient models. The most frequently used activity coefficient models are the Wilson, NRTL, and UNIQUAC models summarized in Table 4.5. The simpler models that are taught in undergraduate thermodynamics classes are rarely adequate for design purposes.

TABLE 4.4 Equations of state

Model	Features	References
Redlich–Kwong Equation (R-K)	Redlich and Kwong (1949)	Extension of Van der Waal's equation, where constants are calculated from critical pressure and temperature. Not suitable for use near the critical pressure ($P_r > 0.8$) or for liquids.
Redlich–Kwong–Soave Equation (R-K-S)	Soave (1972)	Modification to the R-K equation to extend its usefulness to the critical region and for use with liquids.
Benedict–Webb–Rubin Equation (B-W-R)	Benedict et al. (1951)	An eight-parameter empirical model that gives accurate predictions for vapor and liquid phase hydrocarbons. It can also be used for mixtures of light hydrocarbons with carbon dioxide and water.
Lee–Kesler–Plocker Equation (L-K-P)	Lee and Kesler (1975) Plocker et al. (1978)	Lee and Kesler extended the B-W-R equation to a wider variety of substances using the principle of corresponding states. The method was modified further by Plocker et al.
Chao–Seader Equation (C-S)	Chao and Seader (1961)	Gives accurate predictions for hydrogen and light hydrocarbons, but limited to temperatures below 530 K.
Grayson–Streed Equation (G-S)	Grayson and Streed (1963)	Extended the C-S equation for use with hydrogen-rich mixtures and for high-pressure and high-temperature systems. It can be used up to 200 bar and 4700 K.
Peng–Robinson Equation (P-R)	Peng and Robinson (1976)	Extension of the R-K-S equation to overcome instability in the R-K-S equation near the critical point.

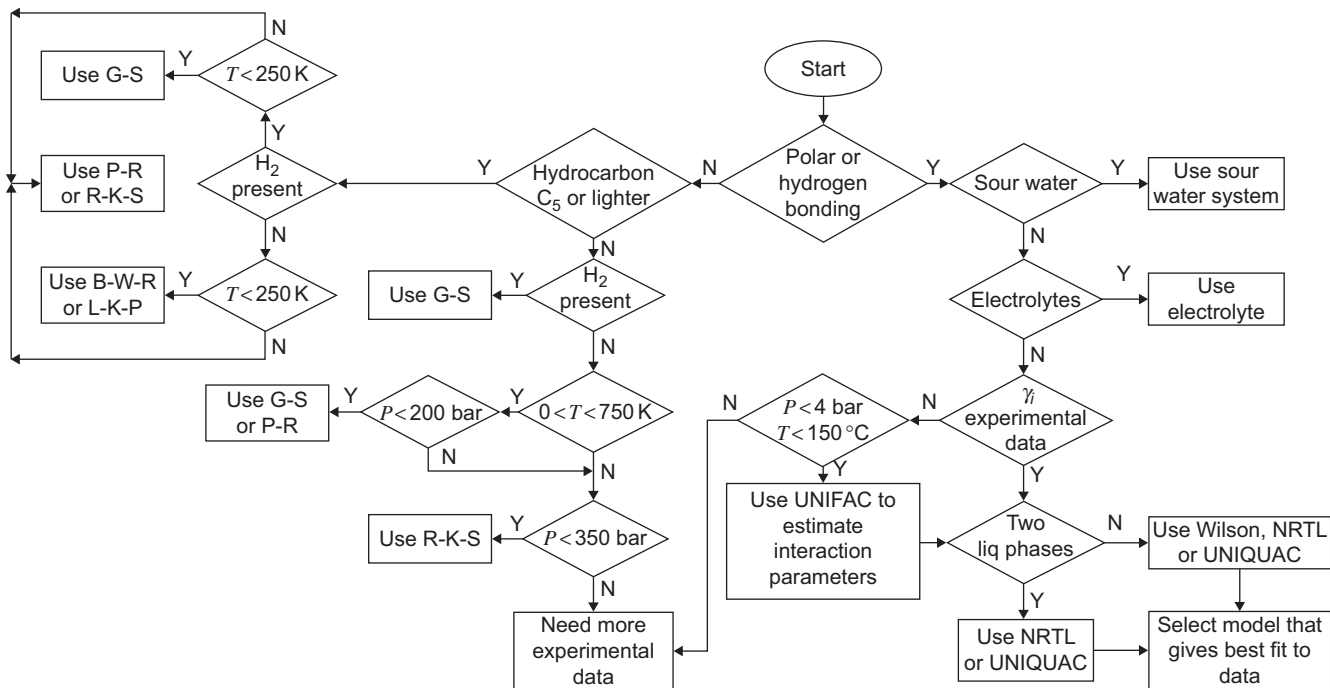


FIG. 4.2 Flow chart for the selection of phase equilibrium model.

TABLE 4.5 Activity coefficient models

Model	Features	References
Wilson Equation	Uses two adjustable parameters to model binary interactions between molecules. Can be extended to multicomponent systems using only binary parameters. Cannot predict formation of a second liquid phase.	Wilson (1964)
NRTL (Nonrandom Two-Liquid) Equation	Uses three parameters for each binary pair, where two are energies of interaction (similar to the Wilson parameters) and the third is a randomness factor that characterizes the tendency of molecules i and j to be distributed randomly in the mixture. Can predict liquid–liquid or vapor–liquid equilibrium.	Renon and Prausnitz (1969)
UNIQUAC (Universal Quasi-Chemical) Equation	Mathematically more complex than NRTL, but uses fewer adjustable parameters. Can predict liquid–liquid and vapor–liquid equilibrium. In the absence of experimental data, the parameters can be predicted by the UNIFAC method. Probably the most widely used model.	Abrams and Prausnitz (1975) Anderson and Prausnitz (1978a)

Activity coefficient models generally give good prediction of liquid phase fugacity for binary mixtures and can be extended to multicomponent mixtures if all the binary interaction parameters are known. The models become less reliable as the number of components increases, and the accuracy can be improved by fitting some data from ternary or higher-order mixtures.

The selection of the most appropriate liquid phase activity coefficient model for a given design is discussed in Section 4.4.5 and illustrated in Fig. 4.2.

The liquid phase activity coefficient, γ_i , is a function of pressure, temperature, and liquid composition. At conditions remote from the critical conditions, it is virtually independent of pressure and, in the range of temperature normally encountered in distillation, can be taken as independent of temperature. For a detailed discussion of the equations for activity coefficients and their relative merits, the reader is referred to the books by Reid et al. (1987), Prausnitz et al. (1998), Walas (1985), and Null (1970).

Most of the commercial process simulation programs contain subroutines that allow the user to enter phase-equilibrium data and perform a localized regression to better tune the binary interaction parameters in any of the

activity coefficient models. The binary interaction parameters are not unique constants, and locally adjusted parameters will provide more accurate prediction of phase equilibrium for a given design problem. Details of how to fit phase equilibrium data are given in the simulation program manuals.

4.4.4 Prediction of phase equilibrium constants

The designer will often be confronted with the problem of how to proceed with the design of a separation process without adequate experimentally determined equilibrium data. Some techniques are available for the prediction of vapor–liquid equilibrium (VLE) data and for the extrapolation of experimental values. The process simulation programs include libraries of measured data and interaction parameters for mixtures, as well as predictive methods. Caution must be used in the application of these techniques in design, and the predictions should be confirmed against experimentally determined values whenever practicable.

Group contribution methods

Group contribution methods have been developed for the prediction of liquid phase activity coefficients. The objective has been to enable the prediction of phase equilibrium data for the tens of thousands of possible mixtures of interest to the process designer to be made from the contributions of the relatively few functional groups that made up the compounds. The UNIFAC method—see [Fredenslund, Gmehling, and Michelsen et al. \(1977\)](#)—is probably the most useful for process design. Its use is described in detail in a book by [Fredenslund, Gmehling, and Rasmussen et al. \(1977\)](#). A method was also developed to predict the parameters required for the NRTL equation: the ASOG method—see [Kojima and Tochigi \(1979\)](#). More extensive work has been done to develop the UNIFAC method, to include a wider range of functional groups; see [Gmehling et al. \(1982\)](#) and [Magnussen et al. \(1981\)](#). The UNIFAC method can be used to estimate binary interaction parameters for the UNIQUAC model and, by extension, the NRTL and Wilson models.

Care must be exercised in applying the UNIFAC method. The specific limitations of the method are:

1. Pressure not greater than a few bar (say, limit to 5 bar)
2. Temperature below 150 °C
3. No noncondensable components or electrolytes
4. Components must not contain more than 10 functional groups

Sour water systems

The term *sour water* is used for water containing carbon dioxide, hydrogen sulfide, and ammonia encountered in refinery operations. Special correlations have been developed to handle the VLE of such systems, and these are incorporated in most design and simulation programs. [Newman \(1991\)](#) gives the equilibrium data required for the design of sour water systems as charts.

Electrolyte systems

When water and salts are present in a mixture, the salts can dissociate into ions in aqueous solution. The phase equilibrium model must account for dissociation and the presence of long-range interactions between charges on ions, as well as vapor–liquid or liquid–liquid equilibrium. Special electrolyte models and databases such as the OLI model have been developed for electrolyte systems. These models are available in the commercial process simulation programs, but sometimes require an additional fee.

Vapor–liquid equilibrium at high pressures

At pressures above a few atmospheres, the deviations from ideal behavior in the gas phase will be significant and must be taken into account in process design by use of a suitable equation of state for the vapor phase. The effect of pressure on the liquid phase activity coefficient must also be considered. A discussion of the methods used to correlate and estimate VLE data at high pressures is beyond the scope of this book. The reader should refer to the texts by [Null \(1970\)](#), [Prausnitz et al. \(1998\)](#), or [Prausnitz and Chueh \(1968\)](#).

Prausnitz and Chueh also discuss phase equilibrium in systems containing components above their critical temperature (supercritical components).

Liquid–liquid equilibrium

Experimental data, or predictions, that give the distribution of components between the two solvent phases are needed for the design of liquid–liquid extraction processes, and mutual solubility limits are needed for the design of decanters and other liquid–liquid separators.

Green and Perry (2007) give a useful summary of solubility data. Liquid–liquid equilibrium (LLE) compositions can be predicted from VLE data, but the predictions are rarely accurate enough for use in the design of liquid–liquid extraction processes. The DECHEMA data collection includes LLE data for several hundred mixtures (DECHEMA, 1977).

The UNIQUAC equation can be used to estimate activity coefficients and liquid compositions for multicomponent liquid–liquid systems. The UNIFAC method can be used to estimate UNIQUAC parameters when experimental data are not available. Some process simulation programs require the user to enable three-phase calculation or switch from a VLE mode to a VLLE mode when solving liquid–liquid equilibrium calculations.

It must be emphasized that extreme caution should be exercised when using predicted values for LLE in design calculations.

4.4.5 Choice of phase equilibrium model for design calculations

There is no universal algorithm for the selection of a phase equilibrium model. Although general rules can be given for the applicability of different equations of state, the models for liquid phase activity coefficients are semi-empirical, and it is often impossible to determine *a priori* which will provide the best fit to a set of experimental phase equilibrium data.

The flow chart shown in Fig. 4.2 has been adapted from a similar chart published by Wilcon and White (1986) and can be used as a preliminary guide to model selection. The abbreviations used in the chart for the equations of state and activity coefficient models correspond to those given in Tables 4.4 and 4.5. It must be emphasized that the best activity coefficient model is the model that provides the best fit to the experimental data over the range of interest. If no experimental data are available, then the best model is probably that for which the fewest interaction parameters must be estimated.

If a phase equilibrium model is created using estimated interaction parameters, the designer should highlight this as a source of uncertainty in the design. Before proceeding to detailed design, the design team should ensure that sufficient data are collected to confirm the model, and an expert on thermodynamics should be consulted to give advice on model selection and parameter estimation.

4.4.6 Validation of physical property models

The physical properties and phase equilibrium predicted by a process simulation program should always be validated by comparison with experimental measurements. It is not necessary to compare every parameter predicted by the model with real data, but any parameter that has a significant influence on the design should be confirmed. In some cases, it may also be necessary to confirm the accuracy of a physical property over a range of temperature or pressure.

In a revamp design, model validation is relatively straightforward, though usually not easy. A simulation model of the existing process can be built and tuned to match the current plant performance. Once the model is successfully benchmarked against the plant data, it can be used to evaluate new cases for the proposed design modifications. Although this sounds simple, the effort involved in matching a model to plant data can be considerable. It is often worthwhile to use a few independent laboratory experiments under more controlled conditions to reduce the number of parameters that are adjusted in the plant-based model.

Pilot plants and laboratory experiments can be a good source of data for model validation. When designing a pilot plant, consideration should be given to the need for collecting data to validate phase equilibrium models. Care must be taken to ensure that samples are taken when streams are at steady state and have had time to equilibrate.

If no experimental data are available, then it is usually a good idea to make an independent estimate of any parameters that have a strong influence on the design to be satisfied that the results from the simulator are credible. If the independent estimate does not agree with the simulation result, it may be worthwhile to conduct some experiments to collect real data. Methods for estimating physical properties are given in the book by Poling et al. (2000).

A flash calculation can be used as a simple technique for validating a phase equilibrium model when no data are available. The designer should set up a simulation model of a flash calculation using the temperature, pressure, and composition of interest. This simulation can then be run using different models for liquid and vapor phase nonideality that might be expected to be applicable to the system of interest. If the model predicts essentially the same stream flows and compositions regardless of the thermodynamic models selected, then the models are equally valid. This does not mean that the models are accurate, but at least they give the same results. If the flash calculation gives substantially different stream flows or compositions with different thermodynamic models, the designer should seek more experimental data with which to determine which model is most applicable.

4.5 Simulation of unit operations

A process simulation is built up from a set of unit operation models connected by mass and energy streams. The commercial simulators include many unit operation subroutines, sometimes referred to as *library models*. These operations can be selected from a palette or menu and then connected together using the simulator graphical user interface. Table 4.6 gives a list of the main unit operation models available in Aspen Plus and UniSim Design. Details of how to specify unit operations are given in the simulator manuals. This section provides general advice on unit operations modeling and modeling of nonstandard unit operations.

TABLE 4.6 Unit operation models in Aspen Plus and UniSim Design suites

Unit Operation	Aspen Plus Models	UniSim Design Models
Stream mixing	Mixer	Mixer
Component splitter	Sep, Sep2	Component Splitter
Decanter	Decanter	3-Phase Separator
Flash	Flash2, Flash3	Separator, 3-Phase Separator
Piping components		
Piping	Pipe, Pipeline	Pipe Segment, Compressible Gas Pipe
Valves & fittings	Valve	Valve, Tee, Relief Valve
Hydrocyclone	HyCyc	Hydrocyclone
Reactors		
Conversion reactor	RStoic	Conversion Reactor
Equilibrium reactor	REquil	Equilibrium Reactor
Gibbs reactor	RGibbs	Gibbs Reactor
Yield reactor	RYield	Yield Shift Reactor
CSTR	RCSTR	Continuous Stirred Tank Reactor
Plug flow reactor	RPlug	Plug Flow Reactor
Columns		
Shortcut distillation	DSTWU, Distl, SCFrac	Shortcut Column
Rigorous distillation	RadFrac, MultiFrac	Distillation, 3-Phase Distillation
Liquid-liquid extraction	Extract	Liquid-Liquid Extractor
Absorption and stripping	RadFrac	Absorber, Refluxed Absorber, Reboiled, Absorber
Fractionation	PetroFrac	3 Stripper Crude, 4 Stripper Crude, Vacuum Resid Column, FCCU Main, Fractionator

TABLE 4.6 Unit operation models in Aspen Plus and UniSim Design suites—cont'd

Unit Operation	Aspen Plus Models	UniSim Design Models
Rate-based distillation	RATEFRAC™	
Batch distillation	BatchFrac	
<i>Heat transfer equipment</i>		
Heater or cooler	Heater	Heater, Cooler
Heat exchanger	HeatX, HxFlux, Hetran, HTRI-Xist	Heat Exchanger
Air cooler	Aerotran	Air Cooler
Fired heater	Heater	Fired Heater
Multi-stream exchanger	MheatX	LNG Exchanger
<i>Rotating equipment</i>		
Compressor	Compr, MCompr	Compressor
Turbine	Compr, MCompr	Expander
Pump, hydraulic turbine	Pump	Pump
<i>Solids handling</i>		
Size reduction	Crusher	
Size selection	Screen	Screen
Crystallizer	Crystallizer	Crystallizer, Precipitation
Neutralization		Neutralizer
Solids washing	SWash	
Filter	Fabfl, CFuge, Filter	Rotary Vacuum Filter
Cyclone	HyCyc, Cyclone	Hydrocyclone, Cyclone
Solids decanting	CCD	Simple Solid Separator
Solids transport		Conveyor
Secondary recovery	ESP, Fabfl, VScrub	Baghouse Filter
User models	User, User2, User3	User Unit Op

4.5.1 Reactors

The modeling of real industrial reactors is usually the most difficult step in process simulation. It is usually easy to construct a model that gives a reasonable prediction of the yield of the main product, but the simulator library models are not sophisticated enough to fully capture all the details of hydraulics, mixing, mass transfer, catalyst and enzyme inhibition, cell metabolism, and other effects that often play a critical role in determining the reactor outlet composition, energy consumption, rate of catalyst deactivation, and other important design parameters.

In the early stages of process design, the simulator library models are usually used with simplistic reaction models that give the design engineer a good enough idea of yields and enthalpy changes to allow design of the rest of the process. If the design seems economically attractive, then more detailed models can be built and substituted into the flowsheet. These detailed models are usually built as user models, as described in Section 4.6 and Section 15.11.

Most of the commercial simulation programs have variants on the following reactor models.

Conversion reactor (stoichiometric reactor)

A conversion reactor requires a reaction stoichiometry and an extent of reaction, which is usually specified as an extent of conversion of a limiting reagent. No reaction kinetics information is needed, so it can be used when the

kinetics are unknown (which is often the case in the early stages of design) or when the reaction is known to proceed to full conversion. Conversion reactors can handle multiple reactions, but care is needed in specifying the order in which they are solved if they use the same limiting reagent.

Equilibrium reactor

An equilibrium reactor finds the equilibrium product distribution for a specified set of stoichiometric reactions. Phase equilibrium is also solved. The engineer can enter the outlet temperature and pressure and let the reactor model calculate the duty needed to reach that condition, or else enter a heat duty and let the model predict the outlet conditions from an energy balance.

An equilibrium reactor only solves the equations specified, so it is useful in situations where one or more reactions equilibrate rapidly while other reactions proceed much more slowly. An example is the steam reforming of methane to hydrogen. In this process, the water–gas shift reaction between water and carbon monoxide equilibrates rapidly at temperatures above 450 °C, whereas methane conversion requires catalysis even at temperatures above 800 °C. This process chemistry is explored in Example 4.5.

In some simulation programs, the equilibrium reactor model requires the designer to specify both liquid and vapor phase products, even though one of the streams may be calculated to have zero flow. If the real reactor has a single outlet, then the two product streams in the model should be mixed back together.

Gibbs reactor

The Gibbs reactor solves the full reaction and phase equilibrium of all species in the component list by minimization of the Gibbs free energy, subject to the constraint of the feed mass balance. A Gibbs reactor can be specified with restrictions such as a temperature approach to equilibrium or a fixed conversion of one species.

The Gibbs reactor is useful when modeling a system that is known to come to equilibrium, in particular, high-temperature processes involving simple molecules. It is less useful when complex molecules are present, as these usually have high Gibbs energy of formation; consequently, very low concentrations of these species are predicted unless the number of components in the model is highly restricted.

The designer must specify the components carefully when using a Gibbs reactor in the model, as the Gibbs reactor can only solve for specified components. If a component that is actually formed is not listed in the component set, then the Gibbs reactor results will be meaningless. Care must be taken to include all isomers, as the absence of isomers can distort the results of a Gibbs reactor. Furthermore, if some of the species have high Gibbs free energy, their concentrations may not be properly predicted by the model. An example is aromatic hydrocarbon compounds such as benzene, toluene, and xylenes, which have Gibbs free energy of formation greater than zero. If these species are in a model component set that also contains hydrogen and carbon, then a Gibbs reactor will predict that only carbon and hydrogen are formed. Although hydrogen and coke are indeed the final equilibrium products, the aromatic hydrocarbons are kinetically stable, and there are many processes that convert aromatic hydrocarbon compounds without significant coke yields. In this situation, the designer must either omit carbon from the component list or use an equilibrium reactor in the model.

Continuous stirred tank reactor

The continuous stirred tank reactor (CSTR) is a model of the conventional well-mixed reactor. It can be used when a model of the reaction kinetics is available and the reactor is believed to be well mixed (i.e., the conditions everywhere in the reactor are the same as the outlet conditions). By specifying forward and reverse reactions, the CSTR model can model equilibrium and rate-based reactions simultaneously. The main drawback of using the CSTR model is that a detailed understanding of kinetics is necessary if by-products are to be predicted properly.

Plug flow reactor

A plug flow reactor (PFR) models the conventional plug flow behavior, assuming radial mixing but no axial dispersion. The reaction kinetics must be specified, and the model has the same limitations as the CSTR model.

Most of the simulators allow heat input or removal from a PFR. Heat transfer can be with a constant wall temperature (as encountered in a fired tube, steam-jacketed pipe, or immersed coil) or with counter-current flow of a utility stream (as in a heat exchanger tube or jacketed pipe with cooling water).

Yield shift reactor

The yield shift reactor overcomes some of the drawbacks of the other reactor models by allowing the designer to specify a yield pattern. Yield shift reactors can be used when there is no model of the kinetics but some laboratory or pilot plant data are available, from which a yield correlation can be established.

Yield shift reactors are particularly useful when modeling streams that contain pseudocomponents, solids with a particle size distribution, or processes that form small amounts of many by-products. These can all be described easily in yield correlations, but can be difficult to model with the other reactor types.

The main difficulty in using the yield shift reactor is in establishing the yield correlation. If a single point—for example, from a patent—is all that is available, then entering the yield distribution is straightforward. If, on the other hand, the purpose is to optimize the reactor conditions, then a substantial set of data must be collected to build a model that accurately predicts yields over a wide enough range of conditions. If different catalysts can be used, the underlying reaction mechanism may be different for each, and each will require its own yield model. The development of yield models can be an expensive process and is usually not undertaken until corporate management has been satisfied that the process is likely to be economically attractive.

Modeling real reactors

Industrial reactors are usually more complex than the simple simulator library models. Real reactors usually involve multiple phases and have strong mass transfer, heat transfer, and mixing effects. The residence time distributions of real reactors can be determined by tracer studies and seldom exactly match the simple CSTR or PFR models; see Section 15.12.

Sometimes a combination of library models can be used to model the reaction system. For example, a conversion reactor can be used to establish the conversion of main feeds, followed by an equilibrium reactor that establishes an equilibrium distribution among specified products. Similarly, reactors with complex mixing patterns can be modeled as networks of CSTR and PFR models, as described in Sections 12.11.2 and 15.11.2 and illustrated in Fig. 12.14.

When using a combination of library models to simulate a reactor, it is a good idea to group these models in a sub-flowsheet. The sub-flowsheet can be given a suitable label, such as “reactor,” that indicates that all the unit operations it contains are modeling a single piece of real equipment. This makes it less likely that someone else using the model will misinterpret it as containing additional distinct operations.

Detailed models of commercial reactors are usually written as user models. User models are described in Section 4.6. Detailed modeling of reactors is discussed in more detail in Section 15.11.

Example 4.4

When heavy oils are cracked in a catalytic or thermal cracking process, lighter hydrocarbon compounds are formed. Most cracking processes on heavy oil feeds form products with carbon numbers ranging from 2 to greater than 20. How does the equilibrium distribution of hydrocarbon compounds with five carbons (C_5 compounds) change as the temperature of the cracking process is increased at 200 kPa?

Solution

This problem was solved using UniSim Design.

The problem asks for an equilibrium distribution, so the model should contain either a Gibbs reactor or an equilibrium reactor.

A quick glance at the component list in UniSim Design shows that there are 22 hydrocarbon species with five carbons. To model the equilibrium among these species, we also need to include hydrogen to allow for the formation of alkenes, dienes, and alkynes. Although it would be possible to enter 21 reactions and use an equilibrium reactor, it is clearly easier to use a Gibbs reactor for this analysis. Fig. 4.3 shows the Gibbs reactor model.

To specify the feed we must enter the temperature, pressure, flow rate, and composition. The temperature, pressure, and flow rate are entered in the stream editor window (Fig. 4.4). The feed composition can be entered as 100% of any of the C_5 paraffin species, for example, normal pentane. The results from a Gibbs reactor would be the same if 100% isopentane was entered. It should be noted, however, that if a mixture of a pentane and a pentene were specified, then the overall ratio of hydrogen to carbon would be different and different results would be obtained.

A spreadsheet was also added to the model, as illustrated in Fig. 4.3, to make it easier to capture and download the results. The spreadsheet was set up to import component mole fractions from the simulation (Fig. 4.5). The simulation was then run for a range of temperatures, and after each run a new column was entered in the spreadsheet (Fig. 4.6).

When the results are examined, many of the individual species are present at relatively low concentrations. It thus makes sense to group some compounds together by molecular type, for example, adding all the dienes together and adding all the alkynes (acetylenes) together.

The spreadsheet results were corrected to give the distribution of C_5 compounds by dividing by one minus the mole fraction of hydrogen and then plotted to give the graph in Fig. 4.7.

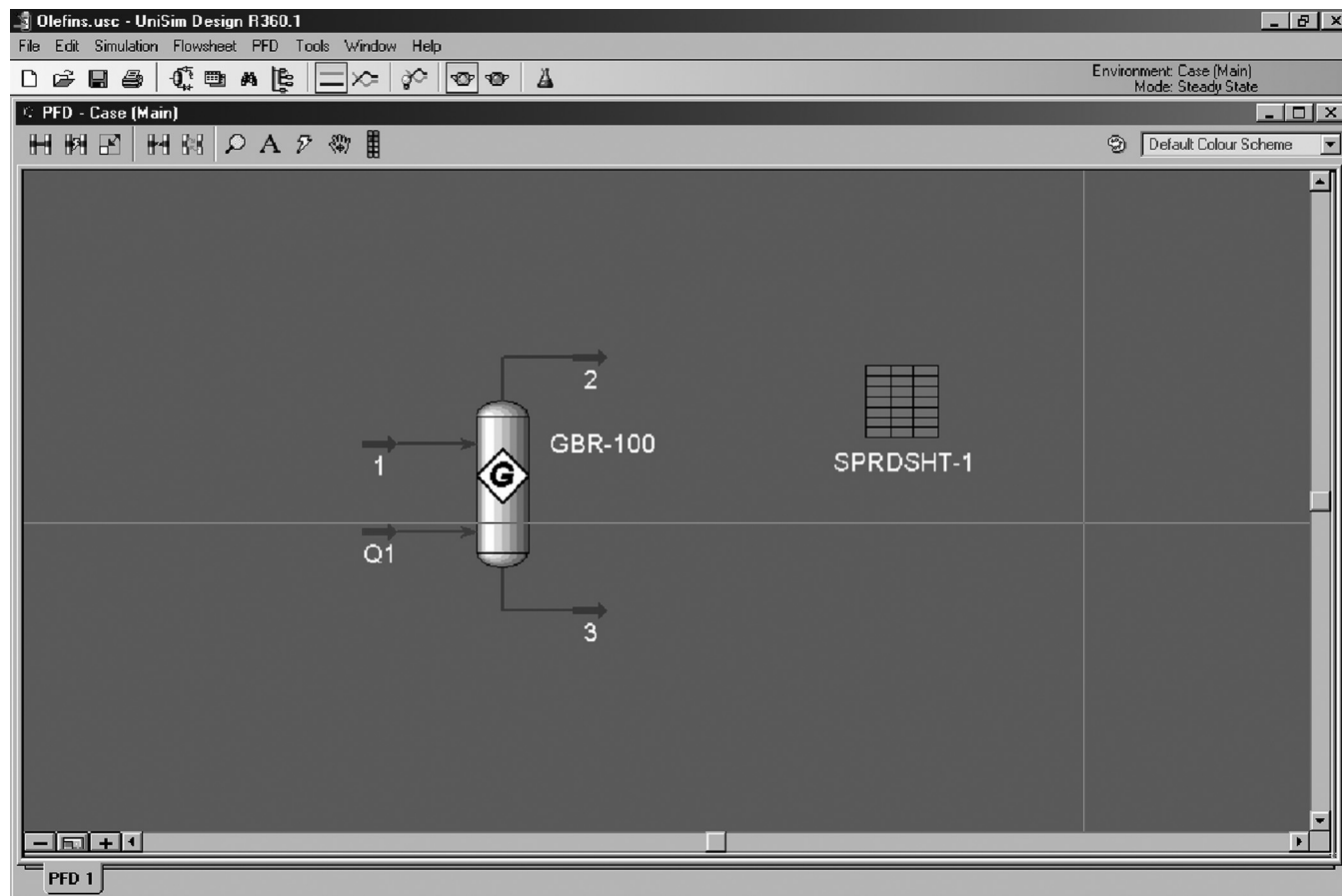


FIG. 4.3 Gibbs reactor model.

It can be seen from the graph that the equilibrium products at temperatures below 500 °C are mainly alkanes (also known as *paraffins* or *saturated hydrocarbons*), with the equilibrium giving roughly a 2:1 ratio of isopentane to normal pentane. As the temperature is increased from 500 °C to 600 °C, there is increased formation of alkene compounds (also known as *olefins*). At 700 °C, we see increased formation of cyclopentene and of dienes, and above 800 °C dienes are the favored product.

Of course this is an incomplete picture, as the relative fraction of C₅ compounds would be expected to decrease as the temperature is raised and C₅ species are cracked to lighter compounds in the C₂ and C₃ range. The model also did not contain carbon (coke) and so could not predict the temperature at which coke would become the preferred product. A more rigorous equilibrium model of a cracking process might include all of the possible hydrocarbon compounds up to C₇ or higher.

A real reactor might give a very different distribution of C₅ compounds from that calculated using the Gibbs reactor model. Dienes formed at high temperatures might recombine with hydrogen during cooling, giving a mixture that looked more like the equilibrium product at a lower temperature. There might also be formation of C₅ compounds by condensation reactions of C₂ and C₃ species during cooling, or loss of dienes and cyclopentene due to coke formation.

Example 4.5

Hydrogen can be made by steam reforming of methane, which is a highly endothermic process:



Steam reforming is usually carried out in fired tubular reactors, with catalyst packed inside the tubes and fuel fired on the outside of the tubes to provide the heat of reaction. The product gas mixture contains carbon dioxide and water vapor, as well as carbon monoxide and hydrogen, and is conventionally known as *synthesis gas* or *syngas*.

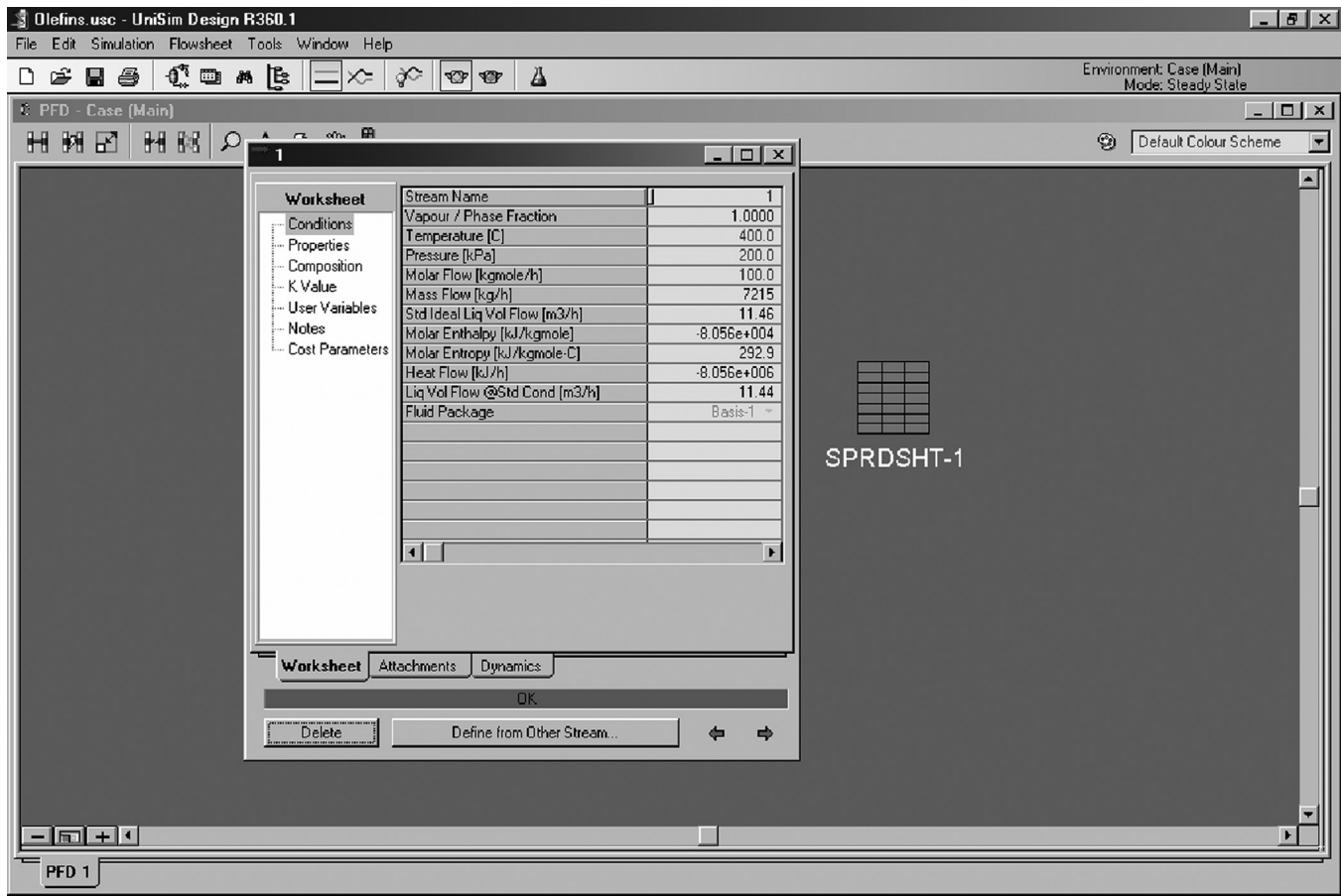


FIG. 4.4 Stream entry.

Hydrogen can also be made by partial oxidation of methane, which is an exothermic process, but yields less product per mole of methane feed:



When steam, oxygen, and methane are combined, heat from the partial oxidation reaction can be used to provide the heat for steam reforming. The combined process is known as *autothermal reforming*. Autothermal reforming has the attraction of requiring less capital investment than steam reforming (because it does not need a fired-heater reactor), but giving higher yields than partial oxidation.

The yield of hydrogen can be further increased by carrying out the water–gas shift reaction:



The water–gas shift reaction equilibrates rapidly at temperatures above about 450 °C. At high temperatures this reaction favors the formation of carbon monoxide, whereas at low temperatures more hydrogen is formed. When hydrogen is the desired product, the shift reaction is promoted at lower temperatures by using an excess of steam and providing a medium- or low-temperature shift catalyst.

In an autothermal reforming process, 1000 kmol/h of methane at 20 °C is compressed to 10 bar, mixed with 2500 kmol/h of saturated steam, and reacted with pure oxygen to give 98% conversion of the methane. The resulting products are cooled and passed over a medium-temperature shift catalyst that gives an outlet composition corresponding to equilibrium at 350 °C.

1. How much heat is required to vaporize the steam?
2. How much oxygen is needed?
3. What is the temperature at the exit of the autothermal reforming reactor?
4. What is the final molar flow rate of each component of the synthesis gas?

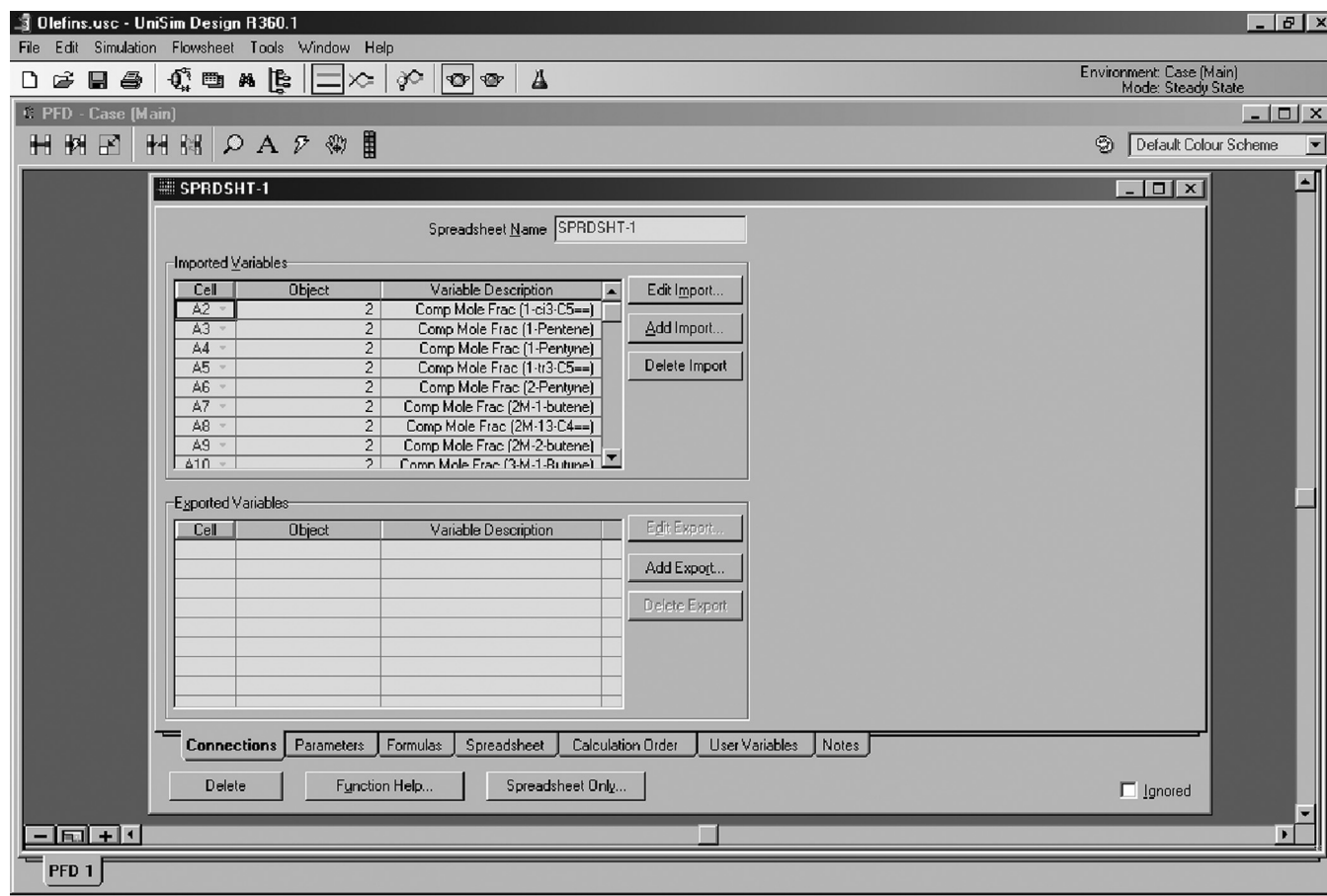


FIG. 4.5 Product composition spreadsheet.

Solution

This problem was solved using Aspen Plus. The model must simulate the high-temperature reforming reaction and the re-equilibration of the water–gas shift reaction as the product gas is cooled. A Gibbs reactor can be used for the high-temperature reaction, but an equilibrium reactor must be specified for the shift reactor, as only the water–gas shift reaction will re-equilibrate at 350 °C. Because the methane compressor supplies some heat to the feed, it should be included in the model. Because the question asks how much heat is needed to vaporize the steam, a steam boiler should also be included. The oxygen supply system can also be included, giving the model shown in Fig. 4.8.

The heat duty to the reforming reactor is specified as zero. The oxygen flow rate can then be adjusted until the desired methane conversion is achieved. For 98% conversion, the flow rate of methane in the autothermal reactor product (stream 502) is 2% of the flow rate in the reactor feed (stream 501) (i.e., 20 kmol/h). For the purpose of this example, the oxygen flow rate was adjusted manually, although a controller could have been used, as described in Section 4.8. The results are shown in Fig. 4.9.

When the simulation model was run, the following values were calculated:

1. The steam heater requires 36MW of heat input.
2. 674 kmol/h of oxygen are needed.
3. The temperature at the exit of the reforming reactor is 893 °C.
4. The molar flow rates at the outlet of the shift reactor (stream 504) are:

H ₂	2504
H ₂ O	1956
CO	68
CO ₂	912
CH ₄	20

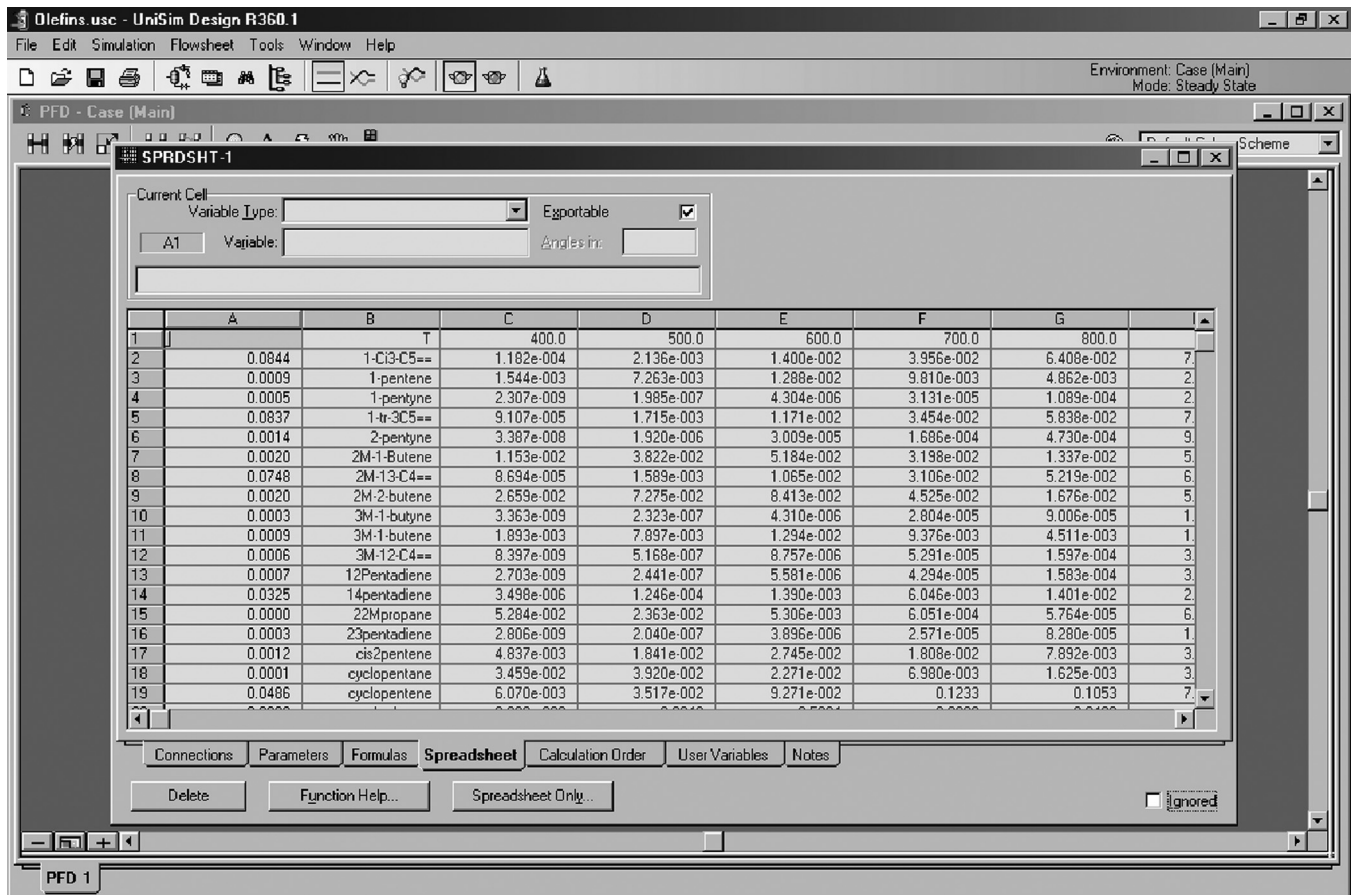


FIG. 4.6 Spreadsheet results.

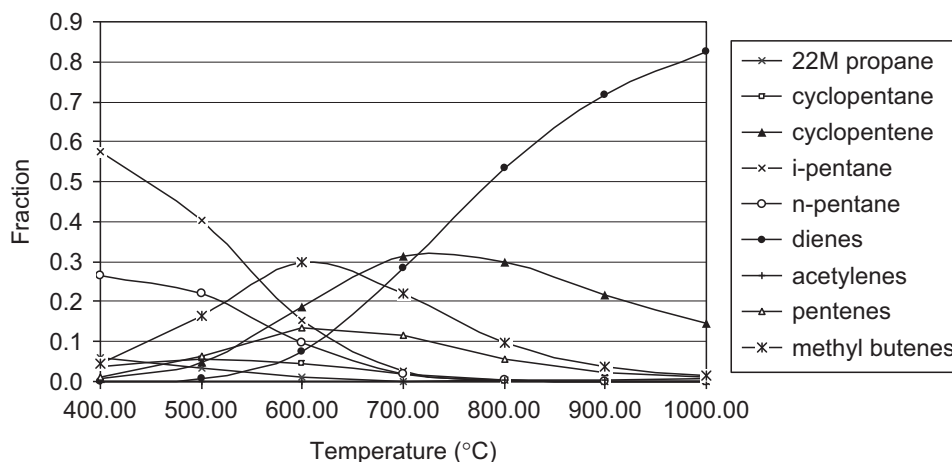


FIG. 4.7 Product distribution.

It should be immediately apparent from the model output that the process as simulated is far from optimal. The oxygen consumption is larger than the 500 kmol/h that would have been needed for partial oxidation. The excess oxygen is needed because the additional steam that is being fed must also be heated to the reactor outlet temperature, which requires more of the feed methane to be burned. The corollary of this result is that the hydrogen yield, at roughly 2.5 moles per mole methane, is not much better than could have been obtained with partial oxidation followed by shift, despite the large excess of steam used.

The designer has several options that could be examined to improve this process:

1. Increase heat recovery from the product gas to the feed streams to pre-heat the reactor feed and reduce the amount of oxygen that is needed.
2. Reduce the amount of steam fed with the methane.

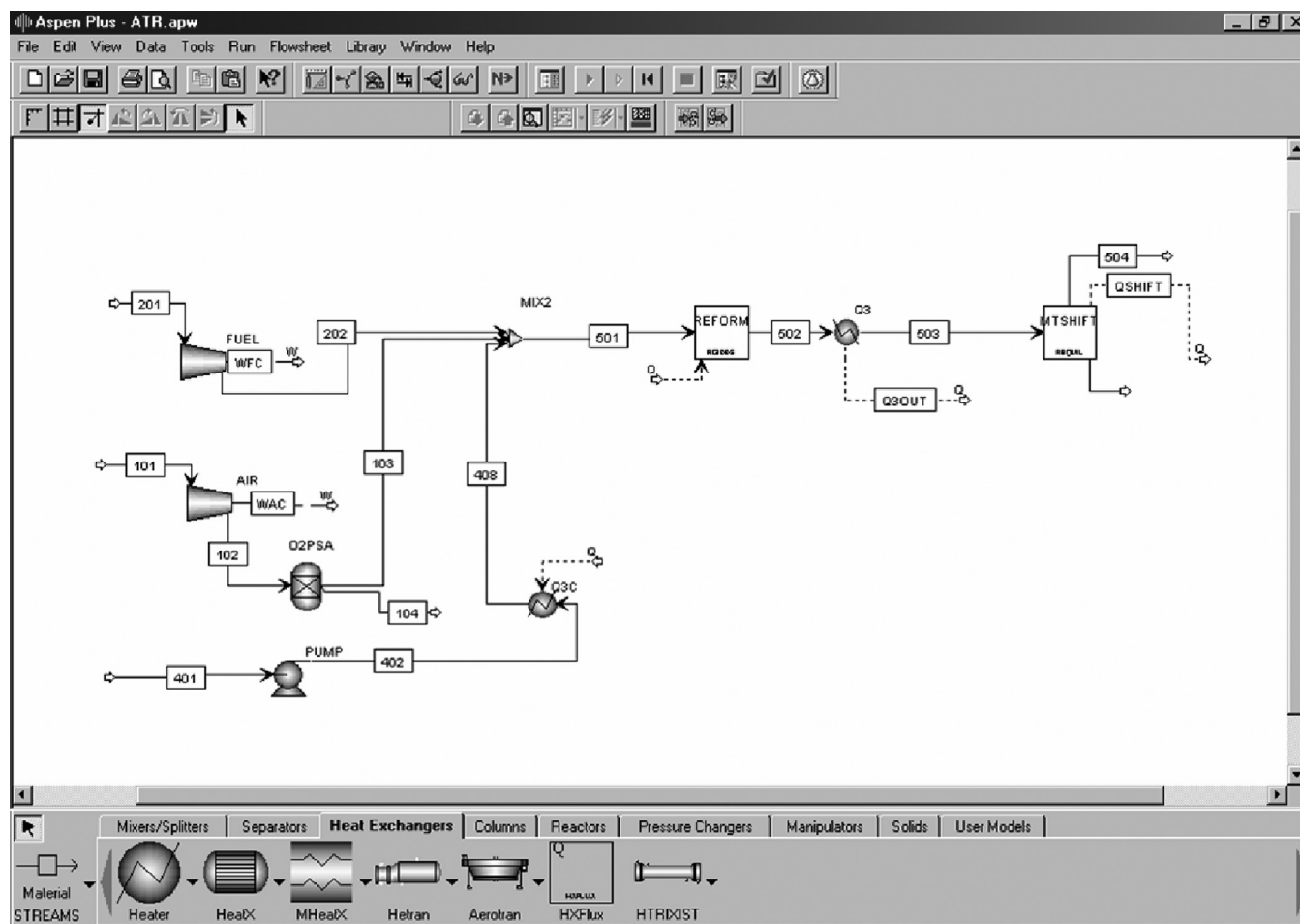


FIG. 4.8 Autothermal reforming model.

3. Bypass a part of the steam from the reformer feed to the shift reactor feed so as to obtain the benefit of driving the equilibrium in the shift reactor without the cost of providing extra heat to the reformer.
4. Reduce the conversion of methane so that a lower reactor conversion and lower outlet temperature are required.

In practice, all of these options are implemented to some extent to arrive at the optimal autothermal reforming conditions. This optimization is explored further in Problem 4.13.

4.5.2 Distillation

The commercial process simulators contain a range of distillation models with different degrees of sophistication. The design engineer must choose a model that is suitable for the purpose, depending on the problem type, the extent of design information available, and the level of detail required in the solution. In some cases, it may make sense to build different versions of the flowsheet, using different levels of detail in the distillation models, so that the simpler model can be used to initialize a more detailed model.

Shortcut models

The simplest distillation models to set up are the shortcut models. These models use the Fenske–Underwood–Gilliland or Winn–Underwood–Gilliland method to determine the minimum reflux and number of stages, or to determine the required reflux given a number of trays, or to determine the required number of trays for a given reflux ratio. These methods are described in Chapter 17. The shortcut models can also estimate the condenser and reboiler duties and determine the optimum feed tray.

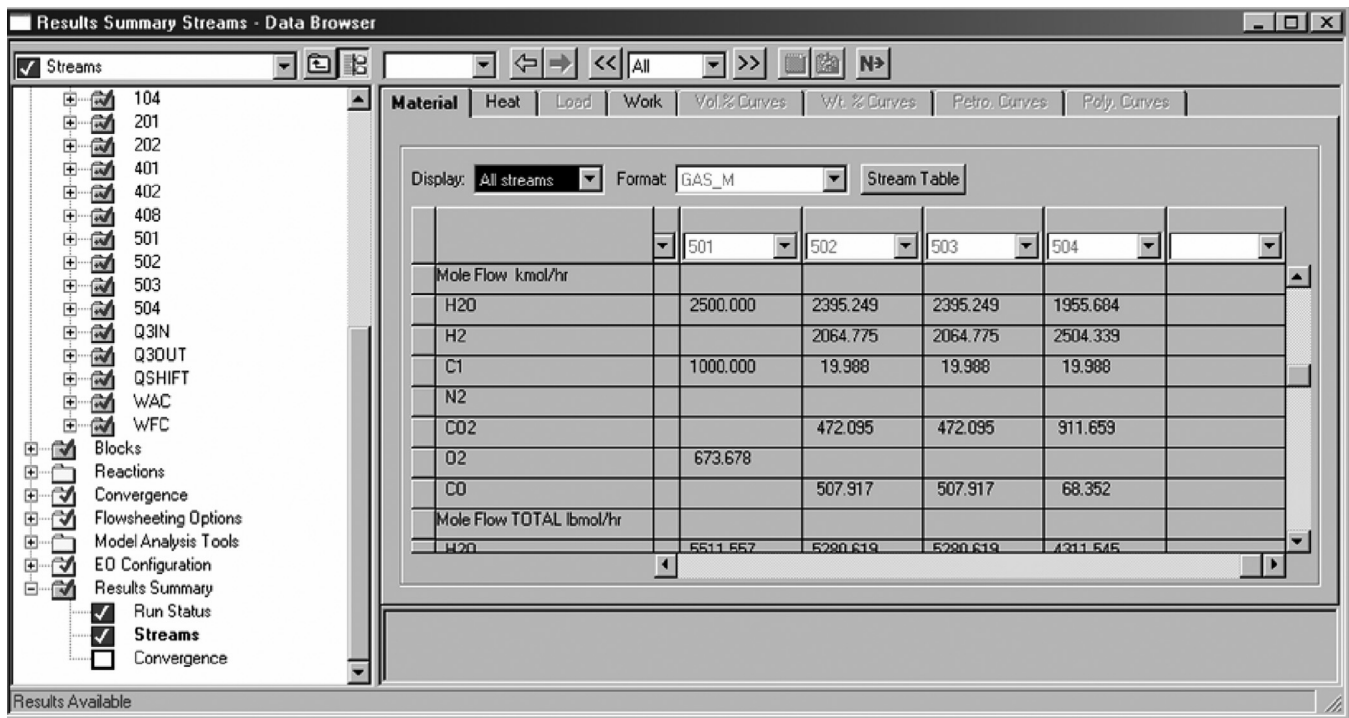


FIG. 4.9 Autothermal reactor model results.

The minimum information needed to specify a shortcut distillation model is:

- The component recoveries of the light and heavy key components
- The condenser and reboiler pressures
- Whether the column has a total or partial condenser

In some cases, the designer can specify the purities of the light and heavy key components in the bottoms and distillate, respectively. Care is needed when using purity as a specification, as it is easy to specify purities or combinations of purity and recovery that are infeasible.

The easiest way to use a shortcut distillation model is to start by estimating the minimum reflux and number of stages. The optimum reflux ratio is usually between 1.05 and 1.25 times the minimum reflux ratio, R_{min} , so $1.15 \times R_{min}$ is often used as an initial estimate. Once the reflux ratio is specified, the number of stages and optimum feed stage can be determined. The shortcut model results can then be used to set up and initialize a rigorous distillation simulation.

Shortcut models can also be used to initialize fractionation columns (complex distillation columns with multiple products), as described next.

Shortcut distillation models are robust and are solved quickly. They do not give an accurate prediction of the distribution of nonkey components, and they do not perform well when there is significant liquid phase nonideality, but they are an efficient way of generating a good initial design for a rigorous distillation model. In processes that have a large number of recycle streams, it is often worthwhile to build one model with shortcut columns and a second model with rigorous columns. The simple model will converge more easily and can be used to provide good initial estimates of column conditions and recycle streams for the detailed model.

The main drawback of shortcut models is that they assume constant relative volatility, usually calculated at the feed condition. If there is significant liquid or vapor phase nonideality, constant relative volatility is a very poor assumption, and shortcut models should not be used.

Rigorous models

Rigorous models carry out full stage-by-stage mass and energy balances. They give better predictions of the distribution of components than shortcut models, particularly when the liquid phase behaves nonideally, as the flash

calculation is made on each stage. Rigorous models allow many more column configurations, including use of side streams, intermediate condensers and reboilers, multiple feeds and side strippers and rectifiers. Rigorous models can be much harder to converge, particularly if poor initial estimates are used or if the column is improperly specified.

The two main types of rigorous distillation models are equilibrium-stage models and rate-based models. Equilibrium-stage models assume either full VLE on each stage or else an approach to equilibrium based on a stage efficiency entered by the designer. When using an equilibrium-stage model for column sizing, the stage efficiencies must be entered. Stage efficiency is typically less than 0.8 and is discussed in more detail in [Chapter 17](#). Rate-based models do not assume phase equilibrium, except at the VLE, and instead solve the interphase mass transfer and heat transfer equations. Rate-based models are more realistic than the idealized equilibrium-stage models, but because it can be difficult to predict the interfacial area and mass transfer coefficients, rate-based models are less widely used in practice.

Rigorous distillation models can be used to model absorber columns, stripper columns, refluxed absorbers, three-phase systems such as extractive distillation columns, many possible complex column configurations, and columns that include reactions such as reactive distillation and reactive absorption columns. The formation of a second liquid phase (usually a water phase) in the column can be predicted if the designer has selected a liquid phase activity model that allows for the prediction of two liquid phases.

One of the most useful features of the rigorous distillation models in the commercial simulation programs is that most include a tool for plotting column profiles. The design engineer can generate plots showing the molar composition of each species in either phase versus tray number. These plots can be helpful in troubleshooting column designs.

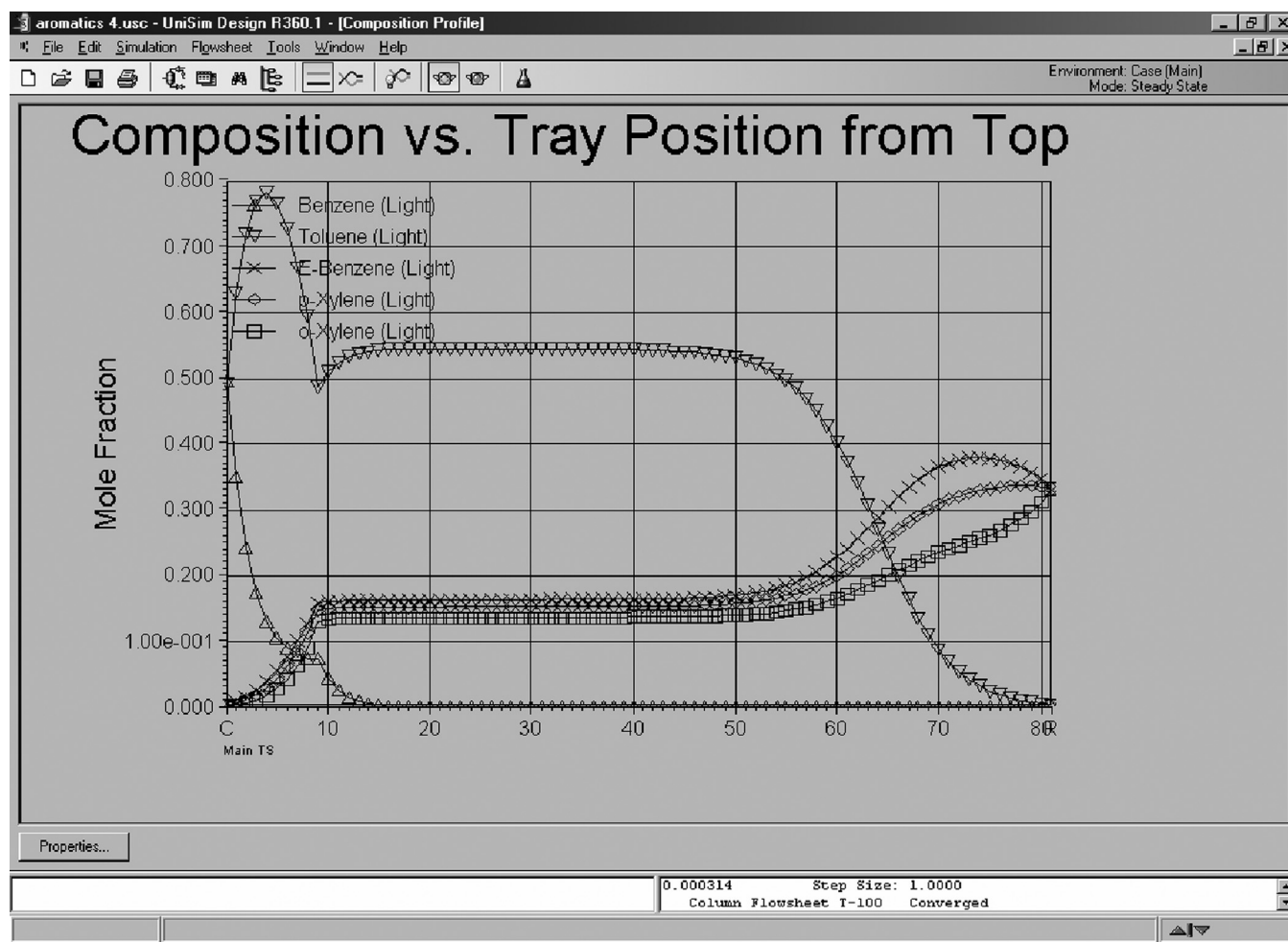


FIG. 4.10 Feed tray too high.

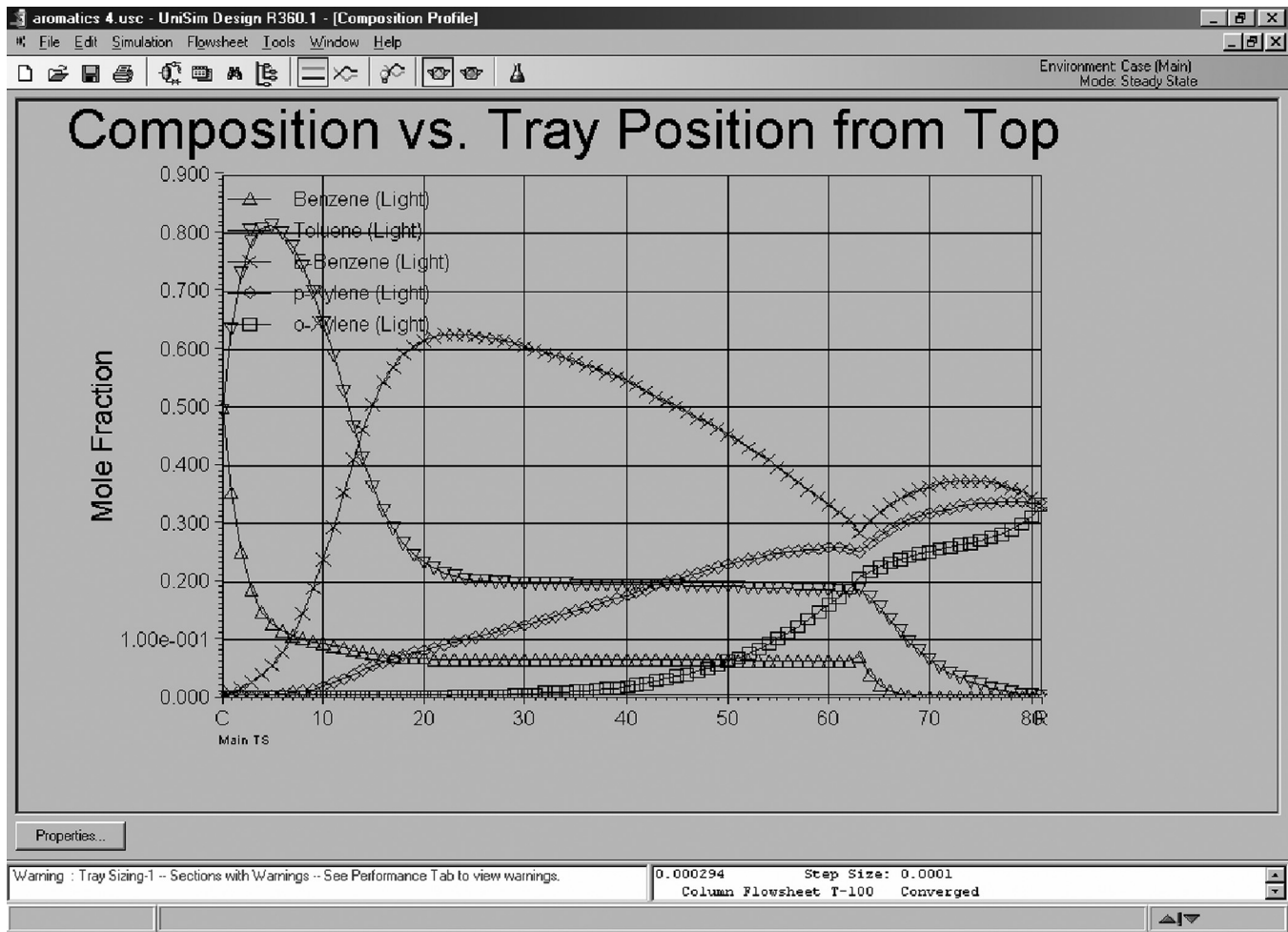


FIG. 4.11 Feed tray too low.

For example, Figs. 4.10 to 4.15 show column profiles for the distillation problem introduced in Examples 4.6 and 4.7, which is optimized in Example 12.1. The column was simulated in UniSim Design.

- In Fig. 4.10, the feed stage was moved up to tray 10, which is too high. The column profiles show a broad flat region between trays 20 and 45, indicating that nothing much is going on over this part of the column. There are too many trays in the stripping section, and the feed tray should be moved lower. Sections with very small change in composition can also be indicative of pinched regions where an azeotropic mixture is being formed.
- In Fig. 4.11, the feed tray has been moved down to tray 63, which is too low. The column profiles for benzene and toluene, the light components, are flat between trays 30 and 60 in the rectifying section, indicating that the feed tray should be moved higher.
- In Fig. 4.12, the column specification was changed from toluene recovery to reflux ratio, and a low value of reflux ratio (2.2) was entered. This is less than the minimum reflux required for the specified separation; consequently, the desired recovery of toluene cannot be achieved. The recovery of toluene is reduced to 72%.
- In Fig. 4.13, the reflux ratio was increased to 4.0. The recovery of toluene is now 100%, which is greater than the 99% required. This represents a suboptimal use of energy and capital.
- Fig. 4.14 shows the column profiles when the number of trays was reduced to 25, with the feed on tray 8. The column profile for toluene shows that there are insufficient stages (and/or reflux). Although the profile is changing smoothly, the recovery in the distillate is only 24.5%.
- The column profiles with the optimum conditions determined in Example 12.1 are shown in Fig. 4.15. The poor features shown in the other profiles are absent.

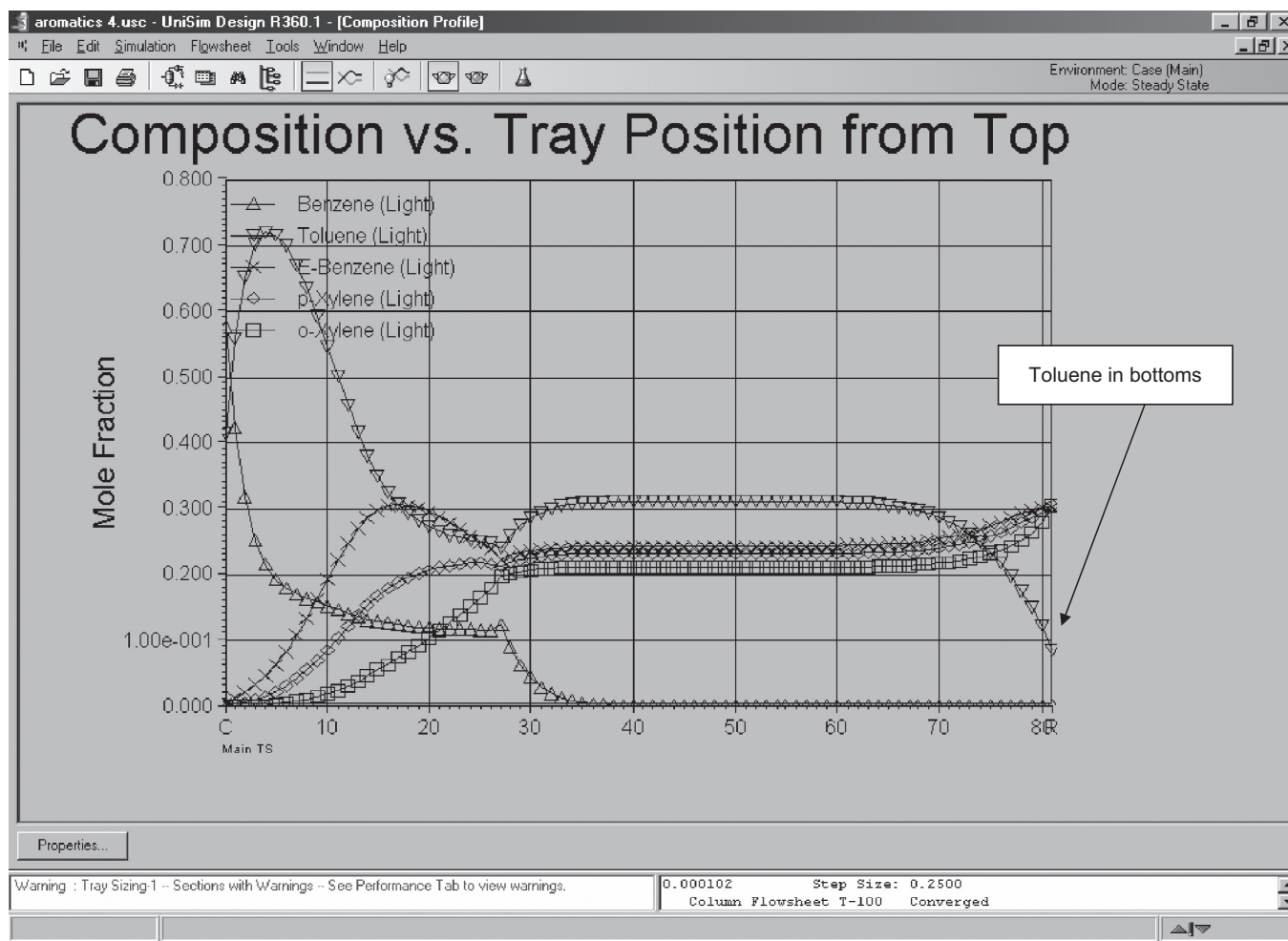


FIG. 4.12 Reflux ratio too low: Toluene recovery 72%.

Column convergence

Convergence of distillation column models is probably the most common problem in process simulation for novice engineers. There are many reasons why rigorous distillation models (and models of other multistage operations) may fail to converge. The most common causes of convergence problems are as follows:

- **Infeasible specifications:** In multicomponent distillation, care must be taken when using purity as a specification. It will not be possible to make a 99% pure distillate if the feed contains 2% of a component that boils lighter than the light key. The choice of column specifications is discussed in more detail in Section 17.6.2.
- **Poor initialization:** If the simulation model is set up with fewer than the minimum number of trays or less than the minimum reflux, it will not converge. Shortcut models can be used to provide an initial design that is feasible, which can then be used as a starting point for a rigorous model.
- **Poor initial estimates:** The faster distillation column solution algorithms (such as the widely used inside-out algorithm, discussed in Section 17.8.2) run much better when provided with a good initial estimate of tray temperatures.

A good practice is to start by using simple hand calculations to check that the column specifications are feasible. A shortcut model can then be used to determine the minimum reflux and the number of trays and feed tray location when the reflux is increased to about 1.1 times minimum reflux. The rigorous model should then be initialized with the reboiler and condenser temperatures found from the shortcut simulation, a linear temperature profile, and a pressure gradient that makes an approximate allowance for the pressure drop per tray (about 2 inches of liquid is typical).

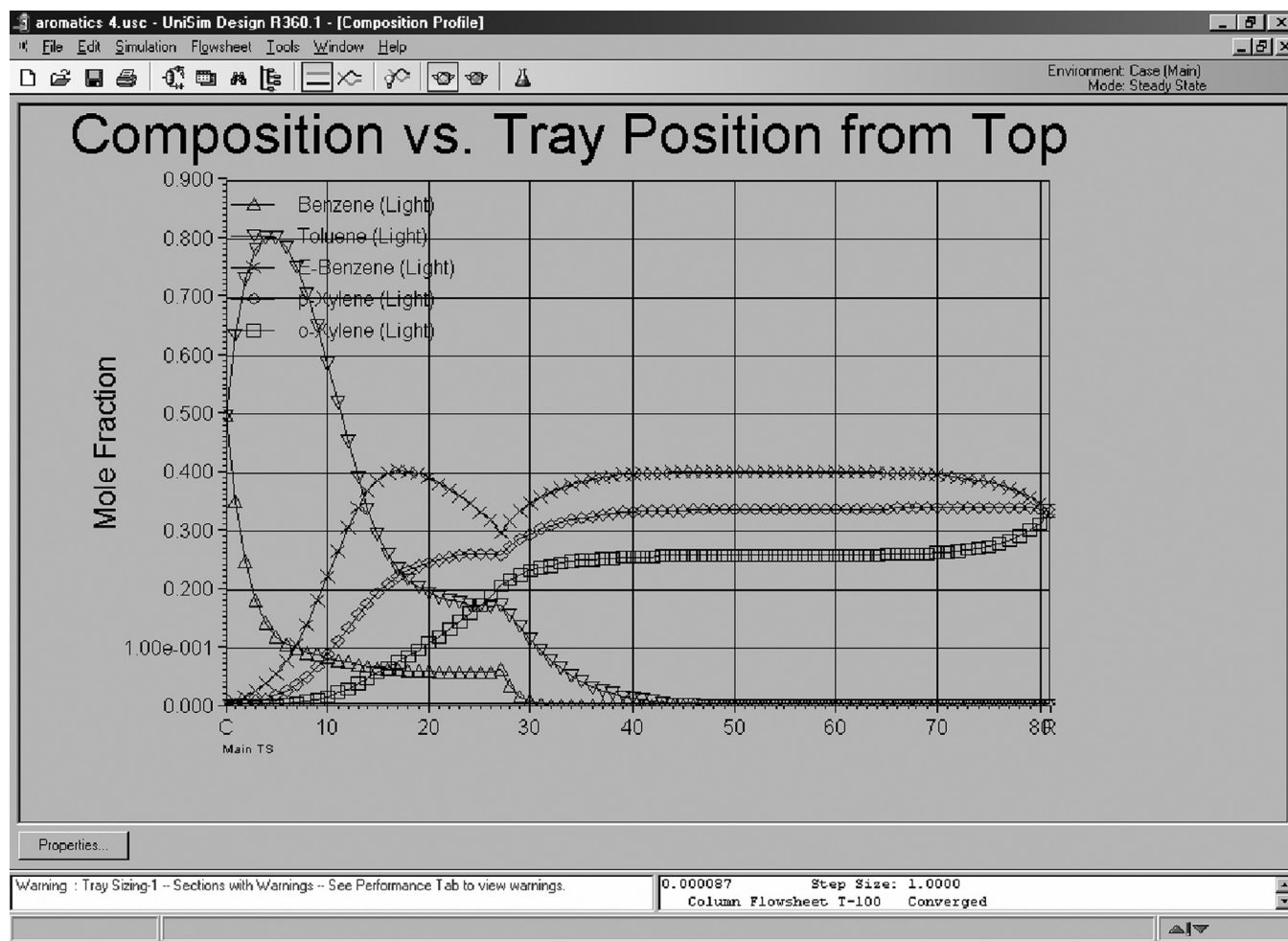


FIG. 4.13 Reflux ratio too high: Toluene recovery 100%.

If the inside-out algorithm is used, the rigorous model should initially be run with easy specifications, such as reflux ratio and either distillate rate or bottoms rate. These will guarantee convergence and generate a realistic column temperature profile that can then be stored as the estimate for future runs. The model specifications can then be changed to the desired design specifications (purity or recovery specifications), and now that the model has a good initialization and set of estimates, convergence will usually be rapid.

In the event that this method fails to give quick convergence, the designer should check that the specifications are feasible, check for existence of azeotropes, try adding trays, and examine the column profiles for clues to the problem.

Complex columns for fractionation

Several of the commercial simulation programs offer preconfigured complex column rigorous models for petroleum fractionation. These models include charge heaters, several side strippers, and one or two pump-around loops. These fractionation column models can be used to model refinery distillation operations such as crude oil distillation, vacuum distillation of atmospheric residue oil, fluidized catalytic cracking (FCC) process main columns, and hydrocracker or coker main columns. Aspen Plus also has a shortcut fractionation model, SCFrac, which can be used to configure fractionation columns in the same way that shortcut distillation models are used to initialize multicomponent rigorous distillation models.

A typical crude oil distillation column is illustrated in Fig. 4.16, which shows a simulation using an Aspen Plus PetroFrac model. The crude oil is pre-heated in a heat-exchange network and charge heater and is then fed to the flash zone at the bottom of the column. Stripping steam is also added at the bottom of the column to provide additional vapor flow. Products with different boiling ranges are withdrawn from the column. The intermediate products

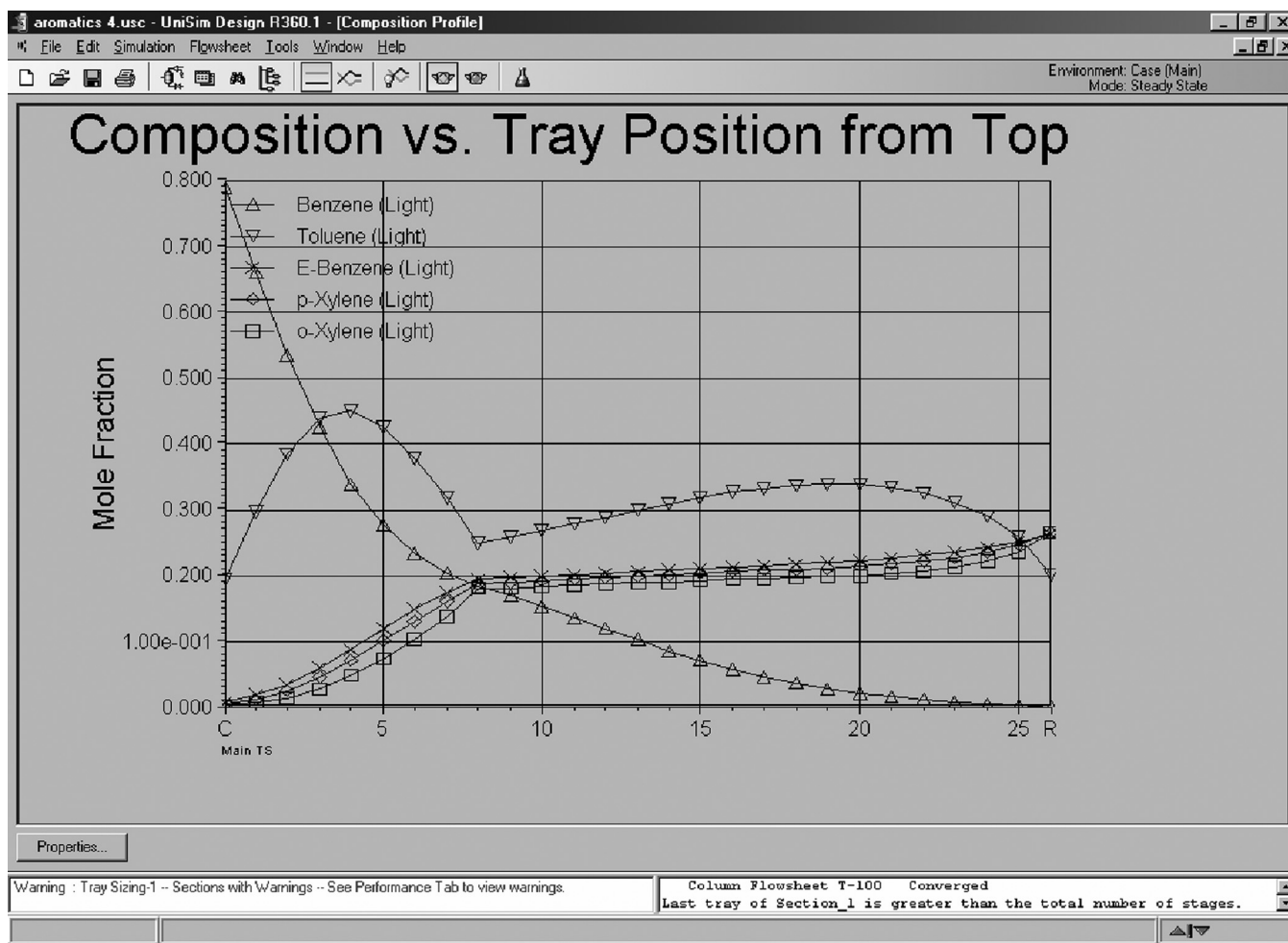


FIG. 4.14 Too few trays: Toluene recovery 24.5%.

are withdrawn from the bottom of side stripper columns, so as to minimize loss of lighter products in the side stream. Although the exact distillation ranges can vary depending on the local fuels specifications and the sophistication of the refinery, the typical products taken in a crude oil distillation unit are (from the bottom up):

1. Atmospheric residue oil (Residue), containing compounds that boil above about 340 °C (650 °F). This is normally sent to a vacuum distillation unit to recover more light products, but parts of it may be blended into high sulfur fuels such as heating oil or bunker fuel (marine fuel).
2. Atmospheric gas oil (AGO), containing compounds that boil in the range 275 °C to 340 °C (530 °F to 650 °F). This material is too high boiling for use as a transportation fuel and is usually sent to a hydrocracker or FCC unit for conversion to lighter products.
3. Heavy distillate (straight-run distillate [SRD]), containing compounds that boil in the range 205 °C to 275 °C (400 °F to 530 °F). This material is hydrotreated to remove sulfur compounds and can then be blended into heating oils and diesel fuels for trucks, railroad engines, and off-road applications such as tractors and mining equipment.
4. Light distillate (straight-run kerosene [SRK]), containing compounds that boil in the range of 175 °C to 230 °C (350 °F to 450 °F). Light distillate is hydrotreated to remove sulfur and can then be blended into jet fuel or sold as kerosene (sometimes called *paraffin*) for lamp and cooking fuel.
5. Naphtha, boiling in the range from 25 °C to 205 °C (80 °F to 400 °F). Naphtha is usually sent to an additional column for separation into a light naphtha boiling below 80 °C (180 °F) and a heavy naphtha. Heavy naphtha has the right boiling range for gasoline but usually has a very low octane number. It is typically upgraded by catalytic reforming using noble metal catalysts to increase the concentration of aromatic hydrocarbons in the naphtha and raise the octane number. Catalytic reforming is also the first step in the production of aromatic hydrocarbons for petrochemicals manufacture. Light naphtha also boils in a suitable range for blending into gasoline and often has

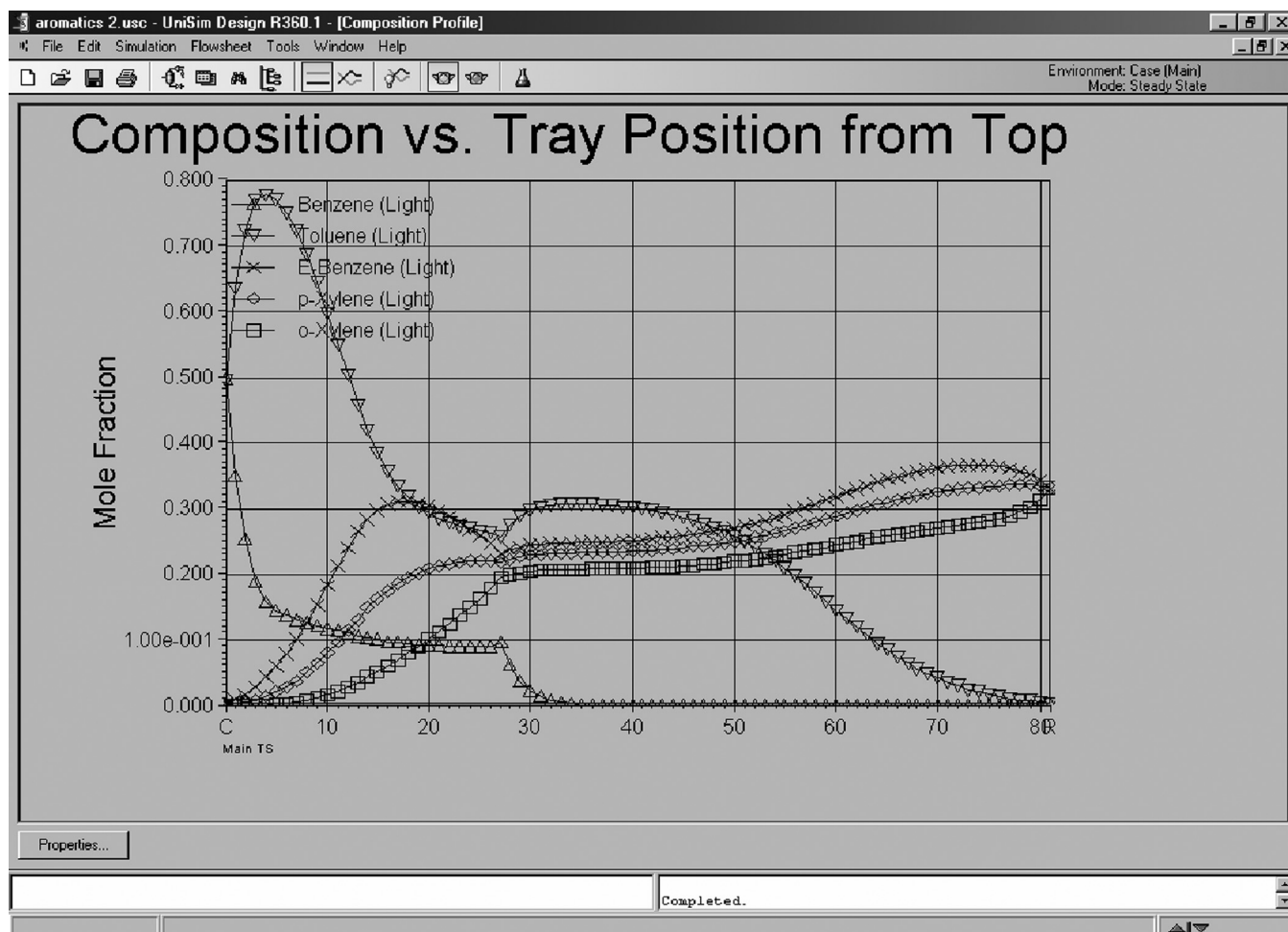


FIG. 4.15 Optimized column profiles.

an acceptable octane number. It is usually treated to oxidize odiferous mercaptan sulfur compounds. Light naphtha is also widely used as a petrochemical feedstock for steam cracking to produce olefin compounds such as ethylene and propylene.

6. The overhead product of the crude unit contains hydrogen, methane, carbon dioxide, hydrogen sulfide, and hydrocarbons up to butanes and some pentanes. It is usually sent to a set of distillation columns known as a *saturate gas plant* for recovery of propane and butane for sale. The lighter gases are then used as refinery fuel.

The design of refinery fractionation columns can be complex. The pump-around streams function as intermediate condensers and remove surplus heat from the column. This heat is usually recovered by heat exchange with the cold crude oil feed. Oil refineries are often designed to handle many different crude oils with different boiling assays. The refinery may make different product slates at different times of the year or in response to market conditions. The crude oil distillation and associated heat-exchange network must be flexible enough to handle all of these variations, while still achieving tight specifications on the boiling point curves of every product.

Column sizing

The rigorous column models allow the design engineer to carry out tray sizing and hydraulics calculations for the basic types of distillation trays and for some types of random and structured packing. Different commercial simulators use different tray sizing correlations, but they all follow a method similar to that described in [Chapter 17](#).

The tray sizing tools are not always enabled when running the distillation models. In some of the simulation programs, the design engineer must enable a tray sizing program and/or enter default values for tray type and tray spacing before the sizing algorithm will work properly. If the column diameter does not change when the reflux rate is significantly changed (or if all the columns in the simulation appear to have the same diameter), then the designer should check to make sure that the tray sizing part of the program is properly configured.

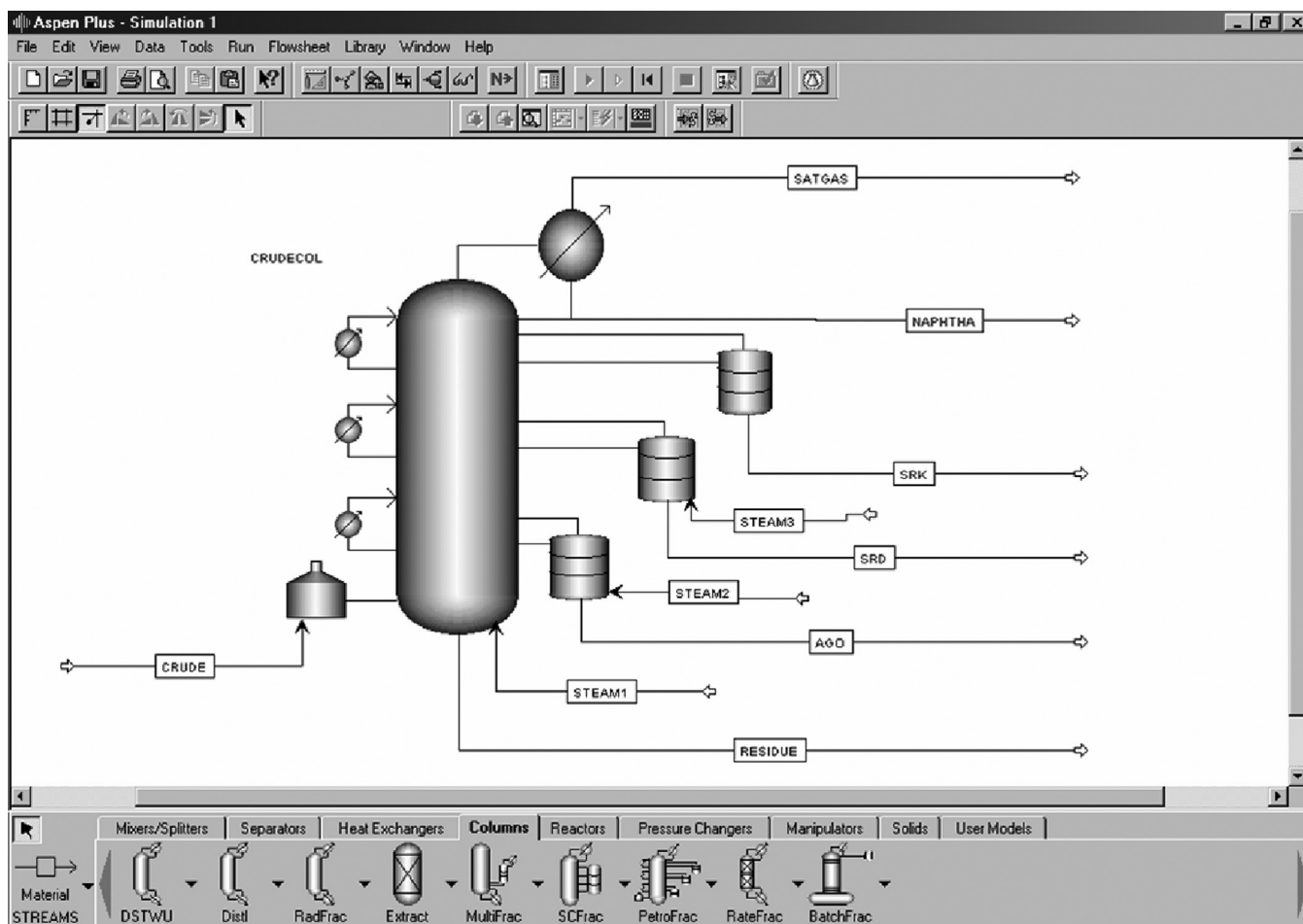


FIG. 4.16 Crude oil fractionation.

The tray sizing options in the simulators are restricted to standard internals such as sieve trays, valve trays, bubble-cap trays, random packings, and structured packings. They do not include high-capacity trays, high-efficiency trays, or the latest packing designs. When designing a column that has many stages or a large diameter, it is always worth contacting the column internals vendors for estimates, as use of high-capacity, high-efficiency internals can lead to substantial savings. Advanced internals are also usually used when revamping an existing column to a higher throughput or tighter product specifications.¹

The design engineer should always allow for tray inefficiency when using column sizing tools in conjunction with an equilibrium-stage model. Failure to do so would underpredict the number of stages and hence have an impact on the column pressure drop and hydraulics. Estimation of stage efficiency is discussed in [Chapter 17](#). For initial design purposes, a stage efficiency of 0.7 to 0.8 is usually used. For detailed design, stage efficiencies depend on the type of tray used and are often provided by the column internals vendor.

The design engineer must remember to allow a suitable design factor or design margin when sizing columns. Design factors are discussed in Section 1.6. It may be necessary to create two versions of the flowsheet. One version will have the design basis flow rates for producing the mass and energy balances, and the second will have flow rates that are 10% larger for purposes of sizing equipment.

The simulation of distillation processes is discussed in more detail by [Luyben \(2013\)](#).

Example 4.6

Design a distillation column to separate 225 metric tons per hour of an equimolar mixture of benzene, toluene, ethylbenzene, paraxylene, and orthoxylene with minimum total annualized cost. The feed is a saturated liquid at 330 kPa. The recovery of toluene in the distillate should be greater than 99%, and the recovery of ethylbenzene in the bottoms should be greater than 99%.

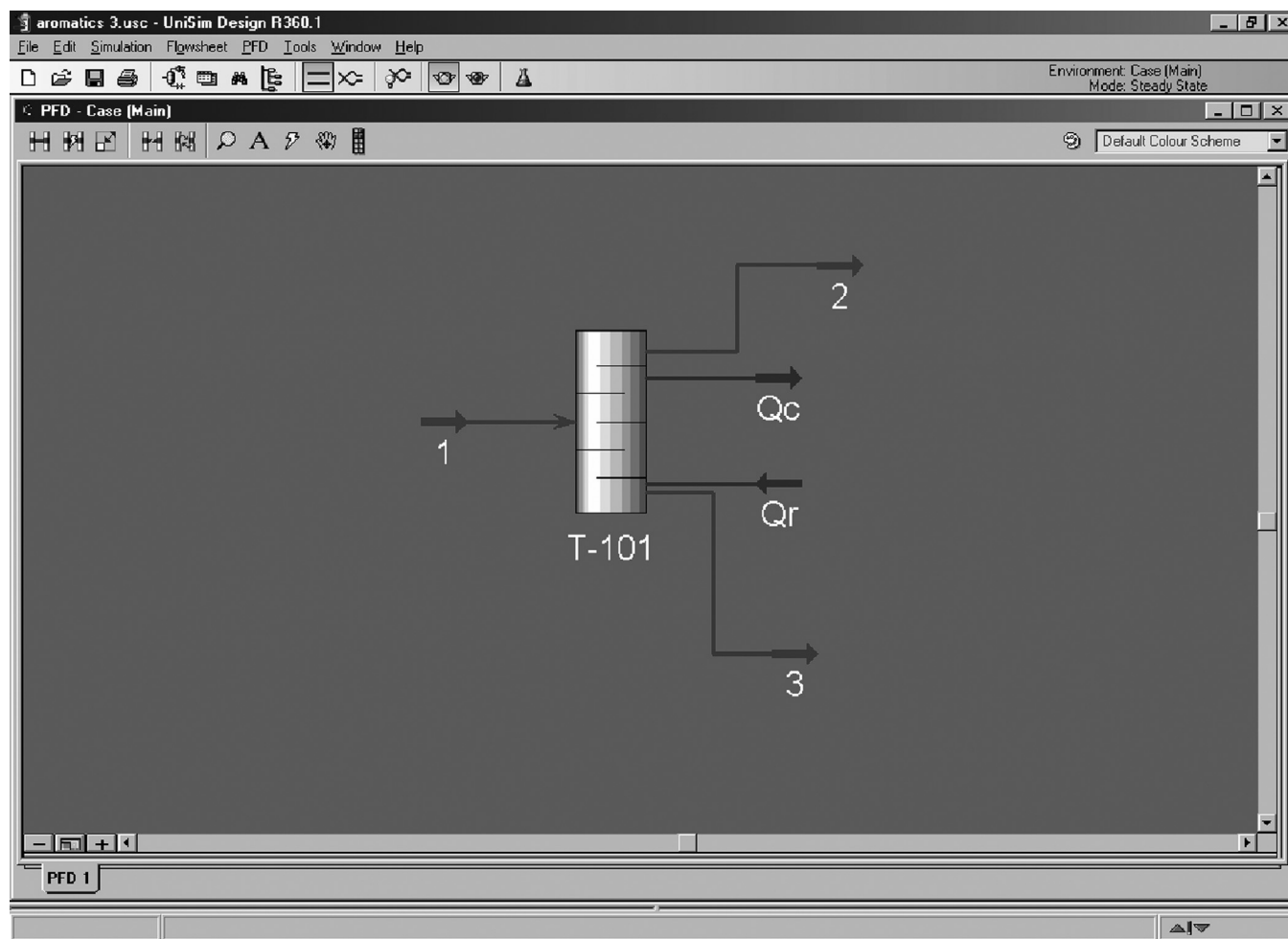


FIG. 4.17 Shortcut distillation.

In this example, a column simulation should be set up using a shortcut model. The shortcut model results will be used to initialize a rigorous model in the example that follows. Determine:

1. The minimum reflux ratio
2. The minimum number of trays
3. The actual number of trays when the reflux is $1.15 R_{min}$
4. The optimum feed tray

Solution

This problem was solved using UniSim Design. The problem was set up as a shortcut column, as shown in Fig. 4.17.

UniSim Design requires the designer to specify the mole fraction of the light key component in the bottoms and the heavy key component in the distillate. We have an equimolar feed, so if we take a basis of 100 mol/h of feed, then the molar flow rate of each component is 20 mol/h. A 99% recovery of each key component corresponds to allowing 0.2 mol/h of that component into the other stream. The mole fractions are then:

$$\text{Ethylbenzene in distillate} = 0.2/40 = 0.005$$

$$\text{Toluene in bottoms} = 0.2/60 = 0.00333$$

When these are entered into the shortcut column as specifications, the minimum reflux is calculated to be $R_{min} = 2.130$. The actual reflux ratio can then be specified as $2.13 \times 1.15 = 2.45$, as shown in Fig. 4.18.

The shortcut column results are shown in Fig. 4.19. The minimum number of stages is calculated as 16.4, which should be rounded up to 17. The actual number of trays required is 39, with feed at stage 18.

T-101

Design

Connections
Parameters
User Variables
Notes

Components

	Component	Mole Fraction
Light Key in Bottoms	Toluene	0.0033
Heavy Key in Distillate	E-Benzene	0.0050

Pressures

Condenser Pressure	290.000 kPa
Reboiler Pressure	330.000 kPa

Reflux Ratios

External Reflux Ratio	2.450
Minimum Reflux Ratio	2.130

Design Rating Worksheet Performance Dynamics

Delete OK ☐ Ignored

FIG. 4.18 Shortcut column specifications.

T-101

Performance

Trays

Minimum Number of Trays	16.376
Actual Number of Trays	38.600
Optimal Feed Stage	18.446

Temperatures

Condenser [C]	132.9
Reboiler [C]	189.8

Flows

Rectify Vapour [kgmole/h]	3183.443
Rectify Liquid [kgmole/h]	2260.706
Stripping Vapour [kgmole/h]	3183.443
Stripping Liquid [kgmole/h]	4562.502
Condenser Duty [kJ/h]	97396297.159
Reboiler Duty [kJ/h]	100983654.209

Design Rating Worksheet **Performance** Dynamics

Delete OK ☐ Ignored

FIG. 4.19 Shortcut column results.

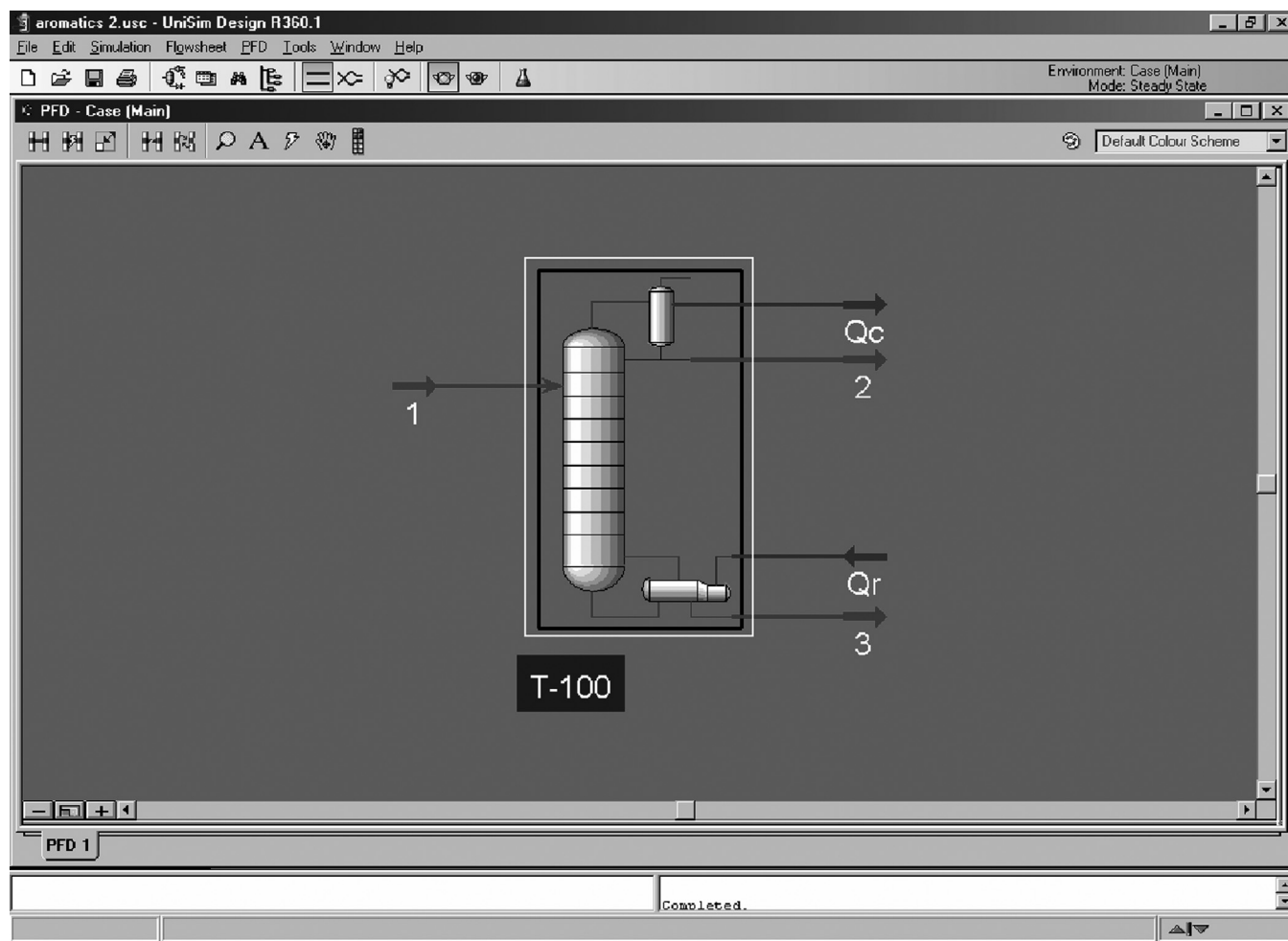


FIG. 4.20 Rigorous distillation.

Example 4.7

Continuing the problem defined in Example 4.6, use a rigorous simulation to carry out tray sizing and estimate the required column diameter.

Solution

Since we are now sizing the column, the first step is to increase the flow rate to allow for a design factor. The process design basis is 225 metric tons per hour of feed. The equipment design should include at least a 10% safety factor, so the equipment design basis was set at 250 metric tons per hour of feed (rounding up from 247.5 for convenience). This flow rate is used in simulating the process for the purpose of sizing equipment, but energy consumption must be based on the reboiler and condenser duties expected for a 225 t/h feed rate.

Fig. 4.20 shows the rigorous column simulation. UniSim Design allows the designer to enter any two specifications for the column, so instead of entering the reflux ratio as a specification, we can enter the required recoveries and provide the value of reflux ratio found in the shortcut model as an initial estimate, as shown in Fig. 4.21.

The column converges quickly with the good estimate provided from the shortcut model. The column profiles can be checked by selecting the “performance” tab in the column environment and then selecting “plots” from the menu on the left and “composition” from the list of possible plots, as shown in Fig. 4.22. This generates composition profiles like those presented in Figs. 4.10 to 4.15.

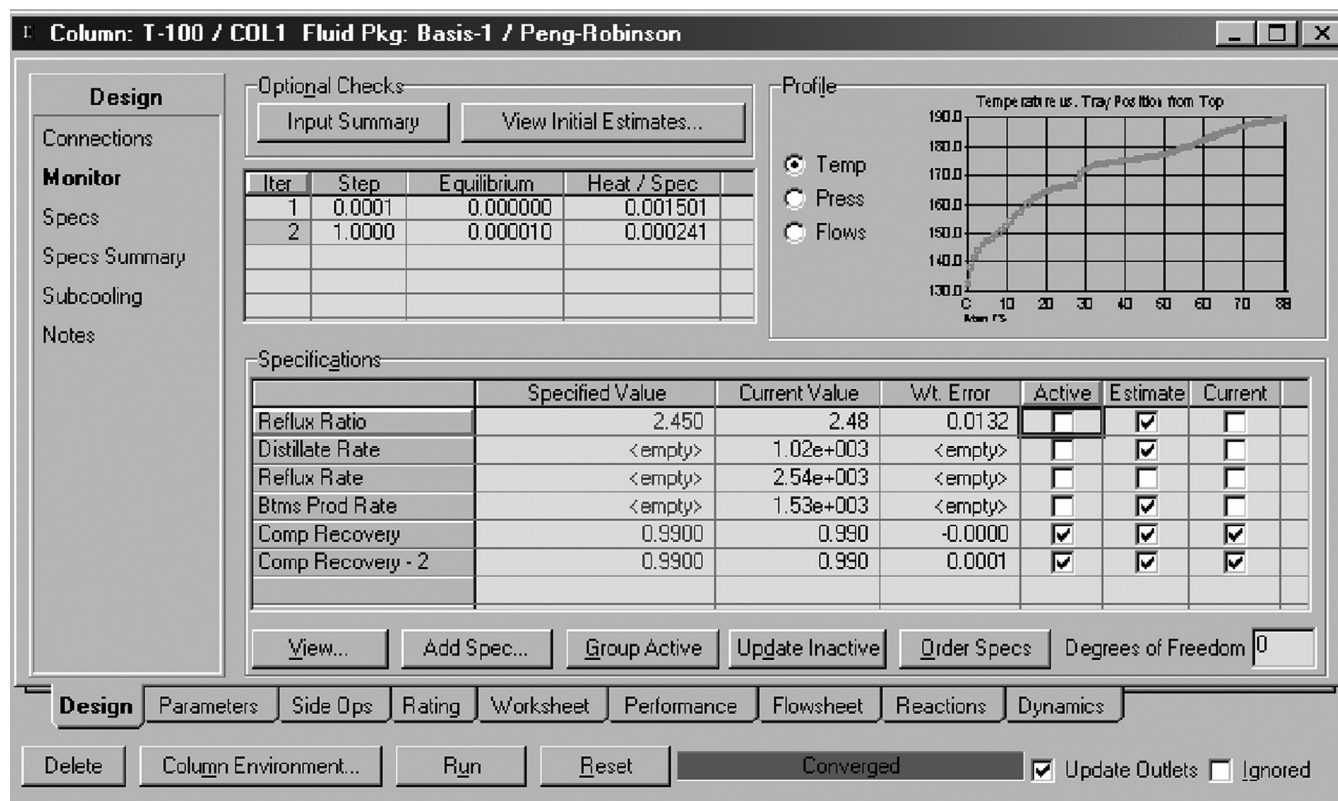


FIG. 4.21 Rigorous column specifications.

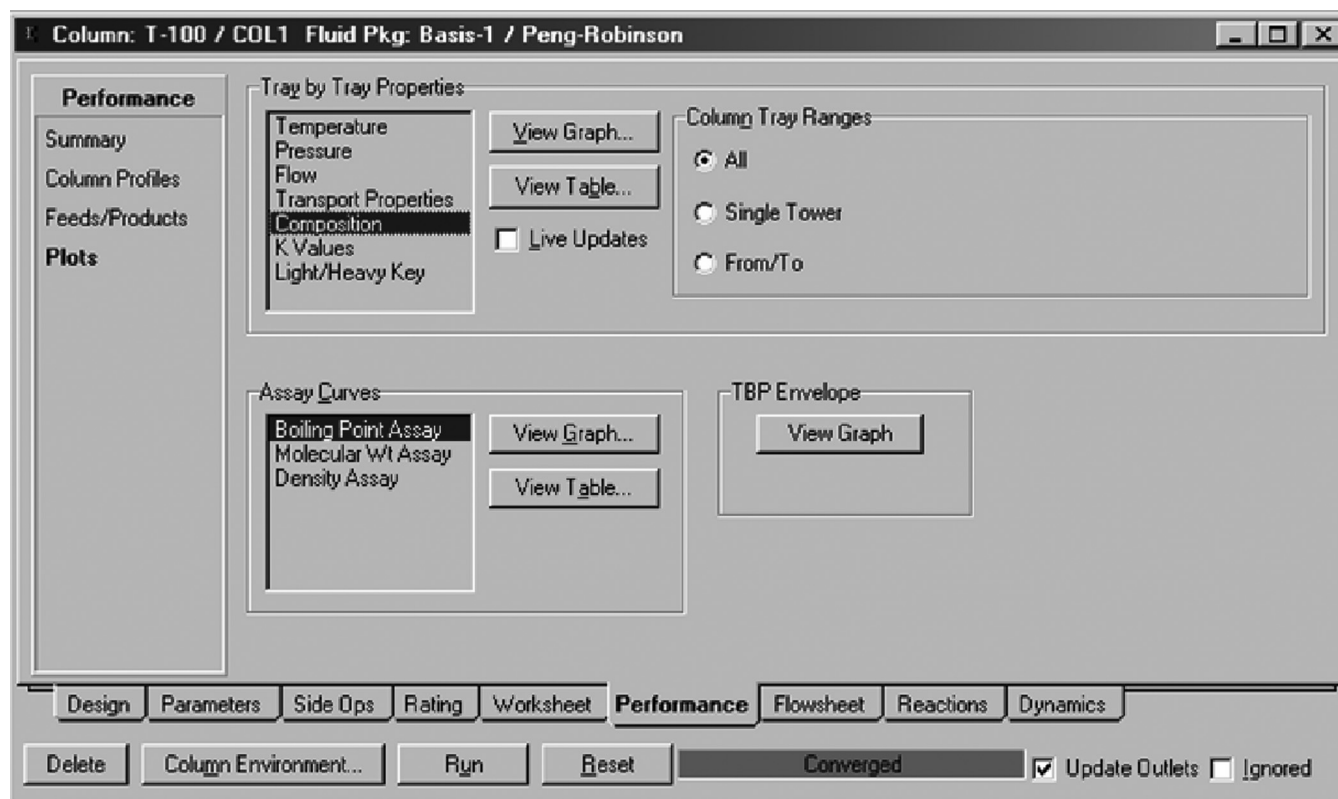


FIG. 4.22 Generating column profiles.

Tray Sizing: Tray Sizing-1

Section Name	Section_1
Start Tray	1_Main TS
End Tray	26_Main TS
Internals	Sieve
Mode	Design
Number of Flow Paths	
Section Diameter [m]	
Tray For Properties	
Tray Spacing [mm]	609.6
Tray Thickness [mm]	3.175
Foaming Factor	1.000
Max Delta P (ht of liq) [mm]	203.2
Max Flooding [%]	85.00
Packing Correlation	
HETP [m]	
Packing Type	<empty>
Riser Hole Area [%]	

Sieve Tray Flooding Method: Minimum Csb

Section Property Mode: First Tray in Section

Buttons: Delete, OK, Ignored

FIG. 4.23 Default tray sizing options.

To size the trays in UniSim Design, the tray sizing utility must be activated (from the tools menu via tools/utilities/tray sizing). When sieve trays are selected with the default spacing of 609.6 mm (2 ft) and the other default parameters shown in Fig. 4.23, then the results in Fig. 4.24 are obtained. The column diameter is found to be 4.42 m (14.5 ft).

The data on column size, number of trays, reboiler, and condenser duty can then be extracted from the simulation and put into a cost model or spreadsheet to carry out optimization of the total annual cost of production. The results of the optimization are described in Example 12.1.

4.5.3 Other separations

Other multistage vapor–liquid separations such as absorption and stripping can be modeled using variations of the rigorous distillation models, as can multistage liquid–liquid extraction.

Single-stage liquid–liquid or vapor–liquid separation can be modeled as a flash vessel, but some caution is needed. The simulation programs assume perfect separation in a flash unless the designer specifies otherwise. If there is entrainment of droplets or bubbles, the outlet compositions of a real flash vessel will be different from those

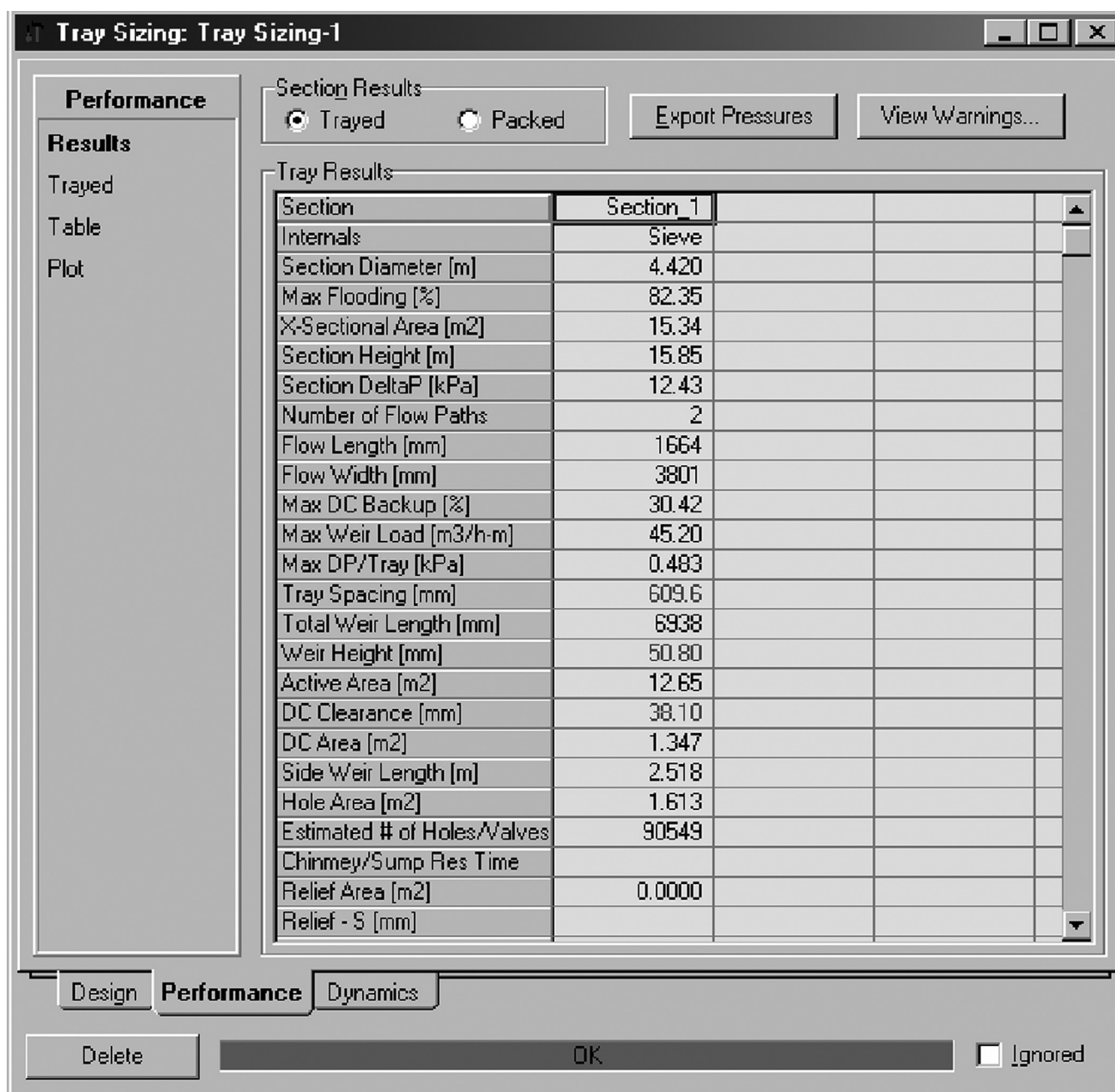


FIG. 4.24 Tray sizing results.

predicted by the simulation. If the flash is critical to process performance, the designer should make an allowance for entrainment. Most of the simulation programs allow the designer to specify a fraction of each phase that is entrained with the other phases. This is illustrated in Fig. 4.25, which shows the data entry sheet for entrained flows for UniSim Design. In UniSim Design, the entrained fractions are entered on the “Rating” tab of the flash model window. Users can also use built-in correlation models with their specified information such as vessel dimensions and nozzle locations. More sophisticated real separator modeling can be found in the three-phase separator model in UniSim Design. The fraction that is entrained depends on the design of the vessel, as described in Chapter 16.

Most of the simulators contain several models for fluid–solid separation. These models can be used to manipulate the particle size distribution when solids are present.

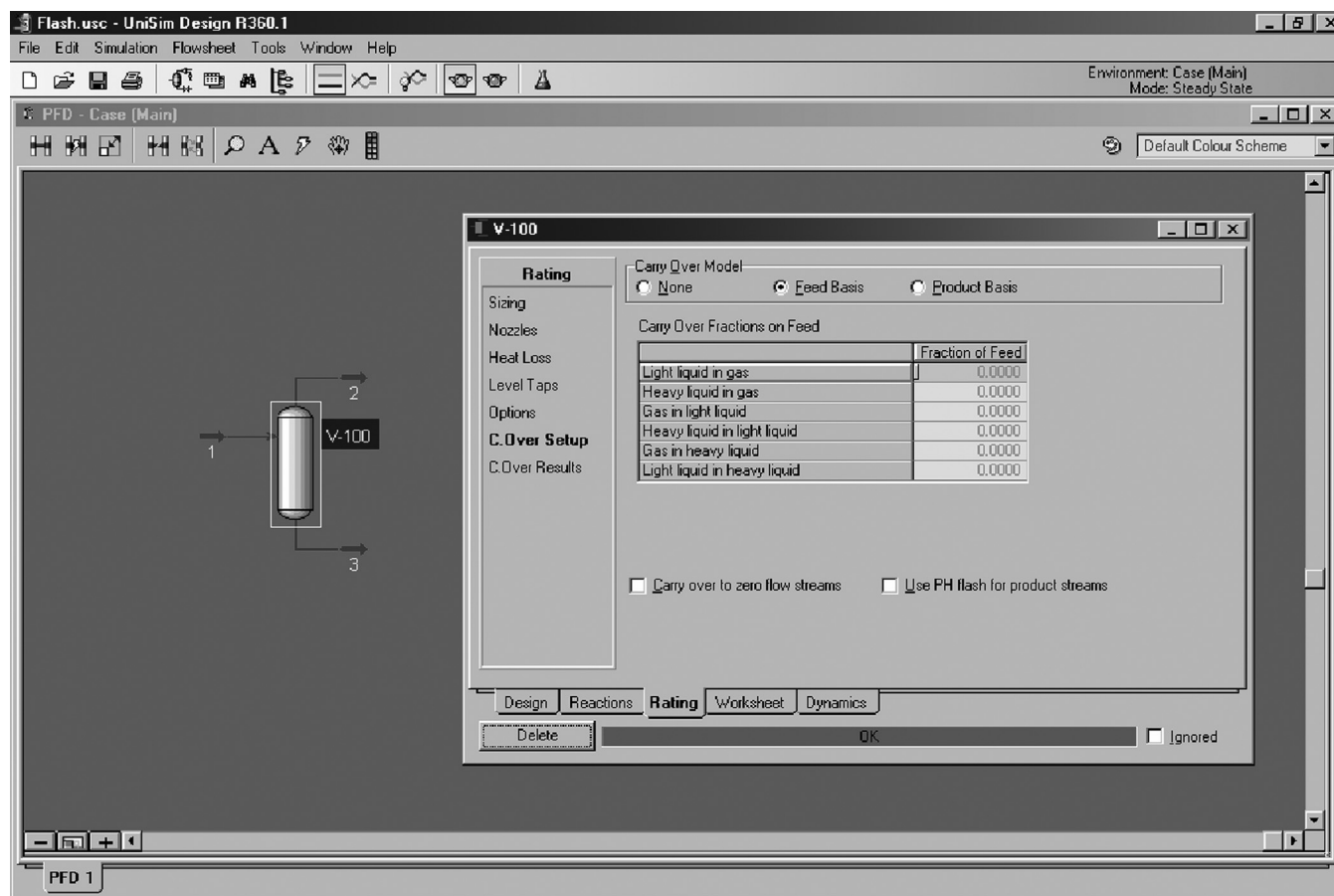


FIG. 4.25 Flash model with entrainment.

None of the commercial process simulators contains a good library model for adsorptive separations or membrane separations at the time of writing. These separation methods are important for gas–gas separations, chromatographic separations, and size exclusion or permeation-based separations and are described in more detail in [Chapter 16](#). All of these processes must be modeled using component splitters, as described next.

Component splitter models

A component splitter is a subroutine in the simulation that allows a set of components from a stream to be transferred into another stream with a specified recovery. Component splitters are convenient for modeling any separation process that cannot be described using one of the library models. Examples of real operations that are usually modeled as component splitters include:

- Pressure swing adsorption
- Temperature swing adsorption
- Chromatography
- Simulated moving bed adsorption
- Membrane separation
- Ion exchange
- Guard beds (irreversible adsorption)

When a component splitter is used in a model, it is a good practice to give the splitter a label that identifies the real equipment that is being modeled.

Component splitters are sometimes used in place of distillation columns when building simple models to provide initial estimates for processes with multiple recycles. There is little advantage to this approach compared with using

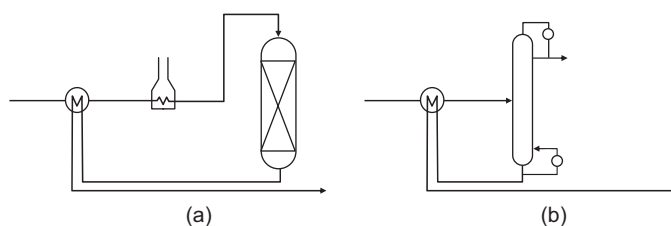


FIG. 4.26 Common feed heating schemes. (a) Feed-effluent exchange. (b) Feed-bottoms exchange.

shortcut distillation models, as the component splitter will not calculate the distribution of nonkey components unless a recovery is entered for each. Estimating and entering the recoveries for every component is difficult and tedious, and poor estimates of recoveries can lead to poor estimates of recycle flows, so the use of component splitters in this context effectively adds another layer of iteration to the model.

4.5.4 Heat exchange

All of the commercial simulators include models for heaters, coolers, heat exchangers, fired heaters, and air coolers. The models are easy to configure, and the only inputs that are usually required on the process side are the estimated pressure drop and either the outlet temperature or the duty. A good initial estimate of pressure drop is 0.3 to 0.7 bar (5 to 10 psi).

The heater, cooler, and heat exchanger models allow the design engineer to enter estimates of film transfer coefficients and hence calculate the exchanger area. As with distillation columns, the designer must remember to add a design factor to the sizes predicted by the model. Design factors are discussed in Section 1.6.

Problems often arise when using heat exchanger models to simulate processes that have a high degree of process-to-process heat exchange. Whenever a process-to-process heat exchanger is included in a simulation, it sets up an additional recycle of information; consequently, an additional loop must be converged. A common situation is where the effluent from a reactor or the bottoms from a distillation column is used to pre-heat the reactor or column feed, as illustrated in Fig. 4.26. If these process flow schemes are simulated using heat exchangers, a recycle of energy is set up between the product and the feed. This recycle must be converged every time the flowsheet is calculated (i.e., at every iteration of any other recycle loop in the process). If more than a few of these exchangers are present, the overall flowsheet convergence can become difficult.

Instead, it is usually a good practice to model the process using only heaters and coolers and then set up subproblems to model the heat exchangers. This facilitates data extraction for pinch analysis, makes it easier for the designer to recognize when exchangers might be internally pinched or have low F factors (see Chapter 19), and improves convergence.

Another problem that is often encountered when simulating heat exchangers and heat-exchange networks is temperature cross. A temperature cross occurs when the cold stream outlet temperature is hotter than the hot stream outlet temperature (see Section 19.6). When temperature cross occurs, many types of shell and tube heat exchangers give a very poor approximation of counter-current flow, and consequently have low F factors and require large surface areas. In some of the commercial simulation programs, the heat exchanger models will indicate if the F factor is low. If this is the case, the designer should split the exchanger into several shells in series so that temperature cross is avoided. Some of the simulation programs allow the designer to plot profiles of temperature versus heat flow in the exchanger. These plots can be useful in identifying temperature crosses and internal pinches.

Example 4.8

A mixture of 100 kgmol/h of 80 mol% benzene and 20 mol% ethylene at 40 °C and 100 kPa is fed to a feed effluent exchanger, where it is heated to 300 °C and fed to a reactor. The reaction proceeds to 100% conversion of ethylene, and the reactor products are withdrawn, cooled by heat exchange with the feed, and sent for further processing. Estimate the outlet temperature of the product after heat exchange and the total surface area required if the average heat transfer coefficient is 200 Wm⁻²K⁻¹.

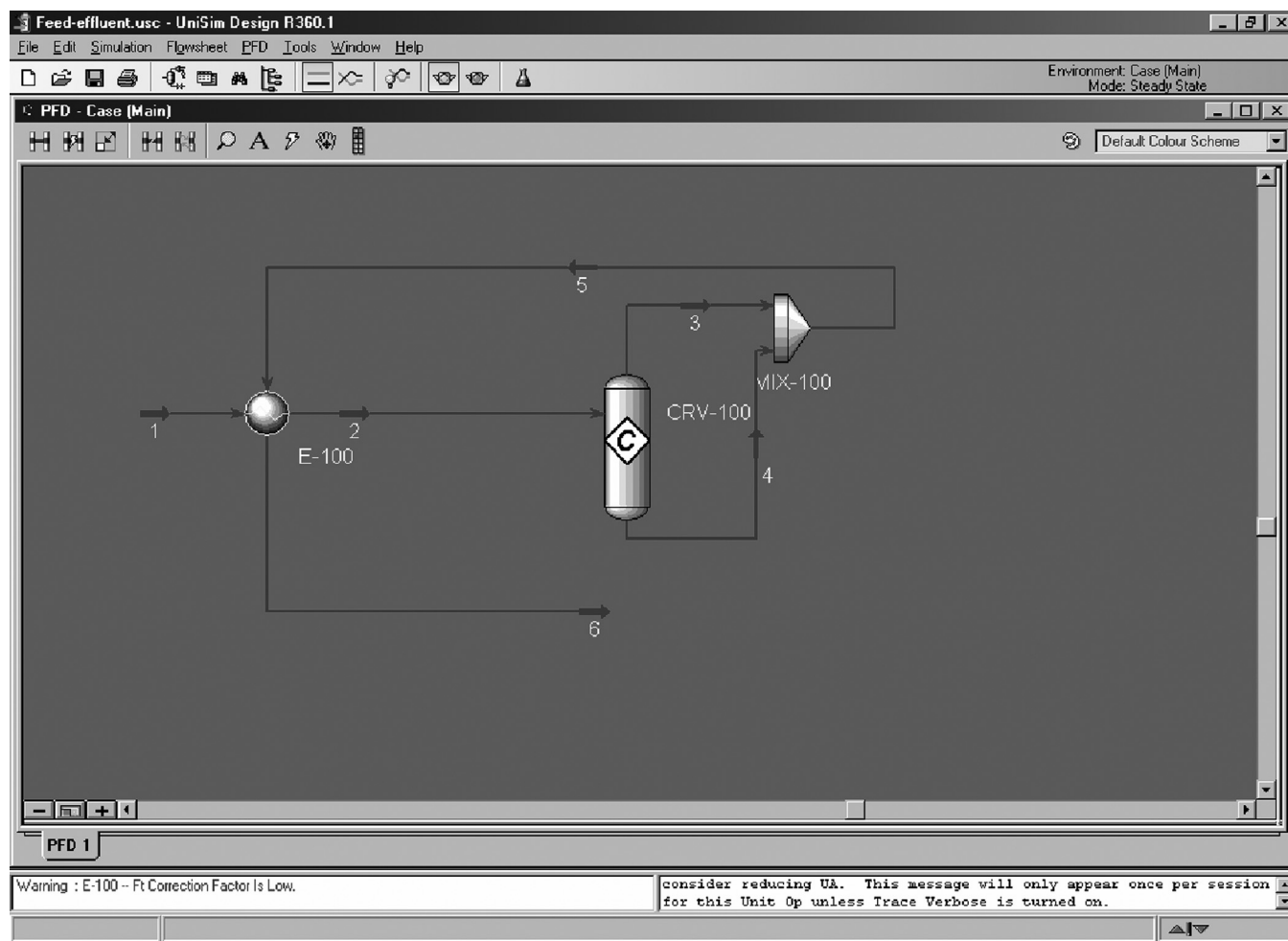


FIG. 4.27 Feed-effluent heat exchange model for Example 4.8.

Solution

This problem was solved using UniSim Design. The reaction goes to full conversion, so a conversion reactor can be used. The simulation model is shown in Fig. 4.27.

When the temperature at the outlet of the exchanger on the feed side is specified, the duty of the exchanger is defined and there is no recycle of information. The model thus solves very quickly, but it is necessary to check the results to see that the exchanger design makes sense.

The outlet temperature of the product (stream 6) is found to be 96.9 °C, so there is enough heat in the product mixture to give an approach temperature of nearly 60 °C, which seems perfectly adequate. If we open the exchanger worksheet, though, there is a warning that the F factor is too low. Fig. 4.28 shows the exchanger worksheet, and the F factor is only 0.2, which is not acceptable. When we examine the temperature–heat duty plot shown in Fig. 4.29 (generated from the Performance tab of the exchanger worksheet), it is clear that there is a substantial temperature cross. This temperature cross causes the exchanger to have such a low F factor and gives a UA value of $78.3 \times 10^3 \text{ WK}^{-1}$, where U is the overall heat transfer coefficient in $\text{Wm}^{-2}\text{K}^{-1}$ and A is the area in m^2 .

If $UA = 78.3 \times 10^3 \text{ WK}^{-1}$ and $U = 200 \text{ Wm}^{-2}\text{K}^{-1}$, then the exchanger area is $A = 392 \text{ m}^2$. This would be a feasible size of exchanger, but is large for the duty and is not acceptable because of the low F factor. We should add more shells in series.

By examining the temperature–heat duty plot in Fig. 4.29, we can see that if we break the exchanger into two shells, with the first shell heating the feed up to the dew point (the kink in the lower curve), then the first shell will not have a temperature cross. This design corresponds to an outlet temperature of about 70 °C for the first exchanger. The second exchanger would

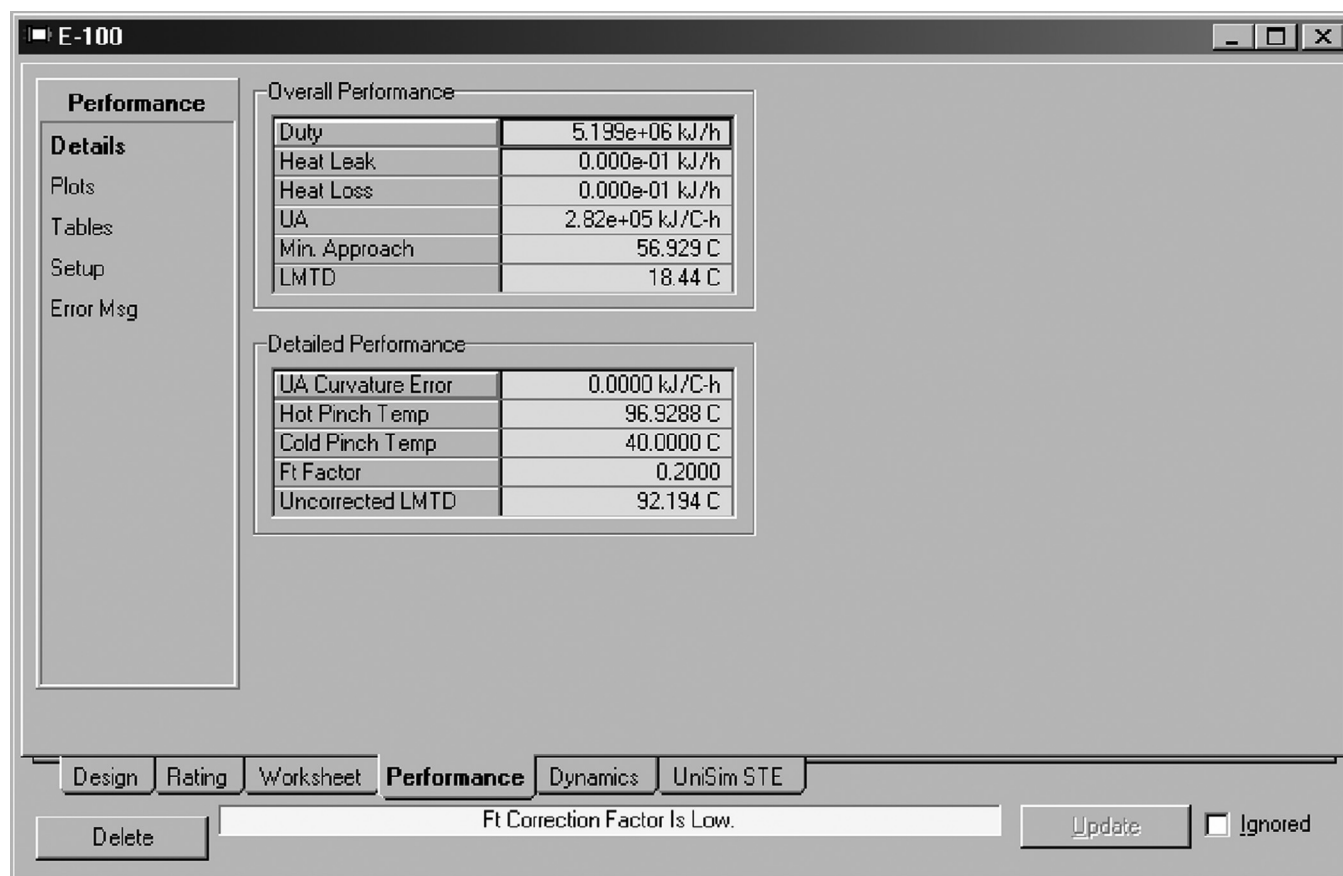


FIG. 4.28 Exchanger worksheet for a single-shell design.

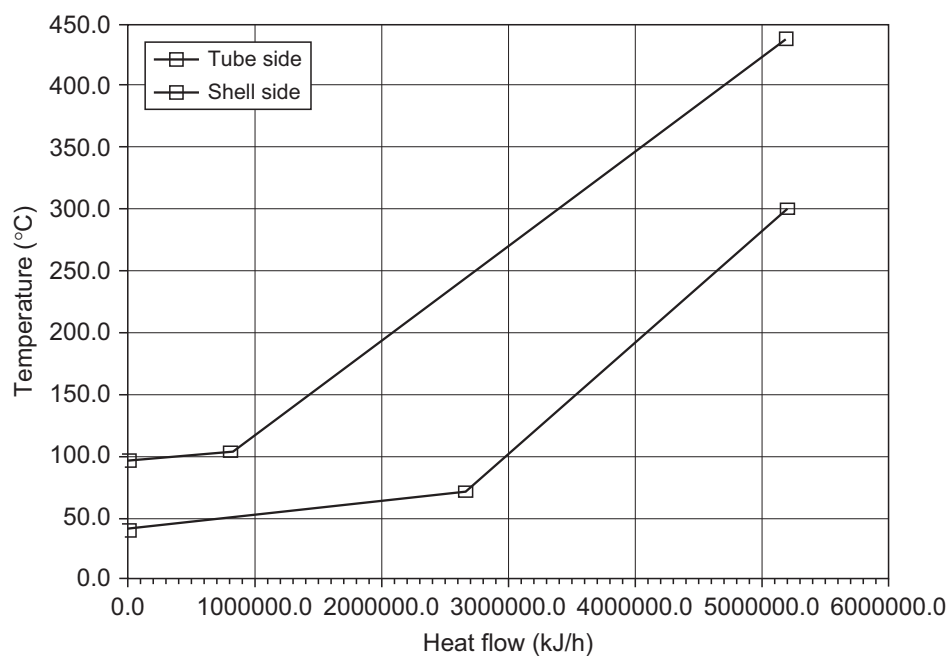


FIG. 4.29 Temperature-heat flow plot for a single-shell design.

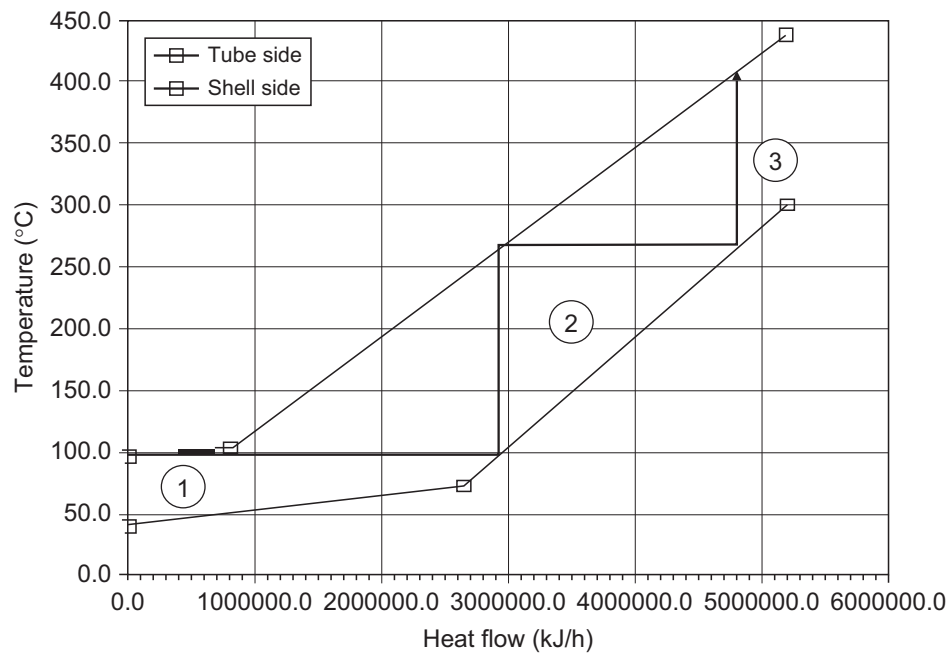


FIG. 4.30 Stepping between heat profiles to avoid temperature cross.

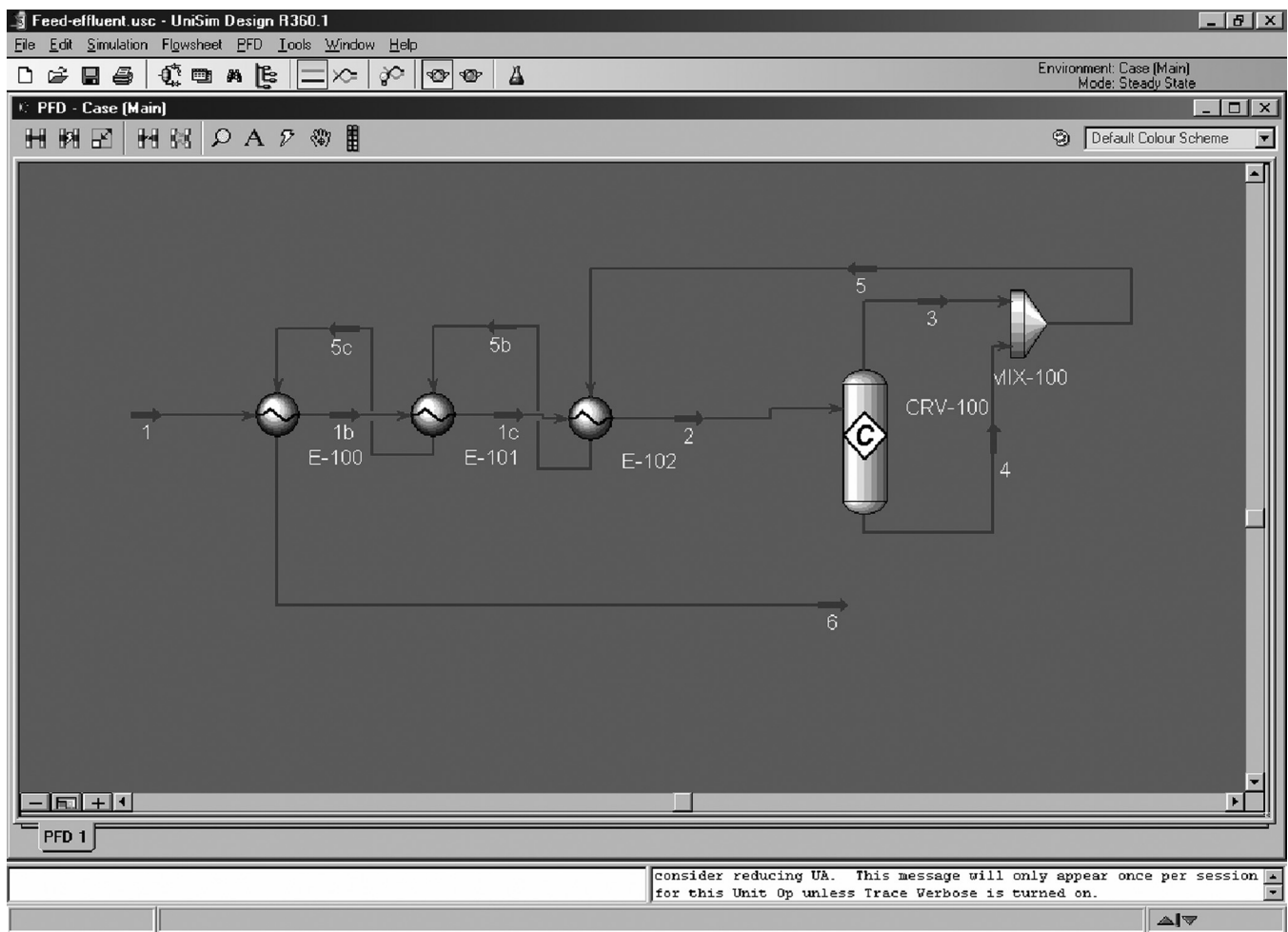


FIG. 4.31 Feed-effluent heat exchange with three shells in series.

still have a temperature cross though. If we break this second exchanger into two more exchangers, then the temperature cross is eliminated. We thus need at least three heat exchangers in series to avoid the temperature cross. This result could have been obtained by “stepping off” between the temperature–heat duty plots, as illustrated in Fig. 4.30.

Fig. 4.31 shows a modified flowsheet with two additional heat exchangers added in series. The outlet temperature of the second exchanger was specified as 200 °C to divide the duty of the second and third exchangers roughly equally. The results are given in Table 4.7. Temperature–heat flow plots for the three exchangers are given in Fig. 4.32.

The modified design achieved a reduction in surface area from 392 m² to 68 m² at the price of having three shells instead of the original one. More importantly, the modified design is more practical than the original design and is less likely to suffer from internal pinch points. The modified design is not yet optimized. Optimization of this problem is explored in Problem 4.11.

TABLE 4.7 Heat exchanger results

Design case	Original (single shell)		Modified (multiple shell)	
Exchanger	E100	E100	E101	E102
Duty (MW)	1.44	0.53	0.57	0.35
UA (W/K)	78,300	6310	4780	2540
F	0.2	0.93	0.82	0.93
$\Delta T_{\min}$	56.9	56.9	134.3	139.7
ΔT_{lmtd}	18.4	83.6	118.7	138.4
A (m ²)	392	32	24	13
Total area (m ²)	392		68	

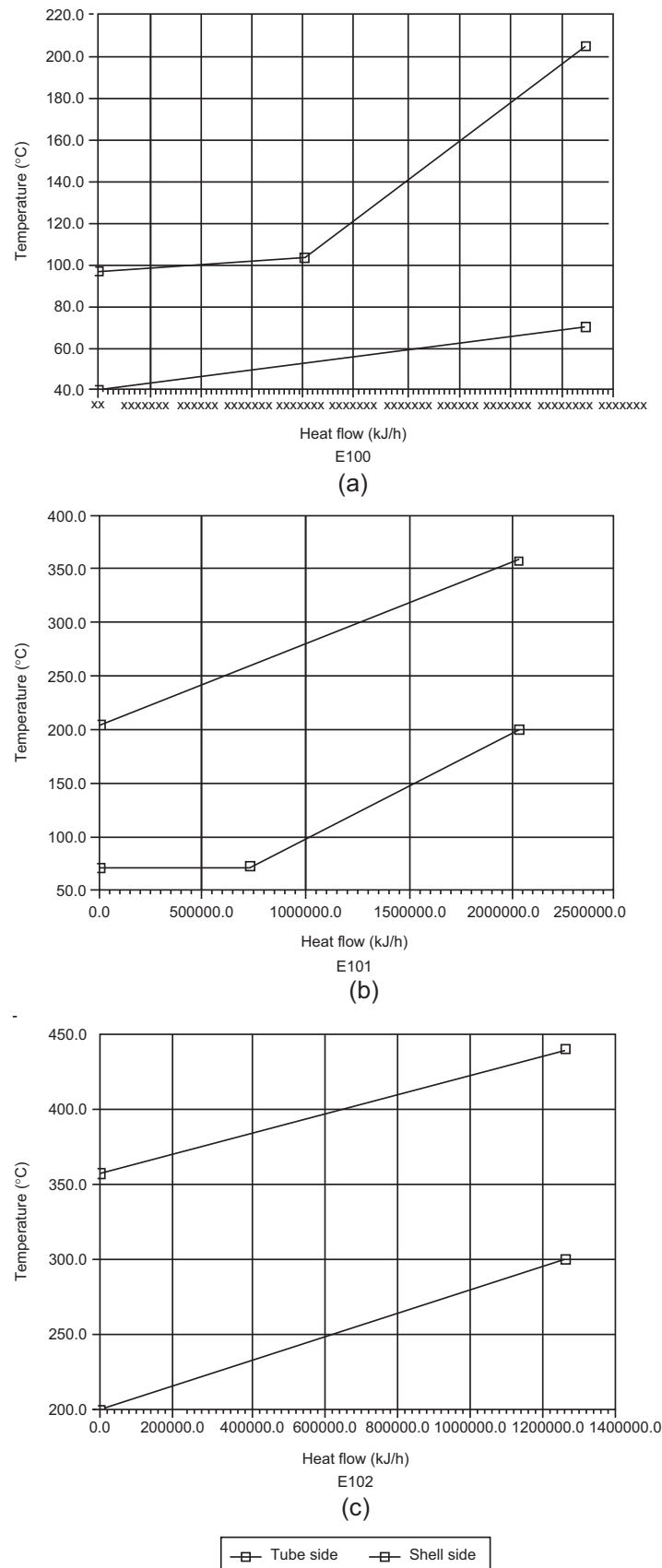


FIG. 4.32 Temperature–heat flow profiles for the three exchangers in series (a) E100, (b) E101, and (c) E102.

4.5.5 Hydraulics

Most of the commercial simulation programs contain models for valves, pipe segments, tees, and elbows. These models can be used to make an initial estimate of system pressure drop for the purposes of sizing pumps and compressors.

If a process hydraulic model is built, care must be taken to specify pressure drop properly in the unit operation models. Rules of thumb are adequate for initial estimates, but in a hydraulic model these should be replaced with rigorous pressure drop calculations. Sufficient allowance must be made for pressure drop across control valves, as discussed in [Chapters 5 and 20](#).

A hydraulic model will not be accurate unless some consideration has been given to plant layout and piping layout. Ideally, the hydraulic model should be built after the piping isometric drawings have been produced, when the designer has a good idea of pipe lengths and bends. The designer should also refer to the piping and instrumentation diagram for isolation valves, flow meters, and other obstructions that cause increased pressure drop. These subjects are discussed in [Chapter 5](#) and [Chapter 20](#).

Care is needed when modeling compressible gas flows, flows of vapor–liquid mixtures, slurry flows, and flows of nonnewtonian liquids. Some simulators use different pipe models for compressible flow. The prediction of pressure drop in multiphase flow is inexact at best and can be subject to very large errors if the extent of vaporization is unknown. In most of these cases, the simulation model should be replaced by a computational fluid dynamics (CFD) model of the important parts of the plant.

4.5.6 Solids handling

The commercial simulation programs were originally developed mainly for petrochemical applications, and none of them has a complete set of solids-handling operations. Although models for filters, crystallizers, decanters, and cyclones are present in most of the simulators, the designer may have to add user models for operations such as:

• hoppers	• washers	• crushers and pulverizers
• belt conveyors	• flocculators	• jet mills
• elevators	• spray driers	• ball mills
• pipe conveyors	• prill towers	• agglomerators
• screw conveyors	• rotary driers	• granulators
• kneaders	• rotary kilns	• tableting presses
• extruders	• belt driers	• paper machines
• slurry pumps	• centrifuges	• classifiers
• fluidized bed heaters	• falling film evaporators	• electrostatic precipitators
• fluidized bed reactors	• moving bed reactors	

Because solids are handled in many commodity chemical processes as well as pharmaceuticals, polymers, and biological processes, the simulation software vendors are under pressure from their customers to enhance the capability of the programs for modeling solids operations. This continues to be an area of evolution of the commercial software.

4.6 User models

When the design engineer needs to specify a unit operation that is not represented by a library model and cannot be approximated by a simple model such as a component splitter or a combination of library models, then it is necessary to construct a user model. All of the commercial simulators allow the user to build add-in models of varying sophistication.

4.6.1 Spreadsheet models

Models that require no internal iteration are easily coded as spreadsheets. Most of the simulators offer some degree of spreadsheet capability, ranging from simple calculation blocks to full Microsoft Excel functionality.

In UniSim Design, spreadsheets can be created by selecting the spreadsheet option on the unit operations palette. The spreadsheet is easy to configure and allows data to be imported from streams and unit operations. The functionality of the UniSim Design spreadsheet is rather basic at the time of writing, but is usually adequate for simple input-output models. Values calculated by the spreadsheet can be exported back to the simulation model. The spreadsheet can thus be set up to act as a unit operation. The use of a spreadsheet as a unit operation is illustrated in Example 4.9. Aspen Plus has a similar simple spreadsheet capability using Microsoft Excel, which can be specified as a calculator block (via Data/Flowsheet Options/Calculator). The Excel calculator block in Aspen Plus requires a little more time to configure than the UniSim Design spreadsheet, but at the time of writing it can perform all of the functions available in Microsoft as discussed in Chapter 5 and Chapter 20.

For more sophisticated spreadsheet models, Aspen Plus allows the user to link a spreadsheet to a simulation via a user model known as a USER2 block. The designer can create a new spreadsheet or customize an existing spreadsheet to interact with an Aspen Plus simulation. The USER2 block is much easier to manipulate when handling large amounts of input and output data, such as streams with many components or unit operations that involve multiple streams. The procedure for setting up a USER2 Microsoft Excel model is more complex than using a calculator block, but avoids having to identify every number required from the flowsheet individually. Instructions on how to build USER2 spreadsheet models are given in the Aspen Plus manuals and online help ([Aspen Technology, 2001](#)).

4.6.2 User subroutines

Models that require internal convergence are best written as subroutines rather than spreadsheets, as more efficient solution algorithms can be used. Most user subroutines are written in FORTRAN or Visual Basic, though some of the simulators allow other programming languages to be used.

It is generally a good practice to compile and test a user model in a simplified flowsheet or as a stand-alone program before adding it to a complex flowsheet with recycles. It is also a good practice to check the model carefully over a wide range of input values, or else constrain the inputs to ranges where the model is valid.

Detailed instructions on how to write user models to interface with commercial simulation programs can be found in the simulator manuals. The manuals also contain specific requirements for how the models should be compiled and registered as extensions or shared libraries (.dll files in Microsoft Windows). In Aspen Plus, user models can be added as USER or USER2 blocks, following the instructions in the Aspen Plus manuals. In UniSim Design, it is easy to add user models using the User Unit Operation, which can be found on the object palette or under the Flowsheet/Add Operation menu. The UniSim Design User Unit Operation can be linked to any program without requiring an extension file to be registered. The User Unit Operation is not documented in the UniSim Design manual, but instructions on setting it up and adding code are given in the online help.

Example 4.9

A gas turbine engine is fueled with 3000 kg/h of methane at 15 °C and 1000 kPa and supplied with ambient air at 15 °C. The air and fuel are compressed to 2900 kPa and fed to a combustor. The air flow rate is designed to give a temperature of 1400 °C at the outlet of the combustor. The hot gas leaving the combustor is expanded in the turbine. Shaft work produced by the turbine is used to power the two compressors and run a dynamo for generating electricity.

If the efficiency of the compressors is 98% and that of the turbine is 88% and 1% of the shaft work is lost due to friction and losses in the dynamo, estimate the rate of power production and the overall cycle efficiency.

Solution

This problem was solved using UniSim Design.

A gas turbine engine should run with a large excess of air to provide full combustion of the fuel so the combustor can be modeled as a conversion reactor. There is no model for a dynamo in UniSim Design, so the dynamo and shaft losses can be modeled using a spreadsheet operation, as shown in [Fig. 4.33](#).

[Fig. 4.33](#) also illustrates the use of an “Adjust” controller to set the air flow rate so as to give the desired reactor outlet temperature. The specifications for the Adjust controller are shown in [Figs. 4.34 and 4.35](#). The Adjust controller was specified with a minimum air flow rate of 60,000 kg/h to ensure that the solver did not converge to a solution in which the air flow did not give full conversion of methane. The stoichiometric requirement is $3000 \times 2 \times (32 / 16) / 0.21 = 57,000$ kg/h of air.

The spreadsheet model of the dynamo is relatively simple, as illustrated in [Fig. 4.36](#). The model takes the turbine shaft work and compressor duties as inputs. The friction losses are estimated as 1% of the turbine shaft work. The friction losses

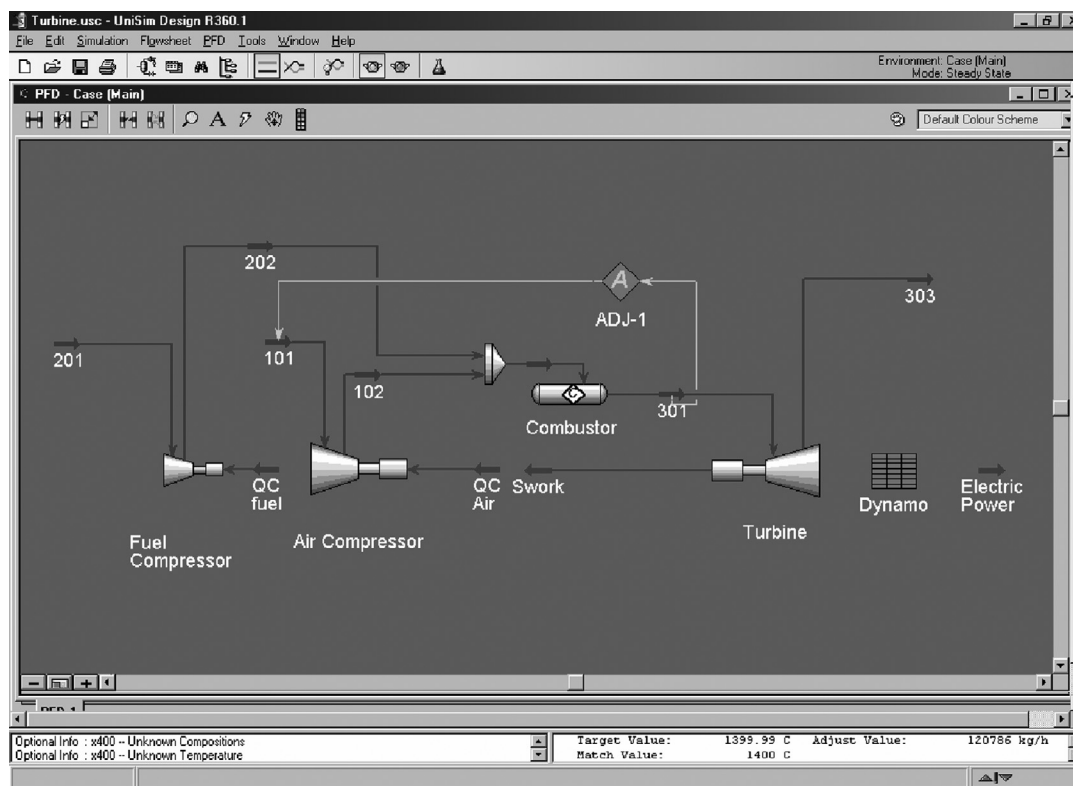


FIG. 4.33 Gas turbine model.

The screenshot shows the ADJ-1 Adjust specifications dialog box. The dialog is titled "ADJ-1" and has the following sections and controls:

- Adjust Name:** ADJ-1
- Adjusted Variable:**
 - Object: 101
 - Variable: Mass Flow
 - Select Var... button
- Target Variable:**
 - Object: 301
 - Variable: Temperature
 - Select Var... button
- Target Value:**
 - Source:
 - ☒ User Supplied
 - ☐ Another Object
 - ☐ SpreadsheetCell Object
 - Specified Target Value:
 - 1400.0000 C
 - 2552.0000 F
- Tabs:** Connections, Parameters, Monitor, User Variables
- Buttons:** OK, Delete, Ignored

FIG. 4.34 Adjust specifications.

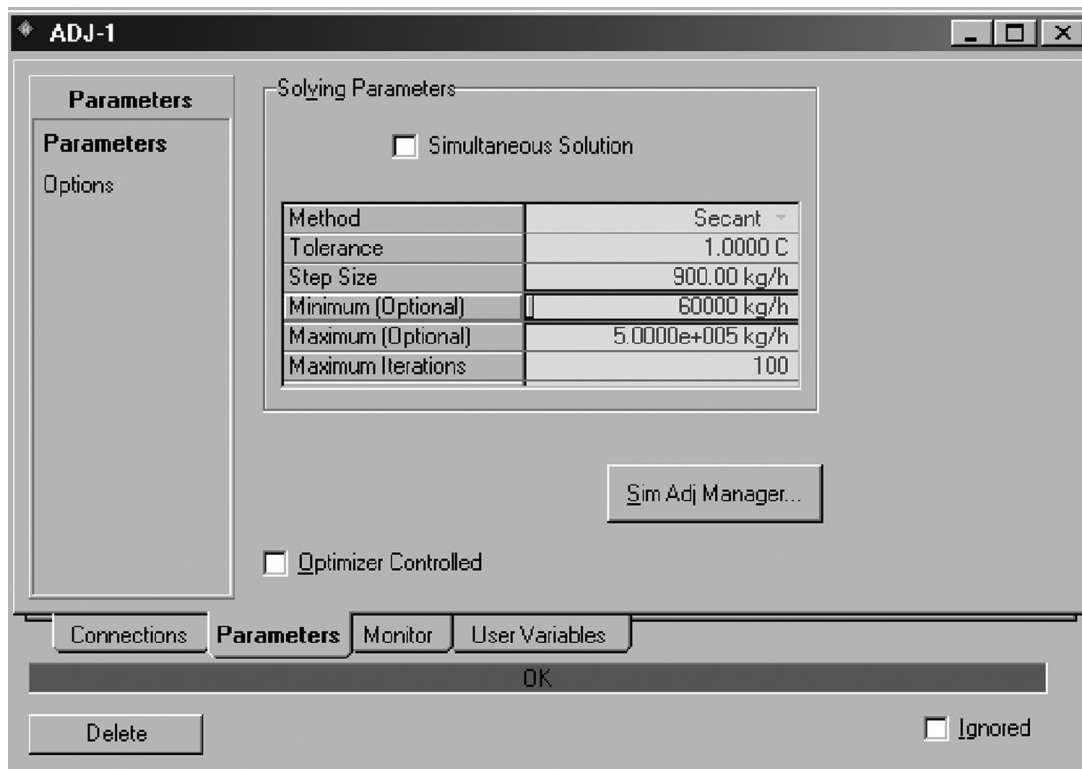


FIG. 4.35 Adjust solving parameters.

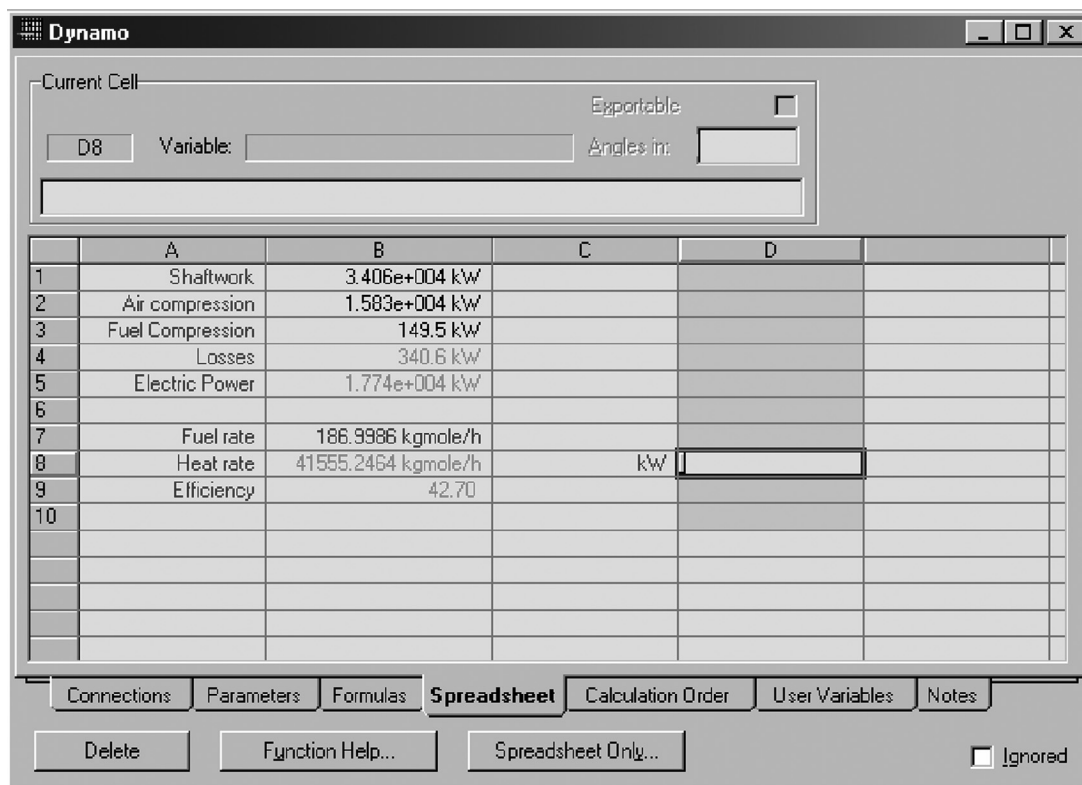


FIG. 4.36 Spreadsheet model of dynamo.

and compressor duties are then subtracted from the shaft work to give the net power from the dynamo, which is calculated to be 17.7 MW.

The cycle efficiency is the net power produced divided by the heating rate of the fuel. The heating rate is the molar flow of fuel multiplied by the standard molar heat of combustion:

$$\text{Heating rate (kW)} = \text{molar flow (mol/h)} \times \Delta H_c^\circ (\text{kJ/mol}) / 3600 \quad (4.13)$$

The cycle efficiency is calculated to be 42.7%.

4.7 Flowsheets with recycle

Recycles of solvents, catalysts, unconverted feed materials, and by-products are found in many processes. Most processes contain at least one material recycle, and some may have six or more. Furthermore, when energy is recovered by process-to-process heat transfer, energy recycles are created, as discussed in Section 4.5.4.

4.7.1 Tearing the flowsheet

For a sequential-modular simulation program to be able to solve a flowsheet with a recycle, the design engineer needs to provide an initial estimate of a stream somewhere in the recycle loop. This is known as a “tear” stream, as the loop is “torn” at that point. The program can then solve and update the tear stream values with a new estimate. The procedure is repeated until the difference between values at each iteration becomes less than a specified tolerance, at which point the flowsheet is said to be converged to a solution.

The procedure for tearing and solving a simulation can be illustrated by a simple example. Fig. 4.37 shows a process in which two feeds, A and B, are combined and fed to a fixed-bed reactor. The reactor product is sent to a stripping column to remove light ends and is then sent to a column that separates heavy product from unreacted feed B. The unreacted feed B is recycled to the reactor.

To solve the reactor model, we need to specify the reactor feeds, streams 2 and 4. Stream 4 is made by adding fresh feed stream 1 to recycle stream 3, so a logical first approach might be to make an estimate of the recycle stream, in which case stream 3 is the tear stream. Fig. 4.38 shows the flowsheet torn at stream 3. The designer provides an initial estimate of stream 3a. The flowsheet then solves and calculates stream 3b. The design engineer specifies a recycle operation connecting streams 3a and 3b, and the simulator then updates stream 3a with the values from stream 3b (or with other values if an accelerated convergence method is used, as discussed later). The calculation is then repeated until the convergence criteria are met.

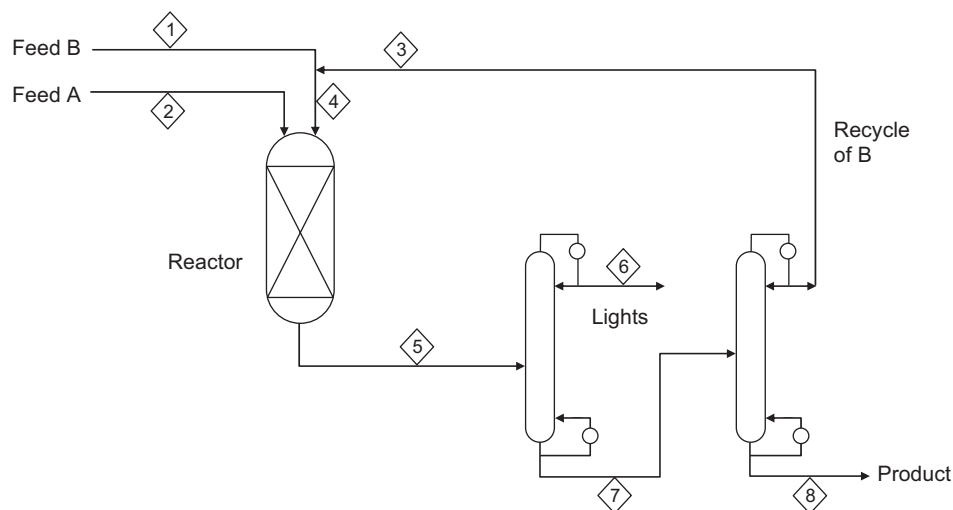
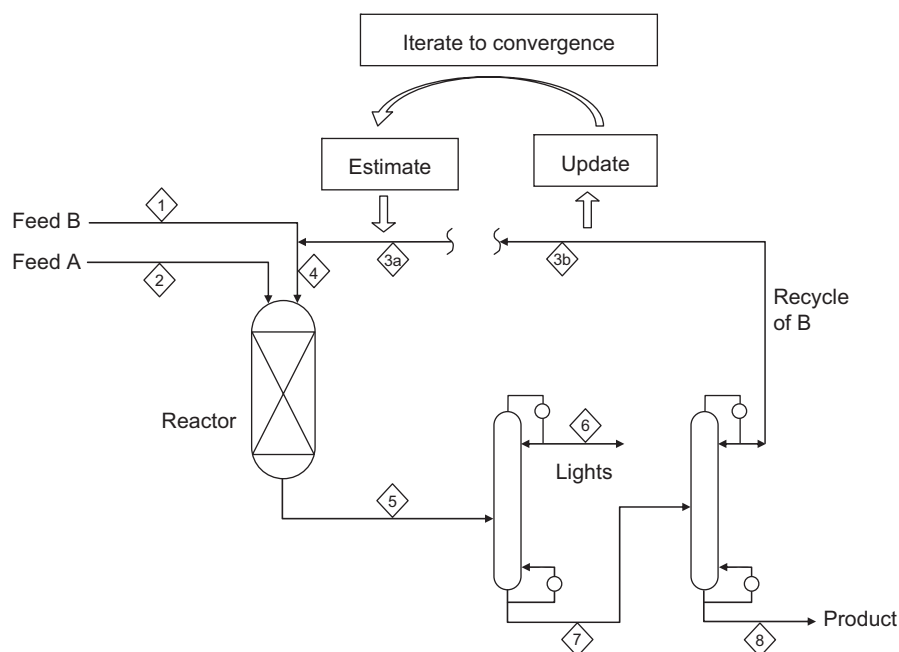


FIG. 4.37 Example process with recycle.



The choice of tear stream can have a significant impact on the rate of convergence. For example, if the process in Fig. 4.37 were modeled with a yield shift reactor, then tearing the flowsheet at stream 5 would probably give faster convergence. Some of the simulation programs automatically identify the best tear stream.

4.7.2 Convergence methods

The methods used to converge recycle loops in the commercial process simulation programs are similar to the optimization methods described in [Chapter 12](#). Most of the commercial simulation programs include the following methods.

Successive substitution (direct substitution)

In this method, an initial estimate, x_k , is used to calculate a new value of the parameter, $f(x_k)$. The estimate is then updated using the calculated value:

$$\begin{aligned} x_{k+1} &= f(x_k) \\ x_{k+2} &= f(x_{k+1}), \text{ etc.} \end{aligned} \quad (4.14)$$

This method is simple to code but is computationally inefficient and convergence is not guaranteed.

Bounded Wegstein

The bounded Wegstein method is the default method in most of the simulation programs. It is a linear extrapolation of successive substitution.

The Wegstein method initially starts out with a direct substitution step:

$$x_1 = f(x_0) \tag{4.15}$$

An acceleration parameter, q , can then be calculated:

$$q = \frac{s}{s-1} \quad (4.16)$$

$$\text{where: } s = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} \quad (4.17)$$

and the next iteration is then:

$$x_{k+1} = q x_k + (1 - q)f(x_k) \quad (4.18)$$

If $q = 0$, the method is the same as successive substitution. If $0 < q < 1$, then convergence is damped, and the closer q is to 1.0, the slower convergence becomes. If q is less than 0, then the convergence is accelerated. The bounded Wegstein method sets bounds on q , usually keeping it in the range $-5 < q < 0$, so as to guarantee acceleration without overshooting the solution too widely.

The bounded Wegstein method is usually fast and robust. If convergence is slow, then the designer should consider reducing the bounds on q . If convergence oscillates, then consider damping the convergence by setting bounds such that $0 < q < 1$.

Newton and quasi-Newton methods

The Newton method uses an estimate of the gradient at each step to calculate the next iteration, as described in Section 12.7.4. Quasi-Newton methods such as the Broyden method use linearized secants rather than gradients. This approach reduces the number of calculations per iteration, although the number of iterations may be increased.

Newton and quasi-Newton methods are used for more difficult convergence problems, for example, when there are many recycle streams or many recycles that include operations that must be converged at each iteration, such as distillation columns. The Newton and quasi-Newton methods are also often used when there are many recycles and control blocks (see Section 4.8.1). The Newton method should not normally be used unless the other methods have failed, as it is more computationally intensive and can be slower to converge for simple problems.

4.7.3 Manual calculations

The convergence of recycle calculations is almost always better if a good initial estimate of the tear stream is provided.

If the tear stream is chosen carefully, it may be easy for the design engineer to generate a good initial estimate. This can be illustrated by returning to the problem of Fig. 4.37. We can tear the recycle loop at the reactor effluent, as shown in Fig. 4.39. We can then state the following about the reactor effluent:

1. The reactor effluent must contain the net production rate of product (which is known), plus any product that is in the recycle. Recycling product to the reactor is not a good idea, as it is likely to lead to by-product formation. A reasonable estimate of product recovery in the separation section is probably 99% or greater, so a good initial estimate of the amount of product in stream 5b is the net production rate divided by the separation recovery, or roughly 101% of the net production rate.
2. Because feed B is recycled and feed A is not, it looks like we are using an excess of B to drive full conversion of A. A good initial estimate of the flow rate of component A in stream 5b is therefore zero. If we have conversion data in terms of A, then we could produce a better estimate.
3. Feed B is supplied to the reactor in excess. The amount of B consumed in the reactor must be equal to the amount required by stoichiometry to produce the product. The amount of B remaining in the reactor effluent is given by:

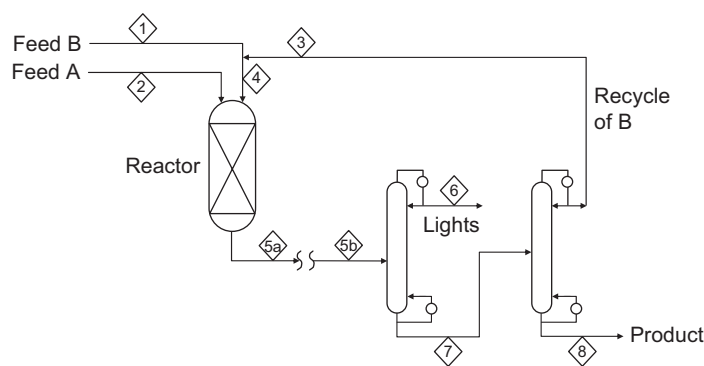


FIG. 4.39 Tearing at the reactor outlet.

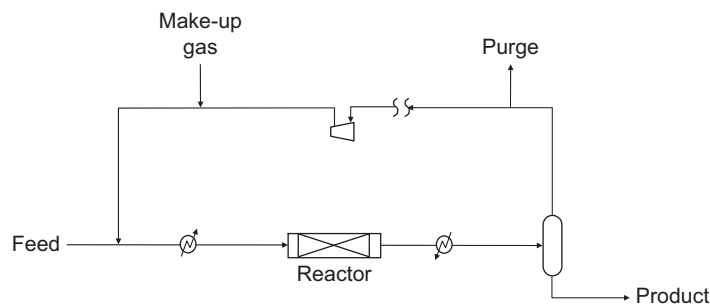


FIG. 4.40 Process with gas recycle and purge.

$$\begin{aligned} \text{moles B remaining per mole product} &= \frac{\text{moles B fed}}{\text{stoichiometric moles B per mole product}} - 1 \\ &= \frac{1}{\text{conversion of B}} - 1 \end{aligned} \quad (4.19)$$

So, knowing the flow rate of product, we can get a good initial estimate of the flow rate of B if we know either the conversion of B or the ratio in excess of the stoichiometric feed rate of B that we want to supply.

We can thus make good estimates of the three major components that are present in stream 5b. If light or heavy by-products are formed in the reactor but not recycled, then a single successive substitution step will provide good estimates for these components, as well as a better estimate of the conversion of B and the amount of A that is required in excess of stoichiometric requirements.

Manual calculations are also useful when solving flowsheets that use recycle and purge. Purge streams are often withdrawn from recycles to prevent the accumulation of species that are difficult to separate, as described in Section 2.3.4. A typical recycle and purge flow scheme is illustrated in Fig. 4.40. A liquid feed and a gas are mixed, heated, reacted, cooled, and separated to give a liquid product. Unreacted gas from the separator is recycled to the feed. A make-up stream is added to the gas recycle to make up for consumption of gas in the process. If the make-up gas contains any inert gases, over time these would accumulate in the recycle, and eventually the reaction would be slowed down when the partial pressure of reactant gas fell. To prevent this situation from occurring, we withdraw a purge stream to maintain the inerts at an acceptable level. We can provide a good initial estimate of the recycle stream by noting:

1. The flow rate of inerts in the purge is equal to the flow rate of inerts in the make-up gas.
2. The required partial pressure of reactant gas at the reactor outlet sets the concentration of reactant gas and inerts in the recycle and the unconverted gas flow rate if the reactor pressure is specified.

We can then write a mass balance on inerts:

$$F_M y_M = F_P y_R \quad (4.20)$$

and on reactant gas:

$$F_M(1 - y_M) = G + F_P(1 - y_R) \quad (4.21)$$

hence:

$$F_M(1 - y_M) = G + F_M \frac{y_M}{y_R}(1 - y_R)$$

where: F_M = make-up molar flow rate

F_P = purge molar flow rate

y_M = mole fraction of inerts in make-up

y_R = mole fraction of inerts in recycle and purge

G = molar rate of consumption of gas in reactor

Hence we can solve for M and P if G is known.

The temperature of the recycle gas at the outlet of the compressor is not easily estimated, so the logical place to tear the recycle is between the purge and the compressor, as indicated in Fig. 4.40.

4.7.4 Convergence problems

If a flowsheet is not converged, or if the process simulation software runs and gives a statement “converged with errors,” then the results *cannot be used* for design. The designer must take steps to improve the simulation so that a converged solution can be found.

The first steps that an experienced designer would usually take are:

1. Make sure that the specifications are feasible.
2. Try increasing the number of iterations.
3. Try a different convergence algorithm.
4. Try to find a better initial estimate.
5. Try a different tear stream.

If one or more unit operations have been given infeasible specifications, then the flowsheet will never converge. This problem also occurs with multicomponent distillation columns, particularly when purity specifications or flow rate specifications are used or when nonadjacent key components are chosen; see Section 17.6. A quick manual mass balance around the column can usually determine whether the specifications are feasible. Remember that all the components in the feed must exit the column somewhere. The use of recovery specifications is usually more robust, but care is still needed to make sure that the reflux ratio and number of trays are greater than the minimum required. A similar problem is encountered in recycle loops if a component accumulates because of the separation specifications that have been set. Adding a purge stream usually solves this problem.

For large problems with multiple recycles, it may be necessary to increase the number of iterations to allow the flowsheet time to converge. This strategy can be effective, but is obviously inefficient if underlying problems in the model are causing the poor convergence.

In some cases, it may be worthwhile to develop a simplified simulation model to arrive at a first estimate of tear stream composition, flow rate, and conditions (temperature and pressure). Models can be simplified by using faster and more robust unit operation models, for example, substituting shortcut column models for rigorous distillation models. Models can also be simplified by reducing the number of components in the model. Reducing the number of components often leads to a good estimate of the bulk flows and stream enthalpies, which can be useful if there are interactions between the mass and energy balances. Another simplification strategy that is often used is to model heat exchangers using a dummy stream on one side (usually the side that is downstream in the process). The recycle of energy from downstream to upstream is then not converged until after the rest of the flowsheet has been converged. Alternatively, heaters and coolers can be used in a simplified model, or even in the rigorous model, as long as the stream data are then extracted and used to design the real exchangers.

Another approach that is widely used is to “creep up on” the converged solution. This entails building up the model starting from a simplified version and successively adding detail while reconverging at each step. As more complexity is added, the values from the previous run are used to initialize the next version. This is a slow but effective method. The design engineer must remember to save the intermediate versions every so often, in case later problems are encountered. A similar strategy is often used when running sensitivity analyses or case studies that require perturbations of a converged model. The designer changes the relevant parameters in small steps to reach the new conditions while reconverging at each step. The results of each step then provide a good initial estimate for the next step, and convergence problems are avoided.

When multiple recycles are present, it is sometimes more effective to solve the model in a simultaneous (equation-oriented) mode rather than in a sequential modular mode. If the simulation program allows simultaneous solution of the equation set, this can be attempted. If the process is known to contain many recycles, the designer should anticipate convergence problems and should select a process simulation program that can be run in a simultaneous mode.

Example 4.10

Light naphtha is a mixture produced by distillation of crude oil. Light naphtha primarily contains alkane compounds (paraffins), and it can be blended into gasoline. The octane value of methyl-substituted alkanes (iso-paraffins) is higher than that of straight-chain compounds (normal paraffins), so it is often advantageous to isomerize the light naphtha to increase the proportion of branched compounds.

A simple naphtha isomerization process has a feed of 10,000 barrels per day (bpd) of a 50 wt% mixture of n-hexane and methyl pentane. The feed is heated and sent to a reactor, where it is brought to equilibrium at 1300 kPa and 250 °C. The reactor

products are cooled to the dew point and fed to a distillation column operated at 300 kPa. The bottoms product of the distillation is rich in n-hexane and is recycled to the reactor feed. An overall conversion of n-hexane of 95% is achieved.

Simulate the process to determine the recycle flow rate and composition.

Solution

This problem was solved using UniSim Design. The first step is to convert the volumetric flow rate into a mass flow rate in metric units. We can set up a stream that has a 50:50 mixture by weight of n-hexane and methyl pentane. This stream has a density of 641 kg/m^3 at 40°C , so the required flow rate is:

$$\begin{aligned} 10,000 \text{ bpd} &= 10,000 \times 641 (\text{kg/m}^3) \times 0.1596 (\text{m}^3/\text{bbl}) / 24 \\ &= 42.627 \text{ metric tons/h} \end{aligned}$$

In a real isomerization process, a part of the feed will be lost due to cracking reactions; however, in our simplified model the only reactions that occur are isomerization reactions. Because we only consider isomerization reactions, all of the product and feed components have the same molecular weight (C_6H_{14} , $M_w = 86$). The feed flow rate of n-hexane is thus $42.627 \times 0.5 = 21.31$ metric tons/h. So for 95% conversion of n-hexane, the amount of n-hexane in the product is $0.05 \times 21.31 = 1.0655$ metric tons/h, or $1065.5 / 86 = 12.39$ kgmol/h. The mole fraction of n-hexane in the product is 5% of 50%, or 2.5 mol%.

To get an initial estimate of the distillation column conditions, the process was first simulated using a shortcut column model, as shown in Fig. 4.41. If we assume that no cyclic compounds are formed in the process, then the component list includes all of the available C_6 paraffin compounds (i.e., n-hexane, 2-methyl pentane, 3-methyl pentane, 2,3-methyl butane, and 2,2-methyl butane). The reactor achieves complete equilibrium between these species and so can be modeled using a Gibbs reactor.

The shortcut column model requires a second specification, given in terms of the heavy key component. We can define either of the methyl pentane species as the heavy key. In the simplified model that we have built, the level of methyl pentane in the recycle is not important to the process performance. Increasing the recycle of methyl pentane species increases the process yield of dimethyl butane species, which would lead to an improvement in the product octane number. In reality, the presence of side reactions that caused cracking to less valuable light hydrocarbons would establish a trade-off that would set the optimum level of methyl pentane recycle. For now, we will assume that the mole fraction of 2-methyl pentane in the bottoms is 0.2.

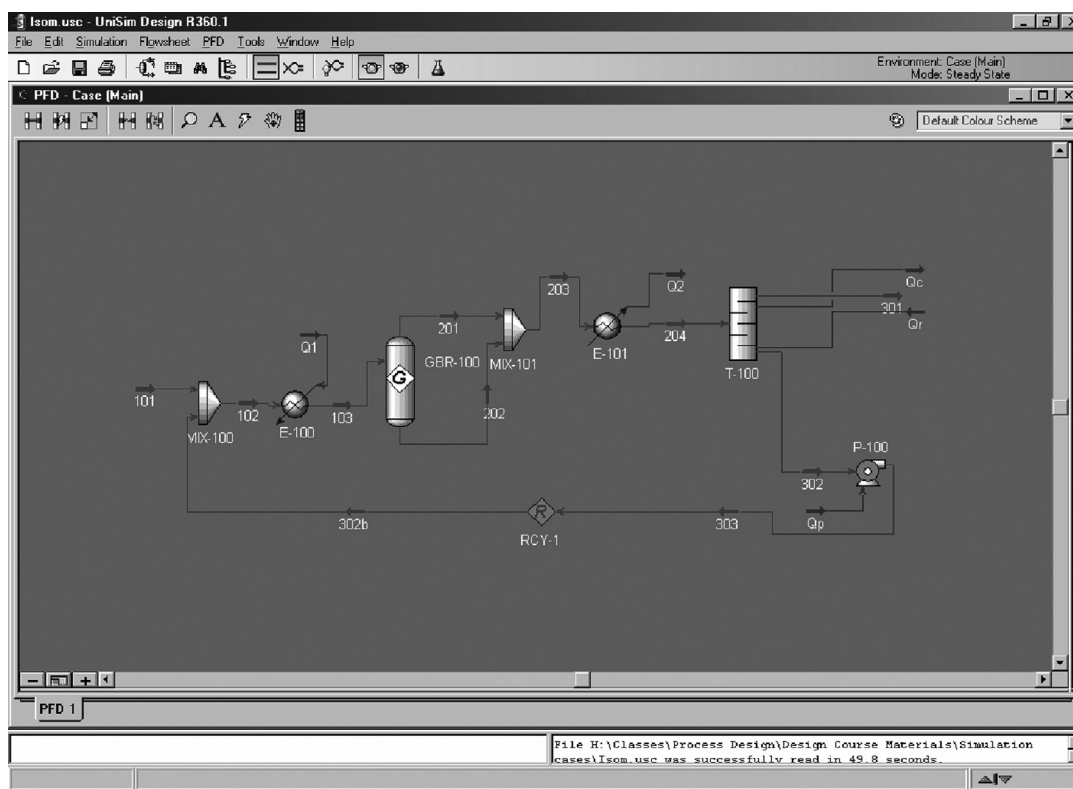


FIG. 4.41 Isomerization process model using shortcut distillation.

T-100

Design

Connections
Parameters
User Variables
Notes

Components

	Component	Mole Fraction
Light Key in Bottoms	2-Mpentane	0.2000
Heavy Key in Distillate	n-Hexane	0.0250

Pressures

Condenser Pressure	300.000 kPa
Reboiler Pressure	300.000 kPa

Reflux Ratios

External Reflux Ratio	4.310
Minimum Reflux Ratio	3.750

Design Rating Worksheet Performance Dynamics

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FIG. 4.42 Shortcut column specifications.

T-100

Performance

Trays

Minimum Number of Trays	17.680
Actual Number of Trays	40.347
Optimal Feed Stage	25.655

Temperatures

Condenser [C]	96.02
Reboiler [C]	104.4

Flows

Rectify Vapour [kgmole/h]	2628.105
Rectify Liquid [kgmole/h]	2133.170
Stripping Vapour [kgmole/h]	2628.033
Stripping Liquid [kgmole/h]	2846.832
Condenser Duty [kJ/h]	66927943.259
Reboiler Duty [kJ/h]	66969425.527

Design Rating Worksheet **Performance** Dynamics

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FIG. 4.43 Shortcut column results.

With these conditions, and with the recycle not closed, the shortcut column model predicts a minimum reflux of 3.75. The reflux ratio is then set at $1.15 \times R_{min} = 4.31$, as shown in Fig. 4.42. The shortcut model then calculates that we need 41 theoretical trays, with optimal feed tray 26, as shown in Fig. 4.43. The column bottoms flow rate is 18,900 kg/h, which can be used as an initial estimate for the recycle flow. The recycle loop can now be closed and run. The converged solution still has $R_{min} = 3.75$, so the reflux ratio does not need to be adjusted. The converged recycle flow rate is 18.85 metric tons/h or 218.7 kgmol/h, as shown in Fig. 4.44. The shortcut column design of the converged flowsheet still has 41 trays with the feed on tray 26.

The results from the shortcut model can now be used to provide a good initial estimate for a rigorous model. The shortcut column is replaced with a rigorous column, as shown in Fig. 4.45. The rigorous column model can be set up with the number

Name	303	302b
Vapour	0.0000	0.0000
Temperature [C]	105.2	105.2
Pressure [kPa]	1300	1300
Molar Flow [kgmole/h]	218.7	218.8
Mass Flow [kg/h]	1.885e+004	1.886e+004
Std Ideal Liq Vol Flow [m3/h]	28.44	28.45
Molar Enthalpy [kJ/kgmole]	-1.844e+005	-1.844e+005
Molar Entropy [kJ/kgmole-C]	84.81	84.81
Heat Flow [kJ/h]	-4.034e+007	-4.036e+007

FIG. 4.44 Converged recycle results for the shortcut column model.

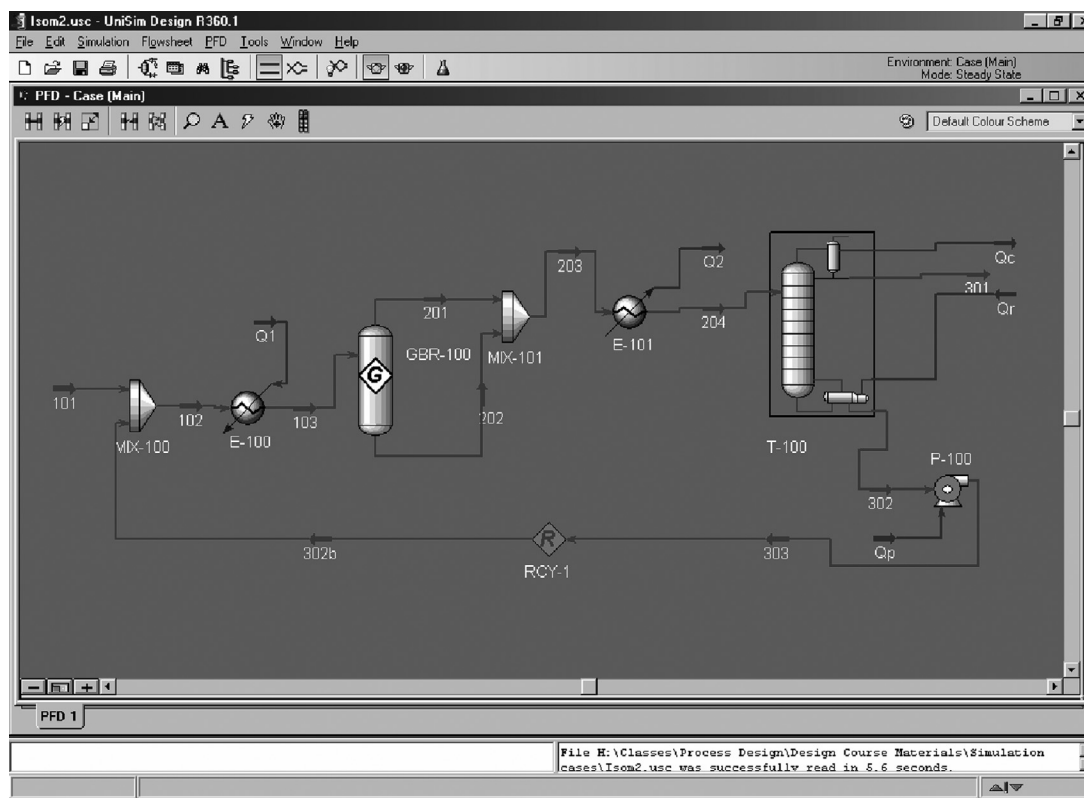


FIG. 4.45 Isomerization process model using rigorous distillation.

of stages and feed stage predicted by the shortcut model (Fig. 4.46). If we specify the reflux ratio and bottoms product rate as column specifications, as in Fig. 4.47, then the flowsheet converges quickly.

The results from the rigorous model with the inputs specified as noted earlier show a flow rate of 1084.5 kg/h of n-hexane in the distillate product. This exceeds the requirements calculated from the problem statement (1065.5 kg/h). The simplest way to get back to the required specification is to use it directly as a specification for the column. From the “Design” tab on the column window we can select “Monitor” and then “Add spec” to add a specification on the distillate flow rate of

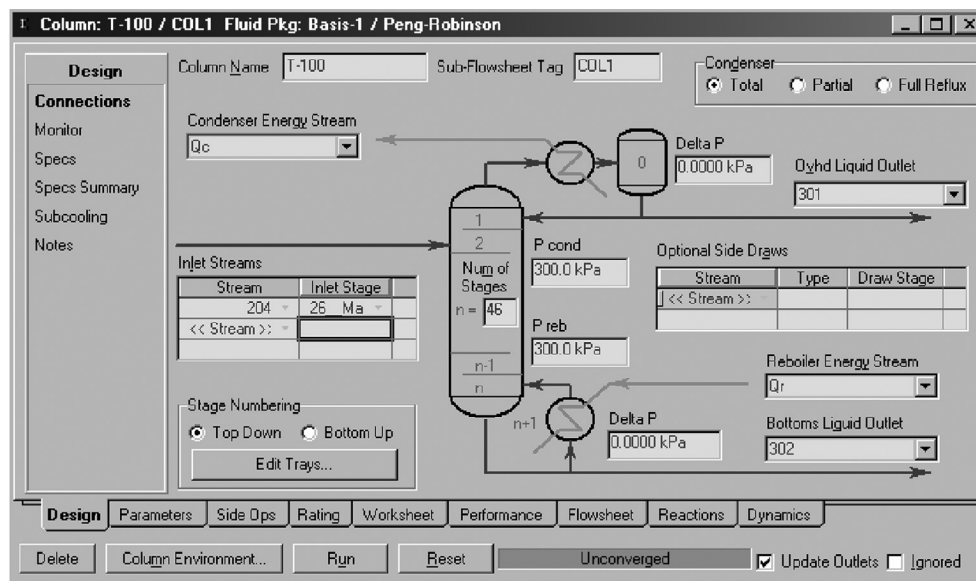


FIG. 4.46 Design parameters for the rigorous distillation column.

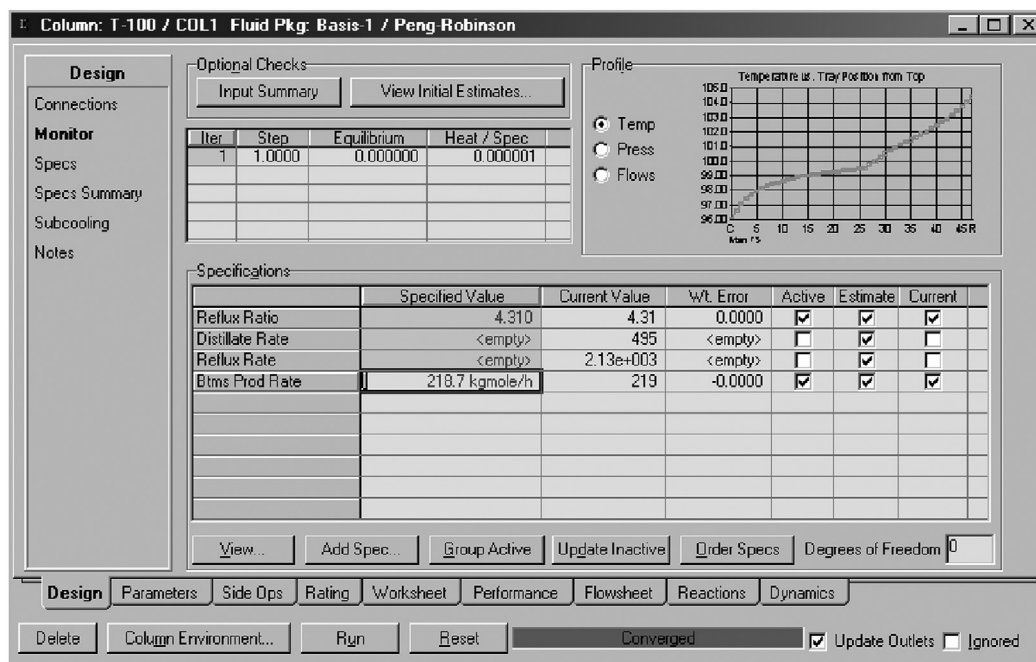


FIG. 4.47 Specifications for the rigorous distillation column.

n-hexane, as shown in Fig. 4.48. This specification can then be made active, and the bottoms flow rate specification can be relaxed. When the simulation is reconverged, the bottoms flow rate increases to 19,350 kg/h and the n-hexane in the distillate meets the specification flow rate of 1065.5 kg/h.

The column profiles for the rigorous distillation model are shown in Fig. 4.49. The profiles do not show any obvious poor design of the column, although the design is not yet optimized.

The simulation was converged to achieve the target conversion of n-hexane with a recycle of 19.35 metric tons/h. The recycle composition is 50.0 mol% n-hexane, 21.1 mol% 2-methyl pentane, 25.1 mol% 3-methyl pentane, 3.6 mol% 2,3-methyl butane, and 0.2 mol% 2,2-methyl butane. This is a converged solution, but it is only one of many possible converged solutions. No attempt has yet been made to optimize the design. The optimization of this process is examined in Problem 4.14. For more realistic information on isomerization process conditions, the reader should consult Meyers (2016).

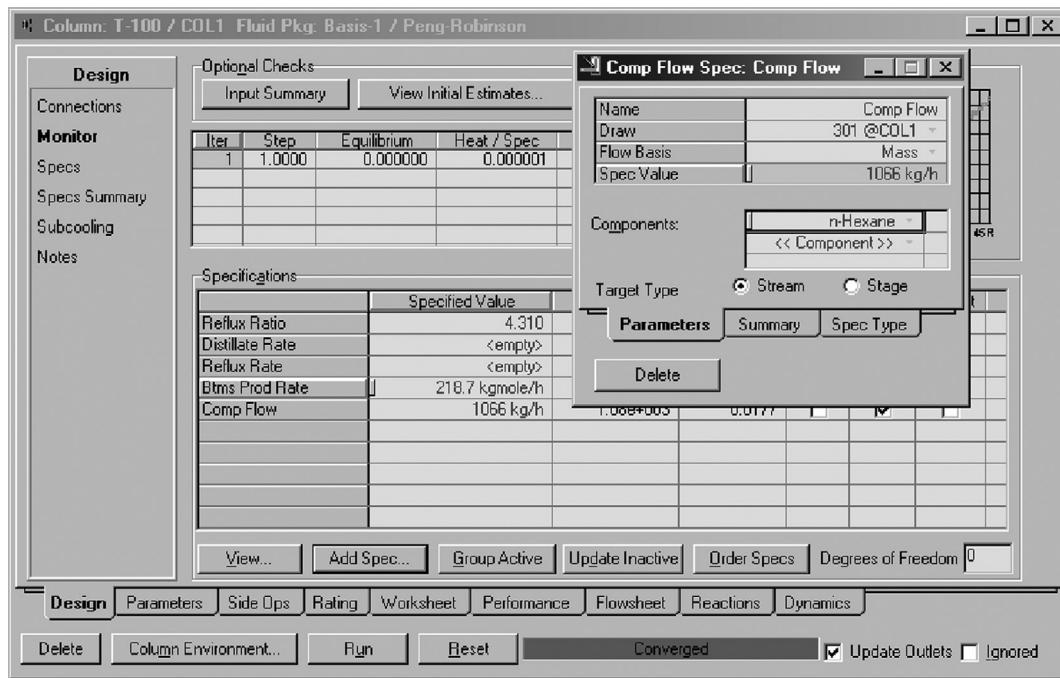


FIG. 4.48 Adding a specification on n-hexane mass flow.

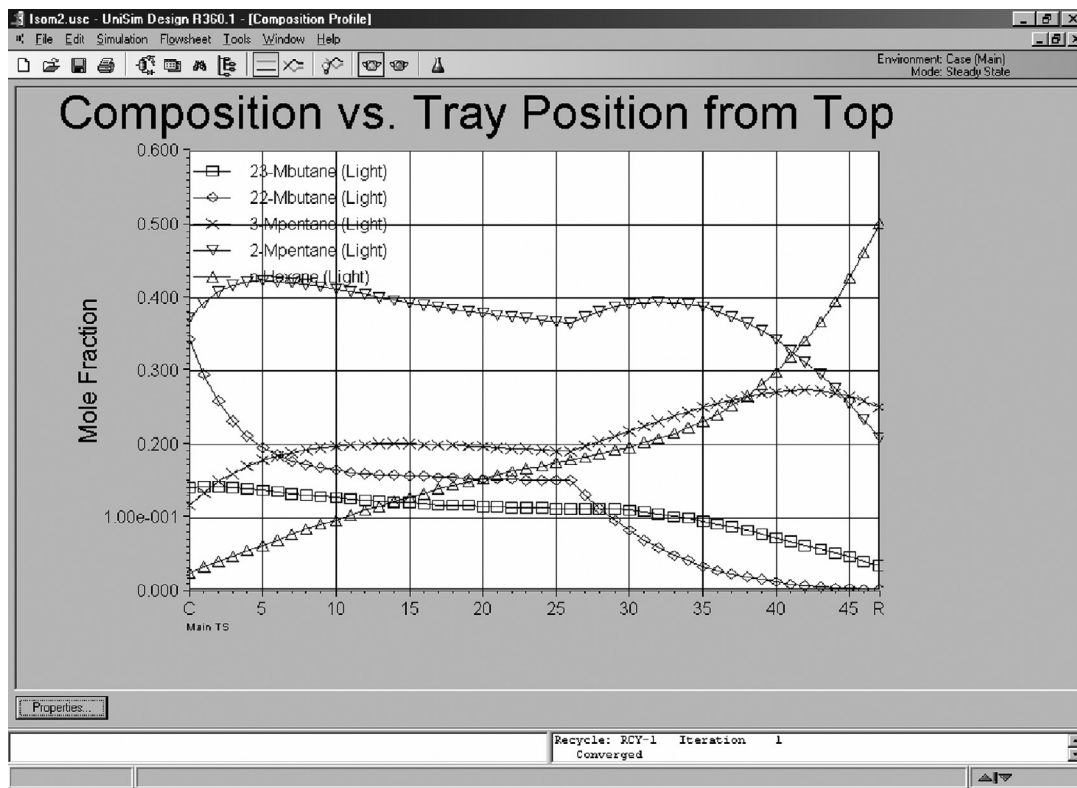


FIG. 4.49 Column profiles for the rigorous distillation model.

4.8 Flowsheet optimization

After achieving a converged simulation of the process, the designer will usually want to carry out some degree of optimization. The commercial simulation programs have a limited optimization capability that can be used with suitable caution.

4.8.1 Use of controllers

The simplest form of optimization is to impose additional constraints on the simulation so that it meets requirements specified by the designer. For example, if the designer made estimates of the feed rates, then the production rate of product that is predicted by the model may be less (or more) than the desired rate. The designer could correct this by calculating the appropriate ratio, multiplying all the feed streams by this ratio, and then reconverging the model, but this approach would soon become tedious.

Instead, the simulation programs allow the designer to impose constraints on the model. In the example noted earlier, this would be a constraint that the product flow rate is equal to a target value. Constraints are imposed using controller functions, known as a “Design Spec” in Aspen Plus or a “Set” or “Adjust” in UniSim Design. Controllers are specified either as:

Set variable x to value z

or

Adjust variable x to value y by manipulating variable z

where z is an unknown variable or set of variables that will be calculated by the simulation, and x is the variable that the designer wants to specify.

Controllers can be used to capture all kinds of design constraints and specifications. They are particularly useful for setting feed ratios and controlling purge rates and recycle ratios to achieve target compositions. Some care is needed to ensure that they are used sparingly; otherwise, too many recycles of information can be introduced and convergence becomes difficult.

Controllers behave much like recycles, and it is usually a good idea to generate a converged simulation to act as a good initial estimate before adding controllers. This does not apply to simple controller functions such as feed ratio controllers.

In a dynamic simulation, controllers are used to model the real control valves of the process. When converting a steady-state simulation to a dynamic simulation, some care is needed to ensure that the controller functions correspond to physically achievable control structures.

4.8.2 Optimization using process simulation software

The commercial process simulation programs all have the ability to solve optimization problems that can be posed as nonlinear programming (NLP) problems. At the time of writing, only Aspen Plus allows the designer to carry out *discrete* optimization using integer variables. It is therefore not possible to optimize integer parameters such as feed tray location while simultaneously optimizing continuous variables in any commercial process simulation program other than Aspen Plus. Likewise, only Aspen Plus can be used to carry out superstructure optimization. Note that this discrete optimization functionality is not included in the general license of Aspen Plus, is only available in versions more recent than release 2006.5, and may not be available to all academic users. The other simulation program vendors are expected to add this capability in future releases.

Optimization of a large process simulation model is intrinsically difficult, particularly if there are multiple recycles. As noted in Section 12.10, the solution algorithms for NLP problems require multiple solutions of the model, which must be converged at each solution.

An additional complication of flowsheet optimization is the formulation of the objective function. The objective function for industrial design is always a measure of economic performance. The design parameters calculated by the simulation program can be used to give relatively good estimates of equipment cost, but this typically requires exporting the parameters into a specialized cost-estimating program, such as Aspen Icarus, as described in Section 7.10. Furthermore, the equipment must usually be oversized by a suitable design factor compared with the design flow rates, as discussed in Section 1.6. The simplest way to address this problem is to generate two or three

simulation runs with variations of the key design parameters. These designs can then be costed to develop approximate cost curves, which can then be used in the optimization tool of the simulation program.

The Aspen Plus manual provides several useful recommendations for specifying optimization problems (Aspen Technology, 2001):

1. Start by converging a simulation of the flowsheet. This helps the designer detect errors, ensures that specifications are feasible, and provides good estimates for tear streams.
2. Carry out a sensitivity analysis to determine which variables have the most impact on the objective function. These are the variables that should be used as decision variables. It is also important to determine reasonable ranges for these variables and set upper and lower bound constraints. If the ranges set are too narrow, then the optimum may not be found. If they are too wide, then convergence may be difficult.
3. While carrying out the sensitivity analysis, see if the optimum is broad or sharp. If there are only small changes in the objective function, further optimization may not be justified.

Another approach that is often used is to carry out optimization using simplified models to fix the process structure and determine the approximate values of key decision variables. A final NLP optimization can then be carried out using a rigorous model.

4.9 Dynamic simulation

Most continuous processes are only simulated in steady-state mode. Some of the simulation programs allow a steady-state simulation to be converted to run in a dynamic mode. Dynamic simulation is useful for:

1. Simulating batch and semicontinuous processes to determine rate-controlling steps and investigate batch-to-batch recycles and heat recovery
2. Simulating process start-up and shut-down
3. Simulating cyclic processes
4. Simulating process disturbances to evaluate control system performance and tune controllers
5. Simulating emergency conditions to evaluate alarm system and relief system responses and ensure that they are adequate

For a good dynamic simulation, the designer must specify the actual control system from the piping and instrumentation diagram (see Chapter 5) and also all of the vessel designs so that hold-ups can be calculated. Mass transfer rates and reaction rates must also be known or assumed.

Dynamic simulation is more computationally intensive than steady-state simulation. Dynamic simulation is usually applied to parts of a process (or even single unit operations) rather than an entire process. Different simulation strategies are needed to give a robust dynamic model. Good introductions to dynamic simulation are given in the books by Luyben (2013), Ingham et al. (2007), Seborg et al. (2016), and Asprey and Machietto (2003) and the paper by Pantelides (1988).

4.10 References

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4.11 Nomenclature

		Dimensions in MLT θ
A	Heat exchanger area	L^2
A	Coefficient in the Antoine equation	—
B	Coefficient in the Antoine equation	θ
C	Coefficient in the Antoine equation	θ
C_p	Specific heat capacity at constant pressure	$L^2 T^{-2} \theta^{-1}$
F	Shell and tube exchanger factor (for noncountercurrent flow)	—
F_M	Make-up gas molar flow rate	MT^{-1}
F_P	Purge gas molar flow rate	MT^{-1}
f_i	Fugacity coefficient for component i	—
f_i^{OL}	Standard state fugacity coefficient of pure liquid	—
G	Molar rate of consumption of gas in reactor	MT^{-1}
K	Equilibrium constant (ratio)	—
L_v	Latent heat of vaporization	$L^2 T^{-2}$
L_{vb}	Latent heat at normal boiling point	$L^2 T^{-2}$
M	Molecular mass (weight)	M
P	Pressure	$ML^{-1} T^{-2}$
P_c	Critical pressure	$ML^{-1} T^{-2}$
p_i^0	Vapor pressure of component i	$ML^{-1} T^{-2}$ or L
P_r	Reduced pressure	—
ΔP_c	Critical constant increment in Lydersen equation (Equation 4.5)	$M^{-1/2} L^{1/2} T$
q	Wegstein method acceleration parameter	—
s	Wegstein method estimate of gradient	—
T	Temperature, absolute scale	θ
T_c	Critical temperature	θ
T_r	Reduced temperature	θ
ΔT_c	Critical constant increment in Lydersen equation (Equation 4.4)	—
U	Overall heat transfer coefficient	$MT^{-3} \theta^{-1}$
ΔV_c	Critical constant increment in Lydersen equation (Equation 4.6)	$M^{-1} L^3$
v_i^L	Liquid molar volume	$M^{-1} L^3$
x	Mol fraction, liquid phase	—
x	Controlled parameter	—
x_k	Estimate of parameter x at k th iteration	—
y	Target value	—
y	Mol fraction, vapor phase	—
y_M	Mole fraction of inerts in make-up	—
y_R	Mole fraction of inerts in recycle and purge	—
z	Target value or unknown variable calculated by simulation program	—
z	Compressibility factor	—
γ	Liquid activity coefficient	—
ϕ	Fugacity coefficient	—
ϕ^s	Fugacity coefficient of pure component	—
Suffixes		
a, b	Components	—
i, j, k		—
1, 2		—
L	Liquid	—
V	Vapor	—

4.12 Problems

4.1. Monochlorobenzene is produced by the reaction of benzene with chlorine. A mixture of monochlorobenzene and dichlorobenzene is produced, with a small amount of trichlorobenzene. Hydrogen chloride is produced as a by-product. Benzene is fed to the reactor in excess to promote the production of monochlorobenzene.

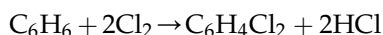
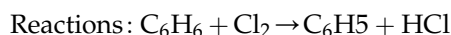
The reactor products are fed to a condenser, where the chlorobenzenes and unreacted benzene are condensed. The condensate is separated from the noncondensable gases in a separator. The noncondensables—hydrogen chloride and unreacted chlorine—pass to an absorption column where the hydrogen chloride is absorbed in water. The chlorine leaving the absorber is recycled to the reactor.

The liquid phase from the separator, containing chlorobenzenes and unreacted benzene, is fed to a distillation column, where the chlorobenzenes are separated from the unreacted benzene. The benzene is recycled to the reactor.

Using the data given here, calculate the stream flows and draw up a preliminary flowsheet for the production of 1.0 ton (metric ton) of monochlorobenzene per day.

Data:

Reactor



mol ratio $\text{Cl}_2 : \text{C}_6\text{H}_6$ at inlet to reactor = 0.9

overall conversion of benzene = 55.3%

yield of monochlorobenzene = 73.6%

yield of dichlorobenzene = 27.3%

Production of other chlorinated compounds can be neglected.

Separator

Assume 0.5% of the liquid stream is entrained with the vapor.

Absorber

Assume 99.99% absorption of hydrogen chloride and that 98% of the chlorine is recycled, with the remainder being dissolved in the water. The water supply to the absorber is set to produce a 30% w/w strength hydrochloric acid.

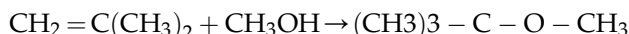
Distillation column

Take the recovery of benzene to be 95% and 99.99% recovery of the chlorobenzenes.

Note:

This problem can be solved without using process simulation software. Start the mass balance at the reactor inlet (after the recycle streams have been added) and assume 100 kgmol/h of benzene at this point.

4.2. Methyl tertiary butyl ether (MTBE) is used as an antiknock additive in gasoline. It is manufactured by the reaction of isobutene with methanol. The reaction is highly selective, and practically any C_4 stream containing isobutene can be used as a feedstock



A 10% excess of methanol is used to suppress side reactions.

In a typical process, the conversion of isobutene in the reactor stage is 97%.

The product is separated from the unreacted methanol and any C_4 compounds by distillation.

The essentially pure, liquid MTBE leaves the base of the distillation column and is sent to storage. The methanol and C_4 compounds leave the top of the column as vapor and pass to a column where the methanol is separated by absorption in water. The C_4 compounds leave the top of the absorption column, saturated with water, and are used as a fuel gas. The methanol is separated from the water solvent by distillation and recycled to the reactor stage. The water, which leaves the base of the column, is recycled to the absorption column. A purge is taken from the water recycle stream to prevent the buildup of impurities.

1. Draw up a block flow diagram for this process.
2. Estimate the feeds for each stage.
3. Draw a flowsheet for the process.

Treat the C_4 compounds other than isobutene as one component.

Data:

1. Feedstock composition, mol%: n-butane = 2, butene-1 = 31, butene-2 = 18, isobutene = 49.
2. Required production rate of MTBE, 7000 kg/h.
3. Reactor conversion of isobutene, 97%.
4. Recovery of MTBE from the distillation column, 99.5%.
5. Recovery of methanol in the absorption column, 99%.
6. Concentration of methanol in the solution leaving the absorption column, 15%.
7. Purge from the water recycle stream to waste treatment, 10% of the flow leaving the methanol recovery column.
8. The gases leave the top of the absorption column saturated with water at 30 °C.
9. Both columns operate at essentially atmospheric pressure.

- 4.3. Ethanol can be produced by fermentation of sugars and is used as a gasoline blending component. Because the sugars can be derived from biomass, ethanol is potentially a renewable fuel. In the fermentation of cane sugar to ethanol, sucrose ($C_{12}H_{22}O_{11}$) is converted by yeast (*Saccharomyces cerevisiae*) to yield ethanol and CO_2 . Some sucrose is also consumed in maintaining the cell culture in the fermentation reactor. The fermentation reaction can be carried out in a continuous reactor as long as the ethanol concentration does not exceed about 8 wt%, at which point the productivity of the yeast declines significantly. The sucrose is fed as a 12.5 wt% solution in water, which must be sterilized before it can be fed to the reactor. The sterilization is usually accomplished by heating with steam. Carbon dioxide is vented from the fermentation reactor. The liquid product of the fermentation reactor is sent to a hydrocyclone to concentrate the yeast for recycle to the reactor. The remaining liquid is sent to a distillation column known as a *beer column*, which concentrates the alcohol to about 40 mol% ethanol and 60 mol% water in the distillate. The recovery of ethanol in the beer column is 99.9%. The bottoms stream from the beer column contains the remaining components of the fermentation broth and can be processed for use as animal feed.

1. Draw a flowsheet for this process.
2. Estimate the stream flow rates and compositions for a production rate of 200,000 U.S. gal/d of dry (100%) ethanol.
3. Estimate the ethanol lost in the CO_2 vent gas.
4. Estimate the reboiler duty of the beer column.

Data:

1. Yield per kg sucrose: ethanol 443.3 g, CO_2 484 g, nonsugar solids 5.3 g, yeast 21 g, fermentation by-products 43.7 g, higher alcohols (fusel oil) 2.6 g.
2. Conversion of sucrose, 98.5%.
3. Yeast concentration in fermentation reactor at steady state, 3 wt%.
4. Fermenter temperature, 38 °C.

- 4.4. In an ethanol plant, the mixture of water and ethanol from the beer column distillate contains about 40% ethanol (molar basis) in water, together with the fusel oils described in the previous problem. This mixture is distilled to give an azeotropic mixture of ethanol and water (89% ethanol) overhead, with 99.9% recovery of ethanol. The fusel oil can cause blending problems if it is allowed to accumulate in the distillate. Fusel oil is a mixture of higher alcohols and ethers that can be approximated as a mixture of n-butanol and diethyl ether. This mixture is usually removed as a side stream from the column. When the side stream is contacted with additional water a two-phase mixture can be formed, and the oil phase can be decanted to leave an ethanol–water phase that is returned to the column.

1. Draw a flowsheet for this process.
 2. Estimate the stream flow rates and compositions for a production rate of 200,000 U.S. gal/d of dry (100%) ethanol.
 3. Optimize the distillation column using the cost correlations given in Section 6.3 and assuming that reboiler heat costs \$5/MMBtu. Minimize the total annualized cost of the column.
- 4.5. Water and ethanol form a low-boiling-point azeotrope; hence water cannot be completely separated from ethanol by conventional distillation. To produce absolute (100%) ethanol, it is necessary to add an entraining agent to break the azeotrope. Benzene is an effective entrainer and is used where the product is not required for food products. Three columns are used in the benzene process.

Column 1. This column separates the ethanol from the water. The bottom product is essentially pure ethanol. The water in the feed is carried overhead as the ternary azeotrope of ethanol, benzene, and water (roughly 24% ethanol, 54% benzene, 22% water). The overhead vapor is condensed and the condensate separated in a decanter into a benzene-rich phase (22% ethanol, 74% benzene, 4% water) and a water-rich phase (35% ethanol, 4% benzene, 61% water). The benzene-rich phase is recycled to the column as reflux. A benzene make-up stream is added to the reflux to make up any loss of benzene from the process. The water-rich phase is fed to the second column.

Column 2. This column recovers the benzene as the ternary azeotrope and recycles it as vapor to join the overhead vapor from the first column. The bottom product from the column is essentially free of benzene (29% ethanol, 51% water). This stream is fed to the third column.

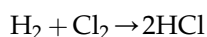
Column 3. In this column, the water is separated and sent to waste treatment. The overhead product consists of the azeotropic mixture of ethanol and water (89% ethanol, 11% water). The overheads are condensed and recycled to join the feed to the first column. The bottom product is essentially free of ethanol.

1. Draw a flowsheet for this process.
2. Estimate the stream flow rates and compositions for a production rate of 200,000 U.S. gal/d of dry (100%) ethanol.
3. Estimate the ethanol lost in the CO₂ vent gas.

Take the benzene losses to total 0.1 kmol/h. All the compositions given are molar percentages.

- 4.6. A plant is required to produce 10,000 tons per year of anhydrous hydrogen chloride from chlorine and hydrogen. The hydrogen source is impure: 90 mol% hydrogen, balance nitrogen. The chlorine is essentially pure chlorine, supplied in rail tankers.

The hydrogen and chlorine are reacted in a burner at 1.5 bar pressure.



Hydrogen is supplied to the burner in 3% excess over the stoichiometric amount. The conversion of chlorine is essentially 100%. The gases leaving the burner are cooled in a heat exchanger. The cooled gases pass to an absorption column, where the hydrogen chloride gas is absorbed in dilute hydrochloric acid. The absorption column is designed to recover 99.5% of the hydrogen chloride in the feed.

The unreacted hydrogen and inerts pass from the absorber to a vent scrubber, where any hydrogen chloride present is neutralized by contact with a dilute, aqueous solution of sodium hydroxide. The solution is recirculated around the scrubber. The concentration of sodium hydroxide is maintained at 5% by taking a purge from the recycle loop and introducing a make-up stream of 25% concentration. The maximum concentration of hydrogen chloride discharged in the gases vented from the scrubber to atmosphere must not exceed 200 ppm (parts per million) by volume.

The strong acid from the absorption column (32% HCl) is fed to a stripping column, where the hydrogen chloride gas is recovered from the solution by distillation. The diluted acid from the base of this column (22% HCl) is recycled to the absorption column.

The gases from the top of the stripping column pass through a partial condenser, where the bulk of the water vapor present is condensed and returned to the column as reflux. The gases leaving the column will be saturated with water vapor at 40 °C.

The hydrogen chloride gas leaving the condenser is dried by contact with concentrated sulfuric acid in a packed column. The acid is recirculated over the packing. The concentration of sulfuric acid is maintained at 70% by taking a purge from the recycle loop and introducing a make-up stream of strong acid (98% H₂SO₄).

The anhydrous hydrogen chloride product is compressed to 5 bar and supplied as a feed to another process.

Using the information provided, calculate the flow rates and compositions of the main process streams, and draw a flowsheet for this process. All compositions are wt%, except where indicated.

- 4.7. Ammonia is synthesized from hydrogen and nitrogen. The synthesis gas is usually produced from hydrocarbons. The most common raw materials are oil or natural gas, though coal and even peat can be used.

When produced from natural gas, the synthesis gas will be impure, containing up to 5% inerts, mainly methane and argon. The reaction equilibrium and rate are favored by high pressure. The conversion is low, about 15%, and so after removal of the ammonia produced, the gas is recycled to the converter inlet. A typical process consists of a converter (reactor) operating at 350 bar, a refrigerated system to condense out the ammonia product from the recycle loop, and compressors to compress the feed and recycle gas. A purge is taken from the recycle loop to keep the inert concentration in the recycle gas at an acceptable level.

Using the data given here, draw a flow diagram of the process and calculate the process stream flow rates and compositions for the production of 600 t/d ammonia.

Data:

Composition of synthesis gas, mol fraction:

N ₂	H ₂	CH ₄	A
24.5	73.5	1.7	0.3

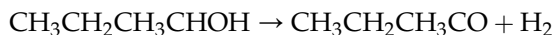
Temperature and operating pressure of liquid ammonia–gas separator, 340 bar and –28 °C.

Inert gas concentration in recycle gas, not greater than 15 mol%.

- 4.8. Methyl ethyl ketone (MEK) is manufactured by the dehydrogenation of 2-butanol.

A simplified description of the process listing the various units used is given here:

1. A reactor in which the butanol is dehydrated to produce MEK and hydrogen, according to the reaction:



The conversion of alcohol is 88%, and the selectivity to MEK can be taken as 100%.

2. A cooler-condenser, in which the reactor off-gases are cooled and most of the MEK and unreacted alcohol are condensed. Two exchangers are used, but they can be modeled as one unit. Of the MEK entering the unit, 84% is condensed, together with 92% of the alcohol. The hydrogen is noncondensable. The condensate is fed forward to the final purification column.
3. An absorption column in which the uncondensed MEK and alcohol are absorbed in water. Around 98% of the MEK and alcohol can be considered to be absorbed in this unit, giving a 10 wt% solution of MEK. The water feed to the absorber is recycled from the next unit, the extractor. The vent stream from the absorber, containing mainly hydrogen, is sent to a flare stack.
4. An extraction column in which the MEK and alcohol in the solution from the absorber are extracted into trichloroethane (TCE). The raffinate, water containing around 0.5 wt% MEK, is recycled to the absorption column.

The extract, which contains around 20 wt% MEK, and a small amount of butanol and water, is fed to a distillation column.

5. A distillation column that separates the MEK and alcohol from the solvent TCE. The recovery of MEK is 99.99%.

The solvent containing a trace of MEK and water is recycled to the extraction column.

6. A second distillation column that produces a 99.9% pure MEK product from the crude product from the first column. The residue from this column, which contains the bulk of the unreacted 2-butanol, is recycled to the reactor.

For a production rate of 1250 kg/h MEK:

1. Draw a flowsheet for the process.
 2. Estimate the stream flow rates and compositions.
 3. Estimate the reboiler and condenser duties of the two distillation columns.
 4. Estimate the number of theoretical trays required in each column.
- 4.9. In the problem of Example 4.4, the feed was specified as pentane (C_5H_{12}) with a hydrogen-to-carbon ratio of 2.4:1. If the feed to the process were a heavy oil, the hydrogen-to-carbon ratio would be more like 2:1. How would the distribution of C_5 carbon compounds change if the feed had a 2:1 carbon ratio?
- 4.10. Example 4.4 examined the equilibrium distribution of hydrocarbon compounds within a single carbon number (C_5). In reality, cracking reactions to ethylene, propylene, and other light alkenes and alkynes will have a significant effect on the yield of a cracking process.
1. What is the effect of including C_2 and C_3 compounds on the equilibrium distribution?
 2. What is the effect of including coke (carbon) as well as the C_2 and C_3 compounds?
 3. What do these results tell you about cracking processes?
- 4.11. Optimize the heat exchanger design of Example 4.8 to minimize the total surface area required.
- 4.12. A stream containing 4 metric tons/h of a 20 wt% mixture of benzene in toluene is heated from 20 °C to the bubble point at 4 atm pressure. The mixture is separated in a distillation column to give 99.9% recovery of benzene overhead and toluene in the bottoms.
1. If the toluene product must be cooled to 20 °C, how much of the feed heat can be supplied by heat exchange with the bottoms?
 2. How many heat exchange shells are needed?
 3. What is the minimum total heat exchange area?
 4. What is the distillation column diameter?
 5. How many sieve trays are needed if the tray efficiency is 70%?
- 4.13. The autothermal reforming of methane to hydrogen was described in Example 4.5. The solution in the example was not optimized, and suggestions were given for how to improve the results. Optimize the process to minimize the cost of production of hydrogen, assuming:
1. Cost of methane = 16 ¢/lb
 2. Cost of oxygen = 2 ¢/lb
 3. Cost of water = 25 ¢/1000 lb
 4. Annualized cost of heat exchangers = \$ 30,000 + 3 A , where A is the area in ft^2
 5. Cost of electric power = 6 ¢/kWh
 6. Reactor and catalyst costs are the same in all cases.
- Hint: First determine the optimal heat recovery and steam and oxygen to methane ratios for a given methane conversion. Repeat for different methane conversions to find the overall optimum.
- 4.14. The light naphtha isomerization process is more complex than the description given in Example 4.10.
1. Hydrogen is flowed through the plant to reduce catalyst deactivation. The hydrogen flow rate is typically about 2 moles per mole of hydrocarbon on a pure hydrogen basis. The hydrogen make-up gas is typically about 90 mol% hydrogen, with the balance comprising methane.
 2. Light hydrocarbon compounds are formed by cracking reactions. These compounds accumulate in the hydrogen recycle and are controlled by taking a purge stream. A stabilizer column is also required, upstream of the distillation column, to remove light hydrocarbons and hydrogen before the distillation.
 3. Each of the C_6 isomers has a different blending octane value. The total octane value of the product can be found by summing the products of the mole fraction of each component and the component blending value. The blending values are n-hexane 60; 2-methyl pentane 78.5; 3-methyl pentane 79.5; 2,2-dimethyl butane 86.3; and 2,3-dimethyl butane 93.

Optimize the design of Example 4.10, subject to the following:

1. The selectivity loss due to cracking reactions can be approximated as 1% conversion of C_6 compounds to propane per reactor pass.
2. The wholesale value of gasoline can be assumed to be $2.0 + 0.05 (\text{octane number} - 87)$ \$/US gal.
3. The cost of hydrogen is \$6/1000 scf, and the fuel value of the hydrogen and propane purge stream is \$5/MMBtu.
4. The reactor plus catalyst total installed cost can be taken as \$0.5 MM per 1000 bpd of liquids processed.
5. Other costs can be estimated using the cost correlations given in Table 7.2.

- 4.15** The composition of the feed to a debutanizer is given here. The column will operate at 14 bar and below 750 K. The process is to be modeled using a commercial simulation program. Suggest a suitable phase equilibrium method to use in the simulation.

Feed composition:

		kg/h
Propane	C3	910
Isobutane	i-C4	180
n-butane	n-C4	270
Isopentane	i-C5	70
normal pentane	n-C5	90
normal hexane	n-C6	20

- 4.16.** The product specifications for Problem 4.15 are 99.5% recovery of C_5+ in the bottoms and 99% recovery of C_4 and lighter compounds in the distillate. Your company's corporate engineering standard is to design for a reflux ratio of 1.15 times the minimum reflux and to assume a stage efficiency of 60%. Using rigorous simulation, determine the number of trays, feed tray, and reboiler duty for this column.
- 4.17.** In the manufacture of methyl ethyl ketone from butanol, the product is separated from unreacted butanol by distillation. The feed to the column consists of a mixture of methyl ethyl ketone, 2-butanol, and trichloroethane. What would be a suitable phase equilibrium correlation to use in modeling this process?

Additional flowsheeting problems are given in the form of design projects in Appendices E and F. Additional reaction and distillation simulation problems are given in [Chapters 15 and 17](#), respectively.

Instrumentation and process control

KEY LEARNING OBJECTIVES

- How to read a piping and instrument diagram drawn using ANSI/ISA 5.1 symbols
- How to design control schemes for common unit operations and whole processes

5.1 Introduction

The process flowsheet shows the arrangement of the major pieces of equipment and their interconnection. It is a description of the nature of the process.

The piping and instrument diagram (P&I diagram or PID) shows the engineering details of the equipment, instruments, piping, valves and fittings, and their arrangement. It is often called the *engineering flowsheet* or *engineering line diagram*. This chapter covers the preparation of the preliminary P&I diagrams at the process design stage of the project.

Some process control information is also indicated on the process flow diagram (PFD). It is common practice to show control valves on the PFD but to omit isolation valves, relief valves, and instrumentation details. Control valves require a significant pressure drop to operate effectively, so the location of control valves will often indicate a need for additional pumps or compressors. In some cases, process control considerations may even lead to the addition of vessels to the flowsheet; for instance, when a surge tank is added to smooth out operation between batch and continuous sections of the plant.

The design of piping systems, and the specification of the process instrumentation and control systems, is usually done by specialist design groups, and a detailed discussion of control systems is beyond the scope of this book. Only general guide rules are given. The piping handbook edited by [Nayyar \(2000\)](#) and the process automation handbook by [Love \(2007\)](#) are particularly recommended for guidance on the detailed design of piping systems and process instrumentation and control. The references cited in the text and listed at the end of the chapter should also be consulted. The detailed design of piping systems, valves, and plant hydraulics and the sizing and selection of control valves are discussed in more detail in [Chapter 20](#).

5.2 The P&I diagram

The P&I diagram shows the arrangement of the process equipment, piping, pumps, instruments, valves, and other fittings. It should include:

1. All process equipment identified by an equipment number. The equipment should be drawn roughly in proportion, and the location of nozzles shown.

2. All pipes identified by a line number. The pipe size and material of construction should be shown. The material may be included as part of the line identification number.
3. All valves—control and block valves—with an identification number. The type and size should be shown. The type may be shown by the symbol used for the valve or included in the code used for the valve number.
4. Ancillary fittings that are part of the piping system, such as inline sight-glasses, strainers, and steam traps, with an identification number.
5. Pumps identified by a suitable code number.
6. All control loops and instruments with an identification number.

For simple processes, the utility (service) lines can be shown on the P&I diagram. For complex processes, separate diagrams should be used to show the service lines so the information can be shown clearly, without cluttering up the diagram. The service connections to each unit should, however, be shown on the P&I diagram.

The P&I diagram will resemble the process flowsheet, but the process information is not shown. The same equipment identification numbers should be used on both diagrams.

5.2.1 Symbols and layout

The symbols used to show the equipment, valves, instruments, and control loops will depend on the practice of the particular design office. The equipment symbols are usually more detailed than those used for the process flowsheet. A typical example of a P&I diagram is shown in Fig. 5.22 at the end of this chapter.

The most widely used international standard symbols for instruments, controllers, and valves are those given by the Instrumentation Systems and Automation Society design code ANSI/ISA 5.1-2009. Some companies use their own symbols, though, and different standards are followed in some countries, such as BS 1646 in the United Kingdom and DIN 19227 and DIN 2429 (now both superseded by DIN 28000-4) in Germany.

When laying out the diagram, it is only necessary to show the relative elevation of the process connections to the equipment where these affect the process operation; for example, the net positive suction head (NPSH) of pumps, barometric legs, siphons, and the operation of thermosiphon reboilers. Full details of pipe layout are usually shown in a different drawing, known as a *piping isometric drawing*. See Fig. 20.21 for an example.

Computer-aided drafting programs are available for the preparation of P&I diagrams. Microsoft Visio Professional edition contains a library of P&I diagram symbols.

5.2.2 Basic symbols

The symbols illustrated in Figs. 5.1 to 5.7 are those given in ANSI/ISA 5.1-2009.

Control valves

Different types of valves are shown in Fig. 5.1 and discussed in Section 20.5.

Actuators

Actuator symbols are illustrated in Fig. 5.2.

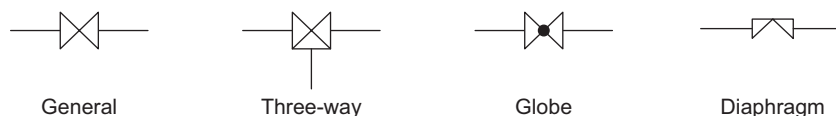


FIG. 5.1 Control valves.

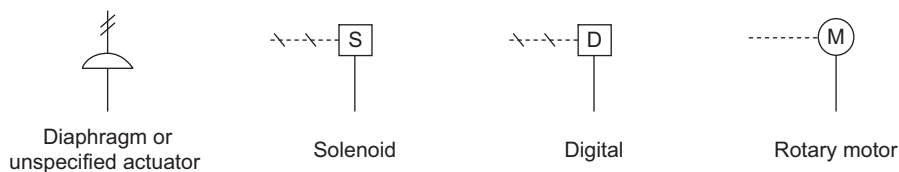
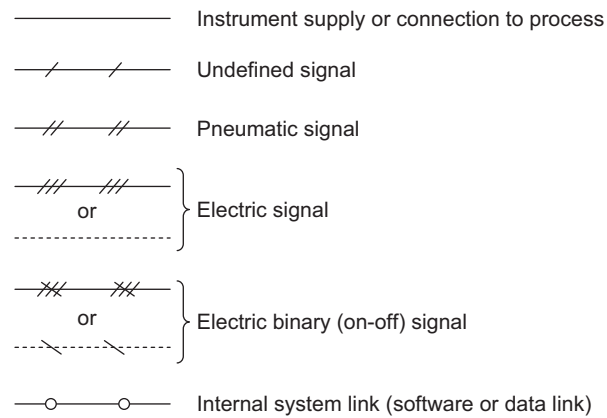


FIG. 5.2 Actuators.



All lines should be drawn fine in relation to process piping lines

FIG. 5.3 Instrument lines.

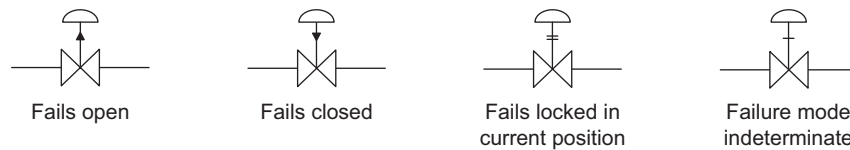


FIG. 5.4 Valve failure modes.

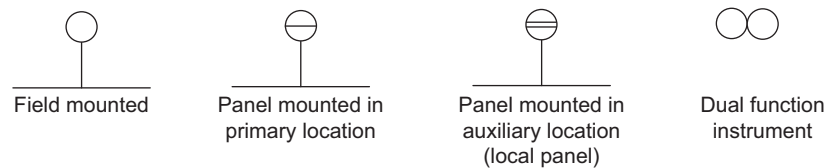


FIG. 5.5 General instrument and controller symbols.

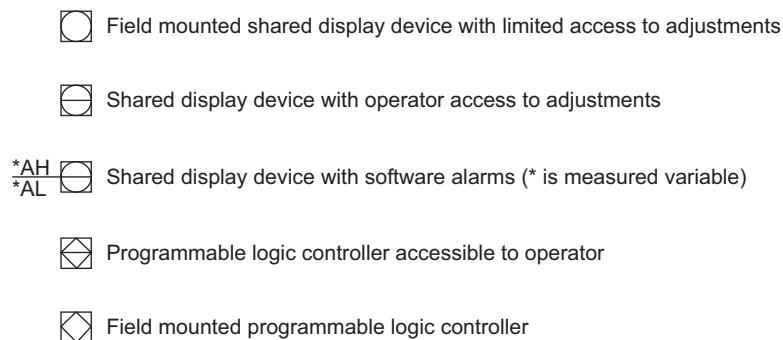


FIG. 5.6 Shared display symbols for distributed control and logic control.